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Software engineering (Pokhara University)

A Manual of

Social and Professional Issues In Information Technology

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Unit 1

History of Computing

Prehistory – the world before 1946

The word "Computing"

Originally, the word computing was synonymous with counting and calculating, and a computer was a person who computes. Since the advent of the electronic computer, it has come to also mean the operation and usage of these machines, the electrical processes carried out within the computer hardware itself, and the theoretical concepts governing them.

Prehistory: Computing related events

750 BC - 1799 A.D.

750 B.C. The abacus was first used by the Babylonians as an aid to simple arithmetic at sometime around this date.

1492 Leonardo da Vinci produced drawings of a device consisting of interlocking cog wheels which could be interpreted as a mechanical calculator capable of addition and subtraction. A working model inspired by this plan was built in 1968 but it remains controversial whether Leonardo really had a calculator in mind

1588 Logarithms are discovered by Joost Buerghi

1614 Scotsman John Napier invents an ingenious system of moveable rods (referred to as Napier's Rods or Napier's bones). These were based on logarithms and allowed the operator to multiply, divide and calculate square and cube roots by moving the rods around and placing them in specially constructed boards.

1622 William Oughtred developed slide rules based on John Napier's logarithms

1623 Wilhelm Schickard of Tübingen, Württemberg (now in Germany), built the first discrete automatic calculator, and thus essentially started the computer era. His device was called the "Calculating Clock". This mechanical machine was capable of adding and subtracting up to 6 digit numbers, and warned of an overflow by ringing a bell. Operations were carried out by wheels, and a complete revolution of the units wheel incremented the tens wheel in much the sameway counters on old cassette decks worked. Schickard was a friend of the astronomer Johannes Kepler since they met in the winter of 1617. Kepler used Schickard's machine for his astronomical studies. The machine and plans were lost and forgotten in the war that was going on, then rediscovered in 1935, only to be lost in war again, and then finally rediscovered in

1956 By the same man (Franz Hammer)! The machine was reconstructed in 1960, and found to be workable.

1642 French mathematician, Blaise Pascal built a mechanical adding machine (the "Pascaline"). Despite being more limited than Schickard's 'Calculating Clock' (see 1623), Pascal's machine became far more well known. He built around fifty, but was only able to sell around a dozen of his machines in various forms, coping with up to 8 digits.

1668 Sir Samuel Morland (1625-1695), of England, produces a non decimal adding machine, suitable for use with English money. Instead of a carry mechanism, it registers carries on auxiliary dials, from which the user must re-enter them as addends.

1671 German mathematician, Gottfried Leibniz designed a machine to carry out multiplication, the 'Stepped Reckoner'. It could multiply numbers of up to 5 and 12 digits to give a 16 digit result. The machine was later lost in an attic until 1879. Leibniz most important contribution to the computing era, however, was the binary number system which is used in all modern machines. He also co-invented calculus.

1726 (1735 ?) Johnathan Swift describes (satirically) a machine ("engine") in his Gulliver's Travels. The "engine" consists of a wooden frame with wooden blocks containing parts of speech. When the engine's 40 levers are simultaneously turned, the machine displays grammatical sentence fragments.

1774 Philip Matthaeus Hahn, somewhere in what will be Germany, also makes a successful multiplying calculator.

1775 Charles Stanhope, 3rd Earl Stanhope, of England, makes a successful multiplying calculator similar to Leibniz's.

1784 Johann H. Müller, of the Hessian army, conceives the idea of what came to be called a "difference engine". That's a special-purpose calculator for tabulating values of a polynomial, given the differences between certain values so that the polynomial is uniquely specified; it's useful for any function that can be approximated by a polynomial over suitable intervals. Müller's attempt to raise funds fails and the project is forgotten.

1800 – 1899 A.D.

1801 Joseph-Marie Jacquard developed an automatic loom controlled by punched cards.

1820 Charles Xavier Thomas de Colmar of France, makes his "Arithmometer", the first mass-produced calculator. It does multiplication using the same general approach as Leibniz's calculator; with assistance from the user it can also do division. It is also the most reliable calculator yet. Machines of this general design, large enough to occupy most of a desktop, continue to be sold for about 90 years.

1822 Charles Babbage designed his first mechanical computer, the first prototype of the decimal difference engine for tabulating polynomials.

1832 Babbage and Joseph Clement produce a prototype segment of his difference engine, which operates on 6-digit numbers and 2nd-order differences (i.e. can tabulate quadratic polynomials). The complete engine, which would be room-sized, is planned to be able to operate both on 6th-order differences with numbers of about 20 digits, and on 3rd-order differences with numbers of 30 digits. Each addition would be done in two phases, the second one taking care of any carries generated in the first. The output digits would be punched into a soft metal plate, from which a plate for a printing press could be made. But there are various difficulties, and no more than this prototype piece is ever assembled.

1834 Babbage conceives, and begins to design, his decimal "Analytical Engine". The program was stored on read-only memory, specifically in the form of punch cards. Babbage continues to work on the design for years, though after about 1840 the changes are minor. The machine would operate on 40-digit numbers; the "mill" (CPU) would have 2 main accumulators and some auxiliary ones for specific purposes, while the "store" (memory) would hold 1000 50-digit numbers. There would be several punch card readers, for both programs and data; the cards would be chained and the motion of each chain could be reversed. The machine would be able to perform conditional jumps. There would also be a form of microcoding: the meaning of instructions would depend on the positioning of metal studs in a slotted barrel, called the "control barrel". The machine would do an addition in 3 seconds and a multiplication or division in 2-4 minutes. It was to be powered by a steam engine.

1835 Joseph Henry invents the electromechanical relay.

1842 Babbage's difference engine project is officially cancelled. (The cost overruns have been considerable, and Babbage is spending too much time on redesigning the Analytical Engine.)

1843 Scheutz and his son Edvard Scheutz produce a 3rd-order difference engine with printer, and the Swedish government agrees to fund their next development.

1847 Babbage designs an improved, simpler difference engine (the Difference Engine No.2), a project which took 2 years. The machine could operate on 7th-order differences and 31-digit numbers, but nobody is interested in paying to have it built.(In 1989-91, however, a team at London's Science Museum did just that. They used components of modern construction, but with tolerances no better than Clement could have provided... and, after a bit of tinkering and detail-debugging, they found that the machine does indeed work. In 2000, the printer has also been completed.)

1848 British Mathematician George Boole devised binary algebra (Boolean algebra) paving the way for the development of a binary computer almost a century later. See 1939

1853 To Babbage's delight, the Scheutzes complete the first full-scale difference engine, which they call a Tabulating Machine. It operates on 15-digit numbers and 4th-order differences, and produces printed output as Babbage's would have. A second machine is later built to the same design by the firm of Brian Donkin of London.

1858 The first Tabulating Machine (see 1853) is bought by the Dudley Observatory in Albany, New York, and the second one by the British government. The Albany machine is used to produce a set of astronomical tables; but the observatory's director is then fired for this extravagant purchase, and the machine is never seriously used again, eventually ending up in a museum. The second machine, however, has a long and useful life.

1869 The first practical logic machine is built by William Stanley Jevons.

1871 Babbage produces a prototype section of the Analytical Engine's mill and printer.

1875 Martin Wiberg produces a reworked difference engine-like machine for preparing logarithmic tables.

1878 Ramon Verea, living in New York City, invents a calculator with an internal multiplication table; this is much faster than the shifting carriage or other digital methods. He isn't interested in putting it into production; he just wants to show that a Spaniard can invent as well as an American.

1879 A committee investigates the feasibility of completing the Analytical Engine and concludes that it is impossible now that Babbage is dead. The project is then largely forgotten, though Howard Aiken is a notable exception.

1884 Dorr E. Felt (1862-1930), of Chicago, makes his "Comptometer". This is the first calculator where the operands are entered merely by pressing keys rather than having to be, for example, dialled in. It is feasible because of Felt's invention of a carry mechanism fast enough to act while the keys return from being pressed.

1885 A multiplying calculator more compact than the Arithmometer enters mass production. The design is the independent, and more or less simultaneous, invention of Frank S. Baldwin, of the United States, and T. Odhner, a Swede living in Russia. The fluted drums are replaced by a "variable-toothed gear" design: a disk with radial pegs that can be made to protrude or retract from it.

1886 Herman Hollerith uses his tabulating system in the Baltimore Department of Health.

1889 Dorr E. Felt invents the first printing desk calculator.

1890 The 1880 census had taken 7 years to complete since all processing had been done by hand off of journal sheets. The increasing population suggested that by the 1890 census the data processing would take longer than the 10 years before the next census - so a competition was held to try to find a better method. This was won by a Census Department employee, Herman Hollerith - who went on to found the Tabulating Machine Company, later to become IBM. Herman used Babbage's idea of using the punched cards from the textile industry for the data storage. His machines used mechanical relays (solenoids) to increment mechanical counters. This method was used in the 1890 census, the result (62,622,250 people) was released in just 6 weeks! This storage allowed much more in-depth analysis of the data and so, despite being more efficient, the 1890 census cost about double (actually 198%) that of the 1880 census. The inspiration for this invention was Hollerith's observation of railroad conductors during a trip in the western US; they encoded a crude description of the passenger (tall, bald, male) in the way they punched the ticket.

1892 William S. Burroughs of St. Louis, invents a machine similar to Felt's (see 1886) but more robust, and this is the one that really starts the mechanical office calculator industry.

1895 "Everything that needed to be invented is now invented.", Lord Kelvin, Royal Society of the UK.

1900 – 1939 A.D.

1906 Henry Babbage, Charles's son, with the help of the firm of R. W. Munro, completes the mill of his father's Analytical Engine, just to show that it would have worked. It does. The complete machine is never produced.

Electronic Tube (or Electronic Valve) developed by Lee De Forest in U.S.A.. Before this it would have been impossible to make digital electronic computers.

1919 W. H. Eccles and F. W. Jordan publish the first flip-flop circuit design.

1924 Walther Bothe builds the first AND logic gate - the coincidence circuit, for use in physics experiments, for which he receives the Nobel Prize in Physics 1954. CPU design will eventually make heavy use of logic gates, 40 years later.

1930 Vannevar Bush builds a partly electronic Difference Engine capable of solving differential equations

1931 Kurt Gödel of Vienna University, Austria, publishes a paper on a universal formal language based on arithmetic operations. He uses it to encode arbitrary formal statements and proofs, and shows that formal systems such as traditional mathematics are either inconsistent in a certain sense or contain unprovable but true statements. This result is often called the fundamental result of theoretical computer science.

E. Wynn-Williams, at Cambridge, England, uses thyratron tubes to construct a binary digital counter for use in connection with physics experiments.

1936 Alan Turing of Cambridge University, England, publishes a paper on "computable numbers" which reformulates Kurt Gödel's results (see related work by Alonzo Church). His paper addresses the famous `Entscheidungsproblem' whose solution is achieved by reasoning (as a mathematical device) about the theoretical simplified universal computer known today as a Turing machine, which in many ways is more convenient than Goedel's arithmetics-based universal formal system.

1937 George Stibitz of the Bell Telephone Laboratories (Bell Labs), New York City, constructs a demonstration 1-bit binary adder using relays. This is one of the first binary computers, although at this stage it was only a demonstration machine improvements continued leading to the 'complex number calculator' of Jan. 1940.

Claude E. Shannon publishes a paper on the implementation of symbolic logic using relays.

1938 Konrad Zuse of Berlin, completes the "Z1", the first mechanical binary programmable computer. It is based on Boolean Algebra and has most of the basic ingredients of modern machines, using the binary system and today's standard separation of storage and control. Zuse's 1936 patent application (Z23139/GMD Nr. 005/021) also suggests a `von Neumann' architecture (re-invented in 1945) with program and data modifiable in storage. Originally the machine was called the "V1" but retroactively renamed after the war, to avoid confusion with the V1 missile. It works with floating point numbers having a 7-bit exponent, 16-bit mantissa, and a sign bit. The memory uses sliding metal parts to store 16 such numbers, and works well; but the arithmetic unit is less successful, occasionally suffering from certain mechanical engineering problems. The program is read from punched discarded 35 mm movie film. Data values can be entered from a numeric keyboard, and outputs are displayed on electric lamps. The machine is not a general purpose computer because it lacks looping capabilities.

1939 John Vincent Atanasoff and graduate student Clifford Berry of Iowa State College (now the Iowa State University), Ames, Iowa, complete a prototype 16-bit adder. This is the first machine to calculate using vacuum tubes.

Konrad Zuse completed the "Z2" (originally "V2"), which combined the Z1's existing mechanical memory unit to a new arithmetic unit using relay logic. Like the Z1, the Z2 lacks looping capabilities. The project is interrupted for a year when Zuse is drafted, but then released.

Helmut Schreyer completes a prototype 10-bit adder using vacuum tubes, and a prototype memory using neon lamps.

1940 – 1949 A.D.

Jan, 1940 At Bell Labs, Samuel Williams and George Stibitz complete a calculator which can operate on complex numbers, and give it the imaginative name of the "Complex Number Calculator"; it is later known as the "Model I Relay Calculator". It uses telephone switching parts for logic: 450 relays and 10 crossbar switches. Numbers are represented in "plus 3 BCD"; that is, for each decimal digit, 0 is represented by binary 0011, 1 by 0100, and so on up to 1100 for 9; this scheme requires fewer relays than straight BCD. Rather than requiring users to come to the machine to use it, the calculator is provided with three remote keyboards, at various places in the building, in the form of teletypes. Only one can be used at a time, and the output is automatically displayed on the same one. On 9th September 1940, a teletype is set up at a Dartmouth College in Hanover, New Hampshire, with a connection to New York, and those attending the conference can use the machine remotely.

Apr 1, 1940 Konrad Zuse founds the world's first computer startup company: the Zuse Apparatebau in Berlin.

12 May 1941 Now working with limited backing from the DVL (German Aeronautical Research Institute), Konrad Zuse completes the "Z3" (originally "V3"): the first operational programmable computer. One major improvement over Charles Babbage's non-functional device is the use of Leibniz's binary system (Babbage and others unsuccessfully tried to build decimal programmable computers). Zuse's machine also features floating point numbers with a 7-bit exponent, 14-bit mantissa (with a "1" bit automatically prefixed unless the number is 0), and a sign bit. The memory holds 64 of these words and therefore requires over 1400 relays; there are 1200 more in the arithmetic and control units. It also featured parallel adders. The program, input, and output are implemented as described above for the Z1. Although conditional jumps are not available, it was shown that Zuse's Z3 is indeed a universal computer. The machine can do 3-4 additions per second, and takes 3-5 seconds for a multiplication. Its rather modern, programmable, binary design makes it the forerunner of today's computers (several later well-known machines such as ENIAC still used the decimal system).

Summer, 1942 Atanasoff and Berry complete a special-purpose calculator for solving systems of simultaneous linear equations, later called the "ABC" ("Atanasoff Berry Computer"). This has 60 50-bit words of memory in the form of capacitors (with refresh circuits -- the first regenerative memory) mounted on two revolving drums. The clock speed is 60 Hz, and an addition takes 1 second. For secondary memory it uses punch cards, moved around by the user. The holes are not actually punched in the cards, but burned. The punch card system's error rate is never reduced beyond 0.001%, and this isn't really good enough. (Atanasoff will leave Iowa State after the US enters the war, and this will end his work on digital computing machines.)

1942 Konrad Zuse develops the S1, the world's first process computer, used by Henschel to measure the surface of wings.

Apr, 1943 Max Newman, Wynn-Williams, Alan Turing and their team at the secret Government Code and Cypher School ('Station X'), Bletchley Park, Bletchley, England, complete the "Heath Robinson". This is a specialized machine for cipher-breaking, not a general-purpose calculator or computer but some sort of logic device, using a combination of electronics and relay logic. It reads data optically at 2000 characters per second from 2 closed loops of paper tape, each typically about 1000 characters long. It was significant since it was the fore-runner of Colossus. Newman knew Turing from Cambridge (Turing was a student of Newman's.), and had been the first person to see a draft of Turing's 1937 paper. Heath Robinson is the name of a British cartoonist known for drawings of comical machines, like the American Rube Goldberg Two later machines in the series will be named after

London stores with "Robinson" in their names.

Sep, 1943 Williams and Stibitz complete the "Relay Interpolator", later called the "Model II Relay Calculator". This is a programmable calculator; again, the program and data are read from paper tapes. An innovative feature is that, for greater reliability, numbers are represented in a biquinary format using 7 relays for each digit, of which exactly 2 should be "on": 01 00001 for 0, 01 00010 for 1, and so on up to 10 10000 for 9. Some of the later machines in this series will use the biquinary notation for the digits of floating-point numbers.)

Dec, 1943 The Colossus was built, by Dr Thomas Flowers at The Post Office Research Laboratories in London, to crack the German Lorenz (SZ42) cipher. It contained 2400 Vacuum tubes for logic and applied a programmable logical function to a stream of input characters, read from punched tape at a rate of 5000 characters a second. Colossus was used at Bletchley Park during WWII - as a successor to April's 'Robinson's. Although 10 were eventually built, unfortunately they were destroyed immediately after they had finished their work - it was so advanced that there was to be no possibility of its design falling into the wrong hands (presumably the Russians). One of the early engineers wrote an emulation on an early Pentium - that ran at half the rate!

Aug 7, 1944 The IBM ASCC (Automatic Sequence Controlled Calculator) is turned over to Harvard University, which calls it the Harvard Mark I It was designed by Howard Aiken and his team, financed and built by IBM - it became the second program controlled machine (after Konrad Zuse's). The whole machine is 51 feet long, weighs 5 tons, and incorporates 750,000 parts. It used 3304 electromechanical relays as on-off switches, had 72 accumulators (each with its own arithmetic unit) as well as mechanical register with acapacity of 23 digits plus sign. Unlike in Zuse's earlier binary machine, the arithmetic is still fixed-point and decimal, with a plugboard setting determining the number of decimal places. I/O facilities include card readers, a card punch, paper tape readers, and typewriters. There are 60 sets of rotary switches, each of which can beused as a constant register - sort of mechanical read-only memory. The program is read from one paper tape; data can be read from the other tapes, or the card readers, or from the constant registers. Conditional jumps are not available. However, in later years the machine is modified to support multiple paper tape readers for the program, with the transfer from one to another being conditional, sort of like a conditional subroutine call. Another addition allows the provision of plugboard-wired subroutines callable from the tape. Used to create ballistics tables for the US Navy.

1945 Konrad Zuse develops Plankalkuel, the first higher-level programming language.

1945 Vannevar Bush develops the theory of the memex, a hypertext device linked to a library of books and films.

The Computing World after 1946

Events from the history

1946-1989

Feb, 1946 ENIAC (Electronic Numerical Integrator and Computer): One of the first totally electronic, valve driven, digital, computers. Development started in 1943 at the Ballistic Research Laboratory, USA, by John W. Mauchly and J. Presper Eckert. It weighed 30 tonnes and contained 18,000 Electronic Valves, consuming around 160 kW of electrical power. It could do 50,000 calculations a second. It was used for calculating Ballistic trajectories and testing theories behind the Hydrogen bomb. Modern computers. however, are conceptually more similar to Konrad Zuse's binary Z3 - ENIAC still used the decimal system and had to be rewired for different programs.

Dec 16, 1947 Invention of the Transistor at Bell Laboratories, USA, by William B. Shockley, John Bardeen and Walter Brattain.

1947 Howard Aikencompletes the Harvard Mark II, built by IBM.

Jan 27, 1948 IBM releases the SSEC (Selective Sequence Electronic Calculator). It is the first computer to modify a stored program.

Jul 21, 1948 SSEM, Small-Scale Experimental Machine or 'Baby' was built at the University of Manchester, It ran its first program on this date. Based on ideas from John von Neumann about stored program computers, it was the first computer to store both its programs and data in RAM, as modern computers do. Interestingly, Konrad Zuse's 1936 patent application (Z23139/GMD Nr. 005/021) earlier already suggested a `von Neumann' architecture with program and data modifiable in storage (not implemented in his 1941 Z1 though). By 1949 the 'Baby' had grown, and acquired a magnetic drum for more permanent storage, and it became the Manchester Mark I (qv.).

1948 IBM introduces the '604', the first machine to feature FRU (Field Replaceable Units), which cust downtime as entire pluggable boards can simply be replaced instead of troubleshot.

Mar, 1949 J. Presper Eckert and John Mauchly construct the BINAC for Northrop.

6 May 1949 Maurice Wilkes and a team at Cambridge University build a stored program computer EDSAC. It used paper tape I/O.

Oct, 1949 The Manchester Mark I final specification is completed; this machine notably being the first computer to use the equivalent of base/index registers, a feature not entering common computer architecture until the second generation around 1955.

1949 CSIR Mk I (later known as CSIRAC), Australia's first computer, ran its first test program. It was a vacuum tube based electronic general purpose computer. Its main memory stored data as a series of acoustic pulses in 5 foot long tubes filled with mercury.

"Computers in the future may weigh no more than 1.5 tons.", Popular Mechanics, forecasting the relentless march of science

1951 EDVAC (electronic discrete variable computer) - First computer to use Magnetic Tape. EDVAC could have new programs loaded off of the tape. Proposed by John von Neumann, it was installed at the Institute for Advance Study, Princeton, USA.

High level language compiler invented by Grace Murray Hopper.

IAS machine completed at the Institute for Advanced Study, Princeton, USA (by Von Neumann and others).

1953 Magnetic core memory developed.

1954 FORTRAN (FORmula TRANslation) development started by John Backus and his team at IBM - continuing until 1957. FORTRAN was the first high-level programming language, still in use for scientific programming. Before being run, a FORTRAN program needs to be converted into a machine program by a compiler, itself a program.

1958 LISP (interpreted language) developed, Finished in 1960. LISP stands for 'LISt Processing', but some call it 'Lots of Irritating and Stupid Parenthesis' due to the huge number of confusing nested brackets used in LISP programs. Used in A.I. development. Developed by John McCarthy at Massachusetts Institute of Technology.

1959 Computers built between 1959 and 1964 are often regarded as 'Second Generation' computers, based on transistors and printed circuits - resulting in much smaller computers. More powerful, the second generation of computers could handle compilers for languages such as FORTRAN (for science) or COBOL (for business), that accepting English-like commands, and so were much more flexible in their applications.

1958 Sep 12 The integrated circuit invented by Jack St Clair Kilby at Texas Instruments. Robert Noyce, who later set up Intel, also worked separately on the invention. Intel later went on to perfect the microprocessor. The patent was applied for in 1959 and granted in 1964. This patent wasn't accepted by Japan so Japanese businesses could avoid paying any fees, but in 1989 - after a 30 year legal battle - Japan granted the patent; so all Japanese companies will pay fees up until the year 2001 - long after the patent became obsolete in the rest of the World!

1960 Compiler compiler - The first compiler is released.

1964 Computers built between 1964 and 1972 are often regarded as 'Third Generation' computers, they are based on the first integrated circuits - creating even smaller machines. Typical of such machines was the IBM System/360 series mainframe, while smaller minicomputers began to open up computing to smaller businesses.

DEC PDP-8 Mini Computer. The first minicomputer, built by Digital Equipment (DEC). It cost $16,000.

1965 Moore's law published by Gordon Moore in the 35th Anniversary edition of Electronics magazine. Originally suggesting processor complexity doubled every year, 1965

BASIC programming language (Beginners All Purpose Symbolic Instruction Code) developed at Dartmouth College, USA, by Thomas E. Kurtz and John Kemeny. Not implemented on microcomputers until 1975. This was the first language designed to be used in a time-sharing environment, such as DTSS (Dartmouth Time-Sharing System), or GCOS.

Packet switching, funded by ARPA was developed. This makes reliable computer networking possible. The first computer-to-computer login does not occur until November 21, 1969, between Stanford and UCLA.the law was revised in 1975 to suggest a doubling in complexity every two years.

The first supercomputer, the Control Data CDC 6600, was developed.

1966 Hewlett-Packard entered the general purpose computer business with its HP-2115 for computation, offering a computational power formerly found only in much larger computers. It supported a wide variety of languages, among them BASIC, Algol, and FORTRAN

1968 Intel founded by Robert Noyce and a few friends.

1969 ARPANET started by the United States Department of Defense for research into networking. It is the original basis for what now forms the Internet. It was opened to non-military users later in the 1970s and many universities and large businesses went on-line.

1970 Development of UNIX operating system started. It was later released as C source code to aid portability, and subsequently versions are obtainable for many different computers, including the IBM PC. It and its clones (such as Linux) are still widely used on network servers and scientific workstations. Originally developed by Ken Thompson and Dennis Ritchie.

1971 Nov 15 First microprocessor, the 4004, developed by Marcian E. Hoff for Intel, was released. It contains the equivalent of 2300 transistors and was a 4 bit processor. It is capable of around 60,000 instructions per second (0.06 MIPS), running at a clock rate of 108 kHz.

1972 Computers built after 1972 are often called 'fourth generation' computers, based on LSI (Large Scale Integration) of circuits (such as microprocessors) - typically 500 or more components on a chip. Later developments include VLSI (Very Large Scale Integration) of integrated circuits 5 years later - typically 10,000 components. The fourth generation is generally viewed as running right up until the present, since although computing power has increased the basic technology has remained virtually the same.

C programming language developed at The Bell Laboratories in the USA by Dennis Ritche (one of the inventors of the Unix operating system), its predecessor was the B programming language - also from Bell. It is a very popular language, especially for systems programming - as it is flexible and fast. C++, allowing for Object-Orientated Programming, was introduced in the early 1980s

The first international connections to ARPANET are established. ARPANET later became the basis for what we now call the Internet.

1974 Apr 1	Introduction of the 8080. An 8 Bit Microprocessor from Intel. Arguably this microprocessor started the microprocessor revolution. First in the line of processors that lead to the currently most used processor architecture, the 80X86 (and pentium) series. It ran at a clock frequency of 2 MHz and did 0.64 MIPS.
1974	Motorola announces the MC6800 8 Bit Microprocessor. It is more easy to implement than the 8080 because it only needs a single power supply to operate and does not need support chips. Unlike the 8080 it is sold not as much as a general purpose "number cruncher / computer" CPU core but more as a control processor for industrial control and as a peripheral processor.
1976	Cray-1, the first commercially developed supercomputer, was invented by Seymour Cray, who left Control Data in 1972 to form his own company. This machine was known as much for its horseshoe-shaped designas it was for being the first super to make vector processing practical. 85 were shipped at a cost of $5 million each.
1978 Jun 8	Introduction of 8086 by Intel, the first commercially successful 16 bit processor. It was too expensive to implement in early computers, so an 8 bit version was developed (the 8088), which was chosen by IBM for the first IBM PC. This ensured the success of the x86 family of processors that succeeded the 8086 since they and their clones are used in every IBM PC compatible computer. The available clock frequencies were 4.77, 8 and 10 MHz. It has an instruction set of about 300 operations. At introduction the fastest processor was the 8 MHz version which achieved 0.8 MIPS and contained 29,000 transistors.
1979	Compact disc was invented.
1981 Aug 12	(MS-DOS 1.0., PC-DOS 1.0.) Microsoft (known mainly for their programming languages) were commissioned by IBM to write the operating system, they bought a program called 86-DOS from Tim Paterson which was loosely based on CP/M 80. The final program from Microsoft was marketed by IBM as PC-DOS and by Microsoft as MS-DOS, collaboration on subsequent versions continued until version 5.0 in 1991. Compared to modern versions of DOS, version 1 was very basic, the most notable difference was the presence of just 1 directory, the root directory, on each disk. Subdirectories were not supported until version 2.0 (March, 1983). MS-DOS (and PC-DOS) was the main operating system for all IBM-PC compatible computers until 1995 when Windows '95 began to take over the market, and Microsoft turned its back on MS-DOS (leaving MS-DOS 6.22 from 1993 as the last version written - although the DOS Shell in Windows '95 calls itself MS-DOS version 7.0, and has some improved features like long filename support). According to Microsoft, in 1994, MS-DOS was running on some 100 million computers world-wide.
1982	MIDI, Musical Instrument Digital Interface, (pronounced "middy") published by International MIDI Association (IMA). The MIDI standard allows computers to be connected to instruments like keyboards through a low-bandwidth (around 30 kbit/s) protocol. The TCP/IP protocol established. This is the protocol that carries most of the

information across the Internet.

Introduction of the BBC Micro. Based on the 6502 processor it was a very popular computer for British schools up to the development of the Acorn Archimedes (in 1987). In 1984 the government offered to pay half the cost of such computers in an attempt to promote their use in secondary education.

1983 Apple introduced its Lisa. The first personal computer with a graphical user interface, its development was central in the move to such systems for personal computers. The Lisa's sloth and high price ($10,000) led to its ultimate failure. The Lisa ran on a Motorola 68000 microprocessor and came equipped with 1 megabyte of RAM, a 12-inch black-and-white monitor, dual 5 1/4-inch floppy disk drives and a 5 megabyte Profile hard drive. The Xerox Star -- which included a system called Smalltalk that involved a mouse, windows, and pop-up menus -- inspired the Lisa's designers

1984 DNS (Domain Name Server) introduced to the Internet, which then consisted of about 1000 hosts.

1987 Microsoft Windows 2 released. It was more popular than the original version but it was nothing special mind you, Windows 3 (see 1990) was the first really useful version.

Fractal Image Compression Algorithm invented by English mathematician Michael F. Barnsley, allowing digital images to be compressed and stored using fractal codes rather than normal image data.

1989 World Wide Web, invented by Tim Berners-Lee who saw the need for a global information exchange that would allow physicists to collaborate on research (he was working at CERN, the European Particle Physics Laboratory in Switzerland, at the time). The Web was a result of the integration of hypertext and the Internet. The hyperlinked pages not only provided information but provide transparent access to older Internet facilities such as ftp, telnet, Gopher, WAIS and Usenet. He was awarded the Institute of Physics' 1997 Duddell Medal for this contribution to the advancement of knowledge. The Web started as a text-only interface, but NCSA Mosaic later presented a graphical interface for it and its popularity exploded as it became accessible to the novice user. This explosion started in ernest during 1993, a year in which web traffic over the Internet increased by 300,000%.

1990 onwards

1990 May 22 Introduction of **Windows** 3.0 by **Microsoft**. It is a true multitasking system (or pretends to be on computers less than an 80386, by operating in 'Real' mode). It maintained compatibility with **MS-DOS**, on an 80386 it even allows such programs to multitask - which they were not designed to do. This created a real threat to the Macintosh and despite a similar product, IBM's OS/2, it was very successful. Various improvements were made, versions 3.1, 3.11 - but the next major step did not come until Windows '95 in 1995 which relied much more heavily on the features of the 80386 and provided support for 32 bit applications.

1990 Aug Linux is born with the following post to the Usenet Newsgroup comp.os.minix: "Hello everybody out there using minix- I'm doing a (free) operating system (just a hobby, won't be big and professional like gnu) for 386(486) AT clones."

The post was by a Finnish college student, Linus Torvalds, and this hobby grew from these humble beginnings into one of the most widely used Unix-like operating systems in the world today. It now runs on many different types of computer, including the Sun SPARC and the Compaq Alpha, as well as many ARM, MIPS, PowerPC and Motorola 68000 based computers.

Feb 1992 The GNU project (www.gnu.org) adopted the Linux kernel for use on GNU systems while they waited for the development of their own (GNU Hurd) kernel to be completed. The GNU project's aim is to provide a complete and free Unix like operating system, combining the Linux or Hurd platform with the a complete suite of free software to run on it. Linus changed the Linux kernel license from "no commercial use" to the GNU General Public License on the 1st of February 1992.

1992 "Windows NT addresses 2 Gigabytes of RAM which is more than any application will ever need" -- Microsoft on the development of Windows NT.

Introduction of CD-I launched by Phillips.

Apr 1992 Introduction of Windows 3.1

1992 Mar 22 Intel Pentium released. At the time it was only available in 60 & 66 MHz versions which achieved up to 100 MIPs, with over 3.1 million transistors.

May 1992 MPC Level 2 specification introduced (see November 1990). This was designed to allow playback of a 15 frame/s video in a window 320x240 pixels. The key difference is the requirement of a CD-ROM drive capable of 300 KB/s (double speed). Also with Level 2 is the requirement for products to be tested by the MPC council, making MPC Level 2 compatibility a stamp of certification.

1994 Mar 7 Intel Release the 90 & 100 MHz versions of the Pentium Processor.

1994 DNA computing proof of concept on toy travelling salesman problem; a method for input/output still to be determined.

Netscape Navigator 1.0 was written as an alternative browser to NCSA Mosaic.

1995 Aug 21 Windows 95 was launched by Bill Gates & Microsoft. Unlike previous versions of Windows, Windows 95 is an entire operating system - it does not rely on MS-DOS (although some remnants of the old operating system still exist). Windows 95 was written specially for the 80386 and compatible computers to make 'full' use of its 32 bit processing and multitasking capabilities, and thus is much more similar to Windows NT than Windows 3.x. Windows 95 and NT 4 are almost indistinguishable in many respects - such as User Interface and API. Unfortunately, in order to maintain backwards compatibility, Windows 95 doesn't impose the same memory protection and security measures that NT does and so suffers from much worse reliability. Despite being remarkable similar in function to OS/2 Warp (produced by IBM and Microsoft several years earlier, but marketed by IBM), Windows 95 has proved very popular.

1995 Nov 1 Pentium Pro released. At introduction it achieved a clock speed of up to 200 MHz (there were also 150, 166 and 180 MHz variants released on the same date), but is basically the same as the Pentium in terms of instruction set and capabilities. It achieves 440 MIPs and contains 5.5 million transistors - this is nearly 2400 times as many as the first microprocessor, the 4004 - and capable of 70,000 times as many instructions per second.

1996 Jun 9 Linux 2.0 released. 2.0 was a significant improvement over the earlier versions: it was the first to support multiple architectures (originally developed for the Intel 386 processor, it now supported the Digital Alpha and would very soon support Sun SPARC and many others). It was also the first stable kernel to support SMP, kernel modules, and much more.

1996 Oct 6 Intel releases the 200 MHz version of the Pentium Processor.

1997 Jan 8 Intel released Pentium MMX (originally 166 and 200 MHz versions), for games and multimedia enhancement. To most people MMX is simply another 3-letter acronym and people wearing coloured suits on Intel ads, and to programmers in meant an even further expanded instruction set that provides, amongst other functions, enhanced 64-bit support - but software needs to be specially written to work with the new functions. A major rival clone, the AMD-K6-MMX containing a similar instruction set, caused a legal challenge from Intel on the right to use the trademarked name MMX — it was not upheld.

May 11 IBM's Deep Blue became the first computer to beat a reigning World Chess Champion, Garry Kasparov, in a full chess match. The computer had played him previously — losing 5/6 games in February 1996.

May 7 Intel Release their Pentium II processor (233, 266 and 300 MHz versions). It featured, as well as an increased instruction set, a much larger on-chip cache.

Aug 6 After 18 months of losses Apple Computer was in serious financial trouble. Microsoft invested in Apple, buying 100,000 non-voting shares worth $150 million — a decision not approved of by many Apple owners! One of the conditions was that Apple was to drop the long running court case — attempting to sue Microsoft for copying the look and feel of their operating system when designing Windows. It must be pointed out that Apple copied the XEROX Star system when designing their WIMP.

1998 Feb Intel released of 333 MHz Pentium II processor. Code-named Deschutes these processors use the new 0.25 micrometre manufacturing process to run faster and generate less heat than before.

Jun 25 Microsoft released Windows 98. Some U.S. attorneys tried to block its release since the new O/S interlaces with other programs such as Microsoft Internet Explorer and so effectively closes the market of such software to other companies. Microsoft has fought back with a letter to the White House suggesting that 26 of its industry allies say that a delay in the release of the new O/S could damage the U.S. economy. The main selling points of Windows '98 were its support for USB and its support for disk partitions greater than 2.1GB.

1999 Feb 22 AMD release K6-III 400 MHz version, 450 to OEMS. In some tests it outperforms soon-to-be released Intel P-III. It contains approximately 23 million transistors, and is based on 100 MHz super socket 7 motherboards, an improvement on the 66 MHz buses their previous chips were based on. This helps its performance when compared to Intel's Pentium II - which also uses a 100 MHz bus speed.

2000 Feb 17	Official Launch of Windows 2000 - Microsoft's replacement for Windows 95/98 and Windows NT. Claimed to be faster and more reliable than previous versions of Windows. It is actually a descendant of the NT series, and so the trade-off for increased reliability is that it won't run some old DOS-based games. To keep the home market happy Microsoft has also released Windows ME, the newest member of the 95/98 series.]
2000 Mar 6	AMD release the Athlon 1GHz
2000 Mar 8	Intel releases very limited supplies of the 1GHz Pentium III chip.
Nov 2000	Intel releases the Pentium IV.
2001 May 30	United Linux officially formed.

History of Hardware, Software and Networking

Predictions / Concepts

'I think there is a world market for maybe five computers.' Thomas J. Watson, chairman of IBM

'Computers in the future may weigh no more than 1.5 tons'

'We'll have to think up bigger problems if we want to keep them busy.' Howard Aiken

(Turing Test) Alan Turing

Moore's law: processor complexity will double every year, revised in 1975: complexity will double every two years. Gordon E. Moore

'There is no reason anyone would want a computer in their home' (Ken Olson, Chairman DEC)

(Quantum Computer) Richard Feynmann

50 60 70 80 90 2000

History of Hardware

First use/(Inventions)

- teletype
- (transistor)
- random access memory
- magnetic drum
- registers
- magnetic core memory
- hard disk
- dot matrix printer
- integrated circuit
- spooling, interrupts, virtual memory paging (ATLAS)
- mouse
- time-sharing
- fuzzy logic
- packet switching
- ARPANET precursor of Internet
- RS-232
- dynamic RAM
- floppy disk
- microprocessor (4004)
- game console (PONG)
- TCP/IP 1982 ??
- ethernet
- single-board computer
- laser printer
- supercomputer (Cray)
- compact disk
- WIMP (GUI)
- CGA video card
- MIDI
- RISC cpu
- portable PC
- math coprocessor
- PC harddisk
- GNU project
- DNS (Domain Name Server)
- IDE interface
- EGA video card
- CD ROM
- SCSI interface
- massive parallel computing (Connection Machine)
- VGA video card
- PC sound card (AD-LIB)
- optical chip
- EISA bus
- ATA interface
- SVGA video card, VESA driver
- ISA bus
- CD-I
- DNA computing
- Expanded Memory
- Postscript

40 50 60 70 80 90 2000

Hardware Systems/(Processors)

- Model I Relay Calculator
- Atanasoff Berry Computer
- Z3
- Harvard Mark I
- Relay Interpolator
- Colossus
- ENIAC
- ATLAS
- IBM 360
- DEC PDP-8
- CDC 6600 1st supercomputer
- electronic calculator TI
- (4004)
- scientific calculator HP
- (8008)
- 1st 32-bit supermini
- (8080)
- (MC 6800)
- Altair 8800 personal computer
- (Z80)
- (6502)
- Cray-1
- Apple II
- TRS-80
- (8086)
- PET
- (68000)
- VIC-20
- ZX80
- 8010 'Star'
- ZX81
- IBM PC
- BBC Micro
- C64
- (80286)
- Compaq Portable
- ZX Spectrum
- Lisa
- IBM XT
- IBM PC jr
- Macintosh
- LaserJet printer
- IBM AT
- Amiga 1000
- Atari ST
- Amstrad 1512
- Acorn Archimedes
- Connection Machine
- IBM PS/2
- (80486)
- (66 MHz Pentium)
- iMac
- PowerMac
- (1 GHz Pentium)
- PowerPC G5
- Athlon 64

40 50 60 70 80 90 2000

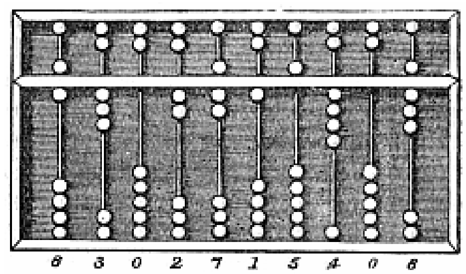
Computing hardware has been an essential component of the process of calculation and data storage since it became necessary for data to be processed and shared. The first known computing hardware was a recording device; the Phoenicians stored clay shapes representing items, such as livestock and grains, in containers. These were used by merchants, accountants, and government officials of the time.

Devices to aid computation have evolved over time. Even today, an experienced abacus user using a device designed thousands of years ago can often complete basic calculations more quickly than an unskilled person using an electronic calculator — though for more sophisticated calculations, computers out-perform even the most skilled human.

This article presents the major developments in the history of computing hardware and attempts to put them in context. For a detailed timeline of events, see the computing timeline article. The history of computing article is a related overview and treats methods intended for pen and paper, with or without the aid of tables.

Earliest devices for facilitating human calculation

Humanity has used devices to aid in computation for millennia. A more arithmetic -oriented machine the abacus is one of the earliest machines of this type was the Roman abacus.



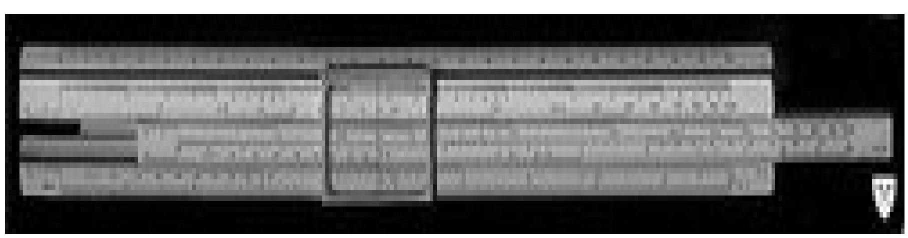
Babylonians and others, frustrated with counting on their fingers, invented the Abacus.

First mechanical calculators

Gears are at the heart of mechanical devices like the Curta calculator.

In 1623 Wilhelm Schickard built the first mechanical calculator and thus became the father of the computing era. Since his machine used techniques such as cogs and gears first developed for clocks, it was also called a 'calculating clock'. It was put to practical use by his friend Johannes Kepler, who revolutionized astronomy.

John Napier noted that multiplication and division of numbers can be performed by addition and subtraction, respectively, of logarithms of those numbers. Since these real numbers can be represented as distances or intervals on a line, the slide rule allowed multiplication and division operations to be carried significantly faster than was previously possible. Slide rules were used by generations of engineers and other mathematically inclined professional workers, until the invention of the pocket calculator.



The slide rule, a basic mechanical calculator, facilitates multiplication and division.

Punched card technology 1801 –

In 1801, Joseph-Marie Jacquard developed a loom in which the pattern being woven was controlled by punched cards. The series of cards could be changed without changing the mechanical design of the loom. This was a landmark point in programmability. Herman Hollerith invented a tabulating machine using punch cards in the 1880s.

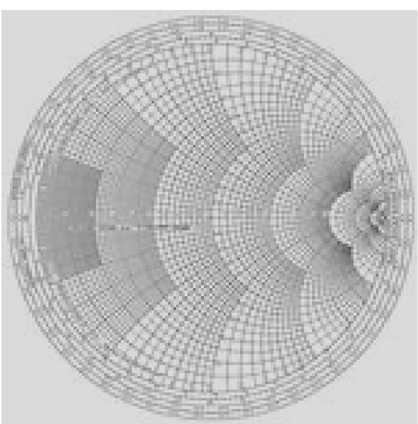
First designs of programmable machines 1835–1900s

The defining feature of a "universal computer" is programmability, which allows the computer to emulate any other calculating machine by changing a stored sequence of instructions. In 1835 Charles Babbage described his analytical engine. It was the plan of a general-purpose programmable computer, employing punch cards for input and a steam engine for power. One crucial invention was to use gears for the function served by the beads of an abacus. In a real sense, computers all contain automatic abaci (technically called the ALU or floating-point unit).

More limited types of mechanical gear computing 1800s–1900s

By the 1900s earlier mechanical calculators, cash registers, accounting machines, and so on were redesigned to use electric motors, with gear position as the representation for the state of a variable. People were computers, as a job title, and used calculators to evaluate expressions. During the Manhattan project, future Nobel laureate Richard Feynman was the supervisor of the roomful of human computers, many of them women mathematicians, who understood the differential equations which were being solved for the war effort. Even the renowned Stanislaw Marcin Ulam was pressed into service to translate the mathematics into computable approximations for the hydrogen bomb, after the war

Analog computers, pre -1940



Before World War II, mechanical and electrical analog computers were considered the 'state of the art', and many thought they were the future of computing. Analog computers use continuously varying amounts of physical quantities, such as voltages or currents, or the rotational speed of shafts, to represent the quantities being processed. An ingenious example of such a machine was the Water integrator built in 1936.

Nomograms, like this Smith chart serve as analog computers for specific classes of problems.

First generation of electrical digital computers 1940s

The era of modern computing began with a flurry of development before and during World War II, as electronic circuits, relays, capacitors and vacuum tubes replaced mechanical equivalents and digital calculations replaced analog calculations. The computers designed and constructed then have sometimes been called 'first generation' computers. First generation computers were usually built by hand using circuits containing relays or vacuum valves (tubes), and often used punched cards or punched paper tape for input and as the main (non-volatile) storage medium. Temporary, or working storage, was provided by acoustic delay lines (which use the propagation time of sound in a medium such as wire to store data) or by Williams tubes (which use the ability of a television picture tube to store and retrieve data).

Electronic computers became possible with the advent of the vacuum tube.

By 1954, magnetic core memory was rapidly displacing most other forms of temporary storage, and dominated the field through the mid-1970s.

In this era, a number of different machines were produced with steadily advancing capabilities. At the beginning of this period, nothing remotely resembling a modern computer existed, except in the long-lost plans of Charles Babbage and the mathematical musings of Alan Turing and others. At the end of the era, devices like the EDSAC had been built, and are universally agreed to be universal digital computers. Defining a single point in the series as the "first computer" misses many subtleties.

Alan Turing's 1936 paper has proved enormously influential in computing and computer science, in two ways. Its main purpose was an elegant proof that there were problems (namely the halting problem) that could not be solved by a mechanical process (a computer). In doing so, however, Turing provided a definition of what a universal computer is: a construct called the Turing machine, a purely theoretical device invented to formalize the notion of algorithm execution, replacing Kurt Gödel's more cumbersome universal language based on arithmetics. Modern computers are Turing-complete (i.e., equivalent algorithm execution capability to a universal Turing machine), except for their finite memory. This limited type of Turing completeness is sometimes viewed as a threshold capability separating general-purpose computers from their special-purpose predecessors.

However, as will be seen, *theoretical* Turing-completeness is a long way from a practical universal computing device. To be a practical general-purpose computer, there must be some convenient way to input new programs into the computer, such as punched tape.

For full versatility, the Von Neumann architecture uses the same memory both to store programs and data; virtually all contemporary computers use this architecture (or some variant). Finally, while it is theoretically possible to implement a full computer entirely mechanically (as Babbage's design showed), electronics made possible the speed and later the miniaturization that characterises modern computers.

There were three, parallel streams of computer development in the World War II era, and two were either largely ignored or were deliberately kept secret. The first was the German work of Konrad Zuse. The second was the secret development of the Colossus computer in the UK. Neither of these had much influence on the various computing projects in the United States. After the war British and American computing researchers cooperated on some of the most important steps towards a practical computing device.

American developments

In 1937, Claude Shannon produced his master's thesis at MIT that implemented Boolean algebra using electronic relays and switches for the first time in history. Entitled *A Symbolic Analysis of Relay and Switching Circuits*, Shannon's thesis essentially founded practical digital circuit design.

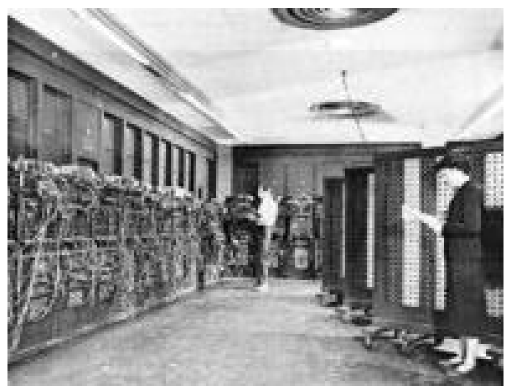
In November of 1937, George Stibitz, then working at Bell Labs, completed a relay-based computer he dubbed the "Model K" (for "kitchen", where he had assembled it), which calculated using binary addition. Bell Labs thus authorized a full research program in late 1938 with Stibitz at the helm. Their Complex Number Calculator, completed January 8, 1940, was able to calculate complex numbers. In a demonstration to the American Mathematical Society conference at Dartmouth College on September 11, 1940, Stibbitz was able to send the Complex Number Calculator remote commands over telephone lines by a teletype.

It was the first computing machine ever used remotely over a phone line. Some participants of the conference who witnessed the demonstration were John Von Neumann, John Mauchly, and Norbert Wiener, who wrote about it in his memoirs.

In 1938 John Vincent Atanasoff and Clifford E. Berry of Iowa State University developed the Atanasoff Berry Computer (ABC), a special purpose computer for solving systems of linear equations, and which employed capacitors fixed in a mechanically rotating drum, for memory. The ABC machine was not programmable, though it was a computer in the modern sense in several other respects.

In 1939, development began at IBM's Endicott laboratories on the Harvard Mark I. Known officially as the Automatic Sequence Controlled Calculator, the Mark I was a general purpose electro-mechanical computer built with IBM financing and with assistance from some IBM personnel under the direction of Harvard mathematician Howard Aiken. Its design was influenced by the Analytical Engine.

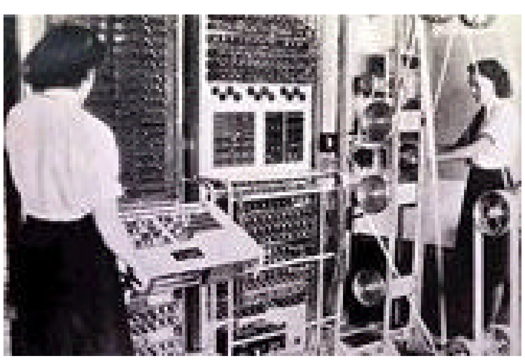
It was a decimal machine which used storage wheels and rotary switches in addition to electromagnetic relays. It was programmable by punched paper tape, and contained several calculators working in parallel. Later models contained several paper tape readers and the machine could switch between readers based on a condition. Nevertheless, this does not quite make the machine Turing-complete. The Mark I was moved to Harvard University to begin operation in May 1944.



ENIAC performed ballistics trajectory calculations with 160kW of power.

The US-built ENIAC (Electronic Numerical Integrator and Computer), often called the first electronic general-purpose computer, publicly validated the use of electronics for large-scale computing. This was crucial for the development of modern computing, initially because of the enormous speed advantage, but ultimately because of the potential for miniaturization. Built under the direction of John Mauchly and J. Presper Eckert, it was 1,000 times faster than its contemporaries. ENIAC's development and construction lasted from 1941 to full operation at the end of 1945

Colossus



Colossus was used to break German ciphers during World War II.

During World War II, the British at Bletchley Park achieved a number of successes at breaking encrypted German military communications. The German encryption machine, Enigma, was attacked with the help of electro-mechanical machines called *bombes*. The bombe, designed by Alan Turing and Gordon Welchman, ruled out possible Enigma settings by performing chains of logical deductions implemented electrically. Most possibilities led to a contradiction, and the few remaining could be tested by hand.

The Germans also developed a series of teleprinter encryption systems, quite different from Enigma. The Lorenz SZ 40/42 machine was used for high-level Army communications, termed "Tunny" by the British. The first intercepts of Lorenz messages begain in 1941. As part of an attack on Tunny, Professor Max Newman and his colleagues helped specify the Colossus. The Mk I Colossus was built in 11 months by Tommy Flowers and his colleagues at the Post Office Research Station at Dollis Hill in London and then shipped to Bletchley Park.

Colossus was the first totally *electronic* computing device. The Colossus used a large number of valuves (vacuum tubes). It had paper-tape input and was capable of being configured to perform a variety of boolean logical operations on its data, but it was not Turing-complete. Nine Mk II Colossi were built (The Mk I was converted to a Mk II making ten machines in total). Details of their existence, design, and use were kept secret well into the 1970s.

Konrad Zuse's Z-Series

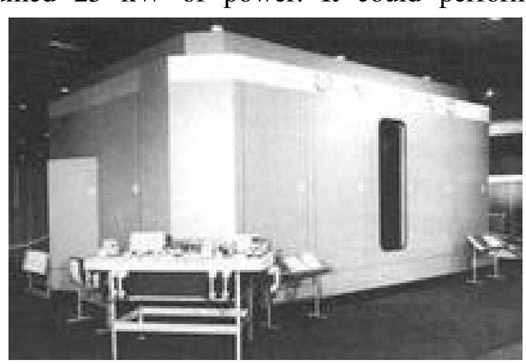
Working in isolation in Nazi Germany, Konrad Zuse started construction in 1936 of his first Z-series calculators featuring memory and (initially limited) programmability. Zuse's purely mechanical, but already binary Z1, finished in 1938, never worked reliably due to problems with the precision of parts.

Postwar von Neumann machines – the first generation

The first working von Neumann machine was the Manchester "Baby" or Small-Scale Experimental Machine, built at the University of Manchester in 1948; it was followed in 1949 by the Manchester Mark I computer which functioned as a complete system using the Williams tube for memory, and also introduced index registers. The other contender for the title "first digital stored program computer" was EDSAC, designed and constructed at the University of Cambridge. Operational less than one year after the Manchester "Baby", it was capable of tackling real problems.

EDSAC was actually inspired by plans for EDVAC (Electronic Discrete Variable Automatic Computer), the successor of ENIAC; these plans were already in place by the time the ENIAC was successfully operational. Unlike the ENIAC, which used parallel processing, EDVAC used a single processing unit. This design was simpler and was the first to be implemented in each succeeding wave of miniaturization, and increased reliability. Some view Manchester Mark I / EDSAC / EDVAC as the "Eves" from which nearly all current computers derive their architecture.

The first universal programmable computer in continental Europe was created by a team of scientists under direction of Sergei Alekseyevich Lebedev from Kiev Institute of Electrotechnology, Soviet Union (now Ukraine). The computer MESM (*? ? ? ?* , *Small Electronic Calculating Machine*) became operational in 1950. It had about 6,000 vacuum tubes and consumed 25 kW of power. It could perform approximately 3,000 operations per second. Another early machine was CSIRAC, an Australian design that ran its first test program in 1949.



UNIVAC I, above, the first commercial electronic computer, achieved 1900 operations per second in a smaller and more efficient package than ENIAC.

In June 1951, the UNIVAC I (Universal Automatic Computer) was delivered to the U.S. Census Bureau. Although manufactured by Remington Rand, the machine often was mistakenly referred to as the "IBM UNIVAC". Remington Rand eventually sold 46 machines at more than $1 million each. UNIVAC was the first 'mass produced' computer; all predecessors had been 'one-off' units. It used 5,200 vacuum tubes and consumed 125 kW of power. It used a mercury delay line capable of storing 1,000 words of 11 decimal digits plus sign (72-bit words) for memory. Unlike earlier machines it did not use a punch card system but a metal tape input.

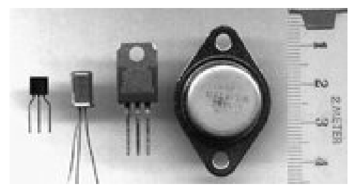
In November 1951, the J. Lyons company began weekly operation of a bakery valuations job on the LEO (Lyons Electronic Office). This was the first business application to go live on a stored program computer.

In 1953, IBM introduced the IBM 701 Electronic Data Processing Machine, the first in its successful 700/7000 series and its first mainframe computer. The first implemented high-level general purpose programming language, Fortran, was also being developed at IBM around this time. (Konrad Zuse's 1945 design of the high-level language Plankalkül was not implemented at that time.

In 1956, IBM sold its first magnetic disk system, RAMAC (Random Access Method of Accounting and Control). It used 50 24-inch metal disks, with 100 tracks per side. It could store 5 megabytes of data and cost $10,000 per megabyte. (As of 2005, disk storage costs less than $1 per gigabyte).

Second generation – late 1950s and early 1960s

The next major step in the history of computing was the invention of the transistor in 1947. This replaced the fragile and power hungry valves with a much smaller and more reliable component. Transistorised computers are normally referred to as 'Second Generation' and dominated the late 1950s and early 1960s. By using transistors and printed circuits a significant decrease in size and power consumption was achieved, along with an increase in reliability



Transistors, above, revolutionized computers as smaller and more efficient replacements for vacuum tubes.

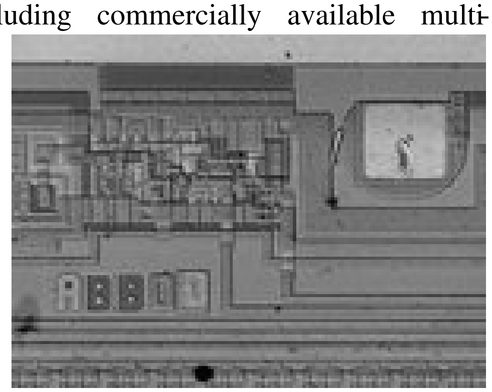
Second generation computers were still expensive and were primarily used by universities, governments, and large corporations.

In 1959 IBM shipped the transistor-based IBM 7090 mainframe and medium scale IBM 1401. The latter was designed around punch card input and proved a popular general purpose computer. Some 12,000 were shipped, making it the most successful machine in computer history at the time. It used a magnetic core memory of 4000 characters (later expanded to 16,000 characters).

In 1964 IBM announced the S/360 series, which was the first family of computers that could run the same software at different combinations of speed, capacity and price. It also pioneered the commercial use of microprograms, and an extended instruction set designed for processing many types of data, not just arithmetic. In addition, it unified IBM's product line, which prior to that time had included both a "commercial" product line and a separate "scientific" line. The software provided with System/360 also included major advances, including commercially available multi-programming, new programming languages, and independence of programs from input/output devices

Third generation and beyond, post-1964

The explosion in the use of computers began with 'Third Generation' computers. These relied on Jack St. Clair Kilby's and Robert Noyce's independent invention of the integrated circuit (or microchip), which later led to Ted Hoff's invention of the microprocessor, at Intel.



The microscopic integrated circuit, above, combined many hundreds of transistors into one unit for fabrication.

The microprocessor led to the development of the microcomputer, small, low-cost computers that could be owned by individuals and small businesses. Microcomputers, the first of which appeared in the 1970s, became ubiquitous in the 1980s and beyond. Computing has evolved with microcomputer architectures, with features added from their larger brethren, now dominant in most market segments.

Fourth Generation (1972-1984)

The next generation of computer systems saw the use of large scale integration (LSI - 1000 devices per chip) and very large scale integration (VLSI - 100,000 devices per chip) in the construction of computing elements. At this scale entire processors will fit onto a single chip, and for simple systems the entire computer (processor, main memory, and I/O controllers) can fit on one chip. Gate delays dropped to about 1ns per gate. Semiconductor memories replaced core memories as the main memory in most systems; until this time the use of semiconductor memory in most systems was limited to registers and cache.

During this period, high speed vector processors, such as the CRAY 1, CRAY X-MP and CYBER 205 dominated the high performance computing scene.

Computers with large main memory, such as the CRAY 2, began to emerge. A variety of parallel architectures began to appear; however, during this period the parallel computing efforts were of a mostly experimental nature and most computational science was carried out on vector processors. Microcomputers and workstations were introduced and saw wide use as alternatives to time-shared mainframe computers.

Fifth Generation (1984-1990)

The development of the next generation of computer systems is characterized mainly by the acceptance of parallel processing. Until this time parallelism was limited to pipelining and vector processing, or at most to a few processors sharing jobs.

The fifth generation saw the introduction of machines with hundreds of processors that could all be working on different parts of a single program. The scale of integration in semiconductors continued at an incredible pace - by 1990 it was possible to build chips with a million components - and semiconductor memories became standard on all computers. Other new developments were the widespread use of computer networks and the increasing use of single-user workstations.

Prior to 1985 large scale parallel processing was viewed as a research goal, but two systems introduced around this time are typical of the first commercial products to be based on parallel processing. The Sequent Balance 8000 connected up to 20 processors to a single shared memory module (but each processor had its own local cache). The machine was designed to compete with the DEC VAX-780 as a general purpose Unix system, with each processor working on a different user's job. However Sequent provided a library of subroutines that would allow programmers to write programs that would use more than one processor, and the machine was widely used to explore parallel algorithms and programming techniques.
The Intel iPSC-1, nicknamed ``the hypercube'', took a different approach. Instead of using one memory module, Intel connected each processor to its own memory and used a network interface to connect processors. This distributed memory architecture meant memory was no longer a bottleneck and large systems (using more processors) could be built. The largest iPSC-1 had 128 processors.
Toward the end of this period a third type of parallel processor was introduced to the market. In this style of machine, known as a data-parallel or SIMD, there are several thousand very simple processors. All processors work under the direction of a single control unit; i.e. if the control unit says ``add a to b'' then all processors find their local copy of a and add it to their local copy of b.

Machines in this class include the Connection Machine from Thinking Machines, Inc., and the MP-1 from MasPar, Inc. Scientific computing in this period was still dominated by vector processing.

Most manufacturers of vector processors introduced parallel models, but there were very few (two to eight) processors in this parallel machines. In the area of computer networking, both wide area network (WAN) and local area network (LAN) technology developed at a rapid pace, stimulating a transition from the traditional mainframe computing environment toward a distributed computing environment in which each user has their own workstation for relatively simple tasks (editing and compiling programs, reading mail) but sharing large, expensive resources such as file servers and supercomputers. RISC technology (a style of internal organization of the CPU) and plummeting costs for RAM brought tremendous gains in computational power of relatively low cost workstations and servers. This period also saw a marked increase in both the quality and quantity of scientific visualization.

Sixth Generation (1990 -)

Transitions between generations in computer technology are hard to define, especially as they are taking place. Some changes, such as the switch from vacuum tubes to transistors, are immediately apparent as fundamental changes, but others are clear only in retrospect.

Many of the developments in computer systems since 1990 reflect gradual improvements over established systems, and thus it is hard to claim they represent a transition to a new ``generation'', but other developments will prove to be significant changes. In this section we offer some assessments about recent developments and current trends that we think will have a significant impact on computational science.

This generation is beginning with many gains in parallel computing, both in the hardware area and in improved understanding of how to develop algorithms to exploit diverse, massively parallel architectures.

Parallel systems now compete with vector processors in terms of total computing power and most expect parallel systems to dominate the future. Combinations of parallel/vector architectures are well established, and one corporation (Fujitsu) has announced plans to build a system with over 200 of its high end vector processors.

Manufacturers have set themselves the goal of achieving teraflops (10 arithmetic operations per second) performance by the middle of the decade, and it is clear this will be obtained only by a system with a thousand processors or more. Workstation technology has continued to improve, with processor designs now using a combination of RISC, pipelining, and parallel processing. As a result it is now possible to purchase a desktop workstation for about $30,000 that has the same overall computing power (100 megaflops) as fourth generation supercomputers.

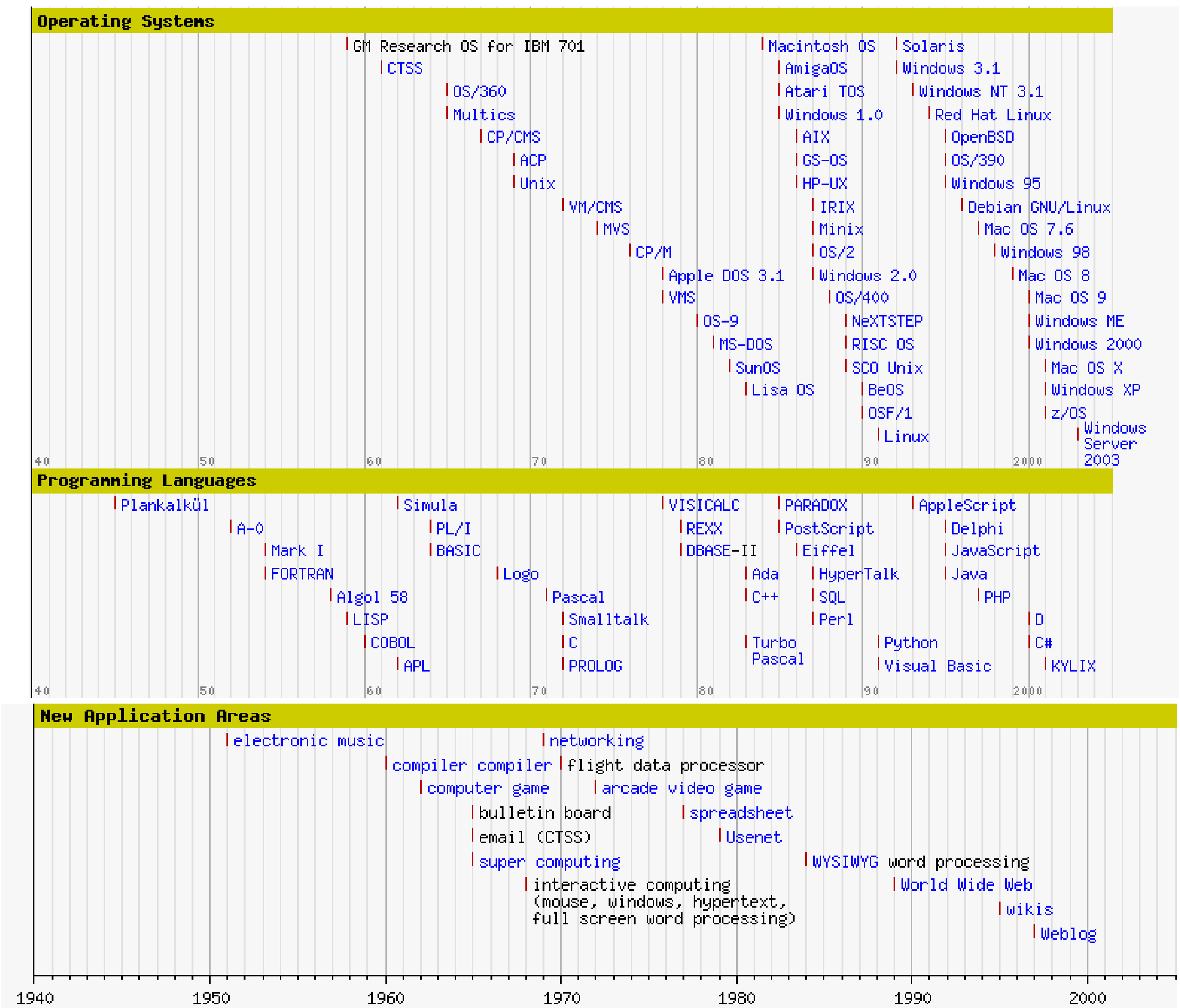
This development has sparked an interest in heterogeneous computing: a program started on one workstation can find idle workstations elsewhere in the local network to run parallel subtasks. One of the most dramatic changes in the sixth generation will be the explosive growth of wide area networking. Network bandwidth has expanded tremendously in the last few years and will continue to improve for the next several years. T1 transmission rates are now standard for regional networks, and the national ``backbone'' that interconnects regional networks uses T3.

Networking technology is becoming more widespread than its original strong base in universities and government laboratories as it is rapidly finding application in K-12 education, community networks and private industry. A little over a decade after the warning voiced in the Lax report, the future of a strong computational science infrastructure is bright.

The federal commitment to high performance computing has been further strengthened with the passage of two particularly significant pieces of legislation: the High Performance Computing Act of 1991, which established the High Performance Computing and Communication Program (HPCCP) and Sen. Gore's Information Infrastructure and Technology Act of 1992, which addresses a broad spectrum of issues ranging from high performance computing to expanded network access and the necessity to make leading edge technologies available to educators from kindergarten through graduate school.

In bringing this encapsulated survey of the development of a computational science infrastructure up to date, we observe that the President's FY 1993 budget contains $2.1 billion for mathematics, science, technology and science literacy educational programs, a 43% increase over FY 90 figures

History of Software



Introduction

A simple question: "What is software?" A very simple answer is: Hardware you can touch, software you can't. But that is too simple indeed.

When talking about software we talk about programming and programming languages and about producing and selling the products made by programming (languages) which are called “Software”.

How It All Started

It shouldn't be a big surprise that the creation or software also went in large but distinguishable steps. Compared with hardware there were fewer developments that went parallel or overlapping. In rare cases developments were reinvented sometimes because the development or invention was not published, even prohibited to be made public (war, secrecy acts etc.) or became known at the same time and after (legal) discussions the "other" party won the honors.

The earliest practical form of programming was probably done by Jaquard (1804, France). He designed a loom that performed predefined tasks through feeding punched cards into a reading contraption.

This picture shows the manufacturing of punched cards for looms

The technology of punched cards will later be adapted by (IBM's) Recording and Tabulating Company to process data
A problem needed to be solved thus a machine was built. (Pascal, Babbage, Scheultz & Son) And it required some sort of instruction; a sequence was designed or written and transferred to either cards or mechanical aids such as wires, gears, shafts actuators etc. This was the birth of programming.
First Ada Lovelace, was writing a rudimentary program (1843) for the Analytical Machine, designed by Charles Babbage in 1827, but the machine never came into operation.

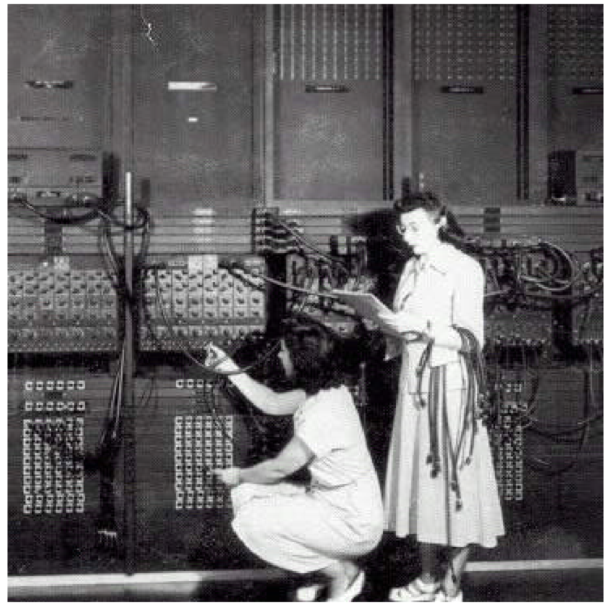
Then there was George Boole (1815-1864), a British mathematician, who proved the relation between mathematics and logic with his algebra of logic (BOOLEAN algebra or binary logic) in 1847. This meant a breakthrough for mathematics. Boole was the first to prove that logic is part of mathematics and not of philosophy.

A big step in thinking

But it took **one hundred years** before this algebra of logic was put to work for computing. It took Claude Shannon (1916-2001) who wrote a thesis (*A Mathematical Theory of Communication* in the *Bell System Technical Journal* -1948) on how binary logic could be used in computing to complete the software concept of modern computing. Now things were in place to start off with the information age.

Enter the Information Age

In the beginning of the so called "Information Age" computers were programmed by "programming" direct instructions into it. This was done by setting switches or making connections to different logical units by wires (circuitry).



Two women wiring the right side of the ENIAC with a new program (US Army photo, from archives of the ARL Technical library, courtesy of Mike Muuss)

Programming like this was nothing else but rewiring these huge machines in order to use all the options, possibilities and calculations. Reprogramming always meant rewiring.

In that way calculations needed days of preparations, handling thousands of wires, resetting switches, plugs etc. (in the most extreme case that is). And the programmed calculation itself just took a few minutes. If the "programming" of wiring did not have a wrong connection, the word bug was not in used yet, for programming errors.

The coding panels very much looked like that a few telephone switchboards hustled together, and in fact many parts actually came from switch boards.

With the invention of vacuum tubes, along with many other inventions, much of the rewiring belonged to the past. The tubes replaced the slow machines based on relays.

When Shannon reinvented or better rediscovered the binary calculus in 1948 and indicated how that could be used for computing a revolution started. The race was on!

First programming was done by typing in 1's or 0's that were stored on different information carriers. Like paper tapes, punched hole cards, hydrogen delay lines (sound or electric pulse) and later magnetic drums and much later magnetic and optical discs.

By storing these 0's and 1's on a carrier (first used by Karl Suze's X1 in 1938) it was possible to have the computer read the data on any later time. But mis typing a single zero or one meant a disaster because all coding (instructions) should absolutely be on the right place and in the right order in memory. This technology was called absolute addressing.

An example:

1010010101110101011

If this is a sequence for switches it means switch one on, switch two off etc. etc.

In the early 50's programmers started to let the machines do a part of the job. This was called automatic coding and made live a lot easier for the early programmers.

The next development was to combine groups of instruction into so called words and abbreviations were thought up called: opcodes (Hopper 1948)

Machine Language

Opcode works like a shorthand and represents as said a group of machine instructions. The opcode is translated by another program into zero's and one's, something a machine could translate into instructions. But the relation is still one to one: one code to one single instruction. However very basically this is already a programming language. It was called: assembly language.

An example:

Label	Opcode	Register
CALC:	**STO**	**R1, HELP0**
	STO	**R2, HELP2**
	LD	**R3, HELP1**
	ADD	**R3, HELP2**
	LD	**R4, HELP1**
	SUB	**R4, HELP2**
	RSR	**SP, 0**
HELP1:	**DS**	**2**
HELP2:	**DS**	**2**

This piece of assembly code calculates the difference between two numbers.

Subroutines

Soon after developing machine languages and the first crude programming languages began to appear the danger of inextricable and thus unreadable coding became apparent. Later this messy programming was called: "spaghetti code".

One important step in unraveling or preventing spaghetti code was the development of subroutines. And it needed Maurice Wilkes, when realizing that "a good part of the remainder of his life was going to be spent in finding errors in ... programs", to develop the concept of subroutines in programs to create reusable modules. Together with Stanley Gill and David Wheeler he produced the first textbook on "The Preparation of Programs for an Electronic Digital Computer". The formalized concept of software development (not named so for another decade) had its beginning in 1951.

Fortran

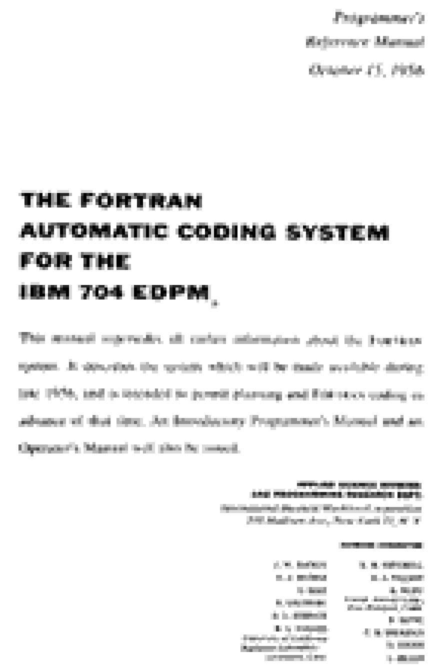
The next big step in programming began when an IBM team under John W. Backus created FORTRAN - FORmula TRANslator 1952. It could only be used on their own machine, the: IBM 704. But later versions for other machines, and platforms were sold soon after. Until long past 1960 different CPU's required an other kind instruction set to add a number, thus for each different machine a different compiler was needed. Typically the manual came years later in 1957!

Programming language

FORTRAN soon was called a programming language. This was because some characteristics of a language are met:

- It must have a vocabulary - list of words
- It must have a grammar - or syntax
- It must be unique, both in words and in context

All the above criteria were easily met for this - strictly defined- set of computer instructions.
An example:


Programmer's Reference Manual October 15, 1956

THE FORTRAN AUTOMATIC CODING SYSTEM FOR THE IBM 704 EDPM

Enter "C"

C came into being in the years 1969-1973 and was developed by Dennis Richey and David Kerningham both working at the Bell laboratories. And the magic word was portability. Parallel at the development of computer languages, Operating Systems (OS) were developed. These were needed because to create programs and having to write all machine specific instructions over and over again was a "waste of time" job. So by introducing the OS 'principle' almost all input and output tasks were taken over by the OS.
Such as:

- writing data to and from memory,
- saving data on disks,
- writing data to screens and printers,
- starting tapes,
- refreshing memory,
- scheduling specific tasks
- etcetera, etcetera.

As the common computer languages had trouble to be translated from one machine to another the OS's had to take the same hurdle every time a new machine was developed. The need and pressure for a common portable language was enormous. There were some unsuccessful projects aimed to solve this problem, like Multics, a joint venture of MIT, General Electric, and Bell Labs. And other developments at DOD's in different countries. But they all came either too late or became too complex to succeed.

But the demise of Multics inspired Dennis Ritchie and Brian Kernigham to develop C in 1972. This was and still is a very strict language that stayed close enough to the machine's internal logic and structure. This new language was reasonable well to read and understand by humans. And because of this combination the language is fast, compact and became very popular amongst system programmers and commercial software manufacturers.

With that language they also developed UNIX, a generic operating system.

The power of C was that the language had a small language base (vocabulary) but leaned heavily on what they called libraries. Libraries contain machine specific instructions to perform tasks, like the OS does. These lbraries were the only parts that had to be redesigned for different machines, or processor families. But, and that was C's strength, the programming interface/language remained the same. Portability was born. Source code could be reused, only to be recompiled when it had to run on other machines.

A classic example of C, printing "Hello World" on your screen:

```
/* helloworld.c */

main()
{
printf('Hello
World\n");
}
```

Soon other language manufacturers sprang on the bandwagon and the software industry leaped ahead. The industry took off like a rocket!

Predecessor(s)	**Year**	**Name**	**Chief Developer, Company**
Pre 1950			
*	~1840	*first program*	Ada Lovelace
*	1945	**Plankalkül** (concept)	Konrad Zuse
1950s			
*	1952	**A-0**	Grace Hopper
*	1954	**Mark I Autocode**	Tony Brooker
A-0	1954-1955	**FORTRAN "0"** (concept)	John W. Backus at IBM
A-0	1954	**ARITH-MATIC**	Grace Hopper
A-0	1954	**MATH-MATIC**	Grace Hopper
*	1954	**IPL V** (concept)	Allen Newell, Cliff Shaw, Herbert Simon
A-0	1955	**FLOW-MATIC**	Grace Hopper
IPL	1956-1958	**LISP** (concept)	John McCarthy
FLOW-MATIC	1957	**COMTRAN**	Bob Bemer
FORTRAN 0	1957	**FORTRAN "I"** (implementation)	John W. Backus at IBM
*	1957	**COMIT** (concept)	
FORTRAN I	1958	**FORTRAN II**	John W. Backus at IBM
FORTRAN	1958	**ALGOL 58 (IAL)**	International effort
*	1958	**IPL V** (implementation)	Allen Newell, Cliff Shaw, Herbert Simon
FLOW-MATIC, COMTRAN	1959	**COBOL** (concept)	The Codasyl Committee
IPL	1959	**LISP** (implementation)	John McCarthy
	1959	**TRAC** (concept)	Mooers
1960s			
ALGOL 58	1960	**ALGOL 60**	

FLOW-MATIC, COMTRAN	1960	**COBOL 61** (implementation)	The Codasyl Committee
*	1961	**COMIT** (implementation)	
FORTRAN II	1962	**FORTRAN IV**	
*	1962	**APL** (concept)	Iverson
ALGOL 58	1962	**MAD**	Arden, *et. al.*
ALGOL 60	1962	**SIMULA**(concept)	
FORTRAN II, COMIT	1962	**SNOBOL**	Griswold, *et al.*
ALGOL 60	1963	**CPL**	Barron, Strachey, *et al.*
SNOBOL	1962	**SNOBOL4** (concept)	Griswold, *et al.*
ALGOL 60	1963	**ALGOL 68** (concept)	van Wijngaarden, *et al.*
ALGOL 58	1963	**JOSS I**	Cliff Shaw, RAND
CPL, LISP	1964	**COWSEL**	Burstall, Popplestone
ALGOL 60, COBOL, FORTRAN	1964	**PL/I** (concept)	IBM
FORTRAN II, JOSS	1964	**BASIC**	Kemeny and Kurtz
	1964	**TRAC** (implementation)	Mooers
	1964?	**IITRAN**	
JOSS	1965	**TELCOMP**	BBN
JOSS I	1966	**JOSS II**	Chuck Baker, RAND
FORTRAN IV	1966	**FORTRAN 66**	
LISP	1966	**ISWIM**	Landin
ALGOL 60	1966	**CORAL66**	
CPL	1967	**BCPL**	Richards
FORTRAN, TELCOMP	1967	**MUMPS**	Massachusetts General Hospital
*	1967	**APL** (implementation)	Iverson
ALGOL 60	1967	**SIMULA 67** (implementation)	Dahl, Myhrhaug,Nygaard at Norsk Regnesentral
SNOBOL	1967	**SNOBOL4** (implementation)	Griswold, *et al.*
COWSEL	1968	**POP-1**	Burstall, Popplestone
	1968	**FORTH** (concept)	Moore
LISP	1968	**LOGO**	Papert
ALGOL 60	1969	**ALGOL 68** (implementation)	van Wijngaarden, *et al.*
ALGOL 60, COBOL, FORTRAN	1969	**PL/I** (implementation)	IBM
1970s			
	1970?	**FORTH** (implementation)	Moore
POP-1	1970	**POP-2**	
ALGOL 60	1971	**Pascal**	Wirth, Jensen
SIMULA 67	1972	**Smalltalk-72**	Xerox PARC
B, BCPL, ALGOL 68	1972	**C**	Ritchie
*	1972	**INTERCAL**	
2-level W-Grammar	1972	**Prolog**	Colmerauer
Pascal, BASIC	1973	**COMAL**	Christensen, Løfstedt
BASIC	1974	**GRASS**	DeFanti
Business BASIC	1974	**BASIC FOUR**	BASIC FOUR CORPORATION
LISP	1975	**Scheme**	Sussman, Steele
Pascal	1975?	**Modula**	Wirth
BASIC	1975	**Altair BASIC**	Gates, Allen
Smalltalk-72	1976	**Smalltalk-76**	Xerox PARC
C, FORTRAN	1976	**Ratfor**	Kernighan
*	1977	**FP**	John Backus
*	1977	**Bourne Shell** (*sh*)	Bourne
MUMPS	1977	**Standard MUMPS**	
FORTRAN IV	1978	**FORTRAN 77**	
Modula	1978?	**Modula-2**	Wirth
*	1978?	**MATLAB**	Moler at the University of New Mexico
*	1978	**VISICALC**	Bricklin, Frankston at VisiCorp
PL/I, BASIC, EXEC 2	1979	**REXX**	Cowlishaw
C, SNOBOL	1979	**Awk**	Aho, Weinberger, Kernighan
*	1979	**Vulcan dBase-II**	Ratliff
ALGOL 68	1979	**Green**	Ichbiah *et al.* at US Dept of Defense
1980s			

C, SIMULA 67	1980	**C with Classes**	Stroustrup
Smalltalk-76	1980	**Smalltalk-80**	Xerox PARC
Smalltalk, C	1982	**Objective-C**	Brad Cox
Green	1983	**Ada 83**	U.S. Department of Defense
C with Classes	1983	**C++**	Stroustrup
Pascal	1983	**Turbo Pascal**	Hejlsberg
BASIC	1983	**True BASIC**	Kemeny, Kurtz at Dartmouth University
sh	1984?	**Korn Shell** (*ksh*)	Dave Korn
*	1984	**Standard ML**	
dBase	1984	**CLIPPER**	Nantucket
LISP	1984	**Common Lisp**	Guy Steele and many others
1977MUMPS	1985	**1984 MUMPS**	
dBase	1985	**PARADOX**	Borland
Interpress	1985	**PostScript**	Warnock
BASIC	1985	**QuickBASIC**	Microsoft
	1986	**Miranda**	David Turner at University of Kent
	1986	**LabVIEW**	National Instruments
SIMULA 67	1986	**Eiffel**	Meyer
	1986	**Informix-4GL**	Informix
C	1986	**PROMAL**	
Smalltalk	1987	**Self** (concept)	Sun Microsystems Inc.
*	1987	**HyperTalk**	Apple
*	1987	**SQL-87**	
C, sed, awk, sh	1987	**Perl**	Wall
MATLAB	1988	**Octave**	
dBase-III	1988	**dBase-IV**	
Awk, Lisp	1988	**Tcl**	Ousterhout
REXX	1988	**Object REXX**	Simon Nash
Ada	1988	**SPARK**	Bernard A. Carré
Turbo Pascal	1989	**Turbo Pascal OOP**	Borland
C	1989	**Standard C89/90**	ANSI X3.159-1989 (adopted by ISO in 1990)
Modula-2	1989	**Modula-3**	Cardeli, et al.
Modula-2	1989	**Oberon**	Wirth

1990s

Oberon	1990	**Object Oberon**	Wirth
APL, FP	1990	**J**	Iverson, R. Hui at Iverson Software
Miranda	1990	**Haskell**	
1984 MUMPS	1990	**1990 MUMPS**	
Fortran 77	1991	**Fortran 90**	
Object Oberon	1991	**Oberon-2**	Wirth
ABC	1991	**Python**	Van Rossum
	1991	**Q**	
QuickBASIC	1991	**Visual Basic**	Alan Cooper at Microsoft
SQL-87	1992	**SQL-92**	
Turbo Pascal OOP	1992	**Borland Pascal**	
ksh	1993?	**Z Shell** (*zsh*)	
Smalltalk	1993?	**Self** (implementation)	Sun Microsystems Inc.
Forth	1993	**FALSE**	Oortmerrsen
FALSE	1993	**Brainfuck**	Mueller
HyperTalk	1993	**Revolution Transcript**	
HyperTalk	1993	**AppleScript**	Apple
APL, Lisp	1993	**K**	Whitney
Smalltalk, Perl	1993	**Ruby**	
	1993	**Lua**	Waldemar Celes *et al.* at Tecgraf, PUC-Rio
C	1993	**ZPL**	Chamberlain *et al.* at University of Washington
Lisp	1994	**Dylan**	many people at Apple Computer
Ada 83	1995	**Ada 95**	ISO
Borland Pascal	1995	**Delphi**	Anders Hejlsberg at Borland
C, SIMULA67 OR C++, Smalltalk	1995	**Java**	James Gosling at Sun Microsystems
1990MUMPS	1995	**1995 MUMPS**	
Self, Java	1995?	**LiveScript**	Brendan Eich at Netscape

Fortran 90	1996	**Fortran 95**	
REXX	1996	**NetRexx**	Cowlishaw
LiveScript	1997?	**JavaScript**	Brendan Eich at Netscape
SML 84	1997	**SML 97**	
PHP 3	1997	**PHP**	
Scheme	1997	**Pico**	Free University of Brussels
Smalltalk-80, Self	1997	**Squeak Smalltalk**	Alan Kay, *et al.* at Apple Computer
JavaScript	1997?	**ECMAScript**	ECMA TC39-TG1
C++, Standard C	1998	**Standard C++**	ANSI/ISO Standard C++
Prolog	1998	**Erlang**	Open Source Erlang at Ericsson
Standard C89/90	1999	**Standard C99**	ISO/IEC 9899:1999
2000s			
FP, Forth	2000	**Joy**	von Thun
C, C++	2000	**D**	Walter Bright at Digital Mars
C, C++, Java	2000	**C#**	Anders Hejlsberg at Microsoft (ECMA)
Whitespace	2003	**Whitespace**	Brady and Morris
Perl, C++	2003	**S2**	Fitzpatrick, Atkins
C#, ML, MetaHaskell	2003	**Nemerle**	University of Wroclaw
J, FL, K	2003	**NGL**	E. Herrera at Tlällian
Joy, Forth, Lisp	2003	**Factor**	Slava Pestov
Fortran 95	2004	**Fortran 2003**	
Python, C#, Ruby	2004	**Boo**	Rodrigo B. de Oliveira

History of Operating System

The history of computer operating systems recapitulates to a degree, the recent history of computing Operating systems provide a set of functions needed and used by most applications, and provide the necessary linkages to control a computer's hardware. Without an operating system, each program would have to have drivers for your video card, sound card, hard drive, and other peripherals.

The mainframe era

Early operating systems were very diverse, with each vendor producing one or more operating systems specific to their particular hardware. This state of affairs continued until the 1960s when IBM developed the System/360 series of machines; although there were enormous performance differences across the range, all the machines ran essentially the same operating system, OS/360.

OS/360 evolved to become successively MFT, MVT, SVS, MVS, MVS/XA, MVS/ESA, OS/390 and z/OS, that includes the UNIX kernel as well as a huge amount of new functions required by modern mission-critical applications running on the zSeries mainframes.

Minicomputers and the rise of UNIX

The UNIX operating system was developed at AT&T Bell Laboratories. Because it was essentially free in early editions, easily obtainable, and easily modified, it achieved wide acceptance. It also became a requirement within the Bell systems operating companies. Since it was written in a high level language, when that language was ported to a new machine architecture it was also able to be ported. This portability permitted it to become the choice for a second generation of minicomputers and the first generation of workstations. By widespread use it exemplified the idea of a operating system that was conceptually the same across various hardware platforms. It still was owned by AT&T and that limited its use to groups or corporations who could afford to license it.

The personal computer era: Apple, DOS and beyond

The development of microprocessors made inexpensive computing available for the small business and hobbyist, which in turn led to the widespread use of interchangeable hardware components using a common interconnection (such as the S-100, SS-50, Apple II, ISA, and PCI buses), and an increasing need for 'standard' operating systems to control them. The most important of the early OSes on these machines was Digital Research's CP/M-80 for the 8080 / 8085 / Z-80 CPUs. It was based on several Digital Equipment Corporation operating systems, mostly for the PDP-11 architecture. MS-DOS (or PC-DOS when supplied by IBM) was based originally on CP/M-80. Each of these machines had a small boot program in ROM which loaded the OS itself from disk.

The decreasing cost of display equipment and processors made it practical to provide graphical user interfaces for many operating systems, such as the generic X Window System that is provided with many UNIX systems, or other graphical systems such as Microsoft Windows, the RadioShack Color Computer's OS-9, Commodore's AmigaOS, Level II, Apple's Mac OS, or even IBM's OS/2. The original GUI was developed at Xerox Palo Alto Research Center in the early '70s (the Alto computer system) and imitated by many vendors.

History of Network

The earliest idea of a computer network intended to allow general communication between users of various computers was the ARPANET, the world's first packet switching network, which first went online in 1969.

The Internet's roots lie within the ARPANET, which not only was the intellectual forerunner of the Internet, but was also initially the core network in the collection of networks in the Internet, as well as an important tool in developing the Internet (being used for communication between the groups working on internetworking research).

Motivation for the Internet

The need for an internetwork appeared with ARPA's sponsorship, by Robert E. Kahn, of the development of a number of innovative networking technologies; in particular, the first packet radio networks (inspired by the ALOHA network), and a satellite packet communication program. Later, local area networks (LANs) would also join the mix.

Connecting these disparate networking technologies was not possible with the kind of protocols used on the ARPANET, which depended on the exact nature of the subnetwork. A wholly new kind of networking architecture was needed.

Early Internet work

Kahn recruited Vint Cerf of University of California, Los Angeles to work with him on the problem, and they soon worked out a fundamental reformulation, where the differences between network protocols were hidden by using a common internetwork protocol, and instead of the network being responsible for reliability, as in the ARPANET, the hosts became responsible. Cerf credits Herbert Zimmerman and Louis Pouzin (designer of the CYCLADES network) with important influences on this design. Some accounts also credit the early networking work at Xerox PARC with an important technical influence.

With the role of the network reduced to the bare minimum, it became possible to join almost any networks together, no matter what their characteristics, thereby solving Kahn's initial problem. (One popular saying has it that TCP/IP, the eventual product of Cerf and Kahn's work, will run over "two tin cans and a string".) A computer called a *gateway* (later changed to *router* to avoid confusion with other types of *gateway*) is provided with an interface to each network, and forwards packets back and forth between them.

Happily, this new concept was a perfect fit with the newly emerging local area ne tworks, which were revolutionizing communication between computers within a site.

Early growth

After ARPANET had been up and running for a decade, ARPA looked for another agency to hand off the network to. After all, ARPA's primary business was funding cutting-edge research and development, not running a communications utility. Eventually the network was turned over to the Defense Communications Agency, also part of the Department of Defense.

In 1983, TCP/IP protocols replaced the earlier NCP protocol as the principal protocol of the ARPANET; in 1984, the U.S. military portion of the ARPANet was broken off as a separate network, the MILNET. At the same time, Paul Mockapetris and Jon Postel were working on what would become the Domain Name System.

The early Internet, based around the ARPANET, was government-funded and therefore restricted to non-commercial uses such as research; unrelated commercial use was strictly forbidden. This initially restricted connections to military sites and universities. During the 1980s, the connections expanded to more educational institutions, and even to a growing number of companies such as Digital Equipment Corporation and Hewlett-Packard, which were participating in research projects, or providing services to those who were.

Another branch of the U.S. government, the National Science Foundation, became heavily involved in Internet research in the mid-1980s. The NSFNet backbone, intended to connect and provide access to a number of supercomputing centers established by the NSF, was established in 1986.
At the end of the 1980s, the U.S. Department of Defense decided the network was developed enough for its initial purposes, and decided to stop further funding of the core Internet backbone. The ARPANET was gradually shut down (its last node was turned off in 1989), and NSF, a civilian agency, took over responsibility for providing long-haul connectivity in the U.S.

In another NSF initiative, regional TCP/IP-based networks such as NYSERNet (New York State Education and Research Network) and BARRNet (Bay Area Regional Research Network), grew up and started interconnecting with the nascent Internet. This greatly expanded the reach of the rapidly growing network. On April 30, 1995 the NSF privatized access to the network they had created. It was at this point that the growth of the Internet really took off.

Commercialization and privatization

Parallel to the Internet, other networks were growing. Some were educational and centrally-organized like BITNET and CSNET. Others were a grass-roots mix of school, commercial, and hobby like the UUCP network.

During the late 1980s the first Internet Service Provider (ISP) companies were formed. Companies like PSINet, UUNET, Netcom, and Portal were formed to provide service to the regional research networks and provide alternate network access (like UUCP-based email and Usenet News) to the public. The first dial-up ISP, world.std.com, opened in 1989.

The interest in commercial use of the Internet became a hotly-debated topic. Although commercial use was forbidden, the exact definition of commercial use could be unclear and subjective. Everyone agreed that one company sending an invoice to another company was clearly commercial use, but anything less was up for debate. The alternate networks, like UUCP, had no such restrictions, so many people were skirting grey areas in the interconnection of the various networks.

Many university users were outraged at the idea of non-educational use of their networks. Ironically it was the commercial Internet service providers who brought prices low enough that junior colleges and other schools could afford to participate in the new arenas of education and research.

By 1994, the NSFNet lost its standing as the backbone of the Internet. Other competing commercial providers created their own backbones and interconnections. Regional NAPs (network access points) became the primary interconnections between the many networks. The NSFNet was dropped as the main backbone, and commercial restrictions were gone.

Early applications

E-mail existed as a message service on early time-sharing mainframe computers connected to a number of terminals. Around 1971 it developed into the first system of exchanging addressed messages between different, networked computers; in 1972 Ray Tomlinson introduced the "name@computer" notation that is still used today. E-mail turned into the Internet "killer application" of the 1980s.

The early e-mail system was not limited to the Internet. Gateway machines connected Internet SMTP e-mail with UUCP mail, BITNET, the Fidonet BBS network, and other services. Commercial e-mail providers such as CompuServe and The Source could also be reached.

The second most popular application of the early Internet was Usenet, a system of distributed discussion groups which is still going strong today. Usenet had existed even before the internet, as an application of Unix computers connected by telephone lines via UUCP. The Network News Transfer Protocol (NNTP), similar in flavor to SMTP, slowly replaced UUCP for the relaying of news articles. Today, almost all Usenet traffic is carried over high-speed NNTP servers.

Other early protocols include the File Transfer Protocol (1985), and Telnet (1983), a networked terminal emulator allowing users on one computer to log in to other computers.

Host naming and the DNS

One of the scalability limitations of the early Internet was that there was no distributed way to name hosts, other than with their IP addresses. The Network Information Centre (NIC) maintained a central hosts file, giving names for various reachable hosts. Sites were expected to download this file regularly to have a current list of hosts.

In 1984, Paul Mockapetris devised the Domain Name System (DNS) as an alternative. Domain names (like "wikipedia.org") provided names for hosts that were both globally unique (like IP addresses), memorable (like hostnames), and distributed -- sites no longer had to download a hosts file.

Domain names quickly became a feature of e-mail addresses -- replacing the older bang path notation -- as well as other services. Many years later, they would become the central part of the World Wide Web's URLs, for which see below.

Standards and control

The Internet has developed a significant subculture dedicated to the idea that the Internet is not owned or controlled by any one person, company, group, or organization. Nevertheless, some standardization and control is necessary for anything to function.

Many people wanted to put their ideas into the standards for communication between the computers that made up this network, so a system was devised for putting forward ideas. One would write one's ideas in a paper called a "Request for Comments" (RFC for short), and let everyone else read it. People commented on and improved those ideas in new RFCs.

With its basis as an educational research project, much of the documentation was written by students or others who played significant roles in developing the network (as part of the original Network Working Group) but did not have official responsibility for defining standards. This is the reason for the very low-key name of "Request for Comments" rather than something like "Declaration of Official Standards".

The first RFC (RFC1) was written on April 7th, 1969. As of 2004 there are over 3500 RFCs, describing every aspect of how the Internet functions.

The liberal RFC publication procedure has engendered confusion about the Internet standardization process, and has led to more formalization of official accepted standards. Acceptance of an RFC by the RFC Editor for publication does not automatically make the RFC into a standard. It may be recognized as such by the IETF only after many years of experimentation, use, and acceptance have proven it to be worthy of that designation. Official standards are numbered with a prefix "STD" and a number, similar to the RFC naming style. However, even after becoming a standard, most are still commonly referred to by their RFC number.

The Internet standards process has been as innovative as the Internet technology. Prior to the Internet, standardization was a slow process run by committees with arguing vendor-driven factions and lengthy delays. In networking in particular, the results were monstrous patchworks of bloated specifications.

The fundamental requirement for a networking protocol to become an Internet standard is the existence of at least two existing, working implementations that inter-operate with each other. This makes sense in retrospect, but it was a new concept at the time. Other efforts built huge specifications with many optional parts and then expected people to go off and implement them, and only later did people find that they did not inter-operate, or worse, the standard was completely impossible to implement.

In the 1980s, the International Organization for Standardization (ISO) documented a new effort in networking called Open Systems Interconnect or OSI. Prior to OSI, networking was completely vendor-developed and proprietary. OSI was a new industry effort, attempting to get everyone to agree to common network standards to provide multi-vendor interoperability. The OSI model was the most important advance in teaching network concepts. However, the OSI protocols or "stack" that were specified as part of the project were a bloated mess. Standards like X.400 for e-mail took up several large books, while Internet e-mail took only a few dozen pages at most in RFC-821 and 822. Most protocols and specifications in the OSI stack, such as token-bus media, CLNP packet delivery, FTAM file transfer, and X.400 e-mail, are long-gone today; in 1996, ISO finally acknowledged that TCP/IP had won and killed the OSI project. Only one OSI standard, X.500 directory service, still survives with significant usage, mainly because the original unwieldy protocol has been stripped away and effectively replaced with LDAP.

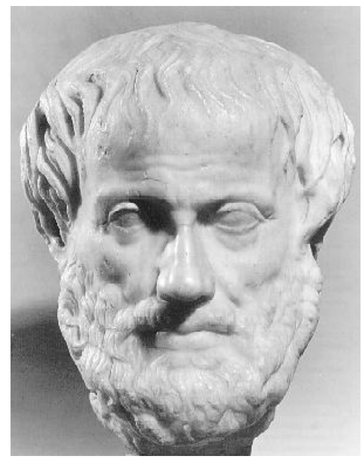
Some formal organization is necessary to make everything operate. The first central authority was the NIC (Network Information Centre) at SRI (Stanford Research Institute in Menlo Park, California).

Pioneers of Computing

Many scientist and engineers have contributed in the development of computing science. This chapter covers all the major scientists and their works, who have played a significant role in the development of computing.

Some of the major Pioneers of Computing and their contributions are described here.

Aristotle (384 BC, Stagira, Greece, 322 BC, Athens, Greece)



He was one of the greatest ancient Greek philosophers of his time. Many of his thoughts have become the back bone of computing and artificial Intelligence. His work in the natural and social sciences greatly influenced virtually every area of modern thinking.

Biography

Aristotle was born in 384 BC in Stagira, on the northwest coast of the Aegean Sea. His father was a friend and the physician of the king of Macedonia, and the lad spent most of his boyhood at the court. At 17, he went to Athens to study. He enrolled at the famous Academy directed by the philosopher Plato.

Aristotle threw himself wholeheartedly into Plato's pursuit of truth and goodness. Plato was soon calling him the "mind of the school." Aristotle stayed at the Academy for 20 years, leaving only when his beloved master died in 347 BC. In later years he renounced some of Plato's theories and went far beyond him in breadth of knowledge. Aristotle became a teacher in a school on the coast of Asia Minor. He spent two years studying marine biology on Lesbos.

In 342 BC, Philip II invited Aristotle to return to the Macedonian court and teach his 13-year-old son Alexander. This was the boy who was to become conqueror of the world (see Alexander the Great). No one knows how much influence the philosopher had on the headstrong youth. After Alexander became king, at 20, he gave his teacher a large sum of money to set up a school in Athens.

The Peripatetic School

In Athens Aristotle taught brilliantly at his school in the Lyceum. He collected the first great library and established a museum. In the mornings he strolled in the Lyceum gardens, discussing problems with his advanced students. Because, he walked about while teaching, Athenians called his school the Peripatetic (which means "to walk about") school. He led his pupils in research in every existing field of knowledge. They dissected animals and studied the habits of insects. The science of observation was new to the Greeks. Hampered by lack of instruments, they were not always correct in their conclusions.

One of Aristotle's most important contributions was defining and classifying the various branches of knowledge. He sorted them into physics, metaphysics, psychology, rhetoric, poetics, and logic, and thus laid the foundation of most of the sciences of today.

Aristotle's Works

After his death, Aristotle's writings were scattered or lost. In the early Middle Ages the only works of his known in Western Europe were parts of his writings on logic. They became the basis of one of the three subjects of the medieval trivium--logic, grammar, and rhetoric. Early in the 13th century other books reached the West. Some came from Constantinople; others were brought by the Arabs to Spain. Medieval scholars translated them into Latin.

Muhammad ibn Musa al-Khwarizmi (800 - 847, Baghdad, Iraq)

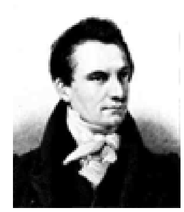
Muhammad ibn Musa al-Khwarizmi was born sometime before 800 A.D. in an area not far from Baghdad and lived at least until 847. He wrote his *Al-jabr wa'l muqabala* (from which our modern word "algebra" comes) while working as a scholar at the House of Wisdom in Baghdad. In addition to this treatise, al-Khwarizmi wrote works on astronomy, on the Jewish calendar, and on the Hindu numeration system. The English word "algorithm" derives from the Latin form of al-Khwarizmi's name.

Henry Briggs (Feb 1561 Worley Wood, Yorkshire)

Briggs is especially known for his publication of tables of logarithms to the base 10, first Logarithmorum chilias prima, 1617, and later Arithmetica logarithmetica, 1624. He also composed a work on trigonometry (basically tables, both of the functions and of the logs of sines and tangents) that was left unfinished at his death; Gellibrand completed and published it. And he left quite a few mathematical manuscripts that remained unpublished.

Charles Babbage (Dec 26, 1791 Teignmouth, Devonshire, UK, Oct 18, 1871, London, UK)

Charles Babbage "Father of computer" was a great scientist who lived in the 18th century. He was the designer of Difference Engine and the Analytical engine.



Biography

Charles Babbage was born in London on December 26, 1792 (3), the son of Benjamin Babbage, a London banker. As a youth Babbage was his own instructor in algebra, of which he was passionately fond, and was well-read in the continental mathematics of his day. Upon entering Trinity College, Cambridge, in 1811, he found himself far in advance of his tutors in mathematics.

With Herschel, Peacock, and others, Babbage founded the Analytical Society for promoting continental mathematics and, reforming the mathematics of Newton, then taught at the university.

In his twenties Babbage worked as a mathematician, principally in the calculus of functions. He was elected a Fellow of the Royal Society, in 1816, and played a prominent part in the foundation of the Astronomical Society (later Royal Astronomical Society) in 1820. It was about this time that Babbage first acquired the interest in calculating machinery that became his consuming passion for the remainder of his life.

Throughout his life Babbage worked in many intellectual fields typical of his day, and made contributions that would have assured his fame irrespective of the Difference and Analytical Engines.

Prominent among his published works are:

- *A Comparative View of the Various Institutions for the Assurance of Lives* (1826); an actuarial paper,
- *Table of Logarithms of the Natural Numbers from 1 to 108, 000* (1827),
- *Reflections on the Decline of Science in England* (1830),
- *On the Economy of Machinery and Manufactures* (1832),
- *Ninth Bridgewater Treatise* (1837), and
- the autobiographical *Passages from the Life of a Philosopher* (1864).

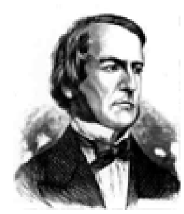
Babbage occupied the Lucasian chair of mathematics at Cambridge from 1828 to 1839. He played an important role in the establishment of the Association for the Advancement of Science and the Statistical Society (later Royal Statistical Society).

Despite his many achievements, the failure to construct his calculating machines, and in particular the failure of the government to support his work, left Babbage in his declining years a disappointed and embittered man. He died at his home in Dorset Street, London, on October 18, 1871.

Honors and Awards

- Elected Fellow of the Royal Society - 1816
- First gold medal of the Astronomical Society of London

George Boole (Nov 2, 1815 - Dec 8, 1864, Lincoln, England)



George Boole laid the groundwork for what we know today as Information Theory through the publication of his masterpiece, *An Investigation of the laws of Thought, on which are founded the Mathematical Theories of Logic and Probabilities.* In this work, published when the author was 39, Boole reduced logic to an extremely simple type of algebra, in which 'reasoning' is carried out through manipulating formulas simpler than those used in second-year traditional algebra. His theory of logic, which recognizes three basic operations - AND, OR and NOT - was to become germane to the development of telephone circuit switching and the design of electronic computers.

Ada Byron (Born: London, England, Dec 10, 1815, Died: London, England, Nov 27, 1852)

Analyst, Metaphysician, and Founder of Scientific Computing

Ada Byron was the daughter of a brief marriage between the Romantic poet Lord Byron and Anne Isabelle Milbanke, who separated from Byron just a month after Ada was born. Four months later, Byron left England forever. Ada never met her father (who died in Greece in 1823) and was raised by her mother, Lady Byron. Her life was an apotheosis of struggle between emotion and reason, subjectivism and objectivism, poetics and mathematics, ill health and bursts of energy.

Lady Byron wished her daughter to be unlike her poetical father, and she saw to it that Ada received tutoring in mathematics and music, as disciplines to counter dangerous poetic tendencies. But Ada's complex inheritance became apparent as early as 1828, when she produced the design for a flying machine. It was mathematics that gave her life its wings.

Lady Byron and Ada moved in an elite London society, one in which gentlemen not members of the clergy or occupied with politics or the affairs of a regiment were quite likely to spend their time and fortunes pursuing botany, geology, or astronomy. In the early nineteenth century there were no "professional" scientists (indeed, the word "scientist" was only coined by William Whewell in 1836)--but the participation of noblewomen in intellectual pursuits was not widely encouraged.

One of the gentlemanly scientists of the era was to become Ada's lifelong friend. Charles Babbage, Lucasian professor of mathematics at Cambridge, was known as the inventor of the Difference Engine, an elaborate calculating machine that operated by the method of finite differences. Ada met Babbage in 1833, when she was just 17, and they began a voluminous correspondence on the topics of mathematics, logic, and ultimately all subjects.

In 1835, Ada married William King, ten years her senior, and when King inherited a noble title in 1838, they became the Earl and Countess of Lovelace. Ada had three children. The family and its fortunes were very much directed by Lady Byron, whose domineering was rarely opposed by King.

Babbage had made plans in 1834 for a new kind of calculating machine (although the Difference Engine was not finished), an Analytical Engine. His Parliamentary sponsors refused to support a second machine with the first unfinished, but Babbage found sympathy for his new project abroad. In 1842, an Italian mathematician, Louis Menebrea, published a memoir in French on the subject of the Analytical Engine. Babbage enlisted Ada as translator for the memoir, and during a nine-month period in 1842-43, she worked feverishly on the article and a set of Notes she appended to it. These are the source of her enduring fame.

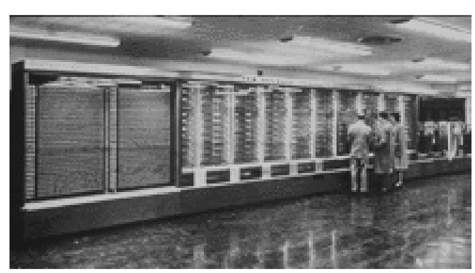
Ada called herself "an Analyst (& Metaphysician)," and the combination was put to use in the Notes. She understood the plans for the device as well as Babbage but was better at articulating its promise. She rightly saw it as what we would call a general-purpose computer. It was suited for "developping [sic] and tabulating any function whatever. . . the engine [is] the material expression of any indefinite function of any degree of generality and complexity." Her Notes anticipate future developments, including computer-generated music.

Ada died of cancer in 1852, at the age of 37, and was buried beside the father she never knew. Her contributions to science were resurrected only recently, but many new biographies* attest to the fascination of Babbage's "Enchantress of Numbers."

Howard Hathaway Aiken (March 8, 1900- March 14, 1973)

Achievement (Invented the Mark I)

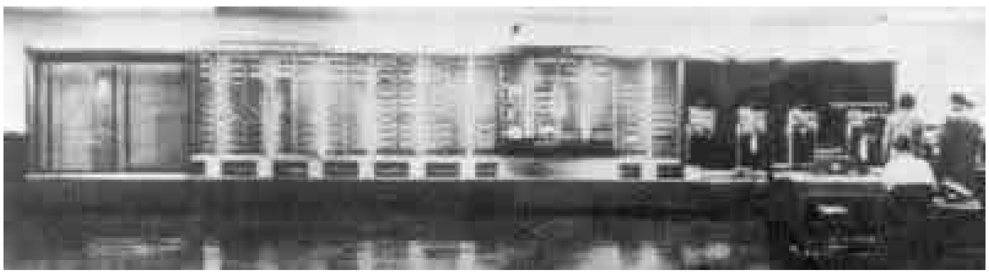
Howard Hathaway Aiken was born March 8, 1900 in Hoboken, New Jersey. However he grew up in Indianapolis, Indiana where he attended the Arsenal Technical High School. After high school he studied at the University of Wisconsin where he received a bachelor's degree in electrical engineering. During college Aiken worked for the Madison Gas Company; after graduation he was promoted to chief engineer there.



Hardware Mark I

In 1935 Aiken decided to return to school. In 1939 he received a Ph.D. from Harvard University. It was while working on his doctoral thesis in physics that Aiken began to think about constructing a machine to help with the more tedious of calculations. Aiken began to talk about his idea and to do some research into what could be done. With some help from colleagues at the university, Aiken succeeded in convincing IBM to fund his project.

The idea was to build an electro mechanical machine that could perform mathematical operations quickly and efficiently and allow a person to spend more time thinking instead of laboring over tedious calculations. IBM was to build the machine with Aiken acting as head of the construction team and donate it to Harvard with the requirement that IBM would get the credit for building it. The constructing team was to use machine components that IBM already had in existence.



It took seven years and a lot of money to finally get the machine operational. Part of the delay was due to the intervention of World War II. Officially the computer was called the IBM Automatic Sequence Controlled Calculator but most everyone called it the Mark I. After completing the Mark I, Aiken went on to produce three more computers, two of which were electric rather than electromechanical.

More important than the actual computer (whose major purpose was to create tables), was the fact that it proved to the world that such a machine was more than just fancy, it was a practical purpose machine. Perhaps more important than the invention of Mark I was Aiken's contribution to academia. He started the first computer science academic program in the world.

Aiken retired from teaching at Harvard in 1961 and moved to Ft. Lauderdale, Florida. He died March 14, 1973 in St. Louis, Missouri.

Charles Ranlett Flint (1850, Maine, USA, 1934, USA)

A venture capitalist. Flint merged several companies together in what will become known as IBM. One of the largest computer manufacturers on the world formed in the 20th century.

Paul Allen (January 21, 1953, Seattle, USA) and Bill Gates

He founded Microsoft together with Bill Gates

Biography

At Lakeside School, Paul Allen (14 years old) and friend Bill Gates (12 years old) became early computer enthusiasts. Allen went on to attend Washington State University, though he dropped out after two years to pursue his and Gates's dream of writing software commercially for the new "personal computers".

Allen and Gates and a small group of other Lakeside students begin programming in BASIC, using a teletype terminal. Helps teach computer course to junior high students at Lakeside.

Graduates from Lakeside; enrolls at Washington State University. Allen and Gates buy an Intel 8008 chip for $360 and build a computer to measure traffic.They launch their first company, Traf-O-Data. He was hired as a programmer by Honeywell in Boston. Allen and Gates write the first microcomputer BASIC for the Altair, a computer kit based on Intel's new 8080 chip. They move to Albuquerque, N.M., where Altair's producer MITS makes Allen its associate director of software. Allen divides his time between MITS and a new company he and Gates have started to develop and market microcomputer languages: Micro Soft.

Apple commissions Microsoft to supply a version of its BASIC for the hot-selling Apple II. Radio Shack buys a Microsoft BASIC for its TRS-80. Microsoft moves from Albuquerque to Bellevue, Wash. Microsoft agrees to develop and license DOS and BASIC to IBM for its new personal computer. Gates and Allen discuss graphical user interfaces, planting the seeds that will become Windows.

Allen and Gates found Microsoft (initially "Micro Soft") in Albuquerque, New Mexico, and begin selling a BASIC interpreter. Allen spearheads a deal for Microsoft to buy an operating system called QDOS for $50,000. Microsoft wins a contract to supply it to for use as the operating system of IBM's new PC. This becomes the foundation for Microsoft's remarkable growth.

Allen was forced to resign from Microsoft in 1983 after being diagnosed with Hodgkin's disease which was successfully treated by several months of radiation therapy. In 1984 he founded Asymetrix, a software development company based in Belleuve, Washington, to make application development tools that nonprogrammers can use. Asymetrix later went on to become Click2learn.com and yet later merged with Docent to become Sum Total System (2004). In the 1990's the company began to specialize in software for developing and delivering computer-based learning.

1992 Allen started Starwave, a producer of online content sites. Starwave did such great work for ESPN SportsZone and ABCNews.com that Disney (NYSE: DIS) bought it for a total of $350 million last year.

1998 In April Allen buys Marcus Cable, the nation's 10th largest cable company, for $2.8 billion--his biggest investment to date. Also this year Allen grabbed a stake of the Internet video-sales market with his purchase of Hollywood Entertainment. And he took another software group public. This time it's Asymetrix Learning Systems, maker of products for online classes.

On September 28, 2000 - Microsoft Corp. announced that Paul Allen is assuming a new role as senior strategy adviser to top Microsoft executives. The company also announced that Allen and Richard Hackborn have decided not to seek re-election to Microsoft's board of directors at the company's November shareholder meeting.

In December 2003 he announced that he was the sponsor behind the SpaceShipOne private rocket plane venture from Scaled Composites, as part of the ANSARI X PRIZE competition. In June 2004, SpaceShipOne became the first sucessful commercial spacecraft when it passed the 100 kilometer threshold of space.

In September 2003, Allen founded the Allen Institute of Brain Science pledging $100 million in seed money to the Seattle-based organization. Its inaugural project is the Allen Brain Atlas, a map of the human brain which will be made publicly accessible. The Brain Atlas is a component of the loosely formed Human Cognome Project.

Starting in 2003, Vulcan Ventures began funding Project Halo, an attempt to apply Artificial Intelligence techniques to the problem of producing a digital Aristotle that might serve as a mentor, providing comprehensive access to the world's knowledge. Allen is a major contributor to the SETI, or Search for Extra-Terrestrial Intelligence project. He is also the founder of the Experience Music Project, originally inspired by his interest in a museum to house his considerable collection of Jimi Hendrix memorabilia. Allen runs a venture capital firm, Vulcan Ventures, and has created the Experience Music Project, a museum of music history, in Seattle, Washington.

He owns (through Rose City Radio Corporation) some Portland radio stations. When he heard Seattle's Cinerama movie theater was about to shut down, he bought, restored, and updated it into a showplace for movies of all formats. He is also one of the principal financiers behind the SETI project, having stepping in to rescue the project when NASA stopped funding it in the 1990s.

Currently Allen is the owner of the Portland Trail Blazers (an NBA basketball team) and the Seattle Seahawks of the National Football League. He also owns Rose Garden Arena, the home court of the NBA Blazers team. Due to declining attendance in 2002 and 2003, as well as difficulties renegotiating the terms of a 1993 loan, the Rose Garden corporation filed for bankruptcy on February 27, 2004.

In June of 2004, Allen opened the Science Fiction Museum and Hall of Fame, located at the Experience Music Project.

Gene Amdahl **(1922, South Dakota, USA)**

Achievement

One of the original architects of the business mainframe computer, including IBM's System/360 computer line, Amdahl started the IBM-compatible market when he left IBM to found Amdahl Corporation. Amdahl's work has been called brilliant and genius by his peers. The Times of London named him one of the "1,000 Makers of the 20th Century" in 1991, and mainframe magazine Computerworld considered Amdahl one of the 25 people "who changed the world." He is the founder of four companies, Amdahl Corporation, Trilogy Systems (now part of Elxsi Corporation), Andor Systems, and Commercial Data Servers (CDS).

Biography

Gene Myron Amdahl was born in South Dakota in 1922. After serving two years in the U.S. Navy during World War II, where he learned electronics, and taking a course in computer programming, he received a bachelor's degree in engineering physics at South Dakota State University in 1948. In 1952 he completed his doctorate in theoretical physics at the University of Wisconsin, where he designed his first computer, the Wisconsin Integrally Synchronized Computer (WISC).

He began his career with IBM in 1952, and became the chief design engineer of the IBM 704. In 1955, Amdahl worked with others to design the Datatron, which led to a computer called the Stretch, and eventually became the IBM 7030, a computer that used the new transistor technology. In 1956, after just four short years with IBM, Amdahl became unhappy with the company and quit. After five years of working for other computer companies, he returned to IBM in 1960.

During the 1960s, Amdahl gained recognition as the principle architect of IBM's impressive System 360 series of mainframe computers. The IBM System 360 was based on the Stretch, which Amdahl had worked on in 1955. The 360 series was one of the greatest success stories in the computer industry and became the main ingredient to IBM's enormous profitability in the late 1960s.

Amdahl became an IBM Fellow and was able to pursue his own research projects. In 1969, he was director of IBM's Advanced Computing Systems Laboratory in Menlo Park, California. He recommended that the laboratory be shut down, which IBM did, and presented his ideas about the internal barriers that prevented IBM from shooting for the high end of computer development. Although his ideas were accepted, IBM executives refused to change policies and Amdahl left IBM again.

In 1970, Amdahl formed his own company, Amdahl Corporation, in Sunnyvale, California. His plan was to compete head-to-head with IBM in the mainframe market. Most industry analysts considered this to be career suicide and gave his start-up company very little chance of surviving. But survive it did, and actually prospered. Instead of creating a rival system to IBM, Amdahl created discounted computers that could be substituted for name brand models and run the same software. Basically, he designed the first computer clones, known then as "plug-to-plug compatibles." Amdahl became the most celebrated entrepreneur in the computer industry for awhile. The only major criticism that was raised about Amdahl Corporation at the time was that Amdahl took start-up money from Fujitsu Ltd. of Japan in exchange for American mainframe technology.

In 1975, Amdahl Corporation shipped its first computer, the Amdahl 470 V/6. Over the next few years, Amdahl and IBM leap-frogged each other with faster, smaller, and cheaper computers. In 1979, Gene Amdahl began moving away from Amdahl Corporation when he resigned his post as chairman. He became chairman emeritus for less than a year, leaving Amdahl Corporation in 1980 to found Trilogy Systems Corporation.

With the success of Amdahl Corporation, Amdahl had no trouble interesting investors in this new company and easily raised $230 million in start-up money. Again, his plan was to compete with IBM, and also Amdahl Corporation, in the high-end mainframe computer market. In addition, Amdahl planned to completely redesign the semiconductor chips that powered the computers. His dream was to combine the functions of 100 separate chips onto one superchip that would work faster and more efficiently than the multiple chips.

Unfortunately, Trilogy was hounded by disasters. Torrential rains delayed construction of the chip plant, then invaded the air conditioning, destroying the clean room atmosphere and all the chips currently being created. At this point, Amdahl had spent one-third of the start-up money with nothing to show for it. To save Trilogy, Amdahl spent the remainder of the money to acquire Elxsi Corporation, a computer manufacturer, in 1985. The new company continued to flounder and never achieved great success. In 1989, Amdahl stepped down as chairman of Elxsi to devote more time to his next venture.

In 1987, Amdahl founded his third company, this one called Andor Systems after the "and" and "or" logic gates of computer circuitry. This time his aim was to build computers that would compete with IBM's smaller mainframes. Industry analysts uniformly gave the company very little chance of success. But Amdahl felt he had an edge -- he could make small mainframe computers more cheaply than IBM. He could use new technology that allowed him to pack the computer's central processor onto one board, rather than the several used by IBM, and he redesigned the compiler to work more quickly and efficiently. These innovations allowed Andor's computers to take up less space and generate less heat, a distinct advantage to customers who no longer would need giant air-conditioned rooms in which to place their computers.

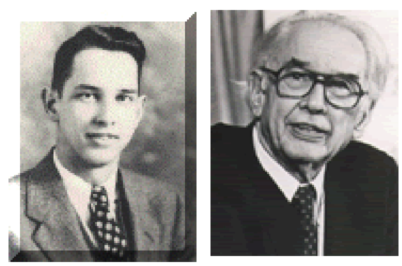
But Andor was plagued by bad chips, causing a delay of almost two years before the first computers hit the market. Meanwhile, IBM came out with its own midsize computer using some of the same technology employed by Andor. To survive, Andor had to come up with other peripheral products that it could quickly get on the market. But Andor never achieved the success it was after with the small mainframes, and in 1991 it had scaled back products to include only a data backup system. By 1994, the company had yet to turn a profit. Eventually, the company declared bankruptcy.

In 1996, at the age of 74, he started his fourth company, this one called Commercial Data Servers (CDS). Through CDS, Amdahl intends to distribute IBM-compatible, PC-based mainframes that use cryogenically-cooled CMOS processors and a new processor design that he created. CDS is targeting its products at companies that need the capabilities and selling price of a smaller mainframe, a market that CDS believes IBM and other manufacturers aren't serving adequately.

Gene Amdahl continues his quest to merge mainframe technologies with the more popular PC technology. Though many find these two areas incompatible (mainframe means centralized, controlled computing; PCs are for individual computing), Amdahl won't give in to those who believe mainframes are dinosaurs that have outlived their usefulness. And, apparently he doesn't intend to ever give up.

John Vincent Atanasoff (October 4, 1903, Hamilton, New York, USA, June 15, 1995, Monrovia, USA)

Achievement



John Vincent Atanasoff was born on 4 October 1903 in Hamilton, New York. He is the inventor of the electronic digital computer. He is, along with being an Inventor, a Mathematical Physicist and a Businessman.

Biography

In any science field, there needs to be a person with the vision to define the future. John Vincent Atanasoff was a genius with such a vision. He developed the first electronic digital computer that has dramatically changed our lives. John Vincent Atanasoff gave birth to the field of electronic computing. In doing so, he also gave birth to a new era, an era of computers.

Today, the computer is an essential part of every person as well as every business. We cannot imagine our lives without a computer being involved. Turning on the TV, making a telephone call, and typing up a report all involves the use of a computer. The invention of the computer meant that technology could improve at a faster rate and our lives became more convenient and more safe.

Take for instance the use of computers in our cars. Anti-lock brakes, air bags, and fuel injections are all controlled by a computer. These advancements make the car safer and more reliable. Computers can also be found in banks, schools, airplanes, businesses, space shuttles, satellites, and numerous other things. In today's society, almost everything involves the use of a computer.

The electronic age is the direct result of the invention of the computer. Never before in the history of humanity has there been an invention that grown so quickly as the computer has. Within the last twenty years, the speed and power of the computer has grown at an exponential rate.

When John Vincent Atanasoff invented the computer, he probably did not know how much of an impact it would have on people's lives. Computers will be involved in every aspect of technology, and it will continue to be a part of technologies to come. The capabilities of computers are advancing every day. Soon, a computer will become more like the human brain than an electronic machine. Computers will take us to Mars, and get us back safely. Computers will always be on the edge of technology and anyone that learns to harness its power will be an important part of the future. Every aspect of our lives has changed because on the computer and its inventor, John Vincent Atanasoff.

The first electronic computer with vacuum tubes is constructed by John Atanasoffen Clifford Berry of the Iowa State College. The Atanasoff-Berry computer was the first digital computer, built during 1937-1942, and introduced the concepts of binary arithmetic, regenerative memory, and logic circuits. This machine will never reach the production stage and remain a prototype.

John Vincent Atanasoff is the first s on of John Atanasoff and Iva Lucena Purdy. At a very early age, John Vincent Atanasoff had a great interest in mathematics. When John Vincent was about ten years old, he was curious in a Dietzgen slide rule that his father had bought. John Vincent read the instructions on how to use the slide rule, and he became more interested in the mathematical principles of the slide rule. With the help of his mother, John Vincent began to study a college algebra book that belonged to his father.

In the years that followed, John Vincent's family moved to Old Chicora, Florida. John Vincent studied at Mulberry High School and graduated in two years. He received A's is all of his science and math courses. John Vincent did not enter into college right away because he wanted to work and save money. In 1921 John Vincent entered the University of Florida as an undergraduate. John Vincent graduated from the University of Florida in 1925 with a Bachelor of Science degree in electrical engineering. He received straight A's as an undergraduate.

John Vincent Atanasoff then went to Iowa State College to pursue his master's degree. At Iowa State, John Vincent met his future wife, Lura Meeks. At the time, John Vincent did not know that she was three years his senior. John Vincent received his master's degree in mathematics from Iowa State College in 1926. Within a few days of receiving his degree, John Vincent and Lura Meeks were married.

After receiving his master's degree, John Vincent went to the University of Wisconsin for his doctorate in theoretic physics. In the same year that John Vincent was accepted as a doctoral candidate, his wife gave birth to their eldest daughter, Elsie. In 1930, John Vincent Atanasoff received his Ph.D. as a theoretic physicist from the University of Wisconsin. Dr. Atanasoff then returned to Iowa State College as an assistant professor in mathematics and physics in 1936.

Dr. Atanasoff had always been interested in finding new ways to perform mathematical computations faster. Dr. Atanasoff examined many of the computational devices that existed at that time. These included the Monroe calculator and the International Business Machines (IBM) tabulator. Dr. Atanasoff concluded that these devices were slow and inaccurate.

After being promoted to associate professor of mathematics and physics, Dr. Atanasoff began to envision a computational device that was "digital." He believed that analog devices were too restrictive and could not get the type of accuracy he wanted. The idea of building an electronic digital computer came to him while he was sitting in a tavern. Dr. Atanasoff came up with four principles for his electronic digital computer.

- He would use electricity and electronics as the medium for the computer.
- In spite of custom, he would use base-two numbers (the binary system for his computer.
- He would use condensers for memory and would use a regenerative or "jogging" process to avoid lapses that might be caused by leakage of power.

- He would compute by direct logical action and not by enumeration as used in analog calculating devices. (Mollenhoff, 34)

As Dr. Atanasoff worked on his computer project, he asked a colleague to recommend a graduate student to assist him with his project. The graduate student that was introduced to him was Clifford Berry. Berry was gifted electrical engineer and had very similar background as Dr. Atanasoff did. They both got along almost immediately.

In December 1939, the first prototype of the Atanasoff Berry Computer (ABC) was ready. The ABC showed some of the potentials of a compute r and it amazed the University. So in 1939, Dr. Atanasoff and his assistant Clifford Berry built the world's first electronic digital computer. With the first prototype working well, Dr. Atanasoff wanted to improve on prototype as well as get patents for the Atanasoff Berry Computer. Obtaining the patents were a slow process that ultimately caused Dr. Atanasoff the recognition that he deserved.

In 1940 Dr. Atanasoff attended a lecture given by Dr. John W. Mauchly. They talked for some time and Dr. Mauchly was very intrigued with Dr. Atanasoff's electronic digital computer. Dr. Mauchly wanted to see the ABC for himself and Dr. Atanasoff agreed. This decision by Dr. Atanasoff would be a mistake since Dr. Mauchly later used many of Dr. Atanasoff's ideas in the design of the ENIAC. The ENIAC is falsely considered by most people as the world's first electronic digital computer designed by Dr. Mauchly and Dr. Eckert. Charges of piracy were later brought against Dr. Mauchly, co-inventor of the ENIAC. A long trial followed and it was not until 1972 that Dr. Atanasoff was given the recognition he so deserved. U.S. District Judge Earl R. Larson ruled that the ENIAC was "derived" from the ideas of Dr. Atanasoff. Although Judge Larson did not explicitly say that Dr. Mauchly "stole" Dr. Atanasoff's ideas, Judge Larson did say that Dr. Mauchly had used many of Dr. Atanasoff's ideas on the ABC to design the ENIAC. When the trial finally ended, Dr. Atanasoff was given credit as the inventor of the electronic digital computer.

Clark Mollenhoff in his book, Atanasoff, Forgotten Father of the Computer, details the design and construction of the Atanasoff-Berry Computer with emphasis on the relationships of the individuals. Alice and Arthur Burks in their book, The First Electronic Computer: The Atanasoff Story, describe the design and construction of the ABC and provide a more technical perspective. Numerous articles provide additional information. In recognition of his achivement, Atanasoff was awarded the National Medal of Technology by President George Bush at the White house on November 13, 1990.

Dr. John Vincent Atanasoff died 15 June 1995 of a stroke at his home in Monrovia, Md. He was 91 years old. Although Dr. Atanasoff was not able to get a patent for the ABC, he held 32 patents for his other inventions

Gordon Bell (August 19, 1934, Kirksville, Missouri)

Achievement

While working for Digital Equipment Corporation (DEC) in the 1960s, Gordon Bell helped build the PDP series of minicomputers, the first minicomputers introduced to the commercial data processing market. He also oversaw the development of one of the industry's most successful computer lines, DEC's VAX series. He is currently one of several experts employed by Microsoft to direct that company's future path.

Biography

Bell was born on August 19, 1934 in Kirksville, Missouri. He received a B.S. and M.S. in electrical engineering from the Massachusetts Institute of Technology (MIT) in 1956 and 1957, respectively. After receiving his master's degree, he worked at MIT's Engineering Speech Communications Laboratory. During his time at MIT, Bell knew two other MIT computer engineers named Ken Olsen and Harlan Anderson and decided to start their own company, Digital Equipment Corporation. By 1959 Olsen and Anderson were starting to build a new, smaller computer that was the precursor to the minicomputer. Bell was asked to join DEC as an engineer in 1960, where he began work on the early Program Data Processor (PDP) minicomputers.

His work on the PDPs contributed to Bell becoming an industry-recognized expert on minicomputers and interactive computing. In 1966, he left DEC to teach computer science at Carnegie-Mellon University. While there, he also worked on minicomputer-related interactive computing. In 1972, he was convinced to return to DEC as vice-president of engineering to oversee the production of the VAX (Virtual Address Extension) line of minicomputers and move the company into the world of semiconductor (chip) technology. The 32-bit VAX minicomputers went on to become some of the most successful computers ever created and completely changed the face of the minicomputer industry.

Bell remained with DEC until his retirement in 1983, which was prompted by a heart attack. In July 1983, he started a new company with Kenneth Fisher and Henry Burkhardt called Encore Computer Corporation. Their intention was to build a new generation of small computers and to help new start-up companies. He then spent the late 1980s leading the National Science Foundation's Information Superhighway initiative and co-writing a book on venture capital called *High Tech Ventures: The Guide for Entrepreneurial Success*. The early 1990s found Bell helping several small start-up companies where he served in various positions, including chief scientist at Stardent Computer in Newton, Massachusetts in 1990 and vice-president of Research and Development at Ardent Corporation in Sunnyvale, California in 1991. He was also a director of the Bell-Mason Group, which programmed computers to analyze the strengths and weaknesses of new businesses.

From 1991 to 1995, Bell acted as an advisor to Microsoft for future development efforts and helped set up its first research laboratory. Microsoft was trying to become the research power that would dictate the next decade's computer technology. To this point, it tempted some of the best minds in the industry to join in its long-term research and development. In August 1995, Bell officially joined this brain trust. Through his association with Microsoft, Bell plans to explore the use of video and high-speed networks to expand and facilitate human-human interactions and to reduce physical travel, fields known as "telepresence" or "telecomputing." He is also exploring what he calls "scalable network and platform computing (SNAP)," a new architecture that would give a single desktop operator the same operating power as a mainframe, something he believes is feasible shortly after the turn of the century.

In addition to his work in the computer field, Bell published several books and helped establish the Computer Museum in Boston. He also received several awards for his contributions to the computer industry, including the National Medal of Technology in 1991 and the Eckert-Mauchly award in 1982, named after the two developers of ENIAC and UNIVAC. In 1986, the National Science Foundation asked him to direct funding for U.S. computer science efforts. As the first assistant director for computing at the National Science Foundation, he led the National Research Network panel that became the National Information Infrastructure/Global Information Infrastructure (NII/GII) and helped write the High-Performance Computer and Communications Initiative.

While many of the pioneers in the computer industry have fallen by the wayside, either of their own volition or because the industry moved away from them, Gordon Bell has remained always at the forefront. He isn't sitting back resting on his laurels or pining for the good old days. One reason he agreed to join Microsoft is to "continue to be a part of the next paradigm shift

Fred Brooks (April 19, 1931, Durham NC, USA)

Achievement

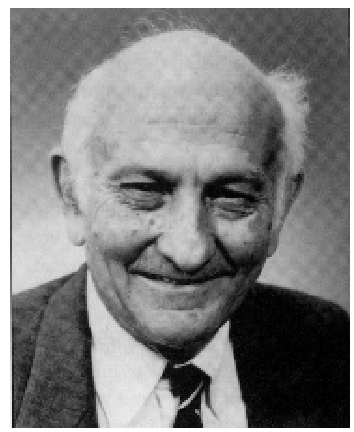
Brooks Jr., Fred; Managed the development of the 360 operating system software for IBM (1960s); wrote *The Mythical Man-Month* about software project management

Edith Clark (1883 - 1959, USA)

Achievement

She translated what many engineers found to be esoteric mathematical methods into graphs or simpler forms during a time when power systems were becoming more complex and when the initial efforts were being made to develop electromechanical aids to problem solving.

John Cocke (30 May 1925, Charlotte, N.C., USA, 16 July 2002, Valhalla, N.Y. USA)



Achievement

Chief designer for the RISC chip architecture at IBM

Biography

Born on May 30, 1925, Mr. Cocke was raised in Charlotte, N.C. His father, Norman Cocke, was the president of the Duke Power Company and a trustee of Duke University. John Cocke's curiosity, which would prove so valuable later in his life, was evident early. As an adult, Mr. Cocke once recalled that when he was given his first bicycle at the age of 6, he dismantled it within a few hours, much to the chagrin of his mother, Mary. Mr. Cocke joined I.B.M.'s research labs in 1956 after he received a doctorate in mathematics from Duke, and he remained with the company until he retired.

Mr. Cocke (rhymes with "sock") was the principal designer of the type of microprocessor that serves as the engine of most of today's large, powerful computers and the Apple Macintosh personal computers. Machines using his chip design — a simplification of the hardware, which opened the door to faster computation — are reduced instruction-set computers, or RISC.Advertisement

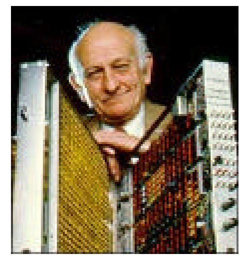
Throughout his long career as a researcher for I.B.M., Mr. Cocke was also responsible for a host of other innovations. He was a leader in the arcane but vital field of designing more efficient software compilers — the software that translates instructions written in a programming language understood by human programmers into the vernacular of all computers, the 1's and 0's of digital code. Mr. Cocke also came up with ideas that helped advance fields as diverse as speech-recognition technology and data storage.

In computer science circles, Mr. Cocke was renowned for the breadth of his intellect, his energy, his insights and his unconventional working methods.

A former colleague, Paul M. Horn, who had joined I.B.M.'s research labs after a career as a physics professor at the University of Chicago, recalled that when he worked on weekends, Mr. Cocke was invariably in the labs. The senior researcher, Mr. Horn recalled, would drop by and engage the newcomer in long discussions of the finer points of unification theory in physics. "John Cocke knew as much about high-energy physics as I did, and it wasn't even his field," said Mr. Horn, who is the director of I.B.M.'s research division.

Even after he retired in 1992, Mr. Cocke always displayed "a wonderful childlike curiosity — he was interested in everything," recalled R. Andrew Heller, who collaborated with Mr. Cocke on the RISC technology, beginning in the late 1960's.

The RISC chip design, experts say, was a striking example of Mr. Cocke's defining attribute. His deep understanding of both the computer hardware and software, and their interaction, often enabled Mr. Cocke to pierce through the complexity of computer problems with fresh insights.



Cocke with prototype server pic IBM, the "Father" of RISC architecture

Among his many achievements, he was named IBM Fellow, the company's highest technical honor, in 1972, he also won National Medal of Technology, the Turing award and was awarded the National Medal of Science in 1991 by President Bush. Over his lifetime as a scientist, he made unique and creative contributions to information technology through his innovative developments in high performance system design.

His expertise in achieving high performance for a broad range of scientific applications led to remarkable advances in compiler design and machine architecture that culminated in his invention of the Reduced Instruction Set Computer (RISC) --- which profoundly affected the broad field of information technology. RISC is the basis for a Unix systems market that last year was $22.3 billion, according to industry analyst group, IDC.

"His tenure at IBM spanned an amazing time, 1956 to 1992," said Peter Capek, John's colleague at IBM and close friend. "His career was unusual in its breadth. He was known for his work in computer architecture, but he was interested in everything -- circuits, storage, compilers -- any technology that could advance the state of the art."

John is a founder and key innovator of the technology of compiler optimization, now used systematically throughout the computer industry to enable computers programmed in higher level languages such as FORTRAN, C, PASCAL and others to reach levels of efficiency comparable to -- and in some cases exceeding -- the levels of efficiency reachable by much more expensive and time-consuming programming techniques closer to the machine's instruction set.

Edgar F. Codd

Citation

For his fundamental and continuing contributions to the theory and practice of database management systems. He originated the relational approach to database management in a series of research papers published commencing in 1970. His paper "A Relational Model of Data for Large Shared Data Banks" was a seminal paper, in a continuing and carefully developed series of papers. Dr. Codd built upon this space and in doing so has provided the impetus for widespread research into numerous related areas, including database languages, query subsystems, database semantics, locking and recovery, and inferential subsystems.

Seymour Cray (1925, Chippewa Falls (Wi), USA-1996, Colorado Springs (CO), USA)

Achievement

Creator of the first super computer Cray I

In 1972, Seymour Cray founded Cray Research to design and build the world's highest performance general-purpose supercomputers. His Cray-1 computer established a new standard in supercomputing upon its introduction in 1976, and his Cray-2 computer system, introduced in 1985, moved supercomputing forward yet again.

In the years following the company's founding, Mr. Cray relinquished the company's management reins to devote more time to computer development. From 1972 to 1977 he served as director, chief executive officer, and president of the company. In October 1977, he left the presidency, but remained chief executive officer and became chairman of the board. In 1980, Mr. Cray resigned as chief executive officer, and in 1981, he stepped aside as chairman of the board to devote himself full time to the Cray-2 project as an independent contractor for Cray Research. For some time, he remained a director and a member of the Policy Committee. Later, Mr. Cray worked to develop a successor to the Cray-2 system and explored gallium arsenide technology.

Mr. Cray spent his entire career designing large-scale computer equipment. He was one of the founders of Control Data Corporation in 1957 and was responsible for the design of that company's most successful large-scale computers, the CDC 1604, 6600, and 7600 systems. He served as a director for CDC from 1957 to 1965 and was senior vice president at the time of his departure in 1972.

Cray supercomputers

From 1950 to 1957, Mr. Cray held several positions with Engineering Research Associates (ERA) of St. Paul, Minnesota. At ERA, he worked on the development of the ERA 1101 scientific computer for the U.S. government. Later, he had design responsibility for a major portion of the ERA 1103, the first commercially successful scientific computer. While with ERA, he worked with the gamut of computer technologies ranging from vacuum tubes and magnetic amplifiers to transistors. Mr. Cray is the inventor of a number of technologies that have been patented by the companies for which he has worked. Among the more significant are the Cray-1 vector register technology, the cooling technologies for the Cray-2 computer, the CDC 6600 freon-cooling system, and a magnetic amplifier for ERA. He also contributed to the Cray-1 cooling technology design.

Mr. Cray earned a bachelor of science degree in electrical engineering in 1950 from the University of Minnesota. In 1951 he earned a master of science degree in applied mathematics from the same institution.

In 1968, he was awarded the W.W. McDowell Award by the American Foundation of Information Processing Societies for his work in the computer field. In 1972 he was presented with the Harry H. Good Memorial Award for his contributions to large-scale computer design and the development of multiprocessing systems.

Mr. Cray was born in 1925 in Chippewa Falls, Wisconsin; he died in 1996 in Colorado Springs.

Edsger Wybe Dijkstra (11 May 1930, Rotterdam, Netherlands - 6 Aug 2002, Nuenen)

Achievement

In 1968 Edsger Dijkstra(7) laid the foundation stone in the march towards creating structure in the domain of programming by writing, not a scholarly paper on the subject, but instead a letter to the editor entitled "GO TO Statement Considered Harmful". (*Comm. ACM*, August 1968) The movement to develop reliable software was underway.

Biography

Edsger Wybe Dijkstra was born in Rotterdam, Netherlands in 1930. Both of his parents were intellectual people and had received good educations. His father was a chemist, and his mother was a mathematician. In 1942, when Dijkstra was 12 years old he entered the Gymnasium Erasminium, a high school for extremely bright students, and he was educated in a number of different subjects including: Greek, Latin, French, German, English, biology, mathematics, and chemistry.

In 1945, Dijkstra thought that he might study law and possibly serve as a representative for the Netherlands at the United Nations. However, due to the fact that he had scored so well in chemistry, mathematics, and physics, he entered the University of Leiden, where he decided to study theoretical physics. He went to summer school on the subject of programming at Cambridge University, during the summer of 1951. He began part-time work at the Mathematical Centre in Amsterdam in March 1952, which further helped fuel his growing interest in programming. He finished the requirements for his theoretical physics degree as quickly as possible and began to pursue his interests in progamming. One of the problems that he ran into, however was that programming still was not officially recognized as a profession. In fact, when he applied for a marriage license in 1957, he had to put down "theoretical physicist" as his profession.

Dijkstra continued to work at the Mathematical Centre until he accepted a job as a research fellow for Burroughs Corporation, in the United States, in the early 1970s. He was awarded the ACM Turing Award in 1972. He was given the AFIPS Harry Goode Memorial Award in 1974. Dijkstra moved to Austin, Texas in the early 1980s. In 1984 he was appointed to a chair in Computer Science at the University of Texas, Austin, where he has been ever since.

Contributions to Computer Science:

In 1956, Dijkstra came up with the "shortest-path algorithm", after he had been assigned the task of showing the powers of ARMAC, the computer that the Mathematical Centre had in it's possession; an algorithm which aids in finding the best way to travel between two points. He also used this to solve the problem of finding a way to "convey electricity to all essential circuits, while using as little expensive copper wire as possible" that the engineers that had designed the ARMAC ran into. He called it the "shortest subspanning tree algorithm." In the early 1960s, Dijkstra applied the idea of mutual exclusion to communications between a computer and its keyboard. He used the letters P and V to represent the two operations that go on in the mutual exclusion problem. This idea has become a part of pretty much all, modern processors and memory board since 1964, when IBM first used it in its 360 architecture. The next problem that computer engineers must deal with that Dijkstra recognized was the "dining philosophers problem." In this problem, five philosophers are sitting at a table with a bowl of rice and a chopstick on either side of the bowl. The problem that arises is how the philosophers will be able to eat without coming to a "deadlock", ending up in a "starvation" situation, or a situation with "lack of fairness." He helped make the computer software industry a lot more disciplined by using one phrase: "GO TO considered harmful. This means that the more GO TO statements there are in a program the harder it is to follow the program's source code.

John Presper Eckert (April 9, 1919 Philadelphia-1995 Bryn Mawr, USA)

Achievement : Co-inventor of the first fully electronical digital computer

Biography

At the university of Pennsylvenia Eckert and the late John. W. Mauchly created ENIAC (Electronic Numerical Integrator and Computer). A 30 ton leviathan that contained less computational power than a single one of today's tiny silicon chips. Designed to figure trajectories for world war II atillery, ENIAC was 1000 times as fast as contemporary calculators , and became an invaluable tool for scientist working to build the first atom bomb. After selling their business in 1950 to Sperry Rand Corp, now Unisys, the partners continued to advance in their field. Eckert was awarded a National Science Medal in 1968.

Chronology

J. Presper Eckert was a graduate of the University of Pennsylvania. He was a researcher at the Moore School of the University of Pennsylvania when he began working with John W Mauchly on ENIAC (Electronic Numerical Integrator and Computer) in 1943. When they completed ENIAC, in 1946, it had 500 000 hand soldered connections and used punched cards to store data. ENIAC was used by the U.S. Army for military calculations.

In 1946 Eckert and Mauchly started a business partnership that become the Eckert-Mauchly Computer Corporation. In 1949 Eckert and Mauchly launched BINAC (Binary Automatic Computer) which used magnetic tape to store data.programming, theoretical

In 1950 the Remington Rand Corporation acquired the Eckert-Mauchly company and changed the name to the Univac Division of Remington Rand. Later research resulted in UNIVAC1 (Universal Automatic Computer), a computer containing many of the elements of today's machines.

In 1955 Remington Rand merged with the Sperry Corporation to form Sperry Rand. Eckert remained with the company and became an executive. He continued with the company as it merged with the Burroughs Corporation to become Unisys. In 1989 Eckert retired from Unisys but continued to act as a consultant for the company. He died of leukemia in Bryn Mawr, Pennsylvenia , USA

Edward A. Feigenbaum (20 january 1936 , Weehawken NJ, USA)

"{Knowledge} is power, and computers that amplify that knowledge will amplify every dimension of power"

"Humans are superb problem solvers; superb learners; superb at coordinating functions of sensing and locomotion and problem solving into an integrated unit. However, computer programs can claim intellectual niches that evolution did not provide for us marvelous creatures."

Achievement

The Expert System (4)

It is Feigenbaum's development of the expert system that has contributed the most to computer science, particularly the field of artificial intelligence. He began hypothesizing about the creation of an expert system as early as 1962, when first he co-edited *Computers and Thought* along with Julien Feldman.

Wanting to use computers to explore induction -- "an informed guess, one that may have to be changed in light of new evidence" (Sasha, Lazare, 215), Feigenbaum considered many possibilities for his first expert system.

Eventually, Feigenbaum along with Joshua Lederberg (chairman of the genetics department) and Carl Djarassi (another colleague, who supplied the needed knowledge of chemistry) created the first expert system -- DENDRAL. DENDRAL was important not only because it helped solve the problem its creators set out to solve (the odds of life on other planets), but also because it became a framework for expert systems to follow.

Feigenbaum tested his framework in other areas, proving successful each time. These days, expert systems are used in everything from airlines to computer manufacturers, the military to hospitals. Feigenbaum's framework of the expert system and theories about it continue to influence the leading computer scientists working in artificial intelligence today. Douglas Lenat's Cyc project, for example, builds upon Feigenbaum's knowledge principle.

The expert system is hardly common place today, knowledge and acceptance is growing.

Biography

From the earliest moments of his life, Feigenbaum had an interest in the sciences -- an interest sparked by his stepfather. With the prodding of his parents, however, Feigenbaum entered the Carnegie Institute of Technology in 1952, in order to earn a degree in electrical engineering. Earning his BS in 1956, Feigenbaum rediscovered science at Carnegie. A professor of his, James March, introduced Feigenbaum to John Von Neumann's ideas of game theory; Edward was intrigued.

After earning his PhD at Carnegie Institute of Technology, Feigenbaum went on to become a member of the faculty at Berkeley's School of Business Administration. The school's lack of a computer science program, however, prompted Feigenbaum to leave the world of psychology and human behavior to the world of artificial intelligence.

With John McCarthy (who had first worked with Marvin Minsky at MIT) recently hired at Stanford to head the artificial intelligence laboratory, Feigenbaum could not help but move on.

At Stanford, Feigenbaum made most of his contributions to the world of AI and computer science in general. Feigenbaum works now as a professor at Stanford, having held numerous positions in the past, including Chief Scientist of the United States Air Force (from 1994-1997). He has written many books, collaborating with various different experts, including his wife H. Penny Nii, who he married in 1975, and has had four children with.

Alan Turing

Alan Turing was born at Paddington, London. His father, Julius Mathison Turing, was a British member of the Indian Civil Service and he was often abroad. Alan's mother, Ethel Sara Stoney, was the daughter of the chief engineer of the Madras railways and Alan's parents had met and married in India. When Alan was about one year old his mother rejoined her husband in India, leaving Alan in England with friends of the family. Alan was sent to school but did not seem to be obtaining any benefit so he was removed from the school after a few months.

Next he was sent to Hazlehurst Preparatory School where he seemed to be an 'average to good' pupil in most subjects but was greatly taken up with following his own ideas. He became interested in chess while at this school and he also joined the debating society. He completed his Common Entrance Examination in 1926 and then went to Sherborne School. Now 1926 was the year of the general strike and when the strike was in progress Turing cycled 60 miles to the school from his home, not too demanding a task for Turing who later was to become a fine athlete of almost Olympic standard. He found it very difficult to fit into what was expected at this public school, yet his mother had been so determined that he should have a public school education. Many of the most original thinkers have found conventional schooling an almost incomprehensible process and this seems to have been the case for Turing. His genius drove him in his own directions rather than those required by his teachers.

He was criticised for his handwriting, struggled at English, and even in mathematics he was too interested with his own ideas to produce solutions to problems using the methods taught by his teachers. Despite producing unconventional answers, Turing did win almost every possible mathematics prize while at Sherborne. In chemistry, a subject which had interested him from a very early age, he carried out experiments following his own agenda which did not please his teacher. Turing's headmaster wrote :-

If he is to stay at Public School, he must aim at becoming educated. If he is to be solely a Scientific Specialist, he is wasting his time at a Public School.

This says far more about the school system that Turing was being subjected to than it does about Turing himself. However, Turing learnt deep mathematics while at school, although his teachers were probably not aware of the studies he was making on his own. He read Einstein's papers on relativity and he also read about quantum mechanics in Eddington's The nature of the physical world.

An event which was to greatly affect Turing throughout his life took place in 1928. He formed a close friendship with Christopher Morcom, a pupil in the year above him at school, and the two worked together on scientific ideas. Perhaps for the first time Turing was able to find someone with whom he could share his thoughts and ideas. However Morcom died in February 1930 and the experience was a shattering one to Turing. He had a premonition of Morcom's death at the very instant that he was taken ill and felt that this was something beyond what science could explain.

Despite the difficult school years, Turing entered King's College, Cambridge, in 1931 to study mathematics. This was not achieved without difficulty. Turing sat the scholarship examinations in 1929 and won an exhibition, but not a scholarship. Not satisfied with this performance, he took the examinations again in the following year, this time winning a scholarship. In many ways Cambridge was a much easier place for unconventional people like Turing than school had been. He was now much more able to explore his own ideas and he read Russell's *Introduction to mathematical philosophy* in 1933. At about the same time he read von Neumann's 1932 text on quantum mechanics, a subject he returned to a number of times throughout his life.

The year 1933 saw the beginnings of Turing's interest in mathematical logic. He read a paper to the Moral Science Club at Cambridge in December of that year of which the following minute was recorded:-

A M Turing read a paper on "Mathematics and logic". He suggested that a purely logistic view of mathematics was inadequate; and that mathematical propositions possessed a variety of interpretations of which the logistic was merely one.

Of course 1933 was also the year of Hitler's rise in Germany and of an anti-war movement in Britain. Turing joined the anti-war movement but he did not drift towards Marxism, nor pacifism, as happened to many.

Turing graduated in 1934 then, in the spring of 1935, he attended Max Newman's advanced course on the foundations of mathematics. This course studied Gödel's incompleteness results and Hilbert's question on decidability. In one sense 'decidability' was a simple question, namely given a mathematical proposition could one find an algorithm which would decide if the proposition was true of false. For many propositions it was easy to find such an algorithm. The real difficulty arose in proving that for certain propositions no such algorithm existed. When given an algorithm to solve a problem it was clear that it was indeed an algorithm, yet there was no definition of an algorithm which was rigorous enough to allow one to prove that none existed. Turing began to work on these ideas.

Turing was elected a fellow of King's College, Cambridge, in 1935 for a dissertation *On the Gaussian error function* which proved fundamental results on probability theory, namely the *central limit theorem.* Although the central limit theorem had recently been discovered, Turing was not aware of this and discovered it independently. In 1936 Turing was a Smith's Prizeman.

Turing's achievements at Cambridge had been on account of his work in probability theory. However, he had been working on the decidability questions since attending Newman's course. In 1936 he published *On Computable Numbers, with an application to the Entscheidungsproblem.* It is in this paper that Turing introduced an abstract machine, now called a "Turing machine", which moved from one state to another using a precise finite set of rules (given by a finite table) and depending on a single symbol it read from a tape.

The Turing machine could write a symbol on the tape, or delete a symbol from the tape. Turing wrote :-

Some of the symbols written down will form the sequences of figures which is the decimal of the real number which is being computed. The others are just rough notes to "assist the memory". It will only be these rough notes which will be liable to erasure.

He defined a computable number as real number whose decimal expansion could be produced by a Turing machine starting with a blank tape. He showed that was computable, but since only countably many real numbers are computable, most real numbers are not computable. He then described a number which is not computable and remarks that this seems to be a paradox since he appears to have described in finite terms, a number which cannot be described in finite terms. However, Turing understood the source of the apparent paradox. It is impossible to decide (using another Turing machine) whether a Turing machine with a given table of instructions will output an infinite sequence of numbers.

Although this paper contains ideas which have proved of fundamental importance to mathematics and to computer science ever since it appeared, publishing it in the *Proceedings of the London Mathematical Society* did not prove easy. The reason was that Alonzo Church published *An* *unsolvable problem in elementary number theory* in the *American Journal of Mathematics* in 1936 which also proves that there is no decision procedure for arithmetic. Turing's approach is very different from that of Church but Newman had to argue the case for publication of Turing's paper before the London Mathematical Society would publish it. Turing's revised paper contains a reference to Church's results and the paper, first completed in April 1936, was revised in this way in August 1936 and it appeared in print in 1937.

A good feature of the resulting discussions with Church was that Turing became a graduate student at Princeton University in 1936. At Princeton, Turing undertook research under Church's supervision and he returned to England in 1938, having been back in England for the summer vacation in 1937 when he first met Wittgenstein. The major publication which came out of his work at Princeton was *Systems of Logic Based on Ordinals* which was published in 1939. Newman writes:-

This paper is full of interesting suggestions and ideas. ... [It] throws much light on Turing's views on the place of intuition in mathematical proof.

Before this paper appeared, Turing published two other papers on rather more conventional mathematical topics. One of these papers discussed methods of approximating Lie groups by finite groups. The other paper proves results on extensions of groups, which were first proved by Reinhold Baer, giving a simpler and more unified approach.

Perhaps the most remarkable feature of Turing's work on Turing machines was that he was describing a modern computer before technology had reached the point where construction was a realistic proposition. He had proved in his 1936 paper that a universal Turing machine existed:-

... which can be made to do the work of any special-purpose machine, that is to say to carry out any piece of computing, if a tape bearing suitable "instructions" is inserted into it.

Although to Turing a "computer" was a person who carried out a computation, we must see in his description of a universal Turing machine what we today think of as a computer with the tape as the program.

While at Princeton Turing had played with the idea of constructing a computer. Once back at Cambridge in 1938 he starting to build an analogue mechanical device to investigate the Riemann hypothesis, which many consider today the biggest unsolved problem in mathematics. However, his work would soon take on a new aspect for he was contacted, soon after his return, by the Government Code and Cypher School who asked him to help them in their work on breaking the German Enigma codes.

When war was declared in 1939 Turing immediately moved to work full-time at the Government Code and Cypher School at Bletchley Park. Although the work carried out at Bletchley Park was covered by the Official Secrets Act, much has recently become public knowledge. Turing's brilliant ideas in solving codes, and developing computers to assist break them, may have saved more lives of military personnel in the course of the war than any other. It was also a happy time for him [13]:-

... perhaps the happiest of his life, with full scope for his inventiveness, a mild routine to shape the day, and a congenial set of fellow-workers.

Together with another mathematician W G Welchman, Turing developed the *Bombe*, a machine based on earlier work by Polish mathematicians, which from late 1940 was decoding all messages sent by the Enigma machines of the Luftwaffe. The Enigma machines of the German navy were much harder to break but this was the type of challenge which Turing enjoyed. By the middle of

1941 Turing's statistical approach, together with captured information, had led to the German navy signals being decoded at Bletchley.

From November 1942 until March 1943 Turing was in the United States liaising over decoding issues and also on a speech secrecy system. Changes in the way the Germans encoded their messages had meant that Bletchley lost the ability to decode the messages. Turing was not directly involved with the successful breaking of these more complex codes, but his ideas proved of the greatest importance in this work. Turing was awarded the O.B.E. in 1945 for his vital contribution to the war effort.

At the end of the war Turing was invited by the National Physical Laboratory in London to design a computer. His report proposing the Automatic Computing Engine (ACE) was submitted in March 1946. Turing's design was at that point an original detailed design and prospectus for a computer in the modern sense. The size of storage he planned for the ACE was regarded by most who considered the report as hopelessly over-ambitious and there were delays in the project being approved.

Turing returned to Cambridge for the academic year 1947-48 where his interests ranged over many topics far removed from computers or mathematics; in particular he studied neurology and physiology. He did not forget about computers during this period, however, and he wrote code for programming computers. He had interests outside the academic world too, having taken up athletics seriously after the end of the war. He was a member of Walton Athletic Club winning their 3 mile and 10 mile championship in record time. He ran in the A.A.A. Marathon in 1947 and was placed fifth.

By 1948 Newman was the professor of mathematics at the University of Manchester and he offered Turing a readership there. Turing resigned from the National Physical Laboratory to take up the post in Manchester. Newman writes that in Manchester:-

... work was beginning on the construction of a computing machine by F C Williams and T Kilburn. The expectation was that Turing would lead the mathematical side of the work, and for a few years he continued to work, first on the design of the subroutines out of which the larger programs for such a machine are built, and then, as this kind of work became standardised, on more general problems of numerical analysis.

In 1950 Turing published Computing machinery and intelligence in Mind. It is another remarkable work from his brilliantly inventive mind which seemed to foresee the questions which would arise as computers developed. He studied problems which today lie at the heart of artificial intelligence. It was in this 1950 paper that he proposed the Turing Test which is still today the test people apply in attempting to answer whether a computer can be intelligent :-

... he became involved in discussions on the contrasts and similarities between machines and brains. Turing's view, expressed with great force and wit, was that it was for those who saw an unbridgeable gap between the two to say just where the difference lay.

Turing did not forget about questions of decidability which had been the starting point for his brilliant mathematical publications. One of the main problems in the theory of group presentations was the question: given any word in a finitely presented groups is there an algorithm to decide if the word is equal to the identity. Post had proved that for semigroups no such algorithm exist. Turing thought at first that he had proved the same result for groups but, just before giving a seminar on his proof, he discovered an error. He was able to rescue from his faulty proof the fact that there was a cancellative semigroup with insoluble word problem and he published this result in 1950. Boone used the ideas from this paper by Turing to prove the existence of a group with insoluble word problem in 1957.

Turing was elected a Fellow of the Royal Society of London in 1951, mainly for his work on Turing machines in 1936. By 1951 he was working on the application of mathematical theory to biological forms. In 1952 he published the first part of his theoretical study of morphogenesis, the development of pattern and form in living organisms.

Turing was arrested for violation of British homosexuality statutes in 1952 when he reported to the police details of a homosexual affair. He had gone to the police because he had been threatened with blackmail. He was tried as a homosexual on 31 March 1952, offering no defence other than that he saw nothing wrong in his actions. Found guilty he was given the alternatives of prison or oestrogen injections for a year. He accepted the latter and returned to a wide range of academic pursuits.

Not only did he press forward with further study of morphogenesis, but he also worked on new ideas in quantum theory, on the representation of elementary particles by spinors, and on relativity theory. Although he was completely open about his sexuality, he had a further unhappiness which he was forbidden to talk about due to the Official Secrets Act.

The decoding operation at Bletchley Park became the basis for the new decoding and intelligence work at GCHQ. With the cold war this became an important operation and Turing continued to work for GCHQ, although his Manchester colleagues were totally unaware of this. After his conviction, his security clearance was withdrawn. Worse than that, security officers were now extremely worried that someone with complete knowledge of the work going on at GCHQ was now labelled a security risk. He had many foreign colleagues, as any academic would, but the police began to investigate his foreign visitors. A holiday which Turing took in Greece in 1953 caused consternation among the security officers.

Turing died of potassium cyanide poisoning while conducting electrolysis experiments. The cyanide was found on a half eaten apple beside him. An inquest concluded that it was self-administered but his mother always maintained that it was an accident.

Ray Holt

born, city, USA

Achievement

invented central processor in 1971

Built what is called a slice computer where he used several elements such as the 74818 and others to make a computer of expandable word width... This is not at MICRO-COMPUTER-BY-ANY MEANS! Once you go outside of one chip for the main control of the organization; all Registers, Program Counter, Arithmetic on one piece of silicon, you are outside the realm of the Microcomputer.

John von Neumann

John von Neumann was born János von Neumann. He was called Jancsi as a child, a diminutive form of János, then later he was called Johnny in the United States. His father, Max Neumann, was a top banker and he was brought up in a extended family, living in Budapest where as a child he learnt languages from the German and French governesses that were employed. Although the family were Jewish, Max Neumann did not observe the strict practices of that religion and the household seemed to mix Jewish and Christian traditions.

It is also worth explaining how Max Neumann's son acquired the "von" to become János von Neumann. In 1913 Max Neumann purchased a title but did not change his name. His son, however, used the German form von Neumann where the "von" indicated the title.

As a child von Neumann showed he had an incredible memory. Poundstone, writes:-

At the age of six, he was able to exchange jokes with his father in classical Greek. The Neumann family sometimes entertained guests with demonstrations of Johnny's ability to memorise phone books. A guest would select a page and column of the phone book at random. Young Johnny read the column over a few times, then handed the book back to the guest. He could answer any question put to him (*who has number such and such?*) *or recite names, addresses, and numbers in order.*

In 1911 von Neumann entered the Lutheran Gymnasium. The school had a strong academic tradition which seemed to count for more than the religious affiliation both in the Neumann's eyes and in those of the school. His mathematics teacher quickly recognised von Neumann's genius and special tuition was put on for him. The school had another outstanding mathematician one year ahead of von Neumann, namely Eugene Wigner.

World War I had relatively little effect on von Neumann's education but, after the war ended, Béla Kun controlled Hungary for five months in 1919 with a Communist government. The Neumann family fled to Austria as the affluent came under attack. However, after a month, they returned to face the problems of Budapest. When Kun's government failed, the fact that i had been largely composed of Jews meant that Jewish people were blamed. Such situations are devoid of logic and the fact that the Neumann's were opposed to Kun's government did not save them from persecution.

In 1921 von Neumann completed his education at the Lutheran Gymnasium. His first mathematics paper, written jointly with Fekete the assistant at the University of Budapest who had been tutoring him, was published in 1922. However Max Neumann did not want his son to take up a subject that would not bring him wealth. Max Neumann asked Theodore von Kármán to speak to his son and persuade him to follow a career in business. Perhaps von Kármán was the wrong person to ask to undertake such a task but in the end all agreed on the compromise subject of chemistry for von Neumann's university studies.

Hungary was not an easy country for those of Jewish descent for many reasons and there was a strict limit on the number of Jewish students who could enter the University of Budapest. Of course, even with a strict quota, von Neumann's record easily won him a place to study mathematics in 1921 but he did not attend lectures. Instead he also entered the University of Berlin in 1921 to study chemistry.

Von Neumann studied chemistry at the University of Berlin until 1923 when he went to Zurich. He achieved outstanding results in the mathematics examinations at the University of Budapest despite not attending any courses. Von Neumann received his diploma in chemical engineering from the Technische Hochschule in Zürich in 1926. While in Zurich he continued his interest in mathematics, despite studying chemistry, and interacted with Weyl and Pólya who were both at Zurich. He even took over one of Weyl's courses when he was absent from Zurich for a time. Pólya said:-

Johnny was the only student I was ever afraid of. If in the course of a lecture I stated an unsolved problem, the chances were he'd come to me as soon as the lecture was over, with the complete solution in a few scribbles on a slip of paper.

Von Neumann received his doctorate in mathematics from the University of Budapest, also in 1926, with a thesis on set theory. He published a definition of ordinal numbers when he was 20, the definition is the one used today.

Von Neumann lectured at Berlin from 1926 to 1929 and at Hamburg from 1929 to 1930. However he also held a Rockefeller Fellowship to enable him to undertake postdoctoral studies at the University of Göttingen. He studied under Hilbert at Göttingen during 1926-27. By this time von Neumann had achieved celebrity status :-

By his mid-twenties, von Neumann's fame had spread worldwide in the mathematical community. At academic conferences, he would find himself pointed out as a young genius.

Veblen invited von Neumann to Princeton to lecture on quantum theory in 1929. Replying to Veblen that he would come after attending to some personal matters, von Neumann went to Budapest where he married his fiancée Marietta Kovesi before setting out for the United States. In 1930 von Neumann became a visiting lecturer at Princeton University, being appointed professor there in 1931.

Between 1930 and 1933 von Neumann taught at Princeton but this was not one of his strong points :-

His fluid line of thought was difficult for those less gifted to follow. He was notorious for dashing out equations on a small portion of the available blackboard and erasing expressions before students could copy them.

In contrast, however, he had an ability to explain complicated ideas in physics :-

For a man to whom complicated mathematics presented no difficulty, he could explain his conclusions to the uninitiated with amazing lucidity. After a talk with him one always came away with a feeling that the problem was really simple and transparent.

He became one of the original six mathematics professors (J W Alexander, A Einstein, M Morse, O Veblen, J von Neumann and H Weyl) in 1933 at the newly founded Institute for Advanced Study in Princeton, a position he kept for the remainder of his life.

During the first years that he was in the United States, von Neumann continued to return to Europe during the summers. Until 1933 he still held academic posts in Germany but resigned these when the Nazis came to power. Unlike many others, von Neumann was not a political refugee but rather he went to the United States mainly because he thought that the prospect of academic positions there was better than in Germany.

In 1933 von Neumann became co-editor of the *Annals of Mathematics* and, two years later, he became co-editor of *Compositio Mathematica.* He held both these editorships until his death.

Von Neumann and Marietta had a daughter Marina in 1936 but their marriage ended in divorce in 1937. The following year he married Klára Dán, also from Budapest, whom he met on one of his European visits. After marrying, they sailed to the United States and made their home in Princeton. There von Neumann lived a rather unusual lifestyle for a top mathematician. He had always enjoyed parties :-

Parties and nightlife held a special appeal for von Neumann. While teaching in Germany, von Neumann had been a denizen of the Cabaret-era Berlin nightlife circuit.

Now married to Klára the parties continued :-

The parties at the von Neumann's house were frequent, and famous, and long.

Ulam summarises von Neumann's work. He writes:-

In his youthful work, he was concerned not only with mathematical logic and the axiomatics of set theory, but, simultaneously, with the substance of set theory itself, obtaining interesting results in measure theory and the theory of real variables. It was in this period also that he began his classical work on quantum theory, the mathematical foundation of the theory of measurement in quantum theory and the new statistical mechanics.

His text *Mathematische Grundlagen der Quantenmechanik* (1932) built a solid framework for the new quantum mechanics. Van Hove writes:-

Quantum mechanics was very fortunate indeed to attract, in the very first years after its discovery in 1925, the interest of a mathematical genius of von Neumann's stature. As a result, the mathematical framework of the theory was developed and the formal aspects of its entirely novel rules of interpretation were analysed by one single man in two years (1927-1929).

Self-adjoint algebras of bounded linear operators on a Hilbert space, closed in the weak operator topology, were introduced in 1929 by von Neumann in a paper in Mathematische Annalen . Kadison explains:-

His interest in *ergodic theory*, *group representations* and quantum mechanics contributed significantly to von Neumann's realisation that a theory of operator algebras was the next important stage in the development of this area of mathematics.

Such operator algebras were called "rings of operators" by von Neumann and later they were called W*-algebras by some other mathematicians. J Dixmier, in 1957, called them "von Neumann algebras" in his monograph *Algebras of operators in Hilbert space (von Neumann algebras).* In the second half of the 1930's and the early 1940s von Neumann, working with his collaborator F J Murray, laid the foundations for the study of von Neumann algebras in a fundamental series of papers.

However von Neumann is know for the wide variety of different scientific studies. Ulam explains how he was led towards game theory:-

Von Neumann's awareness of results obtained by other mathematicians and the inherent possibilities which they offer is astonishing. Early in his work, a paper by Borel *on the minimax property led him to develop ... ideas which culminated later in one of his most original creations, the theory of games.*

In game theory von Neumann proved the minimax theorem. He gradually expanded his work in game theory, and with co-author Oskar Morgenstern, he wrote the classic text *Theory of Games and Economic Behaviour* (1944).

Ulam continues:-

An idea of Koopman on the possibilities of treating problems of classical mechanics by means of operators on a function space stimulated him to give the first mathematically rigorous proof of an ergodic theorem. Haar*'s construction of measure in groups provided the inspiration for his wonderful partial solution of* Hilbert *'s fifth problem, in which he proved the possibility of introducing analytical parameters in compact groups.*

In 1938 the American Mathematical Society awarded the Bôcher Prize to John von Neumann for his memoir *Almost periodic functions and groups.* This was published in two parts in the *Transactions of the American Mathematical Society,* the first part in 1934 and the second part in the following year. Around this time von Neumann turned to applied mathematics :-

In the middle 30's, Johnny was fascinated by the problem of hydrodynamical turbulence. It was then that he became aware of the mysteries underlying the subject of non-linear *partial differential equations*. His work, from the beginnings of the Second World War, concerns a study of the equations of hydrodynamics and the theory of shocks. The phenomena described by these non-linear equations are baffling analytically and defy even qualitative insight by present methods. Numerical work seemed to him the most promising way to obtain a feeling for the behaviour of such systems. This impelled him to study new possibilities of computation on electronic machines ...

Von Neumann was one of the pioneers of computer science making significant contributions to the development of logical design. Shannon writes :-

Von Neumann spent a considerable part of the last few years of his life working in [automata theory]. It represented for him a synthesis of his early interest in logic and proof theory and his later work, during World War II and after, on large scale electronic computers. Involving a mixture of pure and applied mathematics as well as other sciences, automata theory was an ideal field for von Neumann's wide-ranging intellect. He brought to it many new insights and opened up at least two new directions of research.

He advanced the theory of cellular automata, advocated the adoption of the bit as a measurement of computer memory, and solved problems in obtaining reliable answers from unreliable computer components.

During and after World War II, von Neumann served as a consultant to the armed forces. His valuable contributions included a proposal of the implosion method for bringing nuclear fuel to explosion and his participation in the development of the hydrogen bomb. From 1940 he was a member of the Scientific Advisory Committee at the Ballistic Research Laboratories at the Aberdeen Proving Ground in Maryland. He was a member of the Navy Bureau of Ordnance from 1941 to 1955, and a consultant to the Los Alamos Scientific Laboratory from 1943 to 1955. From 1950 to 1955 he was a member of the Armed Forces Special Weapons Project in Washington, D.C. In 1955 President Eisenhower appointed him to the Atomic Energy Commission, and in 1956 he received its Enrico Fermi Award, knowing that he was incurably ill with cancer.

Eugene Wigner wrote of von Neumann's death :-

When von Neumann realised he was incurably ill, his logic forced him to realise that he would cease to exist, and hence cease to have thoughts ... It was heartbreaking to watch the frustration of his mind, when all hope was gone, in its struggle with the fate which appeared to him unavoidable but unacceptable.

von Neumann's death is described in these terms:-

... his mind, the amulet on which he had always been able to rely, was becoming less dependable. Then came complete psychological breakdown; panic, screams of uncontrollable terror every night. His friend Edward Teller said, "I think that von Neumann suffered more when his mind would no longer function, than I have ever seen any human being suffer."

Von Neumann's sense of invulnerability, or simply the desire to live, was struggling with unalterable facts. He seemed to have a great fear of death until the last... No achievements and no amount of influence could save him now, as they always had in the past. Johnny von Neumann, who knew how to live so fully, did not know how to die.

It would be almost impossible to give even an idea of the range of honours which were given to von Neumann. He was Colloquium Lecturer of the American Mathematical Society in 1937 and received the its Bôcher Prize as mentioned above. He held the Gibbs Lectureship of the American Mathematical Society in 1947 and was President of the Society in 1951-53.

He was elected to many academies including the Academia Nacional de Ciencias Exactas (Lima, Peru), Academia Nazionale dei Lincei (Rome, Italy), American Academy of Arts and Sciences (USA), American Philosophical Society (USA), Instituto Lombardo di Scienze e Lettere (Milan, Italy), National Academy of Sciences (USA) and Royal Netherlands Academy of Sciences and Letters (Amsterdam, The Netherlands).

Von Neumann received two Presidential Awards, the Medal for Merit in 1947 and the Medal for Freedom in 1956. Also in 1956 he received the Albert Einstein Commemorative Award and the Enrico Fermi Award mentioned above.

Peierls writes :-

He was the antithesis of the "long-haired" mathematics don. Always well groomed, he had as lively views on international politics and practical affairs as on mathematical problems.

Konrad Zuse (June 22, 1910 in Berlin)

Inventor of the first freely programmable computer

1935 Degree of Civil Engineering at the Polytechnical Institute of Berlin-Charlottenburg (predecessor of the Technische Universität Berlin).

First job at the German aircraft industry. Zuse left this job early to set up an "inventor´s workshop" in the living room of his parents' apartment in Berlin.

First idea to construct a "mechanical brain".

Unit 2

Society and Technological Changes

Society and Technology

Society

Society can be defined as a population that occupies the same territory that is subject to the same political authority and that participates in a common culture. In other words, the following conditions are to be met to be a society: -

- They must occupy a common territory.
- They must share the same government or other political authority.
- They must, to some extent have a common culture and a sense of membership in, and commitment to, the same group.

Society gives content, direction, and meaning to our lives, and we, in turn, in countless ways, reshape the society that we leave to the next generation.

Essential elements of a society

1. ***Plurality:*** society is composed of population of all ages and of the both the sexes.
2. ***Stability:*** a society is permanent in character. The social life is organized mainly on the basis of division of labor.
3. ***Likeness:*** in earlier societies the sense of likeness was focused on kinship i. e., blood relationships. In modem societies the concept has been broadened by the principle of nationality. A society would not be possible without some mutual understanding and that understanding depends on the likeness which each apprehends in the others.
4. ***Differences:*** society also includes differences. All our social systems involve relationship in which differences complement one another, for example, in a family, apart from biological differences of gender, there are other differences of opinions, diversities of interests and etc. In social life, there is an indefinite interplay of likeness and differences, of cooperation and conflict, of agreement and dissent
5. ***Interdependence:*** it is also an essential element to constitute a society. Family is an example of interdependency. Today, even the countries depend on each other.
6. ***Cooperation:*** without cooperation no society can exists. The members of a family cooperate with each other to live happily.

Technology

The term technology refers to how input is transferred to output. One of the most distinctive of all human characteristics is that we are tool-using animals. It is true that a few other animals have learned to employ implements in a rudimentary way, but the tool use of other species has little if any permanent effect on their social of natural environment People, in contrast, have used increasingly sophisticated techniques to act on the social and the natural world for thousands of years-and they have done so in ways that have transformed, and continue to transform, the very conditions of life on the planet

Over the generations, such cultural artifacts as the knife, the wheel, the plow, gunpowder, irrigation, screws, windmills, dams, the compass, clocks, the printing press, steams engines, vaccinations, pesticides, television, rockets, lasers, and nuclear reactors have dramatically influenced our social and sometimes our natural surroundings. These and countless similar items are all examples of technology, the practical application of scientific or other knowledge. Modem civilization depends on our ability to translate knowledge of nature into a multitude of gadgets, contraptions, processes, implements, and techniques: aircraft and antibiotics, computers and plastics, assembly lines and telecommunications, synthetic fabrics and skyscrapers.

Technological and social changes are intimately connected, particularly in the modem world, where rapid technological and social change tends to go hand in hand. Many people in modem societies seem to implicitly assume that technological development and human progress are much the same thing But although some observers see advanced technology as the key to a brighter future, others wonder if it may sometimes be more of a curse than a blessing. Modern technology has certainly oared admirable achievements, but it has also created dismaying problems.

Technological change

Technical change means:

- Modifications
- Alterations, and
- Innovations

Brought about in the:

- Information
- Techniques, and
- Tools

By means of which men utilized the material resources of their environment to satisfy their wants and needs. There is a close relationship between sociology and technical change. Technical change have influenced social institutions, they have brought a metamorphous in our values, norms and attitudes towards more social problems. It has considerable impact on caste system, culture and social change.

Influence of technological change on society

1. Mass production of goods through machines
2. Automation
3. Faster means of transportation
4. Mass communication
5. Availability of labor saving devices
6. Faster pace of life
7. Commercialized recreation
8. Emphasis on high degree of specialization

Family system and technological change:

Technical change has affected traditional family system in the following manner:

- Emergence of nuclear family
- Women's involvement in male dominated area of work
- Change in standard of living
- New way of socialization of the children

- Change in orthodox values

Some demerits brought by the technical changes to the family system:

- Mechanical life-style
- Formal type of relationships
- Change in existing social customs
- Less family ties between family members

Religion and technological change

Followings are the some of the effects of technical change on religion:

- Analysis of religious doctrines and traditions
- The rigidity in caste system has been relaxed
- Men are free from religious rituals
- Religion has become the secondary thing not a primary one

Rural life and technological change

Followings are the some of the effects of technical change on rural life:

- Migration towards urban areas
- Increase in consciousness of rural people
- Change in method of farming
- Life became comfortable than before
- Change in life pattern

Urban life and technological change

Followings are the some of the effects of technical change on urban life:

- Shortage of land and houses
- Increase in slums
- Problem of transportation
- Increase in crimes
- Expensive life
- Money has became the most important thing

Social Implications of IT

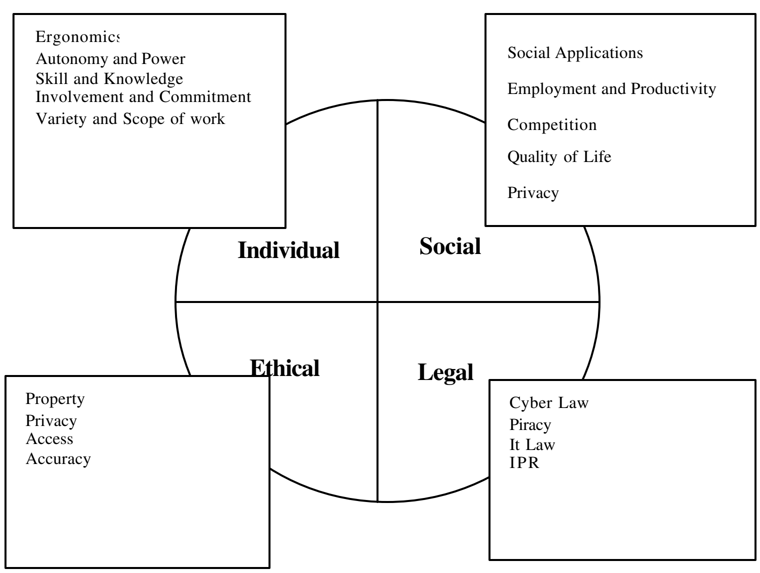
Impact of Information Technology

Concern over Technology has always been present, especially with the advent of IT. As the entire discipline of information technology (here after IT) is less than 50 years old, it is not mature with fundamental concepts, definitions, notations, and tools, as compared to those available in conventional branches of science and engineering. Information Technology does not only effects; technical people but people at various stages but even a common man, a business class man etc. It can change buying habits of a customer, way a service provider provides a service. The need is to think about the effect of IT on social living and the way it can be best utilized.

For example, when a software engineer undertakes a problem to provide a solution, he does not have fundamental axioms to guide him. There is no law in software engineering comparable to the law of thermodynamics, Ohm's laws, etc. Therefore it is high time for a proper education and development of the manpower and engineers. It is also necessary to predict in which way the technology is going and what will be its repercussions, for the role of the visionary is vital. It is the moral responsibility of the sociologist, business people and benign thinkers to think about the way technology is affecting people and in which way can it be, best utilized. It needs greater attention for sustainable growth and development and proper use of information technology.

People working in computer mediated environment will be definitely lack in social contacts and would become alienated. The type of work may also become less interesting and monotonous type as seen in the second example. The modern communications systems such as e-mail and v-mail support additional contact between people separated geographically or organizationally adding a new dimension to the social relationship. The new concepts towards downsizing and telecommuting increase isolation and alienation because the organization is now reducing the number of people and secondly permitting people to work from their homes also has significant affect on the social context of people.

The entire aspects of IT, is covered, in four major areas such as social aspects, participant's aspects, ethical and legal aspects.



Different Aspects of IT

Issues related to IT

The relationship between information technology and society in general is very strong. It needs a greater attention of all classes of people to think how best we can utilize this technology. The first and foremost need is of developing a better manpower in IT, the other issues include Present day Change in Work Environment, Job Design in IT-Enabled Processes, Resistance to Change and its Management and Human-Centric Design.

The Need of Education for a Better IT Manpower Development

In the area of information technology, there is always a gap between the projected manpower and that supplied to this industry. The type of manpower that is produced is, basically of programmers, who know the tits and bits of the program, the engineers who are responsible for the core design of the systems. But the application spectrum of IT is quite vast, ranging from automated control of a stepper motor to stock exchange activity, from nuclear reactor to space sampling robots, thus in this scenario, programmers nor the engineers can do justice to the entire domain of information technology.

However the need is in a systematic process of education and experience so that it is possible to analyze and model the problem, tracing out inconsistencies, ambiguities, and incomplete requirements of the problem in the field of IT. Therefore, the professionals from various fields should involve in the process of IT manpower development. The holistic approach would be to reveal the characteristics of a problem not by an individual component, but by a system engineering approach where people from different fields must participate to develop manpower that meets the current requirements. The students on the other hand are required to practice concept of this type of learning. This type of teaching not only generalizes the problem but removes unnecessity details at the beginning and provides the students the right skills and knowledge to proceed towards the solution in the field of IT.

Today majority of the IT related jobs are in the application areas. One has to understand how businesses are conducted, how operation of a business links with the environment, and so on in order to come up as a competent IT literate. Therefore, IT education needs business professionals who can give a clear and entire picture of the problem and a complete solution to the students. It is required to be a good audience of the abstract subjects, professional subjects, before becoming a good IT solution provider. To achieve this, mere skill in programming may not be sufficient rather the IT literates should acquire knowledge from various associated disciplines.

Present day Change in Work Environment

It is a fact that more amounts of jobs are being done through computers and automated systems, day by day, which is often called computer meditated work. As a consequence, meaningfulness of work is getting reduced. The abstractness of work is a related issue; because computer mediated work does not involve direct physical contact with the object of the task. There are many situations in which working through a computer affects the way workers experience their work. Here the user has to work on the symbols on a computer screen rather than a more reality workplace. If the work is done nicely, the participant user feels it is not his credit but the credit of the machine or the system.

Example: Word processing provides obvious benefits: 1) mistakes can be easily corrected 2) documents and templates can be saved and re-used 3) spell checking. **However, there are problems:** Users make many more revisions than they otherwise would do, but apparently without improving quality; Difficult to master (for average user); Too many features, many of which are never used; Little compatibility between packages.

Consider the example of a bank auditor in the e-Banking system. He also has to do all the work on the data and information, rather than travel to branches and talk with different people, and examine the financial paper work. Although some tasks were quicker and easier, still the auditor tends it more difficult to work with information only and without any human interaction. He feels the job has become abstract for better or worse and he does not like this kind of auditing.

There are also users who feel somewhat different in different types of computer mediated work. Some of the aspects of different systems are as follows.

Intellectual System: In this type of systems the computer is used as a tool for creating ideas, performing analysis, etc. Analysis and manipulation of information is done through a computer, in new and different ways. However, this system may sometimes compel a switch over to a different method after a major investment in one method. This is a major concern in the business practice. This can be tackled by the Re-Engineering concept. The users of this type of system may find it helpful and interesting to work with.

Production System: The worker in a production system enters instructions into a computer terminal attached to a robot or NC machine. The machine does the work based on instructions, instead of the person holding tools and doing the work, in reality. The person feels himself more like a programmer. This type of work accounts for isolation in social relationship and the user may feel alienated.

Office Assistance System: The worker uses a computer terminal to store and retrieve data information instead of writing on paper. The work is done through a key board and display screen. As the computer is the entire work space for the clerk steno, he cannot move here to there, in guise of searching a file or a piece of paper. As a result he feels uncomfortable in his work place.

Control and Supervision: The worker receives instructions through the computer or is monitored based on the rate or accuracy of inputs into a computer Here, Feedback is based on computer recorded data and not on the supervision directed observation Therefore, the worker feels it unnecessary and sometimes neglects it and which more often causes confusion among the line managers and workers.

Job Design in IT-Enabled Processes

There is a basic need for designing scheduling of different jobs in any IT-enabled process. It is clear from the last section, as more numbers of innovations and inventions are coming in the field of Information Technology, the work environments are getting changed with more computerization, advanced communication and intelligent robots, There is a challenge before the operation and production department of any enterprises, to design the work system taking both man and machines together. There is an urge for the division of labor between these two entities, utilizing their respective strengths and weaknesses It is a very difficult task before the production and operation manager and has to be done considering the time and motion study of the man and machine it has been observed that the people are good at jobs involving understanding, imagination, arid the ability to visualize a situation as a whole. Machine is especially good at repetitive tasks involving perseverance, consistency, speed, and execution of unambiguous instructions.

There are two schools of thoughts based on the scheduling of jobs of work systems. They are human-centric design and machine-centric design. The difference in the two groups of thinking is in the design process of technologies and work systems. In human-centric design, the technology or the work process is designed to make participant's work effective and satisfying as far as possible. On the other hand, in machine- centric design, the technology or work process is designed to improve or simplify the procedure of the machine, and participants are expected to adjust to the machine's weaknesses and limitations. Machine-centric design has been the tradition in many computerized systems, where the basic assumption is that participant users will read the procedures, regardless of how confusing or contradictory the technology may be. There has been much progress recently in this direction, considering the human factors after consultation with sociologist, anthropologist and personnel manager.

The rewards and punishments also differ with respect to the two design processes. In the machine centric design, when accidents occur, the user is the one to be blamed, rather than the work system or the technology.

There are many instances, in sophisticated industries, where blame goes to participants who operate equipment or use information within systems. As a result there is a resistance from the people to switch over to an automated work system.

Resistance to Change and its Management

The cause of resistance to change is more or less clear from the last subsection. It can be defined as any action or inaction that obstructs a change in the existing process. It is a complex phenomenon because it involves a combination of motives It can be a highly rational response motivated by a desire to help thy organization, or on the other hand, it can have a selfish or vindictive motive of some individual element, due to some personal rivalry. Resistance may pose in many forms, either in the form of a public debate about the reason and effect of change, or complete closure shutdown of the work system Public debate can take a shape of direct statements about the shortcomings of the proposed system or a statement like "the system is unnecessary or undesirable". Closure/shutdown can occur through conscious misuse and incorrect handling of data or other forms. Even with a lot of effort to make the change process successful, many systems encounter significant resistance from potential participants/ Users.

For managing resistance and overcoming it, what is required is to think about the multiple causes of resistance by looking at various aspects such as characteristics of the people and the characteristics of the system and their interaction. Awareness of the different causes of resistance is helpful in overcoming the implementation problems. Secondly, there must be group discussions and common consensus from different sections of users for the implementations of the new systems.

Human-Centric Design

User-friendliness is one of the main features in the human-centric design of the work system. A system is called user-friendly if most users can use it easily with minimal learning and training time and users find it useful. User friendly work systems are more productive because users waste less time and effort struggling with the system features that come on the way of getting the work done through the computer. At the beginning, computers and computerized systems were more user-hostile rather than user friendly. A technology is user hostile when it is non-interactive and could be programmed only in languages appropriate for professional programmers. But advances in computer languages, interactive computing and graphical interfaces make the computer and the IT-enabled systems more user-friendly systems. Features and characteristics of computer systems can be designed using intelligent tools and other advanced concepts to make them more user friendly. Factors contributing to the user-friendliness of a system are: i) what the user must learn and remember. ii) the nature of applications. iii) The nature of application programs and IV) the nature of the user interfaces. These factors have been discussed below.

i) What the User must learn: The users have to learn basic principles but do not have to remember the exact spelling or grammar for commands. All types of applications have a similar organization and appearance and are therefore easier to learn. ft interacts with the user in readily understandable terms, never forcing the user to learn or pay attention to arbitrary or irrelevant details. In this case, user manual is not required regularly, except as a reference. The users can know how the system works by playing with or modifying one or two examples.

ii) Nature of the Application: A user-friendly system pro ides easy ways to access and reuse previous work by the users, by building templates as starting points. It includes task flexibility, i.e. permitting the user to do the task in more than one way. Secondly, it is designed to minimize errors by users and to make it easy to fix any error that may occur.

iii) Nature of the Interface. Different methods arc combined to make the work more efficient, by using the interface with different applications. The menus are well structured, easy to understand, and consistent with menus in other applications. The system adjusts to, what the user knows. Novices can use basic features, only after a careful observation. Experts need not have to interact the same way as novices.

Social Implications of Information Technology

The implications of IT in the society can be studied under different headings like: -

- Social Applications
- Employment and Productivity
- Competition
- Quality of Life
- Privacy

Social Applications

Computers can have many direct beneficial effects on society when they are used to solve human and social problems through social applications such as medical diagnosis, computer assisted instructions, governmental program planning, environmental quality control and law enforcement. Computers can be used to help diagnose an illness, prescribe necessary treatment, and monitor the progress of hospital patients. Computer-assisted instruction (CAI) allows a computer to serve as "tutor" since it uses conversational computing to tailor instruction to the needs of a particular student. This is a tremendous benefit to students, especially those with learning disabilities.

Computes can be used for crime control through various law enforcement applications that allow police to identify and respond quickly to evidences of criminal activity. Computers have been used to monitor the level of pollution in the air and in bodies of water, to detect the sources of pollution, and to issue early warnings when dangerous levels are reached, Computers are also used for the program planning of many government agencies in such areas as urban planning, population density and land use studies, highway planning and urban transit studies. Computers are being used in job placement systems to help match unemployed persons with available jobs. These and other applications illustrate that computer-based information systems can be used to help solve the problems of society.

Impact on Employment and Productivity

The impact of computers on employment and productivity is directly related to the use of computers to achieve automation. There can be no doubt that the use of computers has created new jobs and increased productivity, while also causing a significant reduction in some types of job opportunities. Computers used for office information processing or for the numerical control of machine tools are accomplishing tasks formerly performed by many clerks and machinists. Also, jobs created by computers within a computer-using organization require different types of skills and education than do the jobs eliminated by computers. Therefore, individuals within an organization may become unemployed unless they can be retrained for new positions or new responsibilities.

However, there can be no doubt that the computer industry has created a host of new opportunities for the manufacture, sale, and maintenance of computer hardware and software, and for other information system services. Many new jobs, such as systems analysts, computer programmers, and computer operators, have been created in computer-using organizations. New jobs have also been created in service industries that provide services to the computer industry and to computer using firms. Additional jobs have been created because computers make possible the production of complex industrial and technical goods and services that would otherwise be impossible to produce. Thus, jobs have been created by activities that are heavily dependent on computers, in such areas as space exploration, microelectronic technology, and scientific.

Impact on Competition

The impact of computers on competition concerns the effect computer systems have on the size and market control of business organizations. Computers allow large firms to become more efficient or gain strategic competitive advantages. This can have several anti-competitive effects. Small business firms that could exist because of the inefficiencies of large firms are now driven out of business or absorbed by the larger firms. The efficiency and technological superiority of the larger firms allows them to continue to grow and combine with other business firms and thus create large corporations or strategic business alliances.

It is undoubtedly true that computers allow large organizations to grow larger and become more efficient. Organizations grow in terms of people, market share, business alliances, productive facilities, and such geographic locations as branch offices and plants. Only computer-based information systems are capable of controlling the complex activities and relationships that occur. However, it should be noted that the cost and size of computer systems continue to decrease; due to the development of microcomputers and minicomputers, and that the availability of computer and telecommunications services continue to increase, due to the offerings of computer service bureaus, time-sharing companies, telecommunications carriers, and cooperative industry ventures. Therefore even small firms can take advantage of the productivity, efficiency, and strategic advantages generated by computer-based systems.

Impact on the Quality of Life

Since computerized business systems increase productivity, they allow the production of better-quality goods and services at lower costs, with less effort and time. Thus, the computer is partially responsible for the high standard pf living and increased leisure time many people enjoy. In addition, the computer has eliminated monotonous or obnoxious tasks in the office and the factory that formerly had to be performed by people. In many instances, this allows people to concentrate on more challenging and interesting assignments, upgrades the skill level of the work to be performed, and creates challenging jobs requiring highly developed skills in the computer industry and within computer-using organizations. Thus, computers can be said to upgrade the quality of life because they can upgrade the quality of working conditions and the content of work activities.

Of course, it must be remembered that some jobs created by the computer - data entry, for example - are quite repetitive and routine and can create an "electronic sweatshop" work environment, especially if computers are used to monitor worker productivity. Also, to the extent that computers are utilized in some types of automation, they must take some responsibility for the criticism of assembly-line operations that require the continual repetition of elementary tasks, thus forcing a worker to work like a "machine" instead of like a skilled craftsperson. Such effects do have a detrimental effect on the quality of life, but they are more than offset by the less burdensome and more creative jobs created by computers.

Impact on Privacy

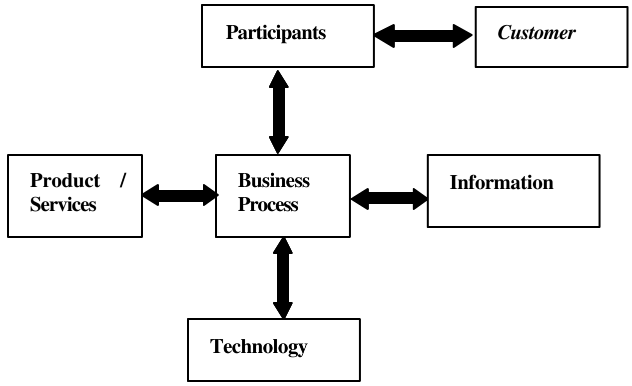
Modem computer systems make it technically and economically feasible to collect, score, integrate, interchange, and retrieve data and information quickly and easily. This characteristic has an important beneficial effect on the efficiency and effectiveness of computer-based information systems. However, the power of the computer to store and retrieve information can have a negative effect on the right to privacy of every individual. Confidential information on individuals contained in centralized computer databases by credit bureaus, government agencies, and private business firms could be m isused and result in the invasion of privacy and other injustices. The unauthorized use of such information would seriously invade the privacy of individuals. Errors in such data files could seriously hurt the credit standing or reputation of an individual.

Such developments were possible before the advent of computers. However, the speed and power of large computers with centralized direct access databases and remote terminals greatly increases the potential for such injustices. The trend toward nationwide information systems with integrated databases by business firms and government agencies substantially increases the potential for the misuse of computer-stored information.

Individual Aspect of Information Technology

People are related indirectly to the Information Technology, but any business is directly related. We can perform a Work Centered Analysis of any Business Process (BP) to reveal four links connected to this and they are: -

1. Technology
2. Information
3. Participants and
4. Product / Services



Four Links of the Business process according to WCA

In this diagram, customers as individual are linked to the BP through Product/Services.

The double headed arrows here indicate that the impacts of BP in all dimensions are both good and bad.

The amount of impact on individuals depends upon type of jobs - the healthy or unhealthy jobs. In healthy jobs, people use their skills in meaningful work by enjoying certain level of autonomy and entertaining social relationships with others. Oppositely in unhealthy jobs, people feel continuous pressure to perform from somebody above.

Impact on individual can be studied in the following headings: -

- Ergonomics and Work Environments
- Solution of RMI
- Autonomy and Power
- Skill and Knowledge
- Involvement and Commitment
- Variety and Scope of Work

Ergonomics and Work Environments

Ergonomics is the scientific study of individuals and their physical relationship with the work environment or in other word, it is the study of the mental and physical capacities of persons with respect to the various kinds of work. The word "ergonomics" comes from the Greek word, "ergos" means work and "nomos" means laws. Impacts of health, related to the physical relationship between people and their work environment are studied in this field. Thus, the science of ergonomics can be defined as the study of the laws of nature and their effects on the work environment. With respect to the office environment, this includes how the body interacts with workspace, computers, tools, and furniture.

Occupational disease/injuries were first studied by Bernardino Ramnazzini during 17th century. He is known as the father of occupational medicine. He identified that certain diseases were due to irregular motions and unnatural postures, which over time led to discomfort, pain or impaired function. This is known as *Repetitive Motion Injuries* (RMI). RMI are also known as Repetitive Strain Injuries (RSI), Cumulative Trauma Disorder (CTD) and Carpal Tunnel Syndrome (CTS), which are the most common musculoskeletal injuries currently reported in the computer related health magazine. Let us know about the origin of this disease.

Human body is made for free movement. Holding the body or a part of it in a particular position causes static muscle contractions. Muscles cannot maintain static contractions for more than a few seconds without experiencing some fatigue. Muscles engaged in static work require more than 12 times longer to recover from fatigue than muscles engaged in dynamic work. Prolonged and excessive static work causes the weakness in joints, ligaments, and tendons. This makes the workers more prone to pain and injuries. On the other hand, dynamic work allows muscles to contract and relax during the work cycle, therefore making muscles more resistant to fatigue and injury. The symptoms of RMIs are as follows:

i) Pain or stiffness in the fingers, hands wrists, forearms, elbows, or shoulders
ii) Pain or stiffness in the back or neck
iii) Tingling or numbness in the hands or fingers
iv) Loss of strength or co-ordination in the hands

v) Pain in the hands or arm that wakes you up at night

vi) Feeling a need to massage the hands, wrists, and forearms

As RIM develop slowly over a period of time, the symptoms of these illnesses can be initially very mild, but it becomes very painful and even leaves the pulses crippled if left untreated. Secondly, it will grow very fast, if the worker does not change the present work habits. The goal of ergonomics is to study work activities and the associated equipment. Secondly, it has to design jobs, tools, equipment, facilities and the environment to prevent injuries and ensure comfort and improve effectiveness.

There are various situations, where the health of intensive users is affected by the IT-enabled systems. Recently the RMI has increased due to the extensive use of the computers. These Individuals often suffer higher stress levels and related physical problems than other workers in the same businesses. This stress has been attributed to a combination of lack of control, feelings of being monitored by the boss.

Work performed at the computer terminal requires that the person hold his body still and in static postures, for considerable periods of time. This involves frequent, repetitive movements of the eyes, head, arms, and fingers. Secondly, retaining a fixed posture over long periods of time requires a significant static holding force, which causes fatigue. It is therefore required to give more emphasis on the proper and safe use of computer equipment to prevent injury. There should be studies and experiments for the safe use of the computer, including body posture, typing and mousing methods, and adjustment of the Video Display Unit (VDU) with the vision. Apart from the musculoskeletal problems, computer users sometimes experience temporary symptoms such as eyestrain, burning eyes, blurred vision, focusing difficulties, and headaches. The major causes of strain are:

i) Poor body alignment with the VDU;

ii) Prolonged sitting in particular positions;

iii) Repetitive movements;

iv) Inadequate vision capability.

A study has been done among an office assistant, who worked on Video Display Unit (VDU), and another assistant who did not work on VDU, and a professional who worked on VDU. It has been revealed that office assistant working on VDUs had the highest stress. He had to follow rigid work procedures and had little control over what he did. He felt that he was being controlled by a machine. On the other hand, the professionals, who used VDU experienced the least stress and found satisfaction in his work and had flexibility in meeting deadline. A number of studied have concluded that there are negative effects of electromagnetic emissions from VDU, which can cause various physiological and psychological problems. World Health Organization reported that *psychosocial factors are as important as the physical ergonomics of workstations and the work environment in influencing health and well being of workers.*

Solutions of RMI

In order to minimize the RMI a number of steps should be taken into considerations. Let us discuss one by one. Proper workstation design is very important in eliminating these types of problems. Some variables of workstation design include the computer table, chair and document holders. The workstation should provide the user with a comfortable sitting position that is sufficiently flexible to reach, to use, and to observe the display screen, keyboard, and related documents. Some general considerations to minimize fatigue include posture support such as back, arms, legs, and feet and adjustable display screens and keyboards. Keyboards have also a lot of design issues that have made them a subject for studies. A variety of keyboard designs are now available to assist in avoiding awkward postures related to keying the characters. Secondly, it is important to consider the typing technique and the location of the keyboard for use. Computer tables or desks should be vertically adjustable to allow for operator adjustment of the screen and keyboard. Furniture, office, and computer accessories can help keep natural postures and reduce static and forceful exertions related to RMIs. Proper chair height and support to the lower region of the back are critical factors in reducing fatigue and related musculoskeletal complaints. Document holders also allow the operator to position and view material without straining the eyes, neck, shoulders, and back muscles.

There are many developments, which can avoid the repetitive use of the keyboard and mouse, by which the user can use their hands for a limited time. This is achieved by speech recognition system. It is tool to assist in getting the job done through dictation. This has been possible by a complex technology, which requires serious dedication and training between the software and the user.

The problems due to regular use of video display unit can be minimized over time by adjusting the VDU, with respect to the body position. Secondly, the user should take a short break after each half-an-hour or so to shift position or stretch the body or walk a while. Next, problem can be minimized by mixing up a second type of physical work, which involves the dynamic work. And lastly, the user should check eyes annually or may have to wear special glasses.

There are software in-built in your computer, which will tell you, to take rest as and when necessary by you, depending on the percentage of mistake you commit on your job. There is also development in the accessories, which has changed the mode of operations from the wrist based operations to foot based operations and the likes.

Autonomy and Power

Autonomy in a job means the degree of discretion of individuals or group in the process of planning, regulation, and control of their own work. *Power* is the ability to get other people to do things. Information technology may increase or decrease autonomy and power in the work systems. It is said to be more autonomous in work system, when the individual can control the use of the tools and techniques to get the output independently. For example, a *Data Analysis System* might give more autonomy to a manager in the analysis work, which required previously, the assistance of a data analyst. Usually, professionals such as engineers and lawyers use IT-enabled systems to do work for them. It gives them more autonomy and flexibility over their work.

On the other hand, sometimes information technology is used to reduce autonomy, as and when it is required. Transaction processing and record keeping systems are examples of such systems, which require limited autonomy. These are designed to use the same rules for processing the same data in the same format by everyone involved in this repetitive process, such as order taking or producing pay checks, etc. If Individuals will be allowed for autonomy, there will be a total collapse of the system. IT-enabled system, which monitor workers closely and decrease autonomy often give rise to threats to the workers. Secondly, systems that increase employee monitoring may lead to resistance and may result in jettisoning of personnel.

Just as an IT-enabled system affects the autonomy, it can also affect power by redistributing information, changing responsibilities, and shifting the balance of power in an organization. It has increased the power of people in the entire organization, who operate on facts, information and technical competence and at the same time has reduced availability of information across the entire organization has made it possible to resolve conflicts based on facts rather than on opinions and power.

An IT-enabled system has another impact in reducing the power of middle managers in the organization. Higher-level executives are now getting information directly, by using MIS or EIS and modern communication system, such as e-mail and v-mail. In addition to this, they can go directly to the individuals who know about a particular. Therefore, middle level managers feel IT-enabled systems pose a threat to their job.

Issue of Skill and Knowledge

Information technology has positive or negative effects on people's skills. Consider the example of the use of a pocket calculator to do arithmetic. One can get the right answer quickly, but the ability to do arithmetic manually deteriorates through repetitive use of calculator. Here, the calculator has the positive impact of calculating more quickly and the negative impact of deteriorating ones skill of manual calculation.

New IT-enabled system has enhanced the skills in a wide range of jobs like MIS and EIS. It helps the manager to learn how to manage an organization, based on analysis of facts and information rather than just on intuition. Decision Support Systems (DSS) and execution systems such as CAD have helped professionals analyze data, define alternatives, and solve problems in new ways. Automating the job components also tends to reduce peoples' skill by encouraging mental dis-engagement.

IT-enabled system also has a negative effect in some cases, as in automated judgment and discretion system. In such systems, an individual's autonomy and power has been replaced with computer- programmed Experts systems for consistency and control. As a result, a less skilled person could do the same task, and the expert in that area has been devalued. Reducing the value of skills is called de-skilling. Therefore while automating the work systems, one must be vary careful in designing. The tasks, which require repetition, perseverance, and speed of operation, should be automated, rather than the tasks, which require flexibility, creativity, and judgment.

IT-enabled systems operate successfully only if participants have the necessary skills and knowledge. It requires new skills to be learnt by the professionals, and the technical staffs. The skill may involve new analytical methods or new ways to obtain information for professionals and may only be literacy for non-professional workers. The system sometimes also requires knowledge about how to use computers for specific tasks and how to interpret information in that particular system.

Involvement and Commitment

People have a tendency to do work in the same fashion they were doing earlier. Therefore, they oppose the change in the work systems. It is known as 'Social Inertia". It is required to overcome this social inertia. The main factor to counter balance the social inertia is involvement and commitment by participants and their managers. The involvement and commitment may be of different degrees, like non-involvement, low level and high-level involvement.

The situation of non-involvement will arise if the users are unable to participate or they are not invited to participate or the system is imposed on them. In this situation, involvement through advice and sign-off helps to initiate input about the priorities and features of the system and therefore reduces political problems. Secondly, if the participants are invited and properly explained and trained about the system, the system will run nicely and it can be converted to a high-level involvement.

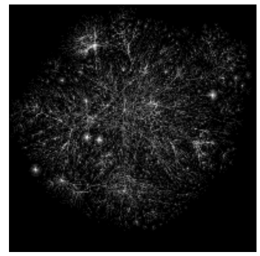
Low level of involvement and commitment makes IT based system prone to failures, although implemented properly. It always leads to overlooking the system shortcomings and organizational issues, which otherwise lead to active participation. The highest level of involvement requires continuous involvement by the participants in the project team. Sometimes a representative of the team may also manage the system. This may lead to chaos after his retirement or departure. Higher level of involvement can solve different issues such as mutually inconsistent requests from different users and different needs that cannot all be supported due to resource constraints.

Variety and Scope of Work

Information technology can either increase or decrease the variety and scope of work. It reduces variety if they force the worker to focus on a small aspect of work. The range of different types of works people do at workplace is called *task variety.* Almost all workers want a variety in their work environments and feel it monotonous if the work becomes too routined and repetitive. *Scope of Work* is the size of the work/tasks relative to the overall purpose of the organization. If the specialization is very narrow, just like a single job in an assembly line, it is a work with minimal scope. Assembling the entire job or a few numbers of jobs is a task with greater scope.

Growth of, Control of and Access to the Internet

The Internet, or simply the Net, is the publicly available worldwide system of interconnected computer networks that transmit data by packet switching using a standardized Internet Protocol (IP) and many other protocols. It is made up of thousands of smaller commercial, academic, and government networks. It carries various information and services, such as electronic mail, online chat and the interlinked web pages and other documents of the World Wide Web. Because this is by far the largest, most extensive internet (with a small i) in the world, it is simply called the Internet (with a capital I).



The Internet as mapped by The Opte Project

Today's Internet

Different Internet applications, such as Web browsers, FTP, and Telnet apart from the incredibly complex physical connections that make up its infrastructure, the Internet is held together by bi- or multi-lateral commercial contracts (for example peering agreements) and by technical specifications or protocols that describe how to exchange data over the network.

Unlike older communications systems, the Internet protocol suite was deliberately designed to be agnostic with regard to the underlying physical medium. Any communications network, wired or wireless, that can carry two-way digital data can carry Internet traffic. Thus, Internet packets flow through wired networks like copper wire, coaxial cable, and fiber optic; and through wireless networks like Wi-Fi. Together, all these networks, sharing the same high-level protocols, form the Internet.

The Internet protocols originate from discussions within the Internet Engineering Task Force (IETF) and its working groups, which are open to public participation and review. These committees produce documents that are known as Request for Comments documents (RFCs). Some RFCs are raised to the status of Internet Standard by the Internet Architecture Board (IAB).

Some of the most used protocols in the Internet protocol suite are IP, TCP, UDP, DNS, PPP, SLIP, ICMP, POP3, IMAP, SMTP, HTTP, HTTPS, SSH, Telnet, FTP, LDAP, SSL, and TLS.

There have been many analyses of the Internet and its structure. For example, it has been determined that the Internet IP routing structure and hypertext links of the World Wide Web are examples of scale-free networks.

Similar to how the commercial Internet providers connect via Internet exchange points, research networks tend to interconnect into large sub networks such as:

- GEANT
- Internet2

- Little GLORIAD

These in turn are built around relatively smaller networks. See also the list of academic computer network organizations

Internet Access

Remote access

The Internet allows computer users easily to connect to other computers and information stores wherever they may be across the world. They may do this with or without the use of security, authentication and encryption technologies, depending on the requirements.

This is encouraging new ways of home-working, collaboration and information sharing in many industries. An accountant sitting at home can audit the books of a company based in another country, on a server situated in a third country that is remotely maintained by IT specialists in a fourth. These accounts could have been created by home-working book-keepers, in other remote locations, based on information e-mailed to them from offices all over the world. Some of these things were possible before the widespread use of the Internet, but the cost of private, leased lines would have made many of them infeasible in practice.

An office worker away from his or her desk, perhaps the other side of the world on a business trip or a holiday, can open a remote desktop session into his or her normal office PC using a secure Virtual Private Network (VPN) connection via the Internet. This gives him or her completely normally access to all their normal files and data, including e-mail and other applications, while they are away.

File-sharing

A computer file can be e-mailed to customers, colleagues and friends as an attachment. It can be uploaded to a web site or FTP server for easy download by others. It can be put into a "shared location" or onto a file server for instant use by colleagues. The load of bulk downloads to many users can be eased by the use of "mirror" servers or peer-to-peer networking.

In any of these cases, access to the file may be controlled by user authentication; the transit of the file over the Internet may be obscured by encryption and money may change hands before or after access to the file is given. The price can be paid by the remote charging of funds from, for example a credit card whose details are also passed - hopefully fully encrypted - across the Internet. The origin and authenticity of the file received may be checked by digital signatures or by MD5 message digests.

These simple features of the Internet, over a world-wide basis, are changing the basis for the production, sale and distribution of many types of product, wherever they can be reduced to a computer file for transmission. This includes all manner of office documents, publications, software products, music, photography, video, animations, graphics and the other arts. This in turn is causing seismic shifts in each of the existing industries that previously controlled the production and distribution of these products. See RIAA - the Recording Industry Association of America has been particularly vocal about the problems this is causing them.

Streaming media and VoIP

Many existing radio and television broadcasters have provided Internet 'feeds' of their live audio and video streams (for example, the BBC (http://www.bbc.co.uk)). They have been joined by a range of pure Internet 'broadcasters' who never had on-air licenses. This means that an Internet-connected device, such as a computer or something more specific, can be used to access on-line media in much the same way as was previously possible only with a TV or radio receiver. The range of material is much wider, from pornography to highly specialized technical web-casts. The simplest equipment can allow anybody, with little censorship or licensing control, to broadcast on a worldwide basis. Time-shift viewing or listening is not a problem as the BBC have shown with their Preview, Classic Clips and Listen Again features.

Web-cams can be seen as an even lower-budget extension of this phenomenon. In this case the picture may update only slowly - perhaps once every few seconds or slower, but Internet users can watch animals around an African waterhole, ships in the Panama Canal or the traffic at a local roundabout live and in real time. Some sex-workers commercially allow web-cam access into their bedrooms-cum-studios, with or without two-way sound, to those who want to pay on line.

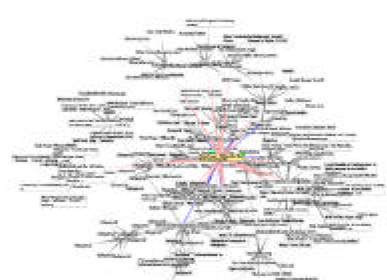
VoIP stands for Voice over IP, where IP refers to the Internet Protocol that underlies all Internet communication. This phenomenon began as an optional two-way voice extension to some of the Instant Messaging systems that took off around the turn of the millenium. In recent years many people and organizations have been working hard to make VoIP systems as easy to use and as convenient as a normal telephone. The benefit is that, as the actual voice traffic is carried by the Internet, VoIP costs much less than an actual telephone call, especially over long distances and especially for those with always-on ADSL or DSL Internet connections anyway.

The disadvantages are that it is still difficult to initiate a call with someone, unless they are at their keyboard and expecting to hear from you, and that there are still a number of competing standards that are mitigating against universal acceptance.

In all of these cases, existing large organizations, that have grown accustomed to regular incomes for their services, are finding increased competition in their service areas, coming directly from the Internet. While newcomers strive to make these inroads, the traditional industries are having to adapt, adopt, complain or suffer. Meanwhile the consumer in each case most probably benefits from the increased range of services and possible price reductions. Some worry about the lack of censorship and control while others see a continuing globalization of culture and norms.

Impact of Internet

Internet culture



The Internet is also having a profound impact on work, knowledge and worldviews. In addition to the creation of electronic commerce and communication with clients by email and related means, the Internet is transforming other aspects of the workplace. Certain companies have adopted the use of blogs, which are largely used as online diaries, for promotional purposes. Since most people search the Web looking for information, these easily-updatable websites can be filled with advice on the company's area of specialization. The company's hope is that, when the visitor finds this free information, they will note the appearance of expert knowledge and may be drawn to the business' site as a result. An example of this practice is Microsoft, which has allowed its developers to publish their own personal blogs in order to pique the public's interest in their work.

Graphic representation of the WWW, a service running over the Internet, as represented by hyperlinks

The World Wide Web

Through keyword-driven Internet research using search engines like Google, millions worldwide have easy, instant access to a vast and diverse amount of online information. Compared to encyclopedias and traditional libraries, the Internet has enabled a sudden and extreme decentralization of information and data.

Cultural awareness

From a cultural awareness perspective, the Internet has both an advantage and a liability. For people who are interested in other cultures and the worldviews of those cultures it provides a significant amount of information and an interactivity that would be unavailable otherwise. However, for people who are not interested in other cultures and worldviews there is some evidence indicating that the Internet enables them to avoid contact to a greater degree than ever before.

Current and potential problems

The Internet, along with its benefits, has a lot of negative publicity associated with it ranging from genuine concerns to tabloid scaremongering.

Child abuse

According to children's charities, the number of annual convictions for child pornography offences have increased by over 1000% since the Internet was first available to the public in the late 1980s. With the recent growth in Chat rooms and instant messaging services in the late 1990s, the potential for a new form of child abuse has emerged: so-called grooming. This involves a pedophile pretending to be a child in a chat room/instant message conversation, to gain the trust of a child before arranging to meet up.

Copyright infringement

Copyright infringement has also been the focus of much media attention, mainly through peer-to-peer file sharing software, but also through private members-only chat rooms, so-called warez sites (which openly offer illegal copies of software or the means to crack copy protection), or even the sale of counterfeit CDs, DVDs and software masquerading as legitimate product. Many ordinary Internet users are less concerned about the actual infringement itself but more about the effect on the Internet as a whole if tighter controls result from the infringement.

Viruses

In the 1980s and early 1990s, when very few people had access to the Internet, viruses were not a huge problem. They did exist and did cause just as much damage to computers as modern viruses can today, but there was no fast-moving epidemic because there was no means for a virus to directly infect other computers. Before the Internet, the only way for a computer to be infected was through use of a removable disc that was itself infected. As a result, virus infections were mercifully rare.

All that changed with the widespread growth of the Internet. With near-universal Internet access among computer users in developed countries, and the proliferation of high-speed broadband Internet connections, a virus on one person's computer can infect thousands of other computers. In fact, much of the disruption from virus outbreaks is caused not by the payload of the virus (e.g. deleting hard drive, shutting down computer every five minutes), but by the Internet congestion caused by the virus spreading itself.

Security cracking

When computers were stand alone machines (or at most connected to a company's internal network), to steal data from a system an intruder had to physically steal it. The Internet means that data from an insecure site could be stolen by someone working two blocks from the site, or just as easily from another country.

Some of the recent high-profile examples of this were when a working version of the source code for Half Life 2 was copied from the developer's computer systems by security crackers and when portions of the Windows NT code base were copied from one of the companies that had access to it via the Microsoft Shared Source initiative. In both cases the Internet was used for dissemination of the leaked code, in particular using P2P networks.

Dated technology

Very few people outside the technical community are aware of the future problems posed by the Internet's archaic technology. It was originally designed for a small number of research institutions to share research data, and was never intended for the multi-billion user behemoth the modern Internet has become.

One serious problem is that the IP address (a unique number assigned to each Internet computer, functioning much like a street address in the real world) will run out eventually. Despite an estimated world population of over six billion, there are only a little over four billion different IP address combinations possible under the current system (see IPv4 address exhaustion for more information.) This also doesn't take into account the fact that there is not a 1:1 person to computer ratio in current computerized countries, where many people will have a desktop machine at home, a laptop machine for on the go, another desktop machine at work, and an e-mail mobile phone, all requiring their own IP address.

This could pose serious problems in the future as more and more nations expand their computer infrastructure (the vast majority of the world's population does not currently use the Internet, that's the so-called digital divide) and even now efforts are proceeding to find new ways of running the Internet. The new version of the Internet Protocol, IPv6, which expands the address space of the Internet, is one proposal for how to deal with some of the technical problems caused by the growth of the Internet.

Self-destructive subcultures

The neutrality of this section is disputed. Since the early 1990s, it has been widely recognized that the Internet enables broader distribution of all ideas, including those considered distasteful by any portion of the population. The most widely condemned of these ideas are those that promote, condone, or justify the infliction of violence upon innocent, non-consenting people. Examples include racism, sexism, and fascism.

Around 2000, The Atlantic Monthly and other publications revealed a similar but distinct issue: The Internet also allows people who exhibit or wish to practice abnormal behavior to find one another easily, due to anonymous search engines and online forums or services. As sparse subpopulations, it was often unlikely or difficult to find willing partners or like-minded individuals prior to the Internet.

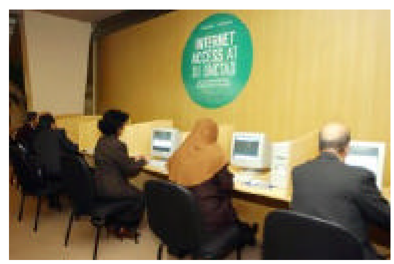
A small number of these subcultures promote self-destructive or mutually destructive behavior. Websites and mailing lists exist that explicitly promote anorexia, apotemnophilia, necrophilia, and suicide. While these activities are easily recognized as abnormal and self-destructive by most adults, many people fear that children or mentally ill persons visiting such sites would lack the maturity necessary to make that discrimination.

In rare cases, people have used the Internet to find willing partners for abnormal activities, but with disastrous or fatal results. In one case, a German named Armin Meiwes (the "German cannibal") made an online arrangement with Bernd Jürgen Armando Brandes to kill and eat him. Meiwes was later convicted of manslaughter. (This is sensitive and a serious impact of internet).

Censorship

Some countries such as Iran and the People's Republic of China restrict what people in their countries can see on the internet. The BBC is proposing to offer its entire range of terrestrial television broadcasting as free downloads, but only to people within the UK. At the moment most internet content is available regardless of where one is in the world, so long as one has the means of connecting to it.

Internet access



Common methods of home access include dial-up, which is the slowest, landline broadband (over coaxial cable, fiber optic or copper wires) and satellite.

Public places to use the Internet include libraries and Internet cafes, where computers with Internet connections are available. There are also Internet access points in public places like airport halls, sometimes just for brief use while standing. Various terms are used, such as "public Internet kiosk", "public access terminal", and "Web payphone".

Wi-Fi provides wireless access to computer networks, and therefore can do so to the Internet itself. Hotspots providing such access include Wifi-cafes, where a would-be user needs to bring their own wireless-enabled devices such as a notebook or PDA. These services may be free to all, free to customers only, or fee-based. A hotspot need not be limited to a confined location. Whole campuses and parks have been enabled, even an entire downtown area. Grassroots efforts have led to wireless community networks.

Aart from Wi-Fi, there have been experiments with proprietary mobile wireless networks like Ricochet, various high-speed data services over cellular or mobile phone networks, and fixed wireless services. These services have not enjoyed widespread success due to their high cost of deployment, which is passed on to users in high usage fees. New wireless technologies such as WiMAX have the potential to alleviate these concerns and enable simple and cost effective deployment of metropolitan area networks covering large, urban areas.

Broadband access over power lines was approved in 2004 in the United States in the face of stiff resistance from the amateur radio community. The problem with modulating a carrier signal over power lines is that an above-ground power line can act as a giant antenna and completely jam long-distance radio frequencies used by amateurs, seafarers and others.

Countries where Internet access is a commodity used by a majority of the population include Iceland, Sweden, Australia, Denmark, the United States, Canada, the UK, The Netherlands, Japan, Singapore, Taiwan, South Korea and Norway. The use of the Internet around the world has been growing rapidly over the last decade, although the growth rate seems to have slowed somewhat after 2000. The phase of rapid growth is ending in industrialized countries, as usage becomes ubiquitous there, but the spread continues in Africa, Latin America, the Caribbean and the Middle East.

However, Internet access is unequally split between low-speed and high-speed accesses. ADSL or other broadband access is rare or nonexistent in most developing countries; even in developed countries, high prices and average performances may limit its penetration (most countries in Eastern Europe, the United States), while low prices and high performances may attract a large number of consumers (Scandinavia, France). Even within the same country, wide differences may exist between larger cities (often having multiple providers of broadband access) and rural areas (where often no broadband access is available).

The expansion of the availability of Internet access is a way to bridge the so-called digital divide.

Internet Explosion / Growth and Access

A survey showed that 74.7% Americans felt television had contributed to the breakdown of the American family and 83.8% felt TV and movies had influenced or caused an increase in "crime". In view of the above facts, a question may be asked who want Internet and for what purposes? In order to answer this "loaded" question, we must first examine the rate of expansion of Internet around the World.

This section presents various statistical data collected from various different surveys carried out by a number of independent organizations.

The interested major groups for Internet are corporate (46%), academic (15%), recreational consumers (27%) and occupational (12%). The users of the Internet are 87% white, 5% African-American, 3% Hispanic and 3% Asian.

The top reasons for using are entertainment (51%), news (49%), travel (30%), computer production (41%) and financial (26%). However, only 28% of consumers considered video on demand highly desirable. Fifty seven percent (57%) Americans would like to participate in interactive, electronic town-hall meetings with political leaders and other citizens; and 46% want to send text or E-mail to elected representatives.

During 1996, more than 6400 articles were published on various subjects such as games (243), education (152), travel (78), entertainment (226), investment (151), finance (81), sport (58), medical (72), security & privacy (900), shopping (ISO), newspaper (25) and on social implications (40).

Growth in the number of hosts station around the world has been un-imaginable. Let us review some of the top growth countries China (1003%), Malaysia (686%), Indonesia (521%), Peru (518%), Brazil (305%), Finland (148%), Japan (211%), Singapore (368%) and England (99%).

23% of the Americans over the age of 16 years have access to the Internet. $23 Billion will be spent worldwide on Internet by year 2000. USA generates 70% of the total traffic. However non-USA traffic is growing more than twice the USA growth rate. Average time on-line is about 2 hour daily. It is interesting to note that there were only 15 Million USA Internet users in 1995. By 1996 this number has increased to 48 Million. The numbers of Web server responses were only 60k in early 1996 but it has jumped to more than 500k by the year-end. It is estimated that 75% of Fortune 1000 companies will have Internet access by mid 1998. The number of Internet users is projected to be 163 Million by year 2000. The number Europeans Internet subscribers by year 2002 may be 200 Million.

Access to Internet is available not only at home and office but also in cafes which will operate like coin telephones. In fact, there are ten coin operated Internet access terminals are in operation in various cafes in Los Angles, USA.

Advertising via Internet is a big business. It is estimated that the advertisement revenue generated by Web for year 1997 to 2001 is $0658, $1.6B, $3, 15B, $4.53B and $59 Billion respectively.

It is interesting note that in computer literacy and availability, Japan is surprisingly defiant. Only 20% of business people use PCs vs. 90%-plus in America and 50% in Europe.

In view of the above factual data, the importance of using PC can not be over emphasized. Our family members including housewives must learn about the PC. Using PC will become as common and important as to know how to drive a car in a western country. With the above rate of expansion of Internet let us examine a number of other key social implications / factors of the future information networks.

Implications of Information Networks

Introduction

The Internet's implications can be studied in five domains:

1) Inequality (the "digital divide");
2) Community and social capital; (**or Social implications**)
3) Political participation;
4) Organizations and other economic institutions; and
5) Cultural participation and cultural diversity.

Here, we are concerned about the 2) Community and Social Capital, which discusses about the social implications of Information Networks.

Social Implications of Information Networks

Privacy

The interactive systems from the home accumulate an awful lot of information on individual behavior, not only what people buy, but how they vote, the books and entertainment they like, etc. Prodigy admitted that their computers could and did collect the information from their subscribers at their will without the knowledge of the subscribers.

For these reasons, Marc Rotenberg of the Electronic Privacy Information Center (EPIC) stated to the US congressional committee "Americans have grown up watching television. Now a new generation would grow up with television watching them".

North Carolina Information Superhighway is working to develop a criminal information network that will be useful throughout the state both to officers on patrol and to officers of the court, such as prosecutors and judges.

The US government favors two initiatives that horrify privacy advocates. One tries to coax industry to use an encryption technology, the so-called Clipper chip, that lets government listen in on otherwise secure communications. The other is proposed legislation that would give federal agents and policemen unprecedented authority to watch every move people make on-line. Some industry analysts believe that these schemes may prevent some companies from putting on the superhighway new services that cannot be engineered to permit access by law enforcers. There is also tremendous pressure on government for deregulation, especially in telecommunication industry.

Security

The National Computer Security Association (NCSA) held a conference in January 1994 to discuss the security issues and a number of questions were discussed such as

- Who will be the traffic cops on the network?
- How will I keep my information private?
- Is this network similar to Internet where hackers, thieves, network surfers and all sorts of undesirables are after my data?
- How can I protect my system if it is connected to the network?

Questions after questions were asked but unfortunately very few real answers were given.

Let us examine some of the real incidents that had happened recently.

- An imposter posing as a Texas A&M professor sent out hate e-mail containing racist comments to thousands of recipients. He was still not found. Unfortunately the poor professor still has to face the "music" even today.
- A student at Dartmouth University sent out Email over the signature of a professor late one night during the exam period. Claiming a family emergency, the forged mail canceled the next day exam, and only a few students showed up.
- 42 pizza delivering boys arrived at a house at the same time.

Firewalls

Internet firewalls offer protection against attacks from outsiders, but they can never provide complete protection for a network. Firewalls force all users to enter and leave at a specific point, allowing the company to closely monitor all traffic.

The firewalls can also prevent unapproved services from entering the network. Firewalls protect the network from spreading a specific problem from one part of the network to another part of the network. Firewalls offer no protection from internal users with malicious intentions. Firewalls are designed to guard against the known threats, but they may be less successful when guarding against new threats. Firewalls do not block viruses.

The electronic messaging association has developed a white paper to help companies develop an e-mail policy, and baseline software's information security policies made easy provides more than 700 policies in addition to policy templates.

According to legal and policy experts, the company has a right to read employee e-mail. In view of the above can we really say that an individual will have privacy or personal security during the course of the Information Superhighway?

Globalization of labor force

Network connection to developing countries is already boosting software development business. Several well-known computer companies in the USA and Europe are taking the advantage of the availability of cheaper and highly skilled workforce in the field of software engineering. These computer companies are working with some local software developing companies in Pakistan and India. Networking has shown clear advantages in this situation.

Software products are developed and files are transferred to contracting companies in USA or Europe. The software development cycle is quite efficient. This type of activity allows those companies to remain competitive in market. This approach is leading towards the globalization of labor force and the large companies are really becoming "truly international" organizations. Another advantage of Globalization of labor force will be less brain drain from the developing countries and at the sometime it will improve the standard of living due to higher wages paid by the International companies as compared to the local employers. However local employers will have difficult time in retaining and hiring of high caliber staff. This will be one the most important challenges for the developing nations to face in the advent of the Information Superhighway.

Children development - excessive information available

Teachers and parents will interactively assess student academic and extracurricular progress. With all boundaries removed, creative educators will be free to explore new ways of teaching and motivating our kids. Inadequate teacher training has been a huge obstacle and may continue in the future. School districts spend 90% of funds on new equipment and only 10% on teacher training.

Seymour Papert, M.I.T., envisions a day when children will have access to what he calls the "Knowledge Machine". This machine will allow children to explore areas of intellectual interest without any barriers such as limitations on the school library, etc. The addition of smell or touch would revolutionize the learning process.

Some educators predict that within the next decade, families will be able to log on to school through their computers. This kind of communication would be particularly useful in the rural areas where transportation can be a difficult problem.

Office of the future

The most visible social change for a white-collar worker will be the disappearing of his normal desk. In addition, there will be no need to be physically present in the office every day. At IBM in New Jersey, there are only 220 desks for the 650 sales and service staff. This concept is called "*hoteling*". The virtual office is the one where can sit anywhere you want and be connected. Portable jobs will allow firms to save on everything from travel to real estate. Personal video conferencing is projected to be doubled over the next three years.

Computer and Telecommunications companies (IBM, AT&T, GTE, US WEST, APPLE, AST, COMPAQ, etc.) have already cut more than 200,000 jobs. This is one of the casualties of the information economy.

Mobility

Telephones will grow smaller, lighter, less expensive and more powerful. Personal digital assistants (PDAs), screen-based phones and "smart" cable boxes that will also be used to access digital data. PDAs are turning from electronic day-timers into wireless communications devices that let you link up with a highway in the sky. The key is a credit card size communications module that fits into many communication systems for various applications that basically based on pager technology. A number of companies plan to introduce smart phones, devices that let you call your mother, catch up on the news, check stock prices, and pay your bills using the devices built-in LCD and touchpad.

Motorola has ordered 66 satellites from Lockheed for its Iridium project which worth about $3.5 billion for mobile communication. Microsoft and McCaw Cellular have proposed a $ 9 billion plan to launch 840 communication satellites by the year 2001.

Paperless society

With customers in control of the information, advertisers will have to rethink what to sell, how to sell it, and how to make consumers pay attention. Sales of computer games and games machines expected to double by 1999 at least $15 billion. It is predicted that by on-line and interactive-TV sales could total closer to $20 billion by 1998. By 1995, it is estimated that 40 percent of all computers purchased in U.S.A. will be used in homes. With the introduction of computers and electronic devices, the normal media (paper) of conveying information and / or messages will be minimized and most likely be totally eliminated. Hence the society will become a paperless society.

This change may not be an easy task for people particularly over 35 - 40 years now. In our present society, physical touch plays an important role. Office staff as well as housewives have to be re-trained and re-developed new criteria for the selection of furniture and decoration.

Impacts on our social life

All these technological and economical developments are bound to have impacts on our social life. At this moment, it is very difficult to determine exactly what will be the fundamental changes in our social behavior. However, some of the obvious expected changes and impacts on the society are projected to be:

- "We will live part of our lives in the world of cyberspace in the world of virtual reality" – Jaron Lanier, *father* of virtual reality.
- "It will revolutionize and expedite the educational process, free up valuable land that can be restored to its natural state, reduce the frequency of ground and air travel (and thus pollution), and deliver three-dimensional pictures and sound, enabling people to spend less time doing mundane things and more time thinking" - Shelly Duvall, Think Entertainment.
- The information age, learning will be a lifelong process.
- In a world of information glut, it becomes progressively harder to sift the useful from the useless, the important from the trivial, the true from the false. In such a world, the database itself supplies no standards for judging one piece of information superior to or more useful than another. Indeed, in such a world, standards cease to matter. Information itself becomes background noise, like the constantly running tap that television has already become. Is this progress?
- Cultural and religious norms will face serious pressure to change the "standards" and may result in family conflicts and communication gap probably will be widen among the family members of "two young generations".
- On the hand a number of public interest groups shudder at the thought of applying this technology to on-line gambling.
- Finally, it will ultimately change the way everybody does business and communicates with each other.

Impacts on Time Use, Community and Social Capital

Initially it was anticipated that the Internet would boost efficiency, making people more productive and enabling them to avoid unnecessary transportation by accomplishing online tasks like banking, shopping, library research, even socializing online. More recently, some studies have suggested that the Internet may induce anomie and erode social capital by enabling users to retreat into an artificial world. We will study these aspects in this section below.

Time Displacement

Much of the debate over social capital is about whether the Internet attenuates users' human relationships, or whether it serves to reinforce them.

Internet users may substitute time online for attention to functionally equivalent social and media activities. Internet users may reduce the time devoted to off-line social interaction and spend less time with print media, as well as with television and other media. Conversely, some different studies have shown that Internet users read more literature, attended more arts events, went to more movies, and watched and played more sports than comparable nonusers.

A research found Internet use substituted for other interactions. Higher levels of Internet use were "associated with declines in communication with family members, declines in social circles, and increased loneliness and depression."

The effect of Internet use may vary with user competence also. Compared to experienced Internet users, the novices engaged in more aimless surfing, were less successful in finding information, and were more likely to report feeling a souring of affect over the course of their sessions. Their negative reactions reflected not the Internet experience but the frustration and sense of impotence of the inexperienced user without immediate access to social support.

Community

Internet has contributed to a shift from a group-based to a network-based society that is decoupling community and geographic propinquity. Moderate to heavy-users' self-reported that there is substitution of email for telephone contact as part of their loss of "contact with their social environment." Some users say that Internet enhances social ties defined in many ways, often by reinforcing existing behavior patterns. Internet puts users in more frequent contact with families and friends, with email being an important avenue of communication.

The Internet makes it easy for people at a distance to assemble and communicate with many others at the same time through chat rooms or online discussion forums like virtual communities, interactive networks. Compared to real-life social networks, online communities are more often based on participants' shared interests rather than shared demographic characteristics or mere propinquity. Nonetheless, issues related to racial, gender, and sexual dynamics do permeate and complicate online interactions.

Internet users maintain community ties through both computer-mediated communication and face-to-face interaction. Although they maintain more long-distance relationships than do non-Internet users, they communicate even more with their neighbors. Although the Internet helps maintain contact over long distances, most email contacts are between people who also interact face-to-face. In other words, the Internet sustains the bonds of community by complementing, not replacing, other channels of interaction.

Social Capital

Internet facilitates the creation of social capital and other public goods by making information flow more efficiently through residential or professional communities. Increased Internet use tended to have a direct positive effect on social capital (by increasing the as participation in community networks and activities) and a positive indirect effect through social capital on political participation. Internet builds social capital by enhancing the effectiveness of community-level voluntary associations.

Internet also provides significant benefits to people with unusual identities or concerns (e.g., rare medical conditions). But there is some evidence that "social capital" produced by less focused networks is rather thin.

As a drawback of the Internet, it has also been described as an inexpensive and effective means of organizing oppositional social movement.

Morals of these Impacts

1) Internet has no intrinsic effect on social interaction and civic participation.

2) Internet use tends to intensify already existing inclinations toward sociability or community involvement.

3) We need to know more than we do about the qualitative characters of online relationships.

4) We know that virtual communities exist in large number, but we know relatively little about their performance.

5) We need more systematic studies of how civic associations and social movements use the Internet

Source:

- Jaleel Syed Hasan, *Information Superhighway and its Social Implications*
- Paul DiMaggio, Eszter Hargittai, W. Russell Neuman and John P. Robinson, *Social Implications of the Internet*

International Issues of Information Society

This section attempts to synthesize issues that have come to be seen as constitutive of a hypothetical information society. These issues have been articulated by government, business and civil society through their input into processes in the UN sphere and other selected bodies. These include declarations to which UN member states were signatories; input into the World Summit on the Information Society (WSIS) by the Coordinating Committee of Business Interlocutors (CCBI), which includes the World Economic Forum; and regional and international statements of civil society formations involved in the WSIS process.

The Foundations of an Information Society

Information systems have long been seen as entities that not only embed society, but also as agents that help to define it. From a global perspective, the major task here has been to develop a shared understanding of the concept of an information society. Such a vision might be best articulated in terms of the fundamental basis of an information society and the roles of societal actors - governments, business and civil society - within it.

Two general perspectives on the fundamental basis of information society seemed to have emerged:-

1. Information society as a consumer-oriented environment containing tools, applications and services; or
2. Information society as a global commons enabled by ICTs in which human needs are central.

Many civil society entities have, on the other hand, articulated positions that would place human at the center of a conception of an information society. In this viewpoint, an information society's purposes, development, operation, and governance would take place within established human rights frameworks and would be evaluated on its ability to meet human needs.

Governments have articulated a spectrum of positions on their roles in an information society. These articulations have been made individually and collectively. The Millennium Declaration of the United Nations states that governments should "...ensure that the benefits of new technologies, especially 'information and communication technologies..." are enjoyed by all people. One implication of this declaration is that ICTs are to be seen as major tools for meeting the series of highly ambitious goals it set forth, including the provision of elementary education to all children by 2015. The G8 nations articulated a similar, but less-specific message. One role of government in an information society that is implicit in these efforts is as a catalyst and organizer for business and civil society processes concerning an information society. For example, the G8 initiated the Digital Opportunity Task Force (DOT Force) to address the digital divide and other societal matters. Positions of member states with respect to these declarations vary. The European Union sees, in part, its role as bringing "the Information Society closer to all citizens of Europe, develop the economic wealth, address growing social needs, and focus on cultural identity and diversity".

A significant number of IT-related NGOs and other NGOs that have added IT issues to their agendas have adopted positions on the role of civil society that might be characterized as running counter to positions articulated by some governments and business or, at the very least, as acting as a balance. Many see civil society as providing oversight for governmental and business activities within an information society that has become global and increasingly privatized. To this end, many entities within civil society have called for greater transparency and citizen involvement in the operation of an information society.

Digital Divides in an Information Society

One of the most critical sets of social, ethical and policy issues that must be dealt with in realizing an information society is the phenomenon known as the digital divide. In domestic contexts this is often defined as a gap between individual citizens having access to Internet technologies and those who lack such access. In a global context, the concept of a digital divide has been expanded to include communities, nations and whole regions. Citizens in many parts of the world still do not have basic access to ICTs. This includes not just Internet, but basic telephony. Democratic access to and use of ICTs, such as digital government services, cannot be truly realized in a society as long as there remain citizens who lack access to both appropriate technologies and technical skills to make use of these services.

It is now being explicitly recognized that a complex of different types of barriers exist, not just the proverbial, monolithic "digital divide". A major focus is on addressing barriers faced by those nations defined by the United Nations Development Program as the least developed countries (LDCs). Given this broader understanding of the digital divide, the other issues that many see as needing to be addressed include: social, economic, and educational barriers; political and social barriers; requirements for achieving universal and equitable access; information as a public good, with due consideration for intellectual property; freedom of expression and of the media; supporting cultural and linguistic diversity in circumventing barriers; and the distinct roles of governments, civil society and the private sector in bridging barriers to an information society.

Access to information (e.g., digital government services) is not only characterized by access to technologies, however. Access to information has been studied on several levels, including the properties and characteristics of access, as well as the means and availability of access. Means and availability of access are not dependent only on the economic status of individuals or their communities, but also on information usage skills and geography. Access to the equipment, software and telecommunication services necessary for Internet access must obviously be accompanied by skills to make use of them.

A digital divide can also be characterized by the information itself: its costs, representations, communication processes used to convey it, and ways it is collected from one's surrounding environment. Of particular concern here are potential cost barriers for information and representations of government information. Certain sources of public information are not free.

Communication researchers have demonstrated that the valuation of information, in general, presents a unique problem for those who are economically disadvantaged. Beyond the basic consideration of whether one can afford to pay for information, the value of information is more uncertain than most other types of goods. Its usefulness cannot be conclusively determined until it is used. Thus, those who are less able to afford information are also less likely to take chances buying it because its usefulness may be highly uncertain to them.

Finally, the cost of technology may present a barrier to information. Policy making to address the digital divide must mandate the leveraging of ICTs to provide access points via more widely available technologies, These include telephony-based applications and low cost Internet appliances such as handled computing devices or Internet appliances. Policies must also encourage innovations that allow electronic government documents to be used in the context of human agent environments, wherein citizens can still communicate directly with government officials.

Developing a Framework for an Information Society

A framework for an information society would define the functional, regulatory and developmental aspects of the society, as well as its relationship to existing human rights and international frameworks. Functional issues would include education, addressing the needs of workers, facilitation of technical literacy, and support for commerce. Regulatory issues are seen as including data protection, privacy and network security, intellectual property rights, public domain and fair use, and the establishment of appropriate policy and market structures. Developmental aspects of an information society would address sustainable and environmentally responsible development of ICTs, appropriate new and traditional ICTs, capacity building in governments, civil society and the private sector, financing and deployment, and examination of social and regulatory impacts of this framework. In this context, participatory design is recognized by many as an indispensable tool for ICT development. An integral part of the developmental processes of an information society would also include a continuing process of implementation and review of the framework itself.

A Knowledge Society Perspective

This issue considers the special relationship between information and knowledge, where knowledge is derived from processes of organizing data and information to convey domain-specific understanding, experience, expertise, and learning. An information society has often been characterized as a knowledge society. From this perspective, an information society is seen as enabling the creation and management of knowledge as the primary benefit to humankind. Social, ethical and policy considerations in this context relate to: the establishment of general educational goals sought through the use of an information society; enabling distance learning; facilitating both formal and lifelong learning: the development of information literacy, including critical appreciation of information and content development skills: access to knowledge; support for cultural and linguistic diversity; and support for needs of young people. This perspective also recognizes that capacity building in academia is necessary to support a knowledge society.

Defining Rights and Governance

A major area of contention in the development of an information society has been in defining and enforcing the rights of all stakeholders, as well as the particulars of its governance. Critical issues in addressing rights and governance are: democratic management of international bodies dealing with ICTs; information and communication rights of governments, business and citizens: privacy and security policies and rights censorship and regulation of content; the role of the media; defining, identifying and responding to criminal activities within an information society; the application of ICTs for government and decentralization; and media ownership and concentration. A major-emphasis here for civil society and some governments has been to establish support for the empowerment of citizens. In addition, many see an information society as enabling the reform and strengthening of democracy.

Infrastructure Development

Infrastructure is a nexus for many of the major technical, social and policy issues in the realization of an information society. The goal here is the evolution from the present technical state to one in which all of the benefits envisioned from an information society can be realized. Key issues here are: the extension of Internet connectivity to areas that are under-served or not served at all: the application of wireless technologies, particularly to realize the economic benefits that technological “leap frogging” affords developing nations; the development of new advanced ICTs to meet outstanding human needs in all societies; the building of bridges between different types of media, including radio, television, print, and the Internet, addressing the needs of rural communities; and the availability of ICTs needed to address emergency situations around the world. Perhaps the most contentious set of social, ethical and policy issues in this category has been in defining the balance of roles between private sector investment, government subsidy and civil society efforts in creating information society infrastructure.

Development and Employment

An information society is seen as having the potential to greatly affect development and create employment. This can be seen, as discussed above, beginning with the communication for development movement among members of the ITU. The key issues here are: the creation of economic opportunities; the role of ICTs in health, agriculture, labor. culture, and other life-critical sectors; the role of ICT-based communications for development; the training of workers for an information society; a consideration of the realities and dangers of labor exploitation in ICT-based sectors; an examination of the roles and impacts of investment and speculation in ICT-based development; and the role and limits of e-commerce in development and employment.

Tools, Services, and Applications

Being enabled mainly by scientific and technological achievements, the dominant focus of conceptions of an information society through the early 1990s was techno-centric, viewing an information society merely in terms of the technical feasibility of classes of tools, services and applications. The increasing influence of social and community informatics perspectives has, however, changed the focus to one of considering which tools, services and applications should be used or developed with regard to their social impacts and human needs that must be addressed. The broad technical issues in this category are: the development of technologies that facilitate active citizenship and improved government; technological support for universal access to knowledge and global communication and cooperation; and the improvement of the standard of living adequate to the health and well-being of all citizens. Specific issues include: the building of bridges between the communication modalities of radio, television, press, and Internet: the development of ICTs for e-government, including citizen input into political processes: support for disaster mitigation and relief operations; support for long-term data retention and archiving for cultural preservation; and tools to facilitate cross-sector co-operation.

Citizens and Communities

A number of issues have been contributed mainly by civil society to the conception of an information society that falls outside of commercial and governmental perspectives. The major issues here are: the creation and preservation of an electronic commons, free public spaces and technical resources that can be used to meet human needs; community control of ICT infrastructures; continuing support for open source technologies; capacity building for communities to participate in an information society: and addressing the multiplicity of dimensions of diversity, including linguistic and cultural diversity. Specific issues here are: the empowerment of communities through ICTs; preservation of culture and language: support for oral information and cultures; support for independent, community controlled media; meeting the needs of people with disabilities; meeting the needs of the elderly; providing support for cross-cultural communications; stemming the technological "brain drain" from developing countries; content dumping, which is the subsidization of information production and its delivery far below cost to culturally vulnerable populations; and geographic-specific issues, such as problems of rural access to ICTs.

Gender Perspectives

It is well understood, "technological designs are also social designs". The social designs in information technologies and processes used to achieve them reflect society's gender biases - among others. Design processes that do not take gender issues into account run the risk of producing information technologies and services that do not adequately address the needs of women.

The promotion of gender equality has been recognized by growing numbers of stakeholders as at that is not only important to women, but a necessary condition for improving all societies given the central roles and responsibilities that women have. The broad issues that have been raised in this context are reducing gender discrimination and improving participation of women in an information society, capacity building and training for women, and the use of ICTs to improve the lives and livelihoods of women worldwide. Specific issues include: supporting wide participation by women and gender specialists in policy and decision making at all levels in the ICT sector; supporting women's greater access and control over resources necessary for their empowerment; improving the participation and representation of women and gender equality advocates in all levels of policy making; reform of decision-making processes in the ICT sector; the development of ICT applications for supporting women's reproductive and productive roles, and in education and literacy programs; the development of ICT applications for reducing violence against women; and addressing issues of pornography and other forms of exploitation that are enabled by ICTs.

<u>Source:</u>

- Linda L. Brennan & Victoria E. Johnson, *Social, Ethical and Policy Implications of Information Technology*

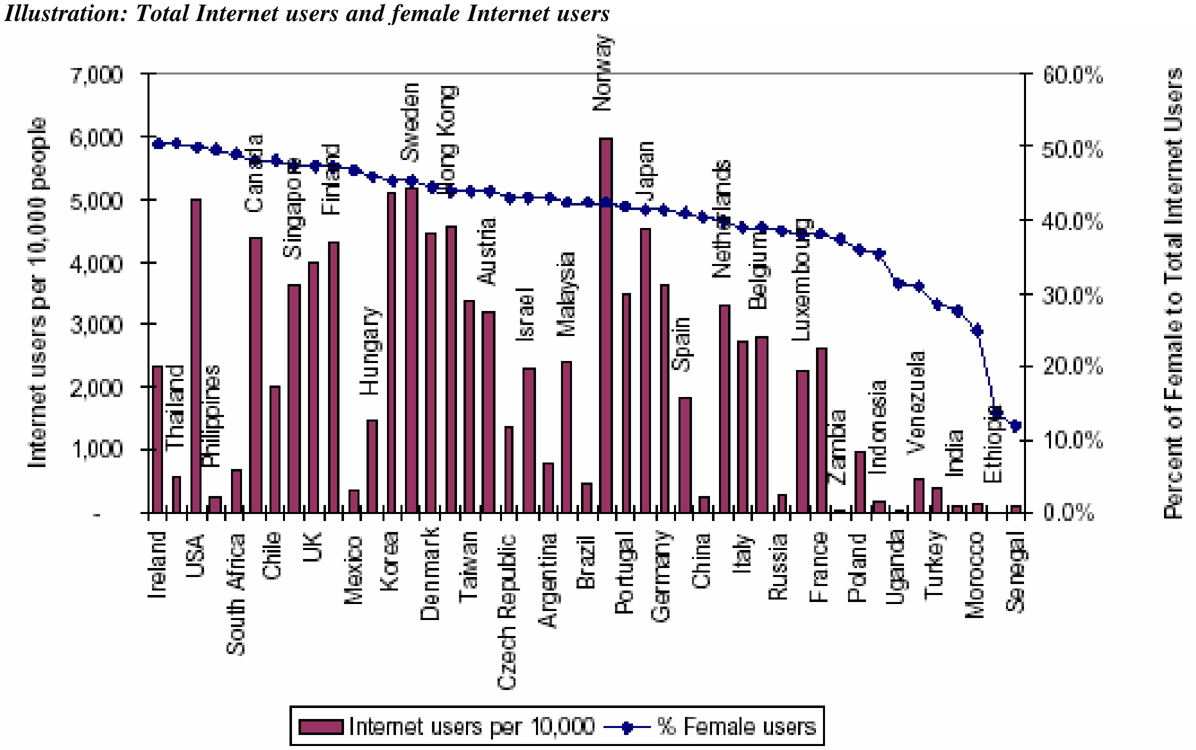
Gender Related Issues in ICT

The gender related issues can be discussed under different headings. But the main theme should be the understanding the difference existed in all aspects between the two genders. Bearing in mind that men have dominated in almost all aspects, the women-centric analysis is carried out in many aspects.

Women's Access to the Internet

The gender divide within the digital divide can be seen in the lower numbers of women users of ICTs compared to men. The majority of the world's women are excluded from the World Wide Web. The digital divide within countries broadly reflects the gender divide. Women are in the minority of users in almost all developed and developing countries. Even in the developing countries where women do make up a high percentage of users, total users themselves constitute very small elite. In some cases where women make up a relatively high percentage of users, the total proportion of the population using the Internet is very small.

Illustration: Total Internet users and female Internet users



Source: Huyer and Mitter, 2003.

Socio-cultural Barriers to Women's Access

There are number of reasons why women have lesser access to ICT. To mention a few, they are socio-cultural attitudes, lack of women's interaction with technology, resource constraints and so on. For the majority of women, specific barriers include illiteracy, unfamiliarity with the dominant languages of the Internet, absence of training in computer skills, domestic responsibilities, and the fact that the information delivered by ICTs is not that valuable to them. Infrastructure itself is also a gender issue: it is concentrated in urban areas and more women live in rural areas. Also, public ICT facilities have a great tendency to become men-only spaces, effectively inhibiting women's access
The following are some socio-cultural factors that impede women's use of ICTs, particularly in rural areas:

- Cultural attitudes discriminate against women's access to technology and technology education.
- Women are less likely to own communication assets – radio, mobile phone.
- Women in poor households do not have the income to use public facilities.
- Information centers may be located in places that women are not comfortable visiting.
- Women's multiple roles and heavy domestic responsibilities limit their leisure time. Centers may not be open when it is convenient for women to visit them.
- It is more problematic for women to use facilities in the evenings and return home in the dark.

Illustration: Pornography in Internet Cafés

Many commercial public Internet access points are appearing in developing countries like India .However, in small towns and rural areas, these Internet facilities are typically entertainment joints, which men frequent for accessing pornographic content. Most often, the Internet café manager, usually a young male, is himself into surfing pornographic content, and the place serves as a hangout for his male friends. This is even true of some community telecenters set-up by non-profit organizations in India. Women and girls are obviously wary of going to such places for accessing the Internet. Additionally, gender insensitive management of many commercial Internet access points poses the threat that personal email addresses of women and girls are accessed and used by boys and men to harass them.

Gender and ICTs Overview Report

Gender in the Information Economy

Women have relatively little ownership and control of the ICT sector. It is clear that women are under-represented on the boards and senior management of IT companies, policy and regulatory organizations, technical standard-setting organizations, industry and professional organizations and within government bodies working in this area.

The new economy rides on the power of ICTs. Job outsourcing is an important business strategy today and has given rise to a new global division of labor. Internationally outsourced jobs, such as medical transcription work or software services, have made a considerable difference to women's work opportunities in developing countries. In software, women enjoy opportunities on a scale that they never experienced in any other field of engineering and science. However, in the information technology sector, women make up only small percentages of managerial, maintenance, and design personnel in networks, operating systems, or software.

Information technology has brought employment gains for women. Here are a few observations: -

- Patterns of gender segregation are being reproduced in the information economy where men hold the majority of high-skilled, high value-added jobs, whereas women are concentrated in the low skilled, lower value-added jobs
- As traditional manufacturing industries that previously employed women gradually disappear, the women finding jobs in the new, often ICT-related industries are rarely the same ones as those who lost their jobs in the traditional sectors. New inequalities are therefore emerging between women with ICT-related job skills versus those without.
- While teleworking has certainly created new employment opportunities for women, the downside is that women can be excluded from better career possibilities, and instead of finding a balance, family responsibilities are combined with paid work, so that women end up acquiring new tasks on top of the old.

Illustration: Gender Ideologies in Information and Communications Work

'While examining gendered processes of work within the ICTs sector it is useful and important to problematic the term "low-skill work" to understand how gender ideology operates within the ICTs arena. Certain kinds of work, historically performed by women, have come to be defined as "unskilled" (and therefore low-pay). Costanza-Chock (2003) highlights how.

- Effective call service often requires a great deal of performative or emotional labor, but such labor is naturalized as "inherent" to women and therefore undervalued.
- Recent studies of women working in call centers in Ireland and Europe found that, contrary to employers' rhetoric about skill development and flexible career advancement, women's info-work is routinized, deskilled, and devalued.
- Women in these centers rarely advance beyond "team leader" roles to managerial positions (Belt, Richardson, and Webster 2000, Breathnach 2002).
- Concerns about the exploitation of low-skill female workers familiar from other sectors (harassment and abuse by male managers, poor health conditions, control of wages by male heads of household) do not disappear in the information and communications sectors.
- New forms of gendered inequality are of particular concern in call services, where 3 out of 4 female call center workers report repeated sexual harassment over the phone (ILO 2001b).'

Source: Costanza-Chock 2003: 11–12.

ICT and Sexual Violence

The global entertainment industry, poised on the power of new ICTs, is a force beyond the grasp of law and regulation. The sex industry markets precisely the violence and oppression that feminists seek to eliminate from the streets, workplaces, and bedrooms. Pornography has assumed mammoth proportions with the Internet. The Internet has made sexual exploitation of and violence against women and children seem more normal, and this is a matter of deep concern. Criminal syndicates violate laws prohibiting sexual exploitation and violence by locating their servers in host countries with less restrictive laws, to avoid regulation. The new technologies have thus enabled the creation of online communities free from interference or standards where any and every type of sexual violence goes and where women-hating is the norm.

While states often take a strong position against websites concerned with the rights of sexual minorities such as lesbians, gays, bisexuals and transgender people, Internet-based sexual violence is not seen as a priority for regulation.

Illustration: The Internet and the Sex Industry

The Internet as a Site of Violence

The Internet has brought in a revival of child pornography, which had more or less been eradicated in developed countries by the 1980s.Today, the business of both adult and child pornography arguably sustains the Internet. It has often been said that pornography is the only profitable entity on the Internet. Over the past ten years, the Internet has emerged as the premier forum of the international sex trade and has facilitated, accelerated, and normalised the sexual exploitation of women and girls. New ICTs have combined with racism, sexism, and capitalism to escalate sexual exploitation worldwide.

The Internet as a Site of Resistance

The Internet was used as a means to form a coalition of activists when thousands of Internet users protested Yahoo's decision to sell pornography. In December 2000, Yahoo created an online store devoted to selling pornographic videos and DVDs. Just a few months later, after receiving over 100,000 emails from Internet users, Yahoo decided to remove the portion of its website that sold pornography and to stop accepting advertisements from pornographic websites. In May 2001, Yahoo decided to make it more difficult to find sexually explicit chat rooms and online clubs (http://cyber.law.harvard.edu/vaw02/module4.html).

Women's Participation and Decision Making

Access to the media can modify the power structure in society. Control over knowledge and information is an important source of power and that is where the media are relevant. It is always striking that women, who make up half the world's population, have to struggle to get their voices in the media. They also need to fight with the increasing globalization that also means patriarchy becoming more powerful, more entrenched. So a few questions are raised for example: -

- Is it possible to use Internet communications and other ICT tools to modify the situation of women's citizenship and the standing of their rights?
- Could we say that ICT opens new channels for participation and decision making in the social and public spheres?

Women worldwide are using ICT to monitor the promotion and protection of their human rights, using the Internet to denounce violations, send alerts and campaign for their rights. They are also using ICT to facilitate communication among organizations, thus empowering the networks that work to ensure that women have equal rights. Gaining access to legal information (law and other legal instruments, new legislation and legal recourse, and accountability procedures) via the Internet enables them to discuss human rights issues with authority and thus further their struggle against any sort of discrimination either under law or practice. Women's access to information sources and interchange channels are crucial for their democratic participation, the respect for their human rights and for intervening with an equal voice in the public sphere.

Illustration: Campaign against violence on Women
In a successful use of the Internet in campaigns against violence on women, the Kenya-based African Women's Development and Communication Network (FEMNET), launched the Men to Men Initiative in 2001 to mark the Sixteen Days of Activism Against Violence on Women. The campaign targeted men to promote male involvement and action to combat gender-based violence at the regional level in Africa.

Old patterns in the new media

Technology has changed, but the fundamentals remain the same. Women's absence from information is crucial because today the media play a decisive role in the building of the public agenda. There is still a sexist and stereotyped portrayal of women in the media, and there is a need to work with media professions to create a media environment that promotes gender equality by fostering positive images of women and women's views.

Illustration: Media's involvement
In Nepal, Sancharika Samuha (Forum of Women Communicators) has focused on using the media to address violations of women's human rights. They challenged the mainstream media campaign against granting women the same rights to inherit property as men. The group was able to place articles on equal property rights in the press, produce and air radio jingles and television advertising, distributes posters and hold workshops with journalists and NGOs. As a result there is a new awareness among journalists and the general public of women's side of the story.

Defining ICT Development

The "intersection approach" that women are in the deepest end of the digital divide has been the main message of gender advocates working in ICT development. Therefore, the definition of ICT development must strike at the root of unequal power relations, not just between men and women, but more fundamentally between rich and poor, North and South, urban and rural, empowered and marginalized.

ICT policy currently rests on the assumption that information and communication technologies are gender neutral and that women must adapt to technologies, rather than have ICT policy specifically formulated to meet the interests and needs of women. Therefore, they need to be actively involved in the definition, design and development of new technologies in order to avoid new forms of exclusion and ensure that women and girls have equal access and opportunities in respect of the developments of science and technology.

There is a need to develop educational projects that stimulate critical and creative skills, and encourage greater participation of women in the design and production of new technologies. This requires a comprehensive set of interventions raging from quality public education for all, through scientific and technological education and research. Instead of trying to make girls fit into the existing computer culture, the computer culture must become more inviting to girls. Girls should be educated to be ICT designers and not just users. Educators and parents should help girls imagine themselves early in life as designers and producers of technology, stimulating deeper interest in ICTs and providing opportunities for girls to express their technological imaginations

Cross Gender Communication in the Cyberspace

Women and men have different ways of conducting themselves electronically. Male users are known to dominate mixed-sex electronic conversations. They have also been found to be the more frequent instigators of online sexual harassment. Men even preempt the women discussants by employing the same techniques they use in face-to-face interaction. When women do attempt to participate on a more equal basis, they risk being ignored by the men. Due to these non-legitimate practices, they sometimes avoid participating.

Women are more cautious in their postings; they also tend to act more like a moral advocate in joining a discussion. The women who deliberately use harsh words do so apologetically. Besides the frequent apologies, the women also tend to write shorter messages and gently reproach those that wrote long messages. They contribute more overt expressions of agreement, appreciation and support.

With the men, a common reason for their participation in online discussions was to avoid face-to-face personal communication. But for the women, they joined online discussions precisely to supplement and enhance their communication with others.

When women's representation is no longer biologically based, as it sometimes happens on cyber space, the Internet can be empowering to women. It allows women to be active and constructive. It allows their voices to be heard, and serves as a mechanism for the consideration of their ideas and insights.

Sources:

- Anita Gurumurthy, *Gender and ICTs Overview Report*
- Dafne Sabanes Plou, *What about Gender Issues in the Information Society?*
- Esther Kuntjara, *Gender Issues in Information Technology Communication*

Unit 3

Methods and Tools of Analysis

Methods and Tools of Analysis

Introduction

From the moment of their invention, computers have generated complex social, ethical, and value concerns. As computer technology evolves and gets deployed in new ways, certain issues persist – issues of privacy, property rights, accountability, and social values. At the same time, seemingly new and unique issues emerge.
The ethical issues can be organized in at least three different ways:

- According to the type of technology
- According to the sector in which the technology is used.
- According to ethical concepts or themes.

According to the type of technology

This is to organize the ethical issues by type of technology and its use. When computers were first invented, they were understood to be essentially sophisticated calculating machines but they seemed to have the capacity to do that which was thought to be uniquely human -- to reason and exhibit a high degree of rationality; hence, there was concern that computers threatened ideas about what it means to be human. In the shadow of World War II, concerns quickly turned to the use of computers by governments to centralize and concentrate power. These concerns accompanied the expanding use of computers for record keeping and the exponential growth in the scale of databases, allowing the creation, maintenance and manipulation of huge quantities of personal information. This was followed by the inception of software control systems and video games, raising issues of accountability-liability and property rights. This evolution of computer technology can be followed through to more recent developments including the Internet, simulation and imaging technologies, and virtual reality systems. Each one of these developments was accompanied by conceptual and moral uncertainty.

- What will this or that development mean for the lives and values of human beings?
- What will it do to the relationship between government and citizen?
- What will it do to the relationship between employer and employee?
- What will it do to the relationship between businesses and consumers?

According to the sector in which the technology is used

This is to organize the issues according to the sector in which they occur. Ethical issues arise in real-world contexts, and computer-ethical issues arise in the contexts in which computers are used. Each context or sector has distinctive issues and if we ignore this context we can miss important aspects of computer-ethical issues. For example, in dealing with privacy protection in general, we might miss the special importance of privacy protection for *medical records* where confidentiality is so essential to the doctor-patient relationship. Similarly, one might not fully understand the appropriate role for computers in education were one not sensitive to distinctive goals of education. Both of these approaches – examining issues by types and uses of particular technologies, and sector by sector – are important and illuminating; however, they take us too far a field of the philosophical issues.

According to ethical concepts or themes

The third approach is to emphasize ethical concepts and themes that persist across types of technology and sectors. In this the issues are divided into two broad categories:

- Meta-theoretical and methodological issues
- Traditional and emerging issues

Meta-theoretical and methodological issues

Perhaps the deepest philosophical thinking on computer-ethical issues has been reflection on the field itself -- its appropriate subject matter, its relationship to other fields, and its methodology. In a seminal piece entitled "What is Computer Ethics?" Moor (1985) recognized that when computers are first introduced into an environment, they make it possible for human beings (individuals and institutions) to do things they couldn't do before and this creates *policy vacuums*. We do not have rules, policies, and conventions on how to behave with regard to the new possibilities.

- Should employers monitor employees to the extent possible with computer software? Should doctors perform surgery remotely?
- Should I make copies of proprietary software?
- Is there any harm in me taking on a pseudo-identity in an on-line chat room?
- Should companies doing business on-line be allowed to sell the transaction generated-information they collect?

These are examples of policy vacuums created by computer technology. Moor's account of computer ethics has shaped the field of computer ethics with many computer ethicists understanding their task to be that of helping to fill policy vacuums. Indeed, one of the topics of interest in computer ethics is to understand this activity of filling policy vacuums.

Philosophical Ethics

Ethics, in general, is categorized in two broad areas – philosophical and professional ethics. We start ethical analysis from philosophical one. To start with, we take unexamined beliefs. But it is important to understand that we can come up with any conclusion after the analysis.

Ethical Analysis

In philosophical ethical analysis, the reasons for moral beliefs are articulated, and then critically evaluated. The reasons you give for holding an ethical belief or taking a position on an ethical issue can be thought of as an argument for a claim. Once stated, we can ascertain whether the argument does, indeed, support the claim being made or the position being taken. This critical evaluation is often done to reject a position, or to adopt another position, but it may also be done simply to explore a claim. When you critically evaluate the argument supporting a claim, you come to understand the claim more fully. A critical examination of the underpinnings of moral beliefs sometimes leads to a change in belief, but it may also simply lead to stronger and better understood beliefs.

Not only must you give reasons for your claims, you are also expected to be consistent from one argument or topic to the next. For example, instead of having separate, isolated views on abortion and capital punishment, philosophical analysis would lead you to recognize that both your views on abortion and your views on capital punishment rest on a claim about the value of human life and what abrogates it. Philosophical analysis would lead you to inquire whether the claim you made about the value of human life in the context of a discussion of capital punishment is consistent with the claim you made about the value of human life in the context of a discussion of abortion. If the claims appeared to be inconsistent from the one context to the next, then you would be expected to change one of your claims or provide an account of how the two positions can be understood as consistent. In other words, you would show that seemingly inconsistent views are in fact consistent.

Philosophical analysis is a dialectic ongoing process

It involves a variety of activities. It involves expressing a claim and putting forward an argument or reasons for the claim, and it involves critical examination of the argument. If the argument does not hold up to critical examination, then it might be reformulated into a revised argument, perhaps rejecting aspects of the original argument but holding on to a core idea. The revised argument, then, has to be critically examined, and so on, with ongoing reformulation and critique. Philosophers often refer to this process as a dialectic (which is related to the word dialogue). We pursue an argument to see where it goes and to find out what you would have to know or assert to defend the argument and establish it on a firm footing.

In addition to moving from claims to reasons and arguments, and from one formulation of an argument to another better formulation, the dialectic also moves back and forth from cases to principles or theory. To illustrate, take the issue of euthanasia. Suppose you start out by making the claim that euthanasia is wrong. You articulate a principle as the reason for this claim. Say, the principle is that human life has the highest value and, therefore, human life should never be intentionally ended. You might then test this principle by seeing how it applies in a variety of euthanasia cases. For example, is it wrong to use euthanasia when the person is conscious but in extreme pain? When the person is unconscious and severely brain damaged? When the person is terminally ill? When the person is young or elderly? Since your principle concerns the value of human life has implications beyond the issue of euthanasia. Hence, you might also test it by applying it to completely different types of cases. Is intentional taking of human life wrong when it is done in a war situation? Is intentional killing wrong when it comes to capital punishment? Given your position on these cases, you may want to qualify the principle or you may hold to the principle and change your mind about the cases. For example, after seeing how the principle applies in various cases, you may want to qualify it so that you now assert that one should never intentionally take a human life except in self-defence or except when taking a life will save another life. Or you might reformulate the principle so that it specifies that the value of human life has to do with its quality. When the quality of life is significantly and permanently diminished, while it is still not permissible to intentionally kill, it is morally permissible to let a person die.

The dialogue continues as the dialectic leads to a more and more precise specification of the principle and the argument. The process clarifies what is at issue and what the possible positions are. It moves from somewhat inchoate ideas to better and better arguments, and more defensible and better articulated positions.

The dialectic (from an initial belief to an argument, from argument to better argument, and from theory to case, and back) does not always lead to definitive conclusions or unanimous agreement. Therefore, it is important to emphasize that understanding can be improved; progress can be made, even when one has not reached definitive conclusions. Through the dialectic we learn which arguments are weaker and stronger and why. We come to understand the ideas that underpin our moral beliefs. We develop deeper and more consistent beliefs and we come to understand how moral ideas are interrelated and interdependent.

A familiarity with traditional ethical theories will help in articulating the reasons for many of your moral beliefs. Ethical theories provide frameworks in which arguments can be cast. Moreover, ethical theories provide some common ground for discussion. They establish a common vocabulary and frameworks within which, or against which, ideas can be articulated.

Descriptive and Normative Claims

In a sense and partly, this is the distinction between facts and values, but the matter of what counts as a fact is very contentious in philosophy. So, it will be better to stay with the terms descriptive and normative. Descriptive statements are statements that describe a state of affairs in the world. For example, “The car is in the driveway.” And “Georgia is south of Tennessee.” In addressing ethical issues and especially the ethical issues surrounding computer and information technology, it is quite common to hear seemingly factual statements about human beings. The following are descriptive statements: “Such and such percentage of the people surveyed admitted to having made at least one illegal copy of computer software.” “The majority of individuals who access pornographic Web sites are males between the ages of 14 and 35.” “Such and such percentages of U.S. citizens use the Internet to obtain information on political candidates.” “In all human societies, there are some areas of life that are considered private.” These statements describe what human beings think and do. They are empirical claims in the sense that they are statements that can be verified or proven false by examining the state of affairs described. To be sure, it may not be easy to verify or disconfirm claims like these, but in principle it is possible. Observations can be made, surveys can be administered, and people can be asked, and so on.

Social scientists gather empirical data and report their findings, both on moral and non moral matters. When it comes to morality, psychologists and sociologists might do such things as identify the processes by which children develop moral concepts and sensibilities. Or they may measure how individuals value and prioritize various goods such as friendship, privacy, and autonomy. When anthropologists go to other cultures, they may describe complex moral rules in that culture. They are describing lived and observed moral systems. Similarly, historians may trace the development of a particular moral notion in an historical period.

All of these social scientific studies are descriptive studies of morality; they examine morality as an empirical phenomenon. They don’t, however, tell us what is right and wrong. They don’t tell us what people should think or do, only what people, in fact, think and do.

In contrast, philosophical ethics is normative. The task of philosophical ethics is to explore what human beings ought to do, or more accurately, to evaluate the arguments, reasons, and theories that are proffered to justify accounts of morality. Ethical theories are prescriptive. They try to provide an account of why certain types of behaviour are good or bad, right or wrong. Descriptive statements may come into play in the dialectic about philosophical ethics, but normative issues cannot be resolved just by pointing to the facts about what people do or say or believe. For example, the fact (if it were true) that many individuals viewed copying proprietary software as morally acceptable would not make it so. The fact that individuals hold such a belief is not an argument for the claim that it is morally permissible to copy proprietary software. You might wish to explore why individuals believe this to see if they have good reasons for the belief. Or you might wish to find out what experiences have led individuals to draw this conclusion. Still, in the end, empirical facts are not alone sufficient to justify normative claims. Figuring out what is right and wrong, what is good and what is bad, involves more than a descriptive account of states of affairs in the world.

The aim of this book is not to describe how people behave when they use computers. For this, the reader should consult social scientists—sociologists, anthropologists, political scientists, and psychologists. Rather the aim of this book is to help you understand how people ought to behave when they use computers and what rules or policies ought to be adopted with regard to computer and information technology.

Ethical Relativism

We can begin our examination of ethical concepts and theories by examining a prevalent, often unexamined moral belief. Many believe that “ethics is relative.” This seems like a good starting place. This claim can be examined carefully and critically. We can begin by formulating the idea as a theory consisting of a set of claims backed by reasons.

The idea of ethical relativism seems to be something like this: “What is right for you may not be right for me,” or “I can decide what is right for me, but you have to decide for yourself.” When we take this idea and formulate it into a more systematic account, it seems to encompass a negative claim (something that it denies), and a positive claim, (something it asserts). The negative claim appears to be: “There are no universal moral norms.” According to this claim, there isn’t a single standard for all human beings. One person may decide that it is right for him to tell a lie in certain circumstances, another person may decide that it is wrong for her to tell a lie in exactly the same circumstances, and both people could be right. So, the claim that “right and wrong are relative” means in part that there are no universal rights and wrongs.

The positive claim of ethical relativism is more difficult to formulate. Sometimes ethical relativists seem to be asserting that right and wrong are relative to the individual, and sometimes they seem to assert that right and wrong are relative to the society in which one lives. I am going to focus on the latter version, and on this version the relativist claims that what is morally right for me, an American living in the twenty-first century, could be different than what is right for a person living, say, in Asia in the fifth century. The positive claim of relativism is that right and wrong are relative to your society.

Ethical relativists often cite a number of descriptive facts to support these claims:

1. They point to the fact that cultures vary a good deal in what they consider to be right and wrong. For example, in some societies, infanticide is acceptable while in other societies it is considered wrong. In some societies, it is considered wrong for women to go out in public without their faces being covered. Polygamy is permissible in some cultures; in others it is not. Examples of this kind abound.

2. Relativists also point to the fact that the moral norms of a given society change over time so that what was considered wrong at one time, in a given society, may be considered right at another time. Slavery in America is a good example of this since slavery was considered morally permissible by many in the United States at one time, but is now illegal and almost universally considered impermissible.

3. Relativists also point to what we know about how people develop their moral ideas. We are taught the difference between right and wrong as children, and what we come to believe is right or wrong is the result of our upbringing. It depends on when, where, how, and by whom we were raised. If I had been born in certain Middle Eastern countries, I might believe that it is wrong for a woman to appear in public without her face covered. Yet because I was raised in the United States in the twentieth century, by parents who had Western ideas about gender roles and public behaviour, I do not believe this, Of course, parents are not the only determinant of morality. A person develops moral ideas from the experiences he or she has in school, at work, with peers, and so on.

It is useful to note that we have already made progress simply by clearly and systematically formulating the idea of ethical relativism, an idea you may have entertained or heard expressed, but never had a chance to examine carefully. Moreover, we have been able to identify and articulate some reasons thought to support ethical relativism. With the idea and supporting evidence now “on the table,” we can carefully and critically examine them.

The facts which ethical relativists point to cannot be denied. For example, I would not want to take issue with the claims that:

1. There is and always has been a good deal of diversity of belief about right and wrong.
2. Moral beliefs change over time within a given society.
3. Social environment plays an important role in shaping the moral ideas you have.

However, there does seem to be a problem with the connection between these facts and the claims of ethical relativism. Do these facts show that there are no universal moral rights or wrongs? Do they show that right and wrong are relative to your society?

On more careful examination, it appears that the facts cited by ethical relativists do not support their claims. To put this another way, we can, without contradiction, accept the facts and still deny ethical relativism. The facts do not necessitate that there are no universal moral standards or that ethics is relative. Lest there be no confusion, you should recognize that "ethics is relative" could be interpreted either as an empirical or a normative claim. As an empirical claim, it asserts that ethical beliefs vary; as a normative claim it asserts that right and wrong (not just beliefs about, but what is actually right and wrong) vary.

If we understand the claim "ethics is relative" to be a description of human behaviour, then it does follow from the facts sited. Indeed, it is redundant of the facts cited, for as a description of human behaviour, it merely repeats what the facts have said. Ethical beliefs vary. Individuals be lieve different things are right and wrong depending on how and by whom they have been raised and where and when they live.

On the other hand, if we understand "ethics is relative" to be a normative claim, a claim asserting the negative and/or positive parts of ethical relativism, then it is not redundant, and the facts do not support the claims. Here the leap from facts to conclusion is problematic for a number of reasons. For one, the argument goes from a set of "is" claims to an "ought" claim and the "ou ght" claim just doesn't follow (in a straightforward way) from the "is" claims.

Utilitarianism

Utilitarianism is an ethical theory claiming that what makes behaviour right or wrong depends wholly on the consequences. In putting the emphasis on consequences, utilitarianism affirms that what is important about human behaviour is the outcome or results of the behaviour and not the intention a person has when he or she acts. On one version of utilitarianism, what is all important is happiness-producing consequences (Becker and Becker, 1992). Crudely put, actions are good when they produce happiness and bad when they produce the opposite, unhappiness. The term utilitarianism derives from the word utility. Ac cording to utilitarianism actions, rules, or policies are good because of their usefulness (their utility) in bringing about happiness.

Lest there be no confusion, philosophers are not always consistent in the way they use the terms utilitarianism and consequentialism. Sometimes, consequentialism is seen as the broadest term referring to ethical theories that claim that what makes an action right or wrong is the consequences and not the internal character of action. Utilitarianism is, then, a particular version of this type of theory with the emphasis specifically on happiness-producing consequences. That is the way I shall use these terms, though I warn readers that the distinction sometimes is made in just the opposite way, that is, with utilitarianism seen as the broadest theory and consequentialism as a particular form of utilitarianism.

In any case, in the version on which I will focus, the claim is that in order to determine what they should do, individuals should follow a basic principle. The basic principle is this: Everyone ought to act so as to bring about the greatest amount of happiness for the greatest number of people.

But, what, you may ask, is the "proof" of this theory? Why should each of us act to bring about the greatest amount of happiness? Why shouldn't we each seek our own interest?

Intrinsic and Instrumental Value

Utilitarians begin by focusing on values and asking what is so important, so valuable to human beings, that we could use it to ground an ethical theory. They note that among all the things in the world that are valued, we can distin guish things that are valued because they lead to something else from things that are valued for their own sake. The former are called instrumental goods and the latter intrinsic goods. Money is a classic example of something that is instrumentally good. It is not valuable for its own sake, but rather has value as a means for acquiring other things. On the other hand, intrinsic goods are not valued because they are a means to something else. They have qualities or characteristics that are valuable in themselves. Knowledge is sometimes said to be intrinsically valuable. So, is art because of its beauty. You might also think about environmental debates in which the value of nature or animal or plant species or ecosystems are said to be valuable independent of their value to human beings. The claim is that these things have value independent of their utility to human beings.

Having drawn this distinction between instrumental and intrinsic goods, utilitarians ask what is so valuable that it could ground a theory of right and wrong? It has to be something intrinsically valuable, for something which is instrumentally valuable is dependent for its goodness on whether it leads to another good. If you want x because it is a means to y, then y is what is truly valuable and x has only secondary or derivative value. Utilitarianism, as I am using the term, claims that happiness is the ultimate intrinsic good, because it is valuable for its own sake. Happiness cannot be understood as simply a means to something else. Indeed, some utilitarians claim that everything else is de sired as a means to happiness and that, as a result, everything else has only secondary or derivative (instrumental) value.

To see this, take any activity that people engage in and ask why they do it. Each time you will find that the sequence of questions ends with happiness. Take, for example, your career choice. Suppose that you have chosen to study computer science so as to become a computer professional. Why do you want to be a computer professional? Perhaps you believe that you have a talent for computing, and you believe you will be able to get a well-paying job in computer science one in which you can be creative and somewhat autonomous. Then we must ask, why are these things important to you? That is, why is it important to you to have a career doing something for which you have a talent? Why do you care about being well paid? Why do you desire a job in which you can be ca and autonomous? Suppose that you reply by saying that being well paid is important to you because you want security or because you like to buy things or because there are people who are financially dependent on you. In turn, we can ask about each of these. Why is it important to be secure? Why do you want security or material possessions? Why do you want to support your dependents? The questions will continue until you point to something that is valuable in itself and not for the sake of something else. It seems that the questions can only stop when you say you want whatever it is because you believe it will make you happy. The questioning stops here because it doesn't seem to make sense to ask why someone wants to be happy.

A discussion of this kind could go off in the direction of questioning whether your belief is right. Will a career as a computer professional make you happy? Will it really bring security? Will security or material possessions, in fact, make you happy? Such discussions centre on whether or not you have chosen the correct means to your happiness. However, the point that utilitarians

Acts versus Rules

As mentioned earlier, there are several formulations of utilitarianism and proponents of various versions disagree on important details. One important and controversial issue of interpretation has to do with whether the focus should be on rules of behaviour or individual acts. Utilitarians have recognized that it would be counter to overall happiness if each one of us had to calculate at every moment what all the consequences of every one of our actions would be. Not only is this impractical, because it is time consuming and because sometimes we must act quickly, but often the consequences are impossible to foresee. Thus, there is a need for general rules to guide our actions in ordinary situations.

Rule-utilitarians argue that we ought to adopt rules that, if followed by everyone, would, in the long run, maximize happiness. Take, for example, telling the truth. If individuals regularly told lies, it would be very disruptive. You would never know when to believe what you were told. In the long run, a rule obligating people to tell the truth has enormous beneficial consequences. Thus, "tell the truth" becomes a utilitarian moral rule. "Keep your promises," and "Don't reward behaviour that causes pain to others," are also rules that can be justified on utilitarian grounds. According to rule-utilitarianism, if the rule can be justified in terms of the consequences that are brought about from people following it, then individuals ought to follow the rule.

Act-utilitarians put the emphasis on individual actions rather than rules. They believe that even though it may be difficult for us to anticipate the consequences of our actions, that is what we should be trying to do. Take, for example, a case where lying may bring about more ha ppiness than telling the truth. Say you are told by a doctor that tentative test results indicate that your spouse may be terminally ill. You know your spouse well enough to know that this knowledge, at this time, will cause your spouse enormous stress. He or she is already under a good deal of stress because of pressures at work and because someone else in the family is very ill. To tell your spouse the truth about the test results will cause more stress and anxiety, and this stress and anxiety may turn out to be unnecessarily if further tests prove that the spouse is not terminally ill. Your spouse asks you what you and the doctor talked about. Should you lie or tell the truth? An Act-utilitarian might say that the right thing to do in such a situation is to lie, for little good would come from telling the truth and a good deal of suffering (per haps unnecessary suffering) will be avoided from lying. A rule-utilitarian would agree that good might result from lying in this one case, but in the long run, if we cannot count on people telling the truth (especially our spouses), more bad than good will come. Think of the anxiety that might arise if spouses routinely

Deontological Theories

In utilitarianism, what makes an action or a rule right or wrong is outside the action; it is the consequences of the action or rule that make it right or wrong. By contrast, deontological theories put the emphasis on the internal character 'of the act itself.' What makes an action right or wrong for deontologists is the principle inherent in the action. If an action is done from a sense of duty, if the principle of the action can be universalized, then the action is right. For ex ample, if I tell the truth (not just because it is convenient for me to do so, but) because I recognize that I must respect the other person, then I act from duty and my action is right. If! tell the truth because I fear getting caught or because I believe I will be rewarded for doing so, then my act is not morally worthy.

I am going to focus here on the theory of Immanuel Kant. If we go back for a moment to the allocation of dialysis machines, Kant's moral theory is applicable -because it proposes what is called a categorical imperative specifying that we should never treat human beings merely as means to an end. We should always treat human beings as ends in themselves. Although Kant is not the only deontologist, I will continue to refer to him as I discuss deontology.

The difference between deontological theories and consequentialist theories was illustrated in the discussion of allocation of dialysis machines. Deontologists say that individuals are valuable in themselves, not because of their social value. Utilitarianism is criticized because it appears to tolerate sacrificing some people for the sake of others. In utilitarianism, right and wrong are dependent on the consequences and therefore vary with the circumstances. By contrast, deontological theories assert that there are some actions that are al ways wrong, no matter what the consequences. A good example of this is killing. Even though we can imagine situations in which intentionally killing one person may save the lives of many others, deontologists insist that intentional killing is always wrong. Killing is wrong even in extreme situations be cause it means using the person merely as a means and does not treat the human being as valuable in and of himself. Deontologists do often recognize self-defence and other special circumstances as excusing killing, but these are cases when, it is argued, the killing is not exactly intentional. (The person attacks me. I would not, otherwise, aim at harm to the person, but I have no other choice but to defend myself.)

At the heart of deontological theory is an idea about what it means to be a person, and this is connected to the idea of moral agency. Charles Fried (1978) put the point as follows:

Substantive contents of the norms of right and wrong express the value of persons, of respect for personality. What we may not do to each other, the things which are wrong, are precisely those forms of personal interaction which deny to our victim the status of a freely choosing, rationally valuing, specially efficacious person, the special status of moral personality.

According to deontologists, the utilitarians go wrong when they fix on happiness as the highest good. Deontologists point out that happiness cannot be the highest good for humans. The fact that we are rational beings, capable of reasoning about what we want to do and then deciding and acting, suggests that our end (our highest good) is something other than happiness. Humans differ from all other things in the world insofar as we have the capacity for rationality. The behaviour of other things is determined simply by laws of nature. Plants turn toward the sun because of photosynthesis. They don't think and decide which way they will turn. Physical objects fall by the law of gravity. Water boils when it reaches a certain temperature. In contrast, human beings are not entirely determined by laws of nature. We have the capacity to legislate for ourselves. We decide how we will behave. As Kant describes this, it is the difference between acting in accordance with law (plants and stones do) and acting in accordance with the conception of law.

The capacity for rationa1 decision making is the most important feature of human beings. Each of us has this capacity; each of us can make choices, choices about what we will do, and what kind of persons we will become. No one else can or should make these choices for us. Moreover, we should recognize this capacity in others.

Notice that it makes good sense that our rationality is connected with morality, for we could not be moral beings at all unless we had this rational capacity. We do not think of plants or fish or dogs and cats as moral beings precisely because they do not have the capacity t9 reason about their actions. We are moral beings because we are rational beings, that is, because we have the capacity to give ourselves rules (laws) and follow them.

Where utilitarians note that all humans seek happiness, deontologists emphasize that humans are creatures with goals who engage in activities directed toward achieving these goals (ends), and that they use their rationality to formulate their goals and figure out what kind of life to live. In a sense, deontologists pull back from fixing on any particular value as structuring morality and instead ground morality in the capacity of each individual to organize his or her own life, make choices, and engage in activities to realize their self-chosen life plans. What morality requires is that we respect each of these beings as valuable in themselves and refrain from valuing them only insofar as they fit into our own life plans.

As mentioned before, Kant put forward what he called the categorical im /Jerutive. While there are several versions of it, I will focus on the second version, which goes as follows: Never treat another human being merely as a means but always as an end. This general rule is derived from the idea that persons are moral beings because they are rational, efficacious beings. Because we each have the Capacity to think and decide and act for ourselves, we should each be treated with respect, that is, with recognition of this capacity.

Note the "merely" in the categorical imperative. Deontologists do not in Sift that we never use another person as a means to an end, only that we never merely use them in this way. For example, if I own a company and hire employees to work in my company, I might be thought of as using those employees as a means to my end (i.e., the success of my business). This, however, is not wrong if I promise to pay a fair wage in exchange for work and the employees agree to work for me. I thereby respect their ability to choose for themselves.

Rights

So far, very little has been said about rights though we often use the language of rights when discussing moral issues. "You have no right to tell me what to do." "I have a right to do that." Ethicists often associate rights with deontological theories. The categorical imperative requires that each person be treated as an end in himself or herself, and it is possible to express this idea by saying that individuals have "a right to" the kind of treatment that is implied in being treated as an end. The idea that each individual must be respected as valuable in himself or herself implies that we each have rights not to be interfered with in certain ways, for example, not to be killed or enslaved, to be given freedom to make decisions about our own lives, and so on.

An important distinction that philosophers often make here is between negative rights and positive rights. Negative rights are rights that require restraint by others. For example, my right not to be killed requires that others refrain from killing me. It does not, however, require that others take positive action to keep me alive. Positive rights, on the other hand, imply that others have a duty to do something to or for the right holder. So, if we say that I have a positive right to life, this implies not just that others must refrain from killing me, but that they must do such things as feed me if! am starving, give me medical treatment if! am sick, swim out and save me if! am drowning, and so on. As you can see, the difference between negative and positive rights is quite significant.

Positive rights are more controversial than negative rights because they have implications that are counter-intuitive. If every person has a positive right to life, this seems to imply that each and every one of us has a duty to do what ever is necessary to keep all people alive. This would seem to suggest that, among other things, it is our duty to give away any excess wealth that we have to feed and care for those who are starving or suffering from malnutrition. It also seems to imply that we have a duty to supply extraordinary life-saving treatment for all those who are dying. In response to these implications, some philosophers have argued that individuals have only negative rights.

While, as I said earlier, rights are often associated with deontological theories, it is important to note that rights can be derived from other theories as well. For example, we can argue for the recognition of a right to property on utilitarian grounds. Suppose we ask why individuals should be allowed to have private property in general and, in particular, why they should be allowed to own computer software. Utilitarians would argue for private ownership of soft ware on grounds that much more and better software will be created if individuals are allowed to own (and then license or sell) it. Thus, they argue that individuals should have a legal right to ownership in software because of the beneficial consequences of acknowledging such a right.

Another important thing to remember about rights is the distinction between legal and moral (or natural or human) rights. Legal rights are rights that are created by law. Moral, natural, or human rights are claims independent of law. Such claims are usually embedded in a moral theory or a theory of human nature.

The utilitarian argument is an argument for creating or recognizing a legal right; it is not an argument to the effect that human beings have a natural right, for example, to own what they create.

Rights and Social Contract Theories

Rights are deeply rooted in the tradition of social contract theories. In this tradition the idea of a social contract (between individuals, or between individuals and government) is hypothesized to explain and justify the obligations that human beings have to one another. Many of these theories imagine human beings in a state of nature and then show that reason would lead individuals in such a state to agree to live according to certain rules, or to give power to a government to enforce certain rules. The depiction of a state of nature in which human beings are in a state of insecurity and uncertainty is used to suggest what human nature is like and to show that human nature necessitates government. That is, in such a state any rational human beings would agree (make a contract) to join forces with others even though this involves giving up some of their natural freedom. The agreement (the social contract) creates obligations and these are the basis of moral obligation.

An argument of this kind is made by several social contract theorists and each specifies the nature and limits of our obligations differently. One important difference, for example, is in whether morality exists prior to the social contract. Hobbes argues that there is no justice or injustice in a state of nature; humans are at war with one another and each individual must do what they must to preserve themselves. Locke, on the other hand, specifies a natural form of justice in the state of nature. Human beings have rights in the state of nature and others can treat individuals unjustly. Government is necessary to insure that natural justice is implemented properly because without government, there s no certainty that punishments will be distributed justly.

Rawlsian Justice

In 1971,John Rawls, a professor at Harvard University, introduced a new version of social contract theory (though some argue it is not a social contract theory in the traditional sense). Rawls introduced the theory in a book entitled simply A Theory of Justice. The theory may well be one of the most influential theories of the twentieth century, for not only did it generate an enormous amount of attention in the philosophical community, it influenced discussion among economists, social scientists, and public policy makers.

Rawls was primarily interested in questions of distributive justice. In the tradition of a social contract theorist, he tries to understand what sort of contract between individuals would be just. Rawls recognizes that we can't arrive at an account of justice and the fairness of social arrangements by reasoning about what rules particular individuals would agree to. He understands that individuals are self-interested and therefore will be influenced by their own experiences and their own situation when they think about fair arrangements. Thus, if some group of us were to get together in something like a state of nature (suppose a group is stranded on an island or a nuclear war occurs and only a few survive), the rules we would agree to would not necessarily be a just system. It would not necessarily exemplify justice.

The problem is that we would each want rules that would favour us. Smart people would want rules that favoured intelligence. Strong people would want a System that rewarded strength. Women would not want rules that were biased against women, and so on. The point is that there is no reason to believe that the outcome of a negotiation in which people expressed their preferences would result in rules of justice and just institutions. In this sense, Rawls believes that justice has to be blind in a certain way.

Rawls specifies, therefore, that in order to get at justice, we have to imagine that the individuals who get together to decide on the rules for society are behind a veil of ignorance. The veil of ignorance is such that individuals do not know what characteristics they will have. They do not know whether they will be male or female, black or white, highly intelligent or moderately intelligent or retarded, physically strong or in ill-health, musically talented, successful at business, indigent and so on.

At the same time, these individuals would be rational and self-interested and would know something about human nature and human psychology. In a sense, what Rawls is suggesting here is that we have to imagine generic human beings. They have abstract features that human beings generally have (i.e., they are rational and self-interested). And, they have background knowledge (i.e., general knowledge of how humans behave and interact and bow they are affected in various ways).

According to Rawls, justice is what individuals would choose in such a situation. Notice that what he has done, in a certain sense, is eliminate bias in the original position. Once a society gets started, once particular individuals have characteristics, their views on what is fair are tainted. They cannot be objective.

So, justice, according to Rawls is what people would choose in the original position where they are rational and self-interested, informed about human nature and psychology but behind a veil of ignorance with regard to their own characteristics. Rawls argues that individuals in the original position would agree to two rules. These are the rules of justice and they are "rules of rules" in the sense that they are general principles constraining the formulation of specific rules. The rules of justice are:

1. Each person should have an equal right to the most extensive basic liberty compatible with a similar liberty for others.

2. Social and economic inequalities should be arranged so that they are both (a) reasonably expected to be to everyone's advantage and (b) attached to positions and offices open to all.

These general principles assure that no matter where an individual ends up in the lottery of life (in which characteristics of intelligence, talents, physical abilities, and so on, are distributed), he or she would have liberty and opportunity. He or she would have a fair shot at a decent life.

While Rawls' account of justice has met with criticism, it goes a long way in providing a framework for envisioning and critiquing just institutions. This discussion of Rawls is extremely abbreviated as were the accounts of Kant and utilitarianism. Perhaps the most important thing to keep in mind as we

Virtue Ethics

Before moving on to the ethical issues surrounding computer and information technology, one other tradition in ethical theory should be mentioned. In recent ears, interest has arisen in resurrecting the tradition of virtue ethics, a tradition going all the way back to Plato and Aristotle. These ancient Greek philosophers pursued the question: What is a good person? What are the virtues associated with being a good person? For the Greeks virtue meant excellence, and ethics was concerned with excellences of human character. A person possessing such qualities exhibited the excellences of human good. To have these qualities is to function well as a human being.

The list of possible virtues is long and there is no general agreement on which are most important, but the possibilities include courage, benevolence, generosity, honesty, tolerance, and self-control. Virtue theorists try to identify e list of virtues and to give an account of each - what is courage? What is honesty? They also give an account of why the virtues are important.

Virtue theory seems to fill a gap left by other theories we considered, because it addresses the question of moral character, while the other theories focused primarily on action and decision making. What sort of character should we be trying to develop in ourselves and in our children? We look to moral heroes, for example, as exemplars of moral virtue. Why do we admire such people? What is it about their character and their motivation that are worthy of tour admiration?

Virtue theory might be brought into the discussion of computer technology and ethics at any number of points. The most obvious is, perhaps, the discussion of professional ethics, where we want to think about the characteristics of a good computer professional. Good computer professionals will, perhaps, exhibit honesty in dealing with clients and the public. They should exhibit courage when aced with situations in which they are being pressured to do something illegal or act counter to public safety. A virtue approach would focus on these characteristics and more, emphasizing the virtues of a good computer professional.

Individual and Social Policy Ethics

One final distinction will be helpful. In examining problems or issues, it is important to distinguish levels of analysis, in particular that between macro and micro level issues or approaches. One can approach a problem from the point of view of social practices and public policy, or from the point of view of individual choice. Macro level problems are problems that arise for groups of people, a community, a state, a country. At this level of analysis, what is sought is a solution in the form of a law or policy that specifies how people in that group or society ought to behave, what the rules of that group ought to be. When we ask the following questions, we are asking macro level questions: Should the United States grant software creators a legal right to own software? Should software engineers be held liable for errors in the software they design? Should companies be allowed to electronically monitor their employees?

On the other hand, micro level questions focus on individuals (in the presence or absence of law or policy). Should I make a copy of this piece of soft ware? Should I lie to my friend? Should I work on a project making military weapons? Sometimes these types of questions can be answered simply by refer ring to a rule established at the macro level. For example, legally I can make a back-up copy of software that I buy, but I shouldn't make a copy and give it to my friend. Other times, there may be no macro level rule or the macro level rule may be vague or an individual may think the macro level rule is unfair. In these cases, individuals must make decisions for themselves about what they ought to do.

The theories just discussed inform both approaches, but in somewhat different ways, so it is important to be clear on which type of question you are asking or answering.

Social Context of a Design

We only have identified the primary factors in a computer system to be performance, reliability, and cost. The factors such as usability and fit-to-task figuring in any software system are not considered by us. A limited effort is included in the analysis of social context in traditional software design. One of the central challenges faced by software designers is how to balance the highly structured nature of computer artifacts with the need to integrate them into different settings. A software designer inevitably faces situations in which design choices are constrained by:

- The conflicting goals and values held by the different parties who have a stake in the changes that new technologies will bring to the work
- Further, Workers and managers have many
    - Common interests
    - Different stakes

The question will be how computers in the workplace change productivity, working conditions, and job satisfaction.

Design for People at Work

Well-designed systems can boost

- Productivity
- Enhance
- Job satisfaction and give both workers and managers a clearer sense of what is going on in the organization.

But a system that interferes with crucial work practices:

- Can result in reduced effectiveness and efficiency,
- Reduced satisfaction and autonomy,
- Increased stress and health problems for the people who use the system.

Design Approaches

The basic design approaches are

- Technology-centered approach
- Work-Oriented Approaches

Work-Oriented approaches are further classified into two design approaches:

- Human-centered design

- Participatory design

Technology-centered approach

This approach is characterized by hard-systems thinking; which involve the imposition of a clear-cut problem definition on a relatively unstable organizational reality and a "fuzzy" system.

In this design a little attention is accorded to the organizational context in which the system is to operate or the social implications of the system.

Human-centered design

This design puts human, social, and organizational considerations on an equal footing with technical considerations in the design process. Well-designed technology should make use of human strengths such as skill, judgment, capacity for learning–to create a robust and flexible production system.

Participatory design

This approach emphasizes the importance of meaningful end-user participation and influence in all phases of the design process. "The employees must have access to relevant information; they must have the possibility for taking an independent position on the problems; and they must in some way participate in the process of decision making."

Systems and Assumptions

Consider the following three cases with a view of assumptions made when bringing these systems into use in a particular context. Many assumptions are ruled out because the output is not what is expected. This gives us more understanding in relation to the topic discussed above that fit to task and usability are also important parameters to be considered in designing a system.

The sense of a hostile, intrusive presence in machinists' work life gave the system its unofficial shop floor name: The Spy in the Sky. The system was tolerated by the machinists because of its benefits as a signaling device, and because their union was able to negotiate an agreement with the company that data from the system would not be used for disciplinary purposes. Still, the system could be used for informal discipline, and it served as a constant reminder of managers' mistrust of the workforce. Productive work relies on the skills of machinists, yet the monitoring system embodies the assumption that, if machines are running at less than 80 percent of the programmed rate, the fault lies with the machinist, rather than with the engineer who wrote the parts program.

The basic intent of bringing a computer system into an existing scenario does not exactly meet all the assumptions with which it is brought. The implementation basically sees the following impact.

- **Representations and Misrepresentations of Work**
- **The High Cost of Bad Design**
- **Displacement.**
- **Intensification.**
- **Reduction or redefinition of skill**

Thus to conclude the social context of a design should also consider the usability and fit to task scenario when designing or redefining the existing systems with computer systems. We should not attach false assumptions and take more realistic measures into consideration.

<u>Source:</u>

- Deborah G. Johnson, *Computer Ethics*, Third Edition

Unit 4

Professional and Ethical Responsibilities

Professional and Ethical Responsibilities

What is Profession?

The subsequent discussion on the terms “job”, “occupation” and “profession” will help define the term “profession”. Let’s have eyes on its attributes first: -

Attributes of Profession

There have been many studies on the nature of professions has been achieved. Attributes of a profession include: -

1. Work that requires sophisticated skills, the use of judgment, and the exercise of discretion. Also, the work is not routine and is not capable of being mechanized;
2. Membership in the profession requires extensive formal education, not simply practical training or apprenticeship;
3. The public allows special societies or organizations that are controlled by members of the profession to set standards for admission to the profession, to set standards of conduct for members, and to enforce these standards; and
4. Significant public good results from the practice of the profession.

In a profession, “judgment” refers to making significant decisions based on formal training and experience. In general, the decisions will have serious impacts on people’s lives and will often have important implications regarding the spending of large amounts of money.

“Discretion” can have two different meanings. The first definition involves being discrete in the performance of one’s duties by keeping information about customers, clients, and patients confidential. This confidentiality is essential for engendering a trusting relationship and is a hallmark of professions. The other definition of discretion involves the ability to make decisions autonomously. Many people are allowed to use their discretion in making choices while performing their jobs. However, the significance of the decision marks the difference between a job and a profession.

Examination of Different Occupations

Let’s examine athletes and carpenters in light of the foregoing definition of professions. An athlete who is paid for her appearances is considered to have sophisticated skills that most people do not possess, and these skills are not capable of mechanization. However, substantial judgment and discretion are not called for on the part of athletes in their “professional” lives, so athletics fails the first part of the definition of professional. Interestingly, though, professional athletes are frequently viewed as role models and are often disciplined for a lack of discretion in their personal lives.

Athletics requires extensive training, not of a formal nature, but more of a practical nature acquired through practice and coaching. No special societies are required by athletes, and athletics does not meet an important public need; although entertainment is a public need, it certainly doesn’t rank highly compared to the needs met by professions such as medicine. So, although they are highly trained and very well compensated, athletes are not professionals.

Similarly, carpenters require special skills to perform their jobs, but many aspects of their work can be mechanized and little judgment or discretions required. Training in carpentry is not formal, but rather is practical by way of apprenticeships. No organizations or societies are required. However carpenter certainly does meet an aspect of the public good—providing shelter is fundamental to society—although perhaps not to the same extent as do professions such as medicine. So, carpentry also doesn’t meet the basic requirements to be a profession.

Let's look at two occupations that are definitely regarded by society as professions: medicine and law. Medicine certainly fits the definition of a profession given previously. It requires very sophisticated skills that cannot be mechanized, it requires judgment as to appropriate treatment plans for individual patients, and it requires discretion. (Physicians have even been granted physician-patient privilege, the duty not to divulge information given in confidence by the patient to the physician). Although medicine requires extensive practical training learned through an apprenticeship called a residency, it also requires much formal training (four years of undergraduate school, three to four years of medical school, and extensive hands-on practice in patient care), Medicine has a special society (e.g. Nepal Medical Council) to which a large fraction of practicing physicians belong and that participates in the regulation of medical schools, sets standards for practice of the profession, and enforces codes of ethical behaviour for its members. Finally, healing the sick and helping to prevent disease clearly involve the public good. By the definition presented previously, medicine clearly qualifies as a profession.

Similarly, law is a profession. It involves sophisticated skills acquired through extensive formal training; has a professional society, the Nepal Bar Association; and serves an important aspect of the public good.

The difference between athletics and carpentry on one hand and law and medicine on the other is clear. The first two really cannot be considered professions, and the latter two most certainly are.

Is Engineering a Profession?

People use terms different terms like "Job", "Occupation" or "Profession" to refer to a work. Any work for hire can be considered a job, regardless of the skill level involved and the responsibility granted. Engineering is certainly a job - engineers are paid for their services - but the skills and responsibilities involved in engineering make it more than just a job.

Similarly, the word "occupation" implies employment through which someone makes a living. Engineering, then, is also an occupation. There are no amateur engineers who perform engineering work without being paid while they train to become professional, paid engineers. Likewise, the length of time one works at an engineering related job, such as air engineering aide or engineering technician, does not confer professional status no matter how skilled a technician one might become.

Engineering is a profession because: -

- It needs extensive and formal education / training
- Engineering design requires sophisticated skills that can not be mechanized. (Only the construction after the design could be mechanized.)
- Engineering design involves a remarkable amount of judgement and discretion.
- Engineers have societies (e.g. IEEE, ACM etc.) which set standards for engineering practice and code of ethics for the professionals.
- They produce publicly important results like bridges, automated machines etc.

Professional Responsibilities

We will begin our discussion of professional rights and responsibilities by first looking more closely at a few of the important responsibilities that engineers have.

Confidentiality and Proprietary Information

A hallmark of the professions is the requirement that the professional keep certain information of the client secret or confidential. Confidentiality is mentioned in most engineering codes of ethics. This is a well-established principle in professions such as medicine, where the patient's medical information must he kept confidential, and in law, where attorney-client privilege is a well-established doctrine. This requirement applies equally to engineers, who have an obligation to keep proprietary information of their employer or client confidential.

Most information about how a business is run, its products and its suppliers, directly affects the company's ability to compete in the marketplace. Such information can be used by a competitor to gain advantage or to catch up. Thus, it is in the company's (and the employee's) best interest to keep such information confidential to the extent possible.

Some information obviously needs to be kept confidential viz. test results and data, information about upcoming unreleased products, and designs or formulas for products. Engineers working for a client are frequently required to sign a nondisclosure agreement. Of course, those engineers working for the government, especially in the defence industries have even more stringent requirements about secrecy placed on them and may even require a security clearance granted after investigation by a governmental security agency before being able to work.

However, there are grey areas that must he considered. For example, a common problem is the question of how long confidentiality extends after an engineer leaves employment with a company. Legally, an engineer is required to keep information confidential even after she has moved to a new employer in the same technical area. In practice, doing so can be difficult. Even if no specific information is divulged to a new employer, an engineer takes with her a great deal of knowledge of what works, what materials to choose, and what components not to choose. This information might be considered proprietary by her former employer. However, when going to a new job, an engineer cannot be expected to forget all of the knowledge already gained during years of professional experience.

The courts have considered this issue and have attempted to strike a balance between the competing needs and rights of the individual and the company. Individuals have the right to seek career advancement wherever they choose, even from a competitor of their current employer. Companies have the right to keep information away from their competitors. The burden of ensuring that both of these competing interests are recognized and maintained lies with the individual engineer.

Conflict of Interest

Avoiding conflict of interest is important in any profession, and engineering is no exception. A conflict of interest arises when an interest, if pursued, could keep a professional from meeting one of his obligations. For example, a civil engineer working for a state department of highways might have a financial interest in a company that has a bid on a construction project. If that engineer has some responsibility in determining which company's bid to accept, then there is a clear conflict of interest. Pursuing his financial interest in the company might lead him not to objectively and faithfully discharge his professional duties to his employer, the highway department. This type of conflict of interest is called **actual conflicts of interest**.

There are two more types of conflicts of interest. The second type is **potential conflicts of interest**, which threaten to easily become actual conflicts of interest. For example, an engineer might find herself becoming friends with a supplier for her company. Although this situation doesn't necessarily constitute a conflict, there is the potential that the engineer's judgement might become conflicted by the needs to maintain the frie ndship.

Finally, there are situations in which there is the **appearance of a conflict of interest**. This might occur when an engineer is paid based on a percentage of the cost of the design. There is clearly no incentive to cut costs in this situation, and it may appear that the engineer is making the design more expensive simply to generate a larger fee. Even eases where there is only an appearance of a conflict of interest can be significant, because the distrust that comes from this situation compromises the engineers ability to do this work and future work and calls into question the engineer's judgement.

A good way to avoid conflicts of interest is to follow the guidelines of company policy. In the absence of such a policy; asking a co-worker or your manager will give you a second opinion and will make it clear that you aren't trying to hide something. In the absence of either of these options, it is best to examine your motives and use ethical problem-solving techniques. Finally, you can look to the statements in the professional ethics codes that uniformly forbid conflicts of interest.

Environmental Ethics

The environmental movement has sought to control the introduction of toxic and unnatural substances into the environment, to protect the integrity of the biosphere, and to ensure a healthy environment for humans. Engineers are responsible in part for the creation of the technology that has led to damage of the environment and are also working to find solutions to the problems caused by modem technology. The environmental movement has led to an increased awareness among engineers that they have a responsibility to use their knowledge and skills to help protect the environment. This duty is even spelled out in the code of ethics of the IEEE.

Fundamental to discussing ethical issues in environmentalism is a determination of the moral standing of the environment. The Western ethical tradition is anthropocentric, meaning that only human beings have moral standing. Animals and plants are important only in respect to their usefulness to humans. If animals, trees, and other components of the environment have no moral standing, then we have no ethical obligations towards them beyond maintaining their usefulness to humans. There are significant numbers of people who feel that the environment, and specifically animals and plants, do have standing beyond their usefulness to humans. In one form, this view holds that humans are just one component of the environment and that all components have equal standing. For those who hold this view, it is an utmost duty of everyone to do what is required to maintain a healthy biosphere for its own sake.

Professional Rights

Engineers also have rights that go along with their responsibilities. Not all of these rights come about due to the professional status of engineering. They are rights that individuals have regardless of professional status, including the right to privacy, the right to participate in activities of one's own choosing outside of work, the right to reasonably object to company policies without fear of retribution, and the right to due process.

The most fundamental right of an engineer is the right of **professional conscience**. This involves the right to exercise professional judgement in discharging one's duties and to exercise this judgement in an ethical manner. This right is basic to an engineer's professional practice.

The right of professional conscience can have many aspects. For example, one of these aspects might he referred to as the **Right of Conscientious Refusal.** This is the right to refuse to engage in unethical behaviour. Put quite simply, no employer can ask or pressure an employee into doing something that she considers unethical and unacceptable. For example, an engineer ought to be allowed to refuse to work on defence projects or environmentally hazardous work if his conscience says that such work is immoral.

Codes of Ethics and Professional Conduct

Codes of ethics serve a variety of functions. A code of ethics may, at one and the same time, be directed to members of the profession, the public and employers and clients of members of the profession.

Perhaps the most important function of a code of ethics is as a statement embodying the collective wisdom of members of the profession. You can read a code of ethics as a statement of what members of the profession, with many years of experience, have found to be the most important things to think about and do when working as a computer professional. The code expresses both the experience of many members and the consensus of many members.

Currently, there are several codes of ethics in computer. Perhaps the most visible codes are those produced by the ACM and the IEEE. The Software Engineering Codes of Ethics and Professional Practice were developed by a joint ACM-IEEE taskforce with an eye to the licensing of software engineers.

The Institute of Electrical and Electronics Engineers, Inc. (IEEE)

We, the members of the IEEE, in recognition of the importance of our technologies affecting the quality of life throughout the world, and in accepting a personal obligation to our profession, its members and the communities we serve, do hereby commit ourselves to the highest ethical and professional conduct and agree:

1. to accept responsibility in making engineering decisions consistent with the safety, health and welfare of the public, and to disclose promptly factors that might endanger the public or the environment;
2. to avoid real or perceived conflicts of interest whenever possible, and to disclose them to affected parties when they do exist;
3. to be honest and realistic in stating claims or estimates based on available data;
4. to reject bribery in all its forms;
5. to improve the understanding of technology, its appropriate application, and potential consequences;
6. to maintain and improve our technical competence and to undertake technological tasks for others only if qualified by training or experience, or after full disclosure of pertinent limitations;
7. to seek, accept, and offer honest criticism of technical work, to acknowledge and correct errors, and to credit properly the contributions of others;
8. to treat fairly all persons regardless of such factors as race, religion, gender, disability age, or national origin;
9. to avoid injuring others, their property reputation, or employment by false or malicious action;
10. to assist colleagues and co-workers in their professional development and to support them in following this code of ethics.

Approved by the IEEE Board of Directors, August 1990.

ACM Code of Ethics and Professional Conduct

Preamble

Commitment to ethical professional conduct is expected of every member (voting members, associate members, and student members) of the Association for Computing Machinery (ACM).This Code, consisting of 24 imperatives formulated as statements of personal responsibility, identifies the elements of such a commitment. It contains many, but not all, issues professionals are likely to face. Section 1 outlines fundamental ethical considerations, while Section 2 addresses additional, more specific considerations of professional conduct. Statements in Section 3 pertain more specifically to individuals who have a leadership role, whether in the workplace or in a volunteer capacity such as with organizations like ACM. Principles involving compliance with this Code are given in Section 4.

The Code shall be supplemented by a set of Guidelines, which provide explanation to assist members in dealing with the various issues contained in the Code. It is expected that the Guidelines will be changed more frequently than the Code.
The Code and its supplemented Guidelines are intended to serve as a basis for ethical decision making in the conduct of professional work. Secondarily, they may serve as a basis for judging the merit of a formal complaint pertaining to violation of professional ethical standards.

It should be noted that although computing is not mentioned in the imperatives of Section 1, the Code is concerned with how these fundamental imperatives apply to one's conduct as a computing professional. These imperatives are expressed in a general form to emphasize that ethical principles which apply to computer ethics are derived from more general ethical principles.
It is understood that some words and phrases in a code of ethics are subject to varying interpretations, and that any ethical principle may conflict with other ethical principles in specific situations. Questions related to ethical conflicts can best be answered by thoughtful consideration of fundamental principles, rather than reliance on detailed regula tions.

1. GENERAL MORAL IMPERATIVES.

As an ACM member, I will....

1.1 Contribute to society and human well-being.

This principle concerning the quality of life of all people affirms an obligation to protect fundamental human rights and to respect the diversity of all cultures. An essential aim of computing professionals is to minimize negative consequences of computing systems, including threats to health and safety. When designing or implementing systems, computing professionals must attempt to ensure that the products of their efforts will be used in socially responsible ways, will meet social needs, and will avoid harmful effects to health and welfare.
In addition to a safe social environment, human well-being includes a safe natural environment. Therefore, computing professionals who design and develop systems must be alert to, and make others aware of, any potential damage to the local or global environment.

1.2 Avoid harm to others.

"Harm" means injury or negative consequences, such as undesirable loss of information, loss of property, property damage, or unwanted environmental impacts. This principle prohibits use of computing technology in ways that result in harm to any of the following: users, the general public, employees, and employers. Harmful actions include intentional destruction or modification of files and programs leading to serious loss of resources or unnecessary expenditure of human resources such as the time and effort required to purge systems of "computer viruses."

Well-intended actions, including those that accomplish assigned duties, may lead to harm unexpectedly. In such an event the responsible person or persons are obligated to undo or mitigate the negative consequences as much as possible. One way to avoid unintentional harm is to carefully consider potential impacts on all those affected by decisions made during design and implementation.

To minimize the possibility of indirectly harming others, computing professionals must minimize malfunctions by following generally accepted standards for system design and testing. Furthermore, it is often necessary to assess the social consequences of systems to project the likelihood of any serious harm to others. If system features are misrepresented to users, co-workers, or supervisors, the individual computing professional is responsible for any resulting injury.

In the work environment the computing professional has the additional obligation to report any signs of system dangers that might result in serious personal or social damage. If one's superiors do not act to curtail or mitigate such dangers, it may be necessary to "blow the whistle" to help correct the problem or reduce the risk. However, capricious or misguided reporting of violations can, itself, be harmful. Before reporting violations, all relevant aspects of the incident must be thoroughly assessed. In particular, the assessment of risk and responsibility must be credible. It is suggested that advice be sought from other computing professionals. See principle 2.5: regarding thorough evaluations.

1.3 Be honest and trustworthy.
Honesty is an essential component of trust. Without trust an organization cannot function effectively. The honest computing professional will not make deliberately false or deceptive claims about a system or system design, but will instead provide full disclosure of all pertinent system limitations and problems. A computer professional has a duty to be honest about his or her own qualifications, and about any circumstances that might lead to conflicts of interest.

Membership in volunteer organizations such as ACM may at times place individuals in situations where their statements or actions could be interpreted as carrying the "weight" of a larger group of professionals. An ACM member will exercise care to not misrepresent ACM or positions and policies of ACM or any ACM units.

1.4 Be fair and take action not to discriminate.
The values of equality, tolerance, respect for others, and the principles of equal justice govern this imperative. Discrimination on the basis of race, sex, religion, age, disability, national origin, or other such factors is an explicit violation of ACM policy and will not be tolerated.

Inequities between different groups of people may result from the use or misuse of information and technology. In a fair society, all individuals would have equal opportunity to participate in, or benefit from, the use of computer resources regardless of race, sex, religion, age, disability, national origin or other such similar factors. However, these ideals do not justify unauthorized use of computer resources nor do they provide an adequate basis for violation of any other ethical imperatives of this code.

1.5 Honour property rights including copyrights and patent.
Violation of copyrights, patents, trade secrets and the terms of license agreements is prohibited by law in most circumstances. Even when software is not so protected, such violations are contrary to professional behaviour. Copies of software should be made only with proper authorization. Unauthorized duplication of materials must not be condoned.

1.6 Give proper credit for intellectual property.
Computing professionals are obligated to protect the integrity of intellectual property. Specifically, one must not take credit for other's ideas or work, even in cases where the work has not been explicitly protected by copyright, patent, etc.

1.7 Respect the privacy of others.
Computing and communication technology enables the collection and exchange of personal information on a scale unprecedented in the history of civilization. Thus there is increased potential for violating the privacy of individuals and groups. It is the responsibility of professionals to maintain the privacy and integrity of data describing individuals. This includes taking precautions to ensure the accuracy of data, as well as protecting it from unauthorized access or accidental disclosure to inappropriate individuals. Furthermore, procedures must be established to allow individuals to review their records and correct inaccuracies.

This imperative implies that only the necessary amount of personal information be collected in a system, that retention and disposal periods for that information be clearly defined and enforced, and that personal information gathered for a specific purpose not be used for other purposes without consent of the individual(s). These principles apply to electronic communications, including electronic mail, and prohibit procedures that capture or monitor electronic user data, including messages, without the permission of users or bona fide authorization related to system operation and maintenance. User data observed during the normal duties of system operation and maintenance must be treated with strictest confidentiality, except in cases where it is evidence for the violation of law, organizational regulations, or this Code. In these cases, the nature or contents of that information must be disclosed only to proper authorities.

1.8 Honour confidentiality.
The principle of honesty extends to issues of confidentiality of information whenever one has made an explicit promise to honour confidentiality or, implicitly, when private information not directly related to the performance of one's duties becomes available. The ethical concern is to respect all obligations of confidentiality to employers, clients, and users unless discharged from such obligations by requirements of the law or other principles of this Code.

2. MORE SPECIFIC PROFESSIONAL RESPONSIBILITIES.

As an ACM computing professional I will ....

2.1 Strive to achieve the highest quality, effectiveness and dignity in both the process and products of professional work.
Excellence is perhaps the most important obligation of a professional. The computing professional must strive to achieve quality and to be cognizant of the serious negative consequences that may result from poor quality in a system.

2.2 Acquire and maintain professional competence.
Excellence depends on individuals who take responsibility for acquiring and maintaining professional compe tence. A professional must participate in setting standards for appropriate levels of competence, and strive to achieve those standards. Upgrading technical knowledge and competence can be achieved in several ways: doing independent study; attending seminars, conferences, or courses; and being involved in professional organizations.

2.3 Know and respect existing laws pertaining to professional work.
ACM members must obey existing local, state, province, national, and international laws unless there is a compelling ethical basis not to do so. Policies and procedures of the organizations in which one participates must also be obeyed. But compliance must be balanced with the recognition that sometimes existing laws and rules may be immoral or inappropriate and, therefore, must be challenged. Violation of a law or regulation may be ethical when that law or rule has inadequate moral basis or when it conflicts with another law judged to be more important. If one decides to violate a law or rule because it is viewed as unethical, or for any other reason, one must fully accept responsibility for one's actions and for the consequences.

2.4 Accept and provide appropriate professional review.
Quality professional work, especially in the computing profession, depends on professional reviewing and critiquing. Whenever appropriate, individual members should seek and utilize peer review as well as provide critical review of the work of others.

2.5 Give comprehensive and thorough evaluations of computer systems and their impacts, including analysis of possible risks.
Computer professionals must strive to be perceptive, thorough, and objective when evaluating, recommending, and presenting system descriptions and alternatives. Computer professionals are in a position of special trust, and therefore have a special responsibility to provide objective, credible evaluations to employers, clients, users, and the public. When providing evaluations the professional must also identify any relevant conflicts of interest, as stated in imperative 1.3.

As noted in the discussion of principle 1.2 on avoiding harm, any signs of danger from systems must be reported to those who have opportunity and/or responsibility to resolve them. See the guidelines for imperative 1.2 for more details concerning harm,including the reporting of professional violations.

2.6 Honour contracts, agreements, and assigned responsibilities.
Honouring one's commitments is a matter of integrity and honesty. For the computer professional this includes ensuring that system elements perform as intended. Also, when one contracts for work with another party, one has an obligation to keep that party properly informed about progress toward completing that work.

A computing professional has a responsibility to request a change in any assignment that he or she feels cannot be completed as defined. Only after serious consideration and with full disclosure of risks and concerns to the employer or client, should one accept the assignment. The major underlying principle here is the obligation to accept personal accountability for professional work. On some occasions other ethical principles may take greater priority.

A judgment that a specific assignment should not be performed may not be accepted. Having clearly identified one's concerns and reasons for that judgment, but failing to procure a change in that assignment, one may yet be obligated, by contract or by law, to proceed as directed. The computing professional's ethical judgment should be the final guide in deciding whether or not to proceed. Regardless of the decision, one must accept the responsibility for the consequences.
However, performing assignments "against one's own judgment" does not relieve the professional of responsibility for any negative consequences.

2.7 Improve public understanding of computing and its consequences.
Computing professionals have a responsibility to share technical knowledge with the public by encouraging understanding of computing, including the impacts of computer systems and their limitations. This imperative implies an obligation to counter any false views related to computing.

2.8 Access computing and communication resources only when authorized to do so.
Theft or destruction of tangible and electronic property is prohibited by imperative 1.2 - "Avoid harm to others." Trespassing and unauthorized use of a computer or communication system is addressed by this imperative. Trespassing includes accessing communication networks and computer systems, or accounts and/or files associated with those systems, without explicit authorization to do so. Individuals and organizations have the right to restrict access to their systems so long as they do not violate the discrimination principle (see 1.4). No one should enter or use another's computer system, software, or data files without permission. One must always have appropriate approval before using system resources, including communication ports, file space, other system peripherals, and computer time.

3. ORGANIZATIONAL LEADERSHIP IMPERATIVES.

As an ACM member and an organizational leader, I will ....
3.1 Articulate social responsibilities of members of an organizational unit and encourage full acceptance of those responsibilities.
Because organizations of all kinds have impacts on the public, they must accept responsibilities to society. Organizational procedures and attitudes oriented toward quality and the welfare of society will reduce harm to members of the public, thereby serving public interest and fulfilling social responsibility. Therefore, organizational leaders must encourage full participation in meeting social responsibilities as well as quality performance.

3.2 Manage personnel and resources to design and build information systems that enhance the quality of working life.
Organizational leaders are responsible for ensuring that computer systems enhance, not degrade, the quality of working life. When implementing a computer system, organizations must consider the personal and professional development, physical safety, and human dignity of all workers. Appropriate human-computer ergonomic standards should be considered in system design and in the workplace.

3.3 Acknowledge and support proper and authorized uses of an organization's computing and communication resources.
Because computer systems can become tools to harm as well as to benefit an organization, the leadership has the responsibility to clearly define appropriate and inappropriate uses of organizational computing resources. While the number and scope of such rules should be minimal, they should be fully enforced when established.

3.4 Ensure that users and those who will be affected by a system have their needs clearly articulated during the assessment and design of requirements; later the system must be validated to meet requirements.
Current system users, potential users and other persons whose lives may be affected by a system must have their needs assessed and incorporated in the statement of requirements. System validation should ensure compliance with those requirements.

3.5 Articulate and support policies that protect the dignity of users and others affected by a computing system.
Designing or implementing systems that deliberately or inadvertently demean individuals or groups is ethically unacceptable. Computer professionals who are in decision making positions should verify that systems are designed and implemented to protect personal privacy and enhance personal dignity.

3.6 Create opportunities for members of the organization to learn the principles and limitations of computer systems.
This complements the imperative on public understanding (2.7). Educational opportunities are essential to facilitate optimal participation of all organizational members. Opportunities must be available to all members to help them improve their knowledge and skills in computing, including courses that familiarize them with the consequences and limitations of particular types of systems. In particular, professionals must be made aware of the dangers of building systems around oversimplified models, the improbability of anticipating and designing for every possible operating condition, and other issues related to the complexity of this profession.

4. COMPLIANCE WITH THE CODE.

As an ACM member I will ....
4.1 Uphold and promote the principles of this Code.
The future of the computing profession depends on both technical and ethical excellence. Not only is it important for ACM computing professionals to adhere to the principles expressed in this Code, each member should encourage and support adherence by other members.

4.2 Treat violations of this code as inconsistent with membership in the ACM.
Adherence of professionals to a code of ethics is largely a voluntary matter. However, if a member does not follow this code by engaging in gross misconduct, membership in ACM may be terminated.
This Code and the supplemental Guidelines were adopted by the ACM Council on October 16, 1992.

Whistleblowing

Whistleblowing is the act by an employee of informing the public or higher management of unethical or illegal behaviour by an employer or supervisor. There are frequent newspaper reports of cases in which an employee of a company has gone to the media with allegations of wrongdoing by his or her employer or in which a government employee has disclosed waste or fraud.

According to the codes of ethics of the professional engineering societies, engineers have a duty to protect the health and safety of the public, so in many cases, an engineer is compelled to blow the whistle on acts or projects that harm these values, Engineers also have the professional right to disclose wrongdoing within their organizations and expect to see appropriate action taken.

Types of Whistleblowing

A distinction is often made between internal and external whistleblowing. **Internal Whistle blowing** occurs when an employee goes over the head of an immediate supervisor to report a problem to a higher level of management. Or, all levels of management are bypassed, and the employee goes directly to the president of the company or the board of directors, However it is done, the whistleblowing is kept within the company or organization. **External Whistleblowing** occurs when the employee goes outside the company and reports wrongdoing to newspapers or law-enforcement authorities. Either type of whistle blowing is likely to be perceived as disloyalty. However, keeping it within the company is often seen as less serious than going outside of the company.

There is also a distinction between acknowledged and anonymous whistleblowing. Anonymous whistleblowing occurs when the employee who is blowing the whistle refuses to divulge his name when making accusations. These accusations might take the form of anonymous memos to upper management or of anonymous phone calls to the police. The employee can also talk to the news media but refuse to let her name be used as the source of the allegations of wrongdoing. Acknowledged whistleblowing, on the other hand, occurs when the employee puts his name behind the accusations and is willing to withstand the scrutiny brought on by his accusations.

Whistleblowing can be very bad from a corporation point of view because it can lead to distrust, disharmony, and an inability of employees to work together. Similarly, in business, whistleblowing is perceived as an act of extreme disloyalty to the company and to co-workers.

When Should Whistleblowing Be Attempted?

Whistleblowing should only be attempted if the following four conditions are met:

Need

There must be a clear and important harm that can be avoided by blowing the whistle. In deciding whether to go public, the employee needs to have a sense of proportion. You do not need to blow the whistle about everything, just the important things Of course, if there is a pattern of many small things that are going on, this can add tip to a major and important matter requiring that the whistle be blown.

Proximity

The whistleblower must be in a very clear position to report on the problem. Hearsay is not adequate. Firsthand knowledge is essential to making an effective case about wrongdoing. This point also implies that the whistleblower must have enough expertise in the area to make a realistic assessment of the situation. This condition stems from the clauses in several codes of ethics that mandate that an engineer not undertake work in areas outside her expertise. This principle applies equally well to making assessments about whether wrongdoing is taking place.

Capability

The whistleblower must have a reasonable chance of success in stopping the harmful activity. You are not obligated to risk your career and the financial security of your family if you can't see the case through to completion or you don't feel that you have access to the proper channels to ensure that the situation is resolved.

Last Resort

Whistleblowing should be attempted only if there is no one else more capable or more proximate to blow the whistle and if you feel that all other lines of action within the context of the organization have been explored and shut off.

Preventing Whistleblowing

There are four ways in which to solve the whistleblowing problem within a corporation.

First, there must he a **strong corporate ethics culture**. This should include a clear commitment to ethical behaviour, starting at the highest levels of management, and mandatory ethics training for all employees. All managers must set the tone for the ethical behaviour of their employees.

Second, there should be **clear lines of communication** within the corporation. This openness gives an employee who feels that there is something that must be fixed a clear path to air his concerns.

Third, all employees must have **meaningful access to high-level managers** in order to bring their concerns forward. This access must come with a guarantee that there will be no retaliation. Rather, employees willing to come forward should be rewarded for their commitment to fostering the ethical behaviour of the company.

Finally, there should be willingness on the part of management to **admit mistakes**, publicly if necessary. This attitude will set the stage for ethical behaviour by all employees.

Source:

- Deborah G. Johnson, *Computer Ethics*, Third Edition
- Charles B. Fleddermann, *Engineering Ethics*, Second Edition

Unit 5

Risks and Liabilities of Software Systems

Risk and Liabilities of Software Systems

Risk

A risk is a potential problem that might or might not happen. It is necessary to identify risk so that potential problems can be avoided.

Two characteristics of risk

- **Uncertainty**
- **Loss**

Associated Definitions and Concepts

There are a number of key terms that should be understood to manage risk. Some of these terms will be defined here because they are used throughout the remainder of the chapter.

Risk: The possibility of suffering harm or loss.

Risk management: The overall decision-making process of identifying threats and vulnerabilities and their potential impacts, determining the costs to mitigate such events, and deciding what actions are cost-effective for controlling these risks.

Risk assessment (or risk analysis): The process of analyzing an environment to identify the threats, vulnerabilities, and mitigating actions to determine (either quantitalively or qualitatively) the impact of an event that would affect a project, program, or business.

Asset: Resource or information an organization needs to conduct its business.

Threat: Any circumstance or event with the potential to cause harm to

Vulnerability: Characteristic of an asset that can be exploited by a threat to cause harm.

Impact: The loss resulting when a threat exploits vulnerability.

Control (also called countermeasure or safeguard): A measure taken to detect, prevent, or mitigate the risk associated with a threat.

Qualitative risk assessment: The process of subjectively determining the impact of an event that affects a project, program, or business. Qualitative risk assessment usually involves the use of expert judgment, experience, or group consensus to compel the assessment, often used to attach a potential monetary loss to a threat.

Quantitative risk assessment: The process of objectively determining the impact of an event that affects a project, program, or business. Quantitative risk assessment usually involves the use of metrics and models to complete the assessment.

Qualitative vs. Quantitative Risk Assessment

It is recognized throughout industry that it is impossible to conduct risk management that is purely quantitative. Usually risk management includes both qualitative and quantitative elements, requiring both analysis and judgment or experience. It is important to note that in contrast to quantitative assessment, it is possible to accomplish purely qualitative risk management.

It is easy to see that it is impossible to define and quantitatively measure all factors that exist in a given risk assessment. It is also easy to see that a risk assessment that measures no factors quantitatively but measures them all qualitatively is possible.

The decision of whether to use qualitative versus quantitative risk management depends on the criticality of the project, the resources available, and the management style. The decision will be influenced by the degree to which the fundamental risk management metrics, such as asset value, exposure factor, and threat frequency can be quantitatively defined.

Examples of Technology Risks

The most common technology risks include:

- Security and privacy
- Information technology operations
- Business systems control and effectiveness
- Business continuity management
- Information systems testing
- Reliability and performance management
- Information technology asset management

Complex Systems: Basic Definition

A Complex System is any system which involves a number of elements, arranged in structure(s) which can exist on many scales. These go through processes of change that are not describable by a single rule nor are reducible to only one level of explanation; these levels often include features whose emergence cannot be predicted from their current specifications.

Previously, when studying a subject, researchers tended to use a reductionist approach, which attempted to summarize the dynamics, processes, and change that occurred in terms of lowest common denominators and the simplest, yet most widely provable and applicable elegant explanations.

But since the advent of powerful computers which can handle huge amounts of data, researchers can now study the complexity of factors involved in a subject and see what insights that complexity yields without simplification or reduction.

Features of complex systems

A complex system has many features. Some of them are listed below

Emergence - more is different

What distinguishes a complex system from a merely complicated one is that some behaviors and patterns emerge in complex systems as a result of the patterns of relationship between the elements. Emergence is perhaps the key property of complex systems and a lot of work is being done to try to understand more about its nature and the conditions which will help it to occur.

The behavior of a complex system can not be understood in terms of a simple extrapolation of the properties of its components, elements and entities. As P.W. Anderson said in his 1972 science article (Science Vol. 177 No. 4047), more is different. At the macroscopic level of the system, new properties appear which can not be predicted by the properties of microscopic components.

Typically, the relationships between elements in a complex system are both short-range and long-range. The direct local interactions are short-range, that is information is normally received from near neighbours. Through top-down feedback and small-world connections agents can indirect influence all other agents. The richness of the connections means that communications will pass across the system but will probably be modified on the way.

Relationships are non-linear

There are rarely simple cause and effect relationships between elements. A small stimulus may cause a large effect (see butterfly effect), or no effect at all. See nonlinearity.

Relationships contain feedback loops

Both negative (damping) and positive (amplifying) feedback are key ingredients of complex systems. The effects of an agent's actions are fed back to the agent and this, in turn, affects the way the agent behaves in the future. This set of constantly adapting nonlinear relationships lies at the heart of what makes a complex system special.

Complex systems are open

Complex systems are open systems - that is, energy and information are constantly being imported and exported across system boundaries. Because of this, complex systems are usually far from equilibrium: even though there is constant change there is also the appearance of stability. The parts cannot contain the whole

There is a sense in which elements in a complex system cannot "know" what is happening in the system as a whole. If they could, all the complexity would have to be present in that element. Yet since the complexity is created by the relationships between elements that is simply impossible. A corollary of this is that no element in the system could hope to control the system.

Complex systems have a history

The history of a complex system is important and cannot be ignored. Even a small change in circumstances can lead to large deviations in the future. This has been referred to as the "Butterfly effect."

Complex systems are nested

Another key aspect of complex adaptive systems is that the components of the system - usually referred to as agents - are themselves complex adaptive systems. For example, an economy is made up of organisations, which are made up of people, who are systems of organs controlled by their nervous systems and endocrine systems, which are made up of cells - all of which, at each level in the hierarchy, are complex adaptive systems.

Boundaries are difficult to determine

It is usually difficult to determine the boundaries of a complex system. The decision is usually based on the observer's needs and prejudices rather than any intrinsic property of the system itself.

For instance, the boundary of an individual human being may appear easy to determine but a little more thought will show some of the ambiguities. For instance, are clothes inside or outside the boundary? If someone stares at you across a room or crowded train, especially in a lustful or aggressive way, have they invaded your boundary? Are your boundaries physical, or emotional? Emotional influence can impact physical existence. At what moment does food become a part of your body?

Complex networks

A complex network forms the backbone of complex systems: the nodes correspond to the agents, entities or parts of the complex system, the edges to the interactions between them. The most complex networks are small-world or scale-free networks at the border between regular and random networks.

Scientists are finding that complexity itself is often characterized by a number of important characteristics:

- **Self-Organization**
- **Non-Linearity**
- **Order/Chaos Dynamic**
- **Emergent Properties.**

From the learning process of analyzing, simulating, and modeling these characteristics of Complex Systems, a number of unique and thought-provoking computer programming approaches have emerged Artificial Life, Genetic Algorithms, Neural Networks, Cellular Automata, Boolean Networks etc.

Self-Organization

Scientists are finding that change occurs naturally and automatically in systems in order to increase efficiency and effectiveness, as long as the systems are complex enough as defined above. This change is accomplished by the elements that make up the system when they respond automatically to feedback from the environment the system inhabits. Environmental feedback can be seen as providing information about the system's efficiency and effectiveness. Elements that survive negative environmental feedback will automatically re-settle themselves, or re-organize themselves and their interactions in order to better accomplish the system's goals. Success at this then assures their continued existence by also protecting or reinforcing the structures of which the elements are a part.

Such responsiveness occurs even when the elements and system are non-organic, unintelligent, and unconscious as long as the system is complex as described above.

Non-Linearity

When change occurs in Complex Systems it occurs in a non-linear fashion. Linear change is where there is a sequence of events that affect each other in order as they appear one after the other. In contrast, in non-linear change, one sees elements being changed by previous elements, but then in turn these changed elements affect the elements that are before it in the sequence. Thus in non-linear analysis, researchers look at how everything in the sequence has the possibility of affecting everything else in the sequence before and after it. Thus often the result ends up being unproportional to the original input. This type of dynamic in a complex system is much closer to how things actually happen in nature. Almost never in nature does a purely linear sequence of events and change occur.

Order/Chaos Dynamic

It is usually fairly easy to predict what will develop in the next stage of system development when one has extensive knowledge of the previous stage. And this knowledge is usually of a range of possibilities that can develop next. But as one begins to deal with stages of development farther and farther down the sequence of developmental stages, it becomes more and more difficult to predict what will develop based only on knowledge of that first stage, even when that knowledge is extensive. Thus even though there is logical development from stage to stage, there is an increasing inability to predict what will actually be the next development. This uncertainty of predictability is called "chaos". Thus, one can then see how a tiny change in a condition can eventually lead to a huge number of different possible results. But yet all these changes are still logical results of that tiny change, it just becomes increasingly difficult to predict exactly which result will actually occur. But since some probability of occurrence for many of them can be known, then statistical analysis is still very important for helping describe the overall situation. The classic illustration for this is the idea of how the flapping of butterfly wings in one part of the world can contribute to the evolution of a hurricane in another part of the world.

Emergent Properties

The unpredictability that is thus inherent in the natural evolution of complex systems then can yield results that are totally unpredictable based on knowledge of the original conditions. Such unpredictable results are called emergent properties. Emergent properties thus show how complex systems are inherently creative ones. Emergent properties are still a logical result, just not a predictable one. This can also include higher level phenomenon that cannot be reduced to it's simpler constitutes or it's origins.

A complex system is emergent. In an emergent system, smaller parts comprise a larger system. This larger system has properties the smaller units lack. For example, the brain is made up of individual neurons that, when functioning together, are capable of tasks no single neuron can perform alone. The new properties only emerge when the neurons work together.

- A complex system is unpredictable.
- A complex system contains many iterations and feedback/feed forward loops.
- In a complex system, decision-making is decentralized.
- Learning is a typically a "complex" activity. Most learning systems contain a number of separate parts that must work together for learning to occur. For example, a typical learning system consists of students, a teacher, a content focus, and resources. This system operates according to a fixed plan--the students follow the teacher's "rules."

Learning environments

A learning environment can be emergent. Working together, a group of learners can collectively build their knowledge of a topic, for instance, the phases of the moon. To do so, each learner might research a particular lunar phase, then share what he or she has learned with the rest of the group. This way, the group amasses a body of knowledge that no one person could have acquired alone.

A learning environment can be unpredictable. An exploration of the phases of the moon could result in the group considering whether planets also have phases. A learning environment can contain many iterations and feedback/feedforward loops. People learn by trial and error--in other words, they learn from their mistakes. Decision-making in a learning environment can be decentralized. Groups can really thrive when students control the learning process, rather than the instructor.

It's quite possible that learning occurs best on the "edge of chaos," where order and chaos meet. To see for yourself, check out these two resources:

What is a non Complex System?

An Example of a Non-Complex System: This approach contrasts with, say, the neo-classical approach to modeling economic systems. Usually, in order to work with expressions and equations that are tractable by mathematical analysis, microeconomic theorists assume that all consumers are identical and never change their preferences or characteristics. (so much for education or advertising!) Consumers either do not communicate at all or they interact in some simplistic random fashion, and the underlying system is always in equilibrium. Rarely do macroeconomic models build naturally on the underlying microeconomic models, as the complex systems approach strives to do.

The Implication of Unrestrained Complexity

"The more complex the system, the more open it is to total breakdown". Rarely would a builder think about adding a new sub-basement to an existing 100-story building; to do so would he very costly and would undoubtedly invite failure. Amazingly, users of software systems rarely think twice about asking for equivalent changes. Besides, they argue, it is only a simple matter of programming.

Our failure to master the complexity of software results in projects that are late, over budget, and deficient in their stated requirements. We often call this condition the software crisis, but frankly, a malady that has carried on this long must he called normal. Sadly, this crisis translates into the squandering of human resources — a most precious commodity — as well as a considerable loss of opportunities. There are simply not enough good developers around to create all the new software that users need. Furthermore, a significant number of the developmental personnel in any given organization must often be dedicated to the maintenance or preservation of geriatric software.

Given the indirect as well as the direct contribution of software to the economic base of most industrialized countries, and considering the ways in which software can amplify the powers of the individual, it is unacceptable to allow this situation to continue. How can we change this dismal picture? Since the underlying problem springs from the inherent complexity of software, our suggestion is to first study how complex systems in other disciplines are organized. Indeed, if we open our eyes to the world about us, we will observe successful systems of significant complexity. Some of these systems are the works of humanity, such as the Space Shuttle, the England/France tunnel, and large business organizations such as Microsoft or General Electric. Many even more complex systems appear in nature, such as the human circulatory system or the structure of a plant.

Unmanaged complexity leads to software difficult to use, maintain, and modify. It causes increased development costs and overrun schedules. But certain elements of software are inherently complex - conformance to arbitrary external interfaces, pressure to adapt to changing user requirements, lack of obvious representations for visualizing software

Controlling Complexities

Controlling and measuring complexity is a challenging engineering, management, and research problem. Metrics have been created for measuring various aspects of complexity such as sheer size, control flow, data structures, and inter-module structure. The well-accepted are probably Source Lines of Code and Cyclomatic complexity.

Reuse & Hidden Problems

Reusability is the process of creation and reuse of software building blocks for the development of a software system. Bennatan suggest four software resource categories that should be considered as planning proceeds regarding the development of a software system.

- Off the shelf components
- Full experience components
- Partial experience components
- New components

The software designers must take care of the following guidelines else a software can have many problems in its development using reusable components.

1. A deeper analysis of the system requirements should be performed in order to use reusable components. A reusable component must be well tested and must fit the deign scenario, else it can add up to errors and lead to system failure.
2. If a software development team is using partial experienced reusable components. A detailed analysis must be carried out and if extensive modification is required in the reusable components, as the cost of modification may sometimes be greater than the cost to develop new components.

Correctness

A program must operate correctly or it provides a little value to its users and degrades the quality of the software. Correctness is the measure of the degree to which the software performs its required functions. The most common measures for correctness are defects per KLOC.

Reliability

The probability that software will not cause the failure of a system for a specified time under specified conditions is the measure of reliability. The probability is a function of the inputs to and use of the system, as well as a function of the existence of faults in the software. The inputs to the system determine whether existing faults, if any, are encountered. Reliability was originally monitored by tracking faults found throughout the lifecycle

Reliability & Safety

Although software reliability and software safety are closely related to one another, it is important to understand the subtle difference between them. Software reliability uses statistical analysis to determine the likelihood that a software failure will occur.

Software safety examines the ways in which failures results in the conditions that can lead to mishap. That is its failures are not considered in a vacuum, but are evaluated in the context of an entire computer based system.

Testing & Reliability

Testing is a critical element of software quality assurance and represents the ultimate review of specifications, designs and coding. It is the process of executing a program with an intent of fining errors. If testing is conducted successfully it will uncover error in the software. As a secondary benefit, testing demonstrates that software functions appear working according to specification and that performance requirements appear to have been met. In addition data collected as testing is conducted provides a good indication of software reliability and some indication of software quality as a whole.

Unit 6

Intellectual Property

Intellectual Property

Definition

Property that can be protected under federal law, including copyrightable works, ideas, discoveries, and inventions. Such property would include novels, sound recordings, a new type of mousetrap, or a cure for a disease.

In law, particularly in common law jurisdictions, intellectual property or IP refers to a legal entitlement which sometimes attaches to the expressed form of an idea, or to some other intangible subject matter. In general terms this legal entitlement sometimes enables its holder to exercise exclusive control over the use of the IP. The term intellectual property reflects the idea that the subject matter of IP is the product of the mind or the intellect, and that once established, such entitlements are generally treated as equivalent to tangible property, and may be enforced as such by the courts.

Types

The most well known forms of intellectual property include

- copyrights,
- patents,
- trademarks, and
- trade secrets.

Patents and trademarks fall into a particular subset of intellectual property known as industrial property.

Copyright

Copyright is a protection that covers published and unpublished literary, scientific and artistic works, whatever the form of expression, provided such works be fixed in a tangible or material form. This means that if you can see it, hear it and/or touch it - it may be protected. If it is an essay, if it is a play, if it is a song, if it is a funky original dance move, if it is a photograph, HTML coding or a computer graphic that can be set on paper, recorded on tape or saved to a hard drive, it may be protected. Copyright laws grant the creator the exclusive right to reproduce, prepare derivative works, distribute, perform and display the work publicly. Exclusive means only the creator of such work, not anybody who has access to it and decides to grab it.

When does Copyright Protection begin, and what is required?

Copyright protection begins when any of the above-described work is actually created and fixed in a tangible form.

For example, my brother is a musician and he lives in the United States. When he writes new lyrics, he prints them out on paper, signs his name at the bottom with the Copyright © symbol to show that he is the author, places it in an envelope and mails it to himself without opening it. His copyright begins at the moment he puts his idea in a tangible form by printing the lyrics out on paper. He creates proof when he mails it to himself - the postmark establishes the date of creation. He then registers his copyright with the U.S. Copyright Office, which is a requirement in order to sue for monetary damages should a violation of his copyright, arise. However, if somebody copies and redistributes his lyrics without permission before his copyright is registered, he still has the right to assert a copyright claim as the true author.

The above applies to digital art and graphics. Open a gif, jpg or png file that you created and look at the properties. It states the date that you saved it to your hard drive as the date of creation. If somebody copies a graphic from your web site, I assure you that the date of creation on your copy of the file is earlier than the copy taken off your web site. If that still doesn't feel like enough proof for you, save everything to a floppy disk and mail it to yourself via certified mail. Keep the envelope sealed, wrap it in protective plastic and put it in a safe place.

Somebody once asked if it was "illegal" to place the copyright © symbol next to your name if you have not registered your copyright. Unless you have stolen the work from somebody else and you are not the true author of the work, it is not illegal to place the copyright © symbol next to your name - it is your right to do so.

The proper way to place a copyright notice is as follows: Copyright © (first date of creation) (name of owner). Like this: Copyright © 2003 John Smith.

When does Copyright Protection end, or expire?

If a copyright statement reads, "© Copyright 1998, 1999 John Smith." does that mean that John Smith's copyright expired in 1999. The dates that you see in a copyright statement do not refer to the dates that the owner's material will expire and become public domain - they actually refer to the dates that the material was created.

When you see several dates in a copyright statement, it simply means that certain things were created in one year and modified later. It could also mean that new things were created and added in a later year. It most definitely does not refer to the date that a copyright will expire. Expiration of a copyright actually takes place much later, and this period of validity begins from the date that you see in the copyright statement. The Berne Convention establishes a general and minimum period that lasts the life of the author and fifty years after his (or her) death. Cinematographic works and photographic works have a minimum period of protection of 50 and 25 years upon the date of creation, respectively. This applies to any country that has signed the Berne Convention, and these are just the minimum periods of protection. A member country is entitled to establish greater periods of protection, but never less than what has been established by the Berne Convention.

So, what does all this mean? This means that if a copyright statement reads, "© Copyright 1998, 1999 John Smith" and John Smith is from a country that has signed the Berne Convention, he created his works in 1998 and 1999, and his copyright is not going to expire until at least fifty years after he dies (this period may be greater - remember that member countries may establish longer periods of protection). Until that time, his works are not in public domain.

I have actually seen copyright statements with future dates, such as "© Copyright 2003, 2004 John Smith", most likely because the copyright holder thought that they could establish an expiration date for the copyright. This is incorrect unless John Smith traveled to the future and created the work in question. These types of copyright statements also mislead others to believe that dates in a copyright statement refer to the date a copyright expire, when the date should really refer to the date of creation.

The Famous © Symbol

The © symbol entitles the author to claim copyright, i.e. to use, reproduce and distribute this material unless someone else who wishes to use it obtains the author's prior written permission to use it as well, and only in the manner that the author previously approve. In other words, it means that nobody may access the author's material and copy it until the author provides a written document that states, "Yes, you can use my work, but only in the manner that I deem appropriate."

International Copyrights

There are no "international copyrights" that enable you protect your work throughout the world. However, most countries are members of the Berne Convention and the Universal Copyright Convention (UCC), which allow you to protect your works in countries of which you are not a citizen or national. Under these treaties, the following works may be protected: (i) both unpublished and published works of an author who is a national or resident of a country that is a member of these treaties; or (ii) published works, with permission, of an author who is not a national or resident of a country that is a member of these treaties. In this case a work may be considered simultaneously published in several countries if it has been published in two or more Berne Union countries within 30 days of its first publication.

To benefit from the above protection, there are no formal requirements established in the Berne Convention other than having the author's name on the work. Under the UCC, a copyright notice is required. This notice should consist of the copyright symbol "©" accompanied by the year of first publication and the name of the copyright owner, for example: Copyright © 2002 John Smith. This notice is to be placed in such manner and location as to give reasonable notice of the claim to copyright.

So, what does this mean? Well, if John Smith is a resident of Canada (member of the treaties), and if somebody in the United States (also a member country) accesses John Smith's website, which complies with (i) and (ii) above and displays the proper notice of Copyright © 2002 John Smith (as required by the UCC) - I am of the opinion that my work is considered to be "published", and therefore protected, in the United States as well as in Canada.

Patent

A patent is a set of exclusive rights granted by a state to a person for a fixed period of time in exchange for the regulated, public disclosure of certain details of a device, method, process or substance (known as an invention) which is new, inventive and useful.

The term "patent" originates from the Latin word patere which means "to lay open" (ie. make available for public inspection) and the term letters patent, which originally denoted royal decrees granting exclusive rights to certain individuals or businesses.

The person applying for a patent generally does not need to be the inventor who created or authored the invention. However, in the United States a patent application must be filed in the name of the actual inventor or inventors, although the application can be assigned to another party, such as the employer of the inventor.

A modern patent provides the right to exclude others from making, using, selling, offering for sale, or importing the patented invention. Generally, patents are enforced only through civil lawsuits. Patent licensing agreements are effectively contracts in which the patent owner (the licensor) agrees not to sue the licensee for infringement of the licensor's patent rights. Governments typically reserve the right to suspend or cancel a patent at will.

Trademark

A trademark (Commonwealth English: trade mark)[1] is a distinctive sign of some kind which is used by a business to identify itself and its products and services to consumers, and to set the business and its products or services apart from those of other businesses. A trademark is a type of intellectual property, and in particular, a type of industrial property.

Conventionally, a trademark comprises a name, word, phrase, logo, symbol, design, image, or a combination of one or more of these elements. There are also a range of non-conventional trademarks which do not fall into these standard categories.

The essential function of a trademark is to uniquely identify the commercial source or origin of products or services, such that a trademark, properly called, indicates source or acts as a badge of origin. The use of a trademark in this way is known as trademark use and a trademark owner seeks to enforce its rights or interests in a trademark by preventing unauthorised trademark use.

Trade secret

A trade secret is a confidential practice, method, process, design, or other information used by a company to compete with other businesses. It is also referred to in some jurisdictions as confidential information.

The concept is that, sometimes, Company A is more successful than Company B, or is successful at all, not so much due to access to markets, resources, or personnel, but due to special knowledge owned by Company A. If others had access to the same knowledge, then Company A's ability to survive in an otherwise equal marketplace would be impaired. Thus, such secrets are guarded zealously.

The precise language by which a trade secret is defined varies by jurisdiction (as do the precise types of information that are subject to trade secret protection). However, there are three factors that (though subject to differing interpretations) are common to all such definitions: a trade secret is some sort of information that

(a) is not generally known to the relevant portion of the public,

(b) confers some sort of economic benefit on its holder (where, note well, this benefit must derive specifically from the fact that it is not generally known, not just from the value of the information itself), and

(c) is the subject of reasonable efforts to maintain its secrecy.

Trade secrets are not protected by law in the same manner as trademarks or patents. Probably one of the most significant differences is that a trade secret is protected without disclosure of the secret.

Copyright, Patent, Design and Trademark in Context of Nepal

Copyright

Copyright is an intellectual property. It is related to literary, artistic, scientific and research works. It protects the authors from unauthorized copying of their works. It extends protection to form or manner of expressing ideas. But the idea itself cannot be copyrighted. The authors are the prime beneficiaries of copyright protection. Technology has greatly broadened the scope of copyright in modem times.

Copyright Act, 2002

The legal provisions about copyright in Nepal are contained in the copyright Act 2002, which has 43 sections. The Copyright Rules guide its implementation. The salient features of the copyright Act are

1. Copyright Protection (Sec 3)

The Act extends copyright protection to any work, including literary, artistic, scientific and other works, which are original and intellectually presented.

Work has been defined as follows

a) Book, pamphlet, article, research paper

b) Drama, lyrical drama, silent acting

c) Musical works with or without words

d) Audio-visual works, including movies

e) Architectural design

f) Paintings, sculpture, wooden arts, lithography

g) Photographic works

h) Photographic works

i) Drawings, maps, plans, three-dimensional geographical works, topographical and scientific articles and works

j) Computer programs

2. Non-protection of Copyright (Sec 4)

Copyright cannot be protected for

a) Ideas, religion, news, concept, principle

b) Rulings of the courts, administrative decisions

c) Folklores, folk stories, old sayings

d) General statistics

3. Economic Beneficiary of copyright (Sec. 6)

The Act vests the ownership of copyright in the author of the work. Exceptions are

a) Coauthor for joint works
b) Persons or institutions commissioning joint works
c) Persons or institutions paying remuneration for producing the work
d) Publisher for anonymous work until the author is identified.

- Registration of work is not essential to enjoy copyright. The copyright author also has ethical rights.

4. Economic Benefits (Sec. 7-13)

The copyright owner has the following economic benefits

a) Reproduction of works
b) Translation of works
c) Revision of-works
d) Changes in the form of works
e) Sale and renting of works
f) Transfer of copyright
g) Public shows and broadcasting, etc.

5. Term of copyright (Sec. 14-15)

The term of copyright granted is the life of the author plus fifty years after death.

- For joint authorship, the term of copyright is fifty years from the death of author who dies last.
- For anonymous work, the term is fifty years from the date of publication.

6. Transfer of Copyright (Sec. 24)

Copyright is transferable in whole or in part.

7. Use of Copyright without Permission (Sec. 16-23)

Copyright materials can be reproduced without permission for:

a) Personal use
b) Reference purposes
c) Teaching purposes
d) Library and archives purposes
e) Public knowledge

8. Unauthorized Publication (Sec. 25-26)

The law prohibits unauthorized publication of work. The following acts are regarded unauthorized publication

a) Sale of copies of other's work for business purpose to make economic gain without permission.

b) Gain advantage from the prestige of other's work through advertisement or publicity.

c) Create work by changing the structure or language of other's work for economic gain.

d) Gain advantage from imitation or advertisements. Import of unauthorized publication is banned.

9. Penalty

Fine ranging from Rs. 10,000 to Rs. 100,000 or six months imprisonment, or both have been prescribed for unauthorized publication. The penalty is double for second and subsequent offences. The unauthorized copies are also confiscated. Compensation can also be claimed by the owner of copyright.

- Unauthorized imports are subject to a fine of Rs. 5000 to Rs. 100,000 plus confiscation of copies imported.

10. Registration

The Registrar of Copyright has been provided to deal with copyright matters.

Conclusions

The copyright Act in Nepal lacks effective implementation.

• The consciousness is lacking about copyright in Nepal. Till 1998, a total of 189 books, 204 audio cassettes, 5 films and 9 paintings were registered for copyright.

4.2 Copyright Rules 1989

Rules are needed to implement the law. The copyright rules in Nepal were formulated for copyright Act of 1965. Now rules have not been passed as yet

The copyright rules deal with the following aspects :

1. Registration and Certificate of Copyright

The format of application and fee has been prescribed for registration of copyright. Copies of the work are also required. The format and fee for certificate are also prescribed.

2. Inspection and Copying

Procedures have been laid down for inspection and copy of copyright register.

3. Corrections in copyright Register

The Salient provisions in the law about patent are

Corrections in copyright register can be made in respect of Name and address of the author

- Name of publisher of the work
- Name of printer

• Transfer of copyright

4. Nepali Translation

Procedures have been laid down tor granting permission for Nepali translation. Format has also been prescribed.

5. Duties, Responsibilities and Powers of the Copyright Advisory Committee

They are as follows

- Give decisions on problems related to any work
- Solve questions related to copyright
- Give decisions on "Yours and mine" type disputes
- Give decisions on plagiarism of work
- Fix number of copies needed for registration of copyright.
- Solve problems related to implementation of Act and rules.

Provisions related to committee meetings have been prescribed.

Patent, Design and Trademark

Patent, design and trademark are intellectual property. They are the creation of human mind.

The Patent, Design and Trademark Act 1965 (amended in 1987), together with Rules, regulate patent, design and trademark in Nepal.

Patent

Patent has been defined as follows

Patent is any new method or way of constructing, conducting or processing any matter or body of matters or any useful invention based on a new principle or formula.

1. Registration

a) A person should register the right of patent in his name. The term of registration is seven years. It can be renewed for two terms of seven years each (total life of patent cannot exceed 21 years)

b) Copying of patent is not allowed without the permission of the patentee.

c) Patent right can be transferred like a movable property. The transfer should be registered.

d) The application for registration of patent should contain

   I. Name, address and occupation of inventor

   II. Manner and nature of right acquired, if applicant is not the inventor

   III. Manner of producing, using or conducting the patent

   IV. Principle or formula on which the patent is based.

V. Map or drawing of the patent with description

e) The registered patents should be published, except those needing confidentiality in the national interests.

2. Non Registration of Patent

The patent cannot be registered in the following conditions

a) Already registered in another person's name

b) Not invented by the applicant or the right to patent not received

c) Causes adverse effects on health, good conduct or morality or is prejudicial to national interests

d) Contravenes existing laws of Nepal

3. Archives

The drawing or sample of patent should be provided to National Archives.

4. Punishment

Infringement of patent laws is subject to fine of upto l(s. 2,000 and confiscation of articles connected with the offence. Compen sation can also be awarded to the aggrieved party.

5. Foreign Patents

Patents registered in foreign countries must be registered in Nepal to claim right of ownership.

Patents for Nepalese products must first be registered in Nepal to get foreign registration.

Design

Design has been defined as follows:

Design means any feature, pattern or shape of a matter prepared and produced in any manner.

The salient legal provisions about de sign are

a) Registration

Registration is needed to obtain right to design. A person can prepare the design himself or through others.

The term of design registration is five years. It can be renewed for two terms of five years each (total life of design cannot exceed 15 years)

- Use or copying of design is not allowed without the permission of the owner. Permission can be given by agreement of both parties—owner and the person desiring to use.
- Map or thawing of design with description plus four copies should be submitted for registration purposes.
- The registered designs can be published for public knowledge. Complaints should be lodged within 35 days of such publication.

b) Non Registration of Design

The design cannot be registered in the following conditions

Already registered in another person's name

- Likely to cause damage to the reputation of an individual or an institution
- Causes adverse effects on good conduct or morality of general public.

c) Infringement of design law is subject to fine upto Rs. 800 and confiscation of articles connected with the offence. Compensation can also be awarded to the aggrieved party.

d) Foreign Design

- Foreign design must be registered in Nepal to claim right to ownership.
- Nepalese designs must first be registered in Nepal to get foreign registration.

Trademark

Trademark has been defined as follows:

Trademark is any sign, picture or word or their combination used by a person, firm or company to distinguish its goods and services from those of others.

A registered brand is a trademark.

The salient legal provisions about trademark are

a) Registration:

Registration is needed to obtain right of trademark. The term of registration is seven years. It can be renewed for unlimited terms of seven years each. It can be cancelled if not used within one year of registration.

- Use or copying of trademark is not allowed without the permission of the owner. Permission can be given by agreement between the owner and the person desiring to use.
- Publication of trademark can be done for public knowledge.

b) Non Registration of Trademark

The trademark cannot be registered in the following conditions

- Already registered in the another's name; or
- Likely to cause damage to the reputation of an individual or an institution; or
- Likely to harm goodwill of trademark of others; or
- Adversely affects the good conduct or morality of general public; or
- Is prejudicial to national interest.

c) Classification of Goods and Services for Trademark

- The government can classify goods and services for the purposes of trademark registration.

- Separate applications are needed for trademark registration in each category of goods and services.
- Same trademark can be registered for various classes of goods and services.

d) Infringement of trademark law is subject to fine of up to Rs. 1000 and confiscation of articles connected with the offence. Compensation can also be awarded to the aggrieved party.

e) Foreign Trademarks

Foreign trademarks must be registered in Nepal to claim right of ownership

- Trademarks for Nepalese products must first be registered in Nepal to get foreign registration.
- The government has not yet laid down rules to facilitate implementation of the design and trademark act.

Technology and IT Policy of Nepal

Technology Policy of Nepal

Policies are guidelines for decision making to achieve goals. Technology is a critical factor for development. Technology policy of the government greatly affects the technological development of the country.

Nepal’s science and technology policy was included in the Ninth Plan (1997-2002). The salient features of this policy are :

1. A conducive environment will be created for imparting standard science and technology education at school and higher education levels. The promotion of technical education will be gradually increased.
2. Improvements in endogenous and traditional technology will be made and special consideration will be given to commercialize it.
3. Advanced technologies will be imported. The selection process for imports will give priority to export promoting and under employment reducing technologies.
4. Production and productivity will be increased by the compulsory adoption of advanced technology in economic and social sectors.
5. Science and Technology Committee and Research and Development (R & D) unit will be formed in all government and semi-government agencies.
6. Private sector will be encouraged to invest a certain percentage of their profits to research. Science and technology sectors will be initiated in districts, municipalities and village development Committees.
7. A national science and technology management system will be developed to ensure efficiency and effectiveness of investment.
8. A separate science and technology service will be developed within the civil service. Incentives will he provided to scientists and technologists involved in R & D. Lateral entry will be allowed.
9. Research findings of science and technology will be disseminated.
10. A twenty year science and technology perspective plan will be prepared.
11. Necessary institutional framework and system will be developed for science and technology.
12. Brain drain of science and technology personnel will be controlled.
13. Technology parks will be established.

Information Technology (IT) Policy of Nepal

Nepal has formulated Information Technology Policy 2000.

Its salient features are

1. Declare information technology as a priority sector.
2. Follow a single door system for the development of iT.
3. Prioritize research in IT.
4. Create conducive environment to attract private sector investment in IT,
5. Provide internet facilities to all village development commit tees.

6. Render assistance to educational institutions and encourage training to develop qualified manpower in IT.
7. Computerize government records and build websites for flow of information.
8. Increase the use of computer in private sector.
9. Develop physical and virtual information technology park for development of TT.
10. Use IT to promote e-commerce, e-education, e-health and transfer of technology to rural areas.
11. Establish National Information Technology Center.
12. Establish a national level fund to contribute to R & D in IT.
13. Establish venture capital fund for 1T.
14. Include computer education from the school level.
15. Establish Nepal in global market through the use of IT.
16. Provide legal sanctions to the use of IT.
17. Gradually use IT in all types of government activities.

The vision of Nepal's Information Technology Policy is to place Nepal on the global map of information technology by 2005.

The objectives of IT policy are

a) Make IT accessible to general public; increase employment through IT.
b) Build knowledge-based society
c) Establish knowledge-based industries.

Source :

- IT Policy of Nepal, 2000, Kathmandu, MOST.
- Dr. Govinda Ram Agrawal, *Dynamics of Business Environment in Nepal*, Edition 2003
- Wekipedia Dictionary

Unit 7

Privacy and Civil Liberties

Privacy and Civil Liberties

Privacy

Computer and information technology, like other new technologies, creates new possibilities; it creates possibilities for behavior and activities that were not possible before the technology. Computers made it possible (and in many cases, cheap and easy) to gather detailed information about individuals to an extent never possible before. Public concern about computers and privacy arises for precisely this reason. Of all the social and ethical concerns surrounding computer technology, the threat to personal privacy was probably the first to capture public attention. And this issue persists in drawing public concern and leading to action by policy makers.

Panopticon: an old age Privacy issue

Panopticon is the word Jeremy Bentham used in 1787 to describe his idea for the design of prisons. In the panopticon, prison cells would be arranged in a circle and the side of each cell facing the inside of the circle would be all glass. The guard tower is placed in the center of the circle, from which every cell is in full view. Everything going on in each cell can be observed. The effect is not two-way; that is, the prisoners can not see the guard in the tower. The idea of the panopticon was picked up by Michel Foucault in 1975 and brought to wider public attention. Bentham and Foucault recognized the power of surveillance to affect the behavior of individuals. In the panopticon, a prison guard need not even be there at every moment; when prisoners believe they are being watched, they adjust their behavior. When individuals believe they are being watched, they are compelled to think of themselves as the observer might think of them. This shapes how individuals see themselves and leads them to behave differently than they might if they weren't being observed.

Privacy then and now

Nowadays, the central, state, and local government agencies maintain extensive records of individual behavior including such things as any interaction with criminal justice agencies, income taxes, employment history , use of human services agencies, motor vehicle registration, and so on. As well, private organizations maintain extensive databases of information on individual purchases, airline travel, credit worthiness, health records, telephone or cellular phone usage,

The only differences between what is now possible through IT and what was envisioned then is that much of the record keeping is done through electronic records instead of by direct human observation or through cameras.

So, it seems like we are building a panopticon in which everything we do is observed and could come hack to haunt us.

Information Technology has changed Record Keeping

Computer technology has changed record-keeping activities in a number of undeniable and powerful ways.

1. The scale of information gathering has changed.
2. The kind of information that can be gathered has changed.
3. The scale/exchange of information has changed enormously.

In the pre-computer 'paper-and-ink' world, there were limitations on the amount of data gathered, who had access, how long records were retained, and so on. Electronic records do not have those imitations. We can collect, store, manipulate, exchange, and retain practically infinite quantities of data.

The kind of information that it is now possible to collect and use is also infinite. Employers can keep records of every keystroke an employee makes. Employers can monitor their employees' uses of the Web, their participation in chat rooms, not to mention their e-mail. Transaction Generated Information (TGI) are also new form of information, which includes purchases made with a credit card, telephone calls, entry and exit from intelligent highways, and so on. TGI can automatically be recorded.

In addition to the scale and kind of information gathered with computer technology, there is yet another element - copy and distribution. Now information can go anywhere in the world through communication media. Hence, the extent to which information can be exchanged is now practically limitless. Once information about an individual is recorded in a machine or on a disk, it can be easily transferred to another machine or disk. It can he bought and sold, given away, traded, and even stolen. The information can spread instantaneously from one company to another or, one sector to another, and from one country to another.

Computers and Privacy Issues

Uses of Information

The computers and privacy issue is often framed as an issue that calls for a balancing of the needs of those who use information about individuals (typically government agencies and private institutions) against the needs or rights of those individuals. Information is created, collected and exchanged because organizations can use it to their interests and activities. Information about individuals is used to make decisions about those individuals, and often the decisions profoundly affect the lives of those individuals whom the information is about.

In general, those who want information about individuals want it because they believe that it will help them to make better decisions. For instance, banks believe that the more information they have about an individual, the better they will be able to make judgments about that individual's ability to pay back a loan or about the size of the credit line the individual can handle.

Personal privacy is generally put on the other side of the balancing scales. The issue is framed so that we have to balance all the good things that are achieved through information gathering and exchange against the desire or need for personal privacy. Some even claim that we have a right to personal privacy for if that were true, the scales would weigh heavily on the side of personal privacy. From a legal and constitutional point of view, however, we have, at most, a limited and complex right to privacy.

This framing of the issue seems to be skewed heavily in favor of information gathering and exchange. The only way to counter the powerful case made on behalf of information gathering and exchange is, it would seem, to make a more powerful case for protecting and ensuring privacy in the lives of individuals. Either we must show that there is a grave risk or danger to these information-gathering activities - a danger so great that it counterbalances the benefits of the activity. Or we must show that there is a greater benefit to be gained from constraining these activities. To put this another way, once the benefits of information gathering and exchange are on the table, the burden of proof is on privacy advocates to show either that there is something harmful about information gathering and exchange or that there is some benefit to be gained from constraining information gathering, Either way, there is a daunting hurdle to overcome.

Many of us feel uncomfortable with the amount of information that is gathered about us. We do not like not knowing who has what information about us and how it is being used. Why are we so uncomfortable? What do we fear? Part of the fear is, no doubt, related to our mistrust of large, faceless organizations, and part of it is related to mistrust of government. The challenge is to translate thus discomfort and fear into all argument that counterbalances the benefits of information gathering.

Personal Privacy

Two big questions have dominated the philosophical literature on privacy:

1. What is it?
2. Why is it valuable?

The term privacy seems to be used to refer to wide range of social practices and domains. For example, what we do in the privacy of our own homes, domains of life in which the government should not interfere, things about ourselves that we tell only our closest friends. Privacy seems, also, to overlap other concepts such as freedom or liberty, seclusion, autonomy, secrecy, controlling information about ourselves. So, privacy is a complex and, in many respects, elusive concept.

As we review several of these, it will be helpful to keep in mind a distinction between privacy as an **instrumental good** and privacy as an **intrinsic good**. When privacy is presented being valuable because it leads to something else, then it is cast as an instrumental good. In such arguments, privacy is presented as a means to an end. Its value lies in its connection to something else. On the other hand, when privacy is presented as good in itself, it is presented as a value in and of itself. As you might predict, the latter argument is harder to make for it requires showing that privacy has value even when it leads to nothing else or even when it may lead to negative consequences.

The arguments on behalf of privacy as an instrumental good begin to cross into privacy as an intrinsic good when they suggest a connection between privacy and autonomy. If privacy were essential to autonomy, then the loss of privacy would be a threat to our most fundamental values. The connection between privacy and autonomy is often presented not exactly as a means-ends relationship. Rather the suggestion is that autonomy is inconceivable without privacy.

Information Mediates Relationships

People need to control information about themselves in order to maintain a diversity of relationships. The insight is that individuals maintain a variety of relationships (e.g., with parents, spouses employers, friends, casual acquaintances, and so on) and each of these relationships is different because of the different information that each party has. These diverse relationships are a function of differing information.

Individual-Individual Relationships

If everything were open to all (that is, if everyone knew the same things about you), then diversity would not be possible. You would have similar relationships with everyone. We control relationships by controlling the information that others have about us. When we lose control over information, we lose significant control over how others perceive and treat us. However, the information gathering and exchange that goes on via computer technology does not seem, on the face of it, to threaten the diversity of personal relationships each of us has. For example, despite the fact that huge quantities of data now exist about purchases, phone calls, medical condition, work history, and so on, one is able to maintain a diversity of personal relationships.

Individual-Organization Relationships

Private and public organizations are powerful actors in the everyday lives of most individuals in our society, and yet it would seem that individuals have very little power in those relationships. One major factor making this possible is that these organizations can acquire, use, and exchange information about us, without our knowledge or consent.

Individuals control what relationships they have and the nature of those relationships by controlling the flow of information to others; when individual have no control over information that others have about them, there is a significant reduction in their autonomy.

Privacy as a Social Good

Priscilla M. Regan, in her book, *Legislating Privacy*, examined three privacy policy debates that took place in the United States in recent years—

- information privacy,
- communications privacy, and
- psychological privacy.

She concludes that while an individual privacy is pitted against social goods such as law enforcement or government efficiency, personal privacy loses. Regan suggests that privacy should be seen not as an individual good but rather as a social good. As an important social good, privacy would be on par with other social goods such as law enforcement or government efficiency. Instead of a social good outweighing an individual good, it would be clear that we have two social goods at stake. In reframing the issue in this way, privacy would be much likely to be treated as equally important, if not more important, than other social goods.

Privacy Protection

Here are a few proposals in favour of the privacy protection.

Broad Conceptual Changes and Legislative Initiatives

One is to think of privacy as a social good at the heart of liberal democratic societies and hence to be given much more weight in balancing of social goods. To make this shift is to move from seeing privacy as simply something individuals want for their personal protection, but not worth the cost in terms of inefficiency to recognizing privacy's- role in democracy and, hence, recognizing that it is worth what it may cost in terms of less efficient institutions,

The United States has taken a piecemeal, ad hoc approach to information privacy with each sector being dealt with separately. Significant improvement could be had by taking a comprehensive approach and developing legislation that lays out the parameters for public anti private information gathering. Such legislation should be made with an eye to global exchange of information.

Legislations of privacy can be made in the constitution. The first one is **the freedom of speech and the press**, while the second will proscribe **unreasonable search and seizure**, and insults security in person, houses, papers, and effects. These two deal with the relationship between the government and the press, and the government and the individual.

Computer Professionals

Computer professionals can play an important role, individually and collectively. First and foremost, individual professionals must not wash their hands of privacy issues, For example, a computer professional can point out privacy matters to clients or employers when building databases containing sensitive information. Whether or not computer professionals should refuse to build systems which they judge to be insecure is a tough question, but certainly one that ought to be considered an appropriate question for a 'professional.'

Individually and collectively, computer professionals can inform the public and public policy makers about privacy/security issues, and they can take positions on privacy legislation as it pertains to electronic records. computer professionals are often in the best position because of their technical expertise, to evaluate the security of databases, and the potential uses and abuses of information.

The original ACM Code of Professional Conduct is also guided by privacy protection concepts. One if the General Moral Imperatives of the 1992 ACM Code of Ethics and Professional Conduct is that an ACM member will 'Respect the privacy of others." The Guidelines explain that: "It is the responsibility of professionals to maintain the privacy and integrity of data describing individuals. This includes taking precautions to ensure the accuracy of data, as well as protecting it from unauthorized access or accidental disclosure to inappropriate individuals."

Technology

Related to the contribution of computer professionals is the potential of technology to protect rather than erode privacy. Privacy enhancing technologies (sometimes referred to as PETs) are now being developed. Computer scientists and engineers have been working on and have developed IT tools that allow one to navigate the Web with anonymity; to send e-mail anonymously through anonymous remailers; or to detect the privacy level of Web sites before one accesses them. More generally cryptographic techniques are being developed with various schemes that could allow one to make various transactions such as banking, purchasing, and so on, with confidentiality or to authenticate mail of various kinds.

Institutional Policies

Where no law applies or the law is unclear, private and public organizations can do a great deal to protect privacy by adopting internal policies with regard to the handling of personal information. Computer professionals working in such organizations can recommend and support such policies. For example, banks, insurance companies, registrars' offices of universities, marketing agencies, and credit agencies should have rules for employees dealing with personal information. They ought to impose sanctions against those who fail to comply. It is not uncommon now to hear of employees who casually reveal interesting information about individuals that they discover while handling their records at work.

Personal Actions

It will not he easy, and may be quite costly, for individuals to achieve a significant degree of personal privacy in our society. Gary Marx (1991) has provided a list of steps that individuals can take. His list includes the following:

(1) Don't give out any more information than is necessary;

(2) don't say things over a cellular or cord less phone that you would mind having overheard by strangers;

(3) ask your bank to sign an agreement that it will not release information about your accounts to anyone lacking legal authorization and that in the event of legal authorization, it will contact you within two days;

(4) obtain copies of your credit, health, and other records and check for accuracy and currency;

(5) if you are refused credit, a job, a loan, or an apartment ask why (there may be a file with inaccurate, incomplete, irrelevant information);

(6) remember that when you respond to telephone or door- to-door surveys, the information will go into a databank; and

(7) realize that when you purchase a product or service and file a warranty card or participate in a rebate program, your name may well be sold to a mailing-list company.

Source:

- Deborah G. Johnson, *Computer Ethics*, Third Edition

Case Discussions

Case Discussion: Internet in Schools

Over the past decade, the number of schools with Internet access has grown exponentially, and the number of children going online from school has followed suit. In 1999, the National Center for Education Statistics reported, 95 percent of public schools were connected to the Internet, and, more importantly, 63 percent of public school classrooms had an Internet connection.

Now, some 14 million children access the Internet from school, a figure that is expected to grow to more than 30 million by 2003 as schools continue to build their networks. By that year, it is anticipated that more students will access the Internet from their classrooms than the number who will access it from home

Concern

The Internet truly is like "a vast library including millions of readily available and indexed publications," containing content "as diverse as human thought," as the U.S. Supreme Court described it in a 1997 decision. **But just as in any city frequented by millions of people, there are neighborhoods on the Internet that are inappropriate for children to visit alone and strangers they would be better off not meeting.**

Throughout the past decade, policy makers, industry advocates, parents and teachers have tried to address these concerns.

Meanwhile, a number of initiatives have been launched that were designed to help educate adults about how to protect children when they go online. These projects and tools have included "Child Safety on the Information Highway," published by the National Center for Missing and Exploited Children and the Interactive Services Association in 1994, Project OPEN (the Online Public Education Network), organized by the ISA and the National Consumers' League in 1996, the Direct Marketing Association's "Get CyberSavvy" program in 1997, the American Library Association's "KidsConnect," as well as "America Links Up" and "Get Net Wise," both sponsored by a broad coalition of industry and non-profit groups. All of these initiatives have made useful contributions that educators and parents can continue to use to teach children "the rules of the road" when they go online.

What Is the Source of the Concern?

In the early 1990s, as the number of subscribers to proprietary online systems grew, transforming those communities from the moral equivalent of small towns to large cities or even states, children began to meet people in online chat rooms who would engage in inappropriate conversations or encourage them to divulge information about themselves. Then as proprietary systems such as America Online, Prodigy and CompuServe connected their users to the Internet, and more households accessed the Internet directly through Internet service providers, online users began moving easily to the wide open, unregulated spaces of the World Wide Web.

As the number of Internet users grew, the Internet changed as well. No longer was it a small, tightly knit community of academics, researchers and scientists, where all users generally supported an unwritten code of good behavior. With the advent of the World Wide Web, anyone could open up a store or publish a magazine. Some experts argue that the Web is expanding so fast that it is virtually impossible to track every site that could be objectionable.

A related debate rages over what percentage of Web sites would truly be considered objectionable. Some advocates argue that sites that would be considered harmful to minors represent only a very small proportion of the Web. What is of greater concern, they say, is that perfectly benign and possibly very useful information could be blocked when software is used to screen inappropriate material.

For others, however, the actual extent to which adult-oriented materials are available on the Internet is irrelevant. Because it is possible for children to access such materials without the traditional-world protections of brown-paper wrappers or adult-bookstore doorways, they argue, the existence of any pornographic material is sufficient cause for concern. Those who support government-mandated content controls tend to argue that any amount of inappropriate content is too much, when children are concerned

Pornography is not the only issue involved. Many adults are concerned about Web sites that are created by hate groups or devoted to topics such as bomb-making and weaponry, gambling or alcohol and smoking. Although pornography on the Internet has captured the greatest attention on the part of policy-makers, it is not the only area of potential concern for parents and educators.

How Big Is the Problem?

Data

According to some estimates, the World Wide Web now includes 1.5 billion pages and new Web sites spring up at the rate of 4,400 per day. Cyveillance, a company that monitors the Internet for business clients, estimated in July 2000 that the Web included 2.1 billion pages, with 7 million new pages created each day. Many Web sites are abandoned, or rarely updated, but continue to persist on servers around the world, where anyone in the world can still access them.

In 1995, a study conducted by Marty Rimm, an engineering student at Carnegie-Mellon University, asserted that 83.5 percent of the pictures transmitted through UseNet newsgroups were sexually explicit. His findings were given additional credence when Time magazine featured the study in a cover story headlined "On a Screen Near You: Cyberporn!" Subsequently, the study was discredited and Time backed down from its story. But average Americans, not to mention members of Congress, were left with the perception that pornography could be easily found on the Internet, just a mouse click away. Advocacy groups from the conservative end of the political spectrum contend that there are between 72,000 and 100,000 sexually explicit sites on the Internet and that 85 percent of the 3,900 new sites that are created each day, sell commercial pornography.

Yet another study, by researchers at the University of Pennsylvania's Annenberg School for Communications, reviewed a random selection of Web pages turned up by a search engine and found that no more than 3.6 percent had "highly objectionable" material, only 2.4 percent had "provocative sexual content" and only 0.7 percent had "notable violent content."

The World Wide Web, however, is not the only source of concern. Children can receive e-mail messages with pornographic file attachments and e-mail from UsetNet groups, which communicate through an older Internet protocol, can contain postings from users that would be considered inappropriate. Of special concern, too, are Internet chat rooms and so-called Instant Messaging, where children can communicate online in real-time with adult strangers who may not have their best interests at heart.

How Concerned Are Parents?

If children apparently *are* accessing materials that could be considered inappropriate, how concerned are their parents?

America Online officials say that about two-thirds of their user households with children make use of the online service's parental controls features. However, the National Center for Missing and Exploited Children found in its survey that only about one-third of the families used filtering or blocking software, including software offered by an Internet service provider. A survey conducted by the Annenberg Public Policy Center found a similar percentage of families that used filters, despite the fact that 76 percent of those parents said they were concerned that their children might view sexually explicit images on the Internet. (Interestingly, a greater percentage of parents (82 percent) were worried about what their children might encounter online when their families did not have internet access

Another recent survey of the general public found that while 71 percent believed the Internet could enhance their educational level and 86 percent said it would help their children learn more, there were still concerns about the kind of content that could be accessed. Seventy-six percent said they thought that "inappropriate content" could be a barrier to Internet adoption and 61 percent were worried about the potential impact of "dangerous ideas."

<u>The conclusion that some researchers have drawn from these surveys is that parents believe that the Internet can be an unsafe place, but are not particularly worried that their own children are getting into trouble. The children, it appears, are not always talking about what they are finding when they go online.</u>

The Legal and Regulatory Backdrop

Despite the development of new software products, rating systems, and industry-supported online safety campaigns, Congress continued to regard the situation with alarm. In 1996, it passed the Communications Decency Act, which prohibited the posting of materials on the World Wide Web that would be considered "indecent" or "patently offensive." The legislation did not address the dissemination of child pornography or child stalking on the Internet, which, it was generally agreed, would be prohibited by existing laws. However, in 1997, in a decision known as *Reno v. ACLU* (or commonly as *Reno I*), the Supreme Court unanimously ruled that the measure was much too broad and amounted to an unconstitutional restriction on free speech.

Congress tried again the next year, passing the Child Online Protection Act, which made it a crime to communicate through the Web information that would be considered "harmful to minors" unless access was restricted through the use of a credit card. Violators could be subject to criminal penalties of up to $50,000 a day. In February 1999, a federal district judge blocked enforcement of the act, a decision that was supported by the U.S. Court of Appeals for the 3rd Circuit in a June 2000 decision known as *ACLU v. Reno III*. Once again, the judge ruled that the measure amounted to an impermissible restriction on free speech. Critics of the law noted that the legislation would have had no bearing on Web sites hosted on servers outside the United States, or on other kinds of Internet communications such as e-mail. In May 2001, as expected, the Supreme Court agreed to review the decision.

In December 2000, Congress passed the Children's Internet Protection Act and Neighborhood Internet Protection Act (commonly referred to as CIPA or "the CHIP Act") as amendments to the fiscal 2001 appropriations bill for the U.S. Department of Education. This measure will require schools that accept federal E-rate discounts to purchase Internet access or internal connections or that use funding under Title III of the Elementary and Secondary Education Act to purchase Internet access or computers that access the Internet to adopt an "Internet safety policy" and a technology protection measure that blocks or filters content that is obscene, child pornography or "harmful to minors."

An Opportunity for Learning

Some educators argue that instead of relying on filtering technology to block inappropriate content, schools should focus on teaching students how to evaluate Internet content and to form their own judgments on what is inappropriate. By relying on filtering software, they argue, schools may miss an opportunity to prepare students for what they are likely to encounter in an unfiltered environment, such as through their home computer, a friend's computer, a university network when they go off to college or a company's computer when they enter the workforce. Further, the rapidly evolving nature of the Internet virtually ensures that no filtering technology can be 100 percent perfect. Thus, even when a school uses a particular solution, children should be taught how to respond if they still manage to access something that it is inappropriate for them.

Further, the dangers that children can encounter online are not limited to off-color Web sites. Unsupervised chat rooms and so-called Instant Messaging functionality can enable adults to contact children, sometimes posing as children or offering enticements for further real-world contact. When schools begin accessing the Internet, with or without content controls, they should make sure that children are armed with the same kind of safety advice that parents and teachers would normally provide about the dangers of the traditional world: "don't talk with strangers and don't give out information about yourself."

Whether or not a school district decides to use a technological approach to manage content, it would be well advised to promote "information literacy," that is, teaching children how to find good sources of information online and how to evaluate online information, as well as how to protect themselves when they go online.

What the Future Holds

As with all aspects of technology, the tools and solutions for managing Internet content are continuing to evolve. But so too are the methods that unscrupulous Internet users and merchants can use to lure children to places and materials that could be considered inappropriate for minors.

So far, most of the focus of Internet filtering products—and government policy makers—has been on controlling access to obscene or pornographic Web sites. But that is only part of the content management problem. Children can receive e-mail with pornographic attachments. In addition, functions like chat and Instant Messaging, which children enjoy and which can promote interactivity, can also expose them to potentially dangerous adults, posing as children or sympathetic mentors, if the network is not restricted to children. Many products are beginning to address these problems and providing parents and educators with tools to limit children's access to them—or to provide greater assurances that the "friends" they meet in chat rooms are indeed children. Unsolicited e-mail, or "spam" as it is commonly called, continues to be a problem for adults as well as children, not to mention system operators.

Another source of concern is about a practice known as "mouse-trapping," in which an Internet user requests a certain site, but is routed to another one, often featuring adult-oriented content. What makes this practice particularly onerous is that the user finds that the only way he can leave the site is to shut down his browser. The

COPA Commission, for one, has recommended that government regulators try to fight this practice through existing consumer-protection laws aimed at deceptive advertising.

Questions to Ask When Deciding Whether to Manage Content

✓**How will students use the Internet?**

The planning process for every school technology initiative should address this question. Is it anticipated that students will be using the Internet unsupervised, as a research tool? Or will teachers be managing their experiences more closely? How many computers are in classrooms, labs and media centers and how will that impact the ability of school staff to closely monitor how students are using the Internet? Answering these questions will help determine whether students' Internet access should be supervised and whether staff will be in a position to do it on their own.

✓**Do you want students to be able to direct their own learning or is it more important for teachers to retain control of what goes on in the classroom?**

To what extent does your school or school district foster a classroom culture in which students are independent learners? Or are you more comfortable with a more structured, more formal classroom model? How your school or school district answers this question will provide guidance on how comfortable you will be with giving students unrestricted access to the Internet.

✓**Should different standards be applied, based on the age of the student?**

Your district may decide that different approaches are appropriate, depending, say, on whether a student is in high school or elementary school. If so, you will want to make sure that your proposed solution will give you that flexibility. You may decide to adopt content controls only in certain schools, or you may decide to choose a product that will let you set different levels of restriction for different age groups or classes.

✓**Should school employees be subject to the same rules as students, to their own set of rules or to no rules?**

Are you concerned about how your employees will use the school district's online resources? If so, what kind of rules or limits do you want to impose? Should staff members be required to follow the same rules that students do, or is it more appropriate to adopt a separate policy for adults? Should you distinguish between the kind of online activities they pursue during school hours and those that they pursue outside of school hours?

✓**Would you prefer to simply monitor how students and employees use the Internet, rather than blocking their access to sites?** Would this approach raise any privacy concerns? Will your staff have time to monitor these logs and respond to potential abuses?

✓**Are there other issues that you want to address at the same time?**

Some content management solutions address other concerns about the Internet or network operations. These include protecting a school network against hackers or viruses, protecting the privacy of students, and restricting children's exposure to advertising messages. If so, you may wish to evaluate whether a certain approach will provide a cost-effective solution to more than one problem.

✓**How will school officials respond if students are found to be accessing inappropriate material?**

This issue should be addressed in your Acceptable Use Policy. Students should know how they are expected to respond if they access a clearly inappropriate site, whether or not it was intentional.

✓**What strategies will your school district use to teach "information literacy?"**

No matter what approach a school district takes, it should ensure that its students understand the "rules of the road" when they go online and how to evaluate the content that they find there. These lessons can be imparted either as part of regular online classwork, or as special activities that must be completed before a student can go online.

Questions to Consider When Evaluating Content Management Products

✓ **Who should make the decision on what kind of sites are blocked or accessed?**

If school personnel will make the decision on which sites students will be allowed to access, will they have enough time to devote to that task? How frequently will they be able to update their lists? Who will be responsible for updating the list of approved sites? Will that give students access to a wide enough variety of sites?

If a third party will make the decision on which sites will be blocked or accessed, do you understand the criteria it uses to evaluate Web sites? Does the organization or company have any particular bias? How easy will it be for school personnel to override those decisions if they disagree with them? How frequently does the organization or company update its lists of sites, and how easily is that update accomplished?

✓ **What kinds of content are you concerned about?**

Are you primarily concerned about children's access to pornography and obscenity, or are you concerned about their access to materials on topics such as weapons, hate groups and alternative lifestyles?

✓ **What has the experience been with the solution you propose to use?**

To what extent are children able to access inappropriate sites, either directly or through search engines and other links? If a product appears to be effective in blocking problem sites, does it go too far in also blocking sites that would be considered benign or needed by a class studying a sensitive topic? A team of staff members may want to test proposed solutions to see what kind of results they turn up. Research papers and testimony compiled by the COPA Commission provides information about the methods that have been used by other researchers to test the effectiveness of blocking software.

✓ **How are users notified when they try to access a blocked site?**

Some products provide a clear notice when a site has been blocked. Some allow this message to be tailored to the network's needs. Some products block in a more invisible way. Is it important for your Internet users, both students and staff, to know if they have tried to access sites that were blocked? Or would you prefer that this information be withheld?

✓ **Does the proposed solution address other forms of content besides just Web sites?**

Does it provide tools for controlling such things as e-mail, access to chat rooms, and Instant Messaging? Is it important for your solution to include that kind of functionality?

✓ **How easy would it be for a child to hack into and disable a particular filtering solution?**

Generally, tools are more difficult to disable when installed on a server, whether it belongs to a school, an Internet service provider or a filtering company, than they are when installed on a desktop computer.

✓ **Does the proposed solution incorporate advertising messages? Will third parties be able to collect information about how your students are accessing the Internet?**

Some products incorporate these features, sometimes in exchange for reduced fees, or no fees at all for the product or service. School officials should understand whether advertising or marketing messages are incorporated into a product and what information, if any, will be gathered about users, either individually or in aggregate. Sites that gather information about children are now subject to the Children's Online Privacy Protection Act and the role schools are expected to play in the administration of this law is still under discussion.

✓ **If your students speak many different languages, does your proposed solution control access to sites written in languages other than English?**

Some children learn to subvert content controls by making their requests in a foreign language. Further, if a child's native language is something other than English, will he receive the same level of protection that a child typing in Web site names in English would?

✓ **How will the proposed solution serve your district in the future?**

Will the solution still work as the number of Internet-accessible computers grows? How will that change the price? What future enhancements are on the drawing boards? If your district plans to let children access the Internet through other kinds of devices, how will you extend the controls to those products?

Case Discussion: Computer Crime

USPA & IRA Company

A former programmer in Fort Worth, Texas, has been convicted of planting a computer virus in his employer's system that wiped out 168,000 records and was activated like a time bomb, doing its damage two days after he was fired. Tarrant County Assistant District Attorney Davis McCown said he believes he is the first prosecutor in the country to convict someone for destroying computer records using a virus. Donald Gene Burleson, 40, was convicted of Charges of harmful access to a computer, a third-degree felony that carries up to 10 years in prison and up to $5,000 in fines. Burleson planted the virus in revenge for its firing from an insurance company, McCown said.

Jurors were told during a technical and sometimes complicated three-week trial that Burleson planted a rogue program in the computer system used to store records at USPA and IRA Co., a Fort Worth-based insurance and brokerage firm. The virus, McCown said, was activated two days after Burleson had been fired because of personality conflicts with other employees. "There were a series of programs built into the system as early as Labor Day," McCown said. Once he got fired, those programs went off. The virus was discovered two days later, after it had eliminated 168,000 payroll records, holding up company paychecks for more than a month.

Questions for discussion:

a. Why is "Donald Burleson virus attack" a crime? List the harms it did to the person, the employer and the society.
b. What social issues, in addition to crime, can be related to the case? And how?

   (Employment, Social Application, Privacy, Competition, Quality of Life etc.)
c. Are there any individual issues, which caused the crime to occur?

   (Ergonomics, Skill and Knowledge, Autonomy and Power, Scope of Work, Involvement and Commitment etc.)
d. What are ethical and legal issues involved with it?
e. Suggest the way the crime could have been stopped.

   *(**Note:** the questions may be distributed among groups for discussion.)*

Issues and Discussions:

Summary of discussions (an example)

It is a crime because

- it was a deed that is against the law
- it seized the employees' right to get salary for a month
- it resulted in the loss of data and information (it wiped out 168,000 records)
- it had made malicious access to corporate data

It did harm the person individually in the following ways:

- Burleson, the programmer, had to pay $5,000 as a fine and 10 years of imprisonment.
- He developed bad impression in the organization and the society
- There was negative impact in his career.
- He had to bear mental torture also

It did harm the employer in the following ways:

- It affected the Payroll management
- Extra efforts, time and resources needed to be paid in recovering from the disaster
- Misunderstanding among the employees could be created
- Security of the system of the organization was challenged.

It did harm to the society in the following ways:

- The dependents of the employees were affected for a month because they did not get paid
- Could have established a bad impression in the society and the credibility of the company could have become an issue
- Could be a source for committing such crime for others in the society too.

There seemed to have following Social and Individual aspects involved in the case:

- Personal conflict between the programmer and his employer (?)
- Work environment of the company
- Unhealthy competition, ego, revenge etc developed in the company
- Quality of life of the programmer
- Attitude of the person
- Skill and knowledge of Burleson
- Autonomy and Power given to him
- Scope of Work of the criminal
- Level of Involvement in the work and level of Commitment to the organization

There are some legal and ethical issues, too, in the case and they are:

- Burleson should not be fired just because of the personal conflict.
- He, the programmer should not take revenge to the whole organization
- Destroying information of the company is illegal and making employees payless is unethical.

Different attempts could have been carried out to prevent from such crimes well in advance or safety measures could have been taken to recover from the disaster such as:

- Physical and data security of the information
- Regular data backup
- Conflict resolution of the programmer and the employer
- Strict enforcement of rules and regulations of the company.

References for the discussion

What is a computer crime?

Computer crime is defined by the laws and the current legislations, which is different for different nations and also may change at different times.

The Data Processing Management Association **(DPMA)** defines computer crime more specifically. Its Model Computer Crime Act defines computer crime as

(1) the unauthorized use, access, modification, and destruction of hardware, software, or data resources,

(2) the unauthorized release of information,

(3) the unauthorized copying of software,

(4) denying an end user access to his or her own hardware, software or data resources,

(5) using or conspiring to use computer resources to commit a felony

(6) the *intent* to illegally obtain information or tangible property through the use of computers, and

(7) the *intent* to establish control for the purpose of unauthorized experimentation with computer resources.

Types of Crime (examples): -

- Money theft
- Computer viruses
- Service theft
- Program and data theft
- Program copying
- Data alteration
- Program damage
- Data destruction
- Malicious access
- Violation of Privacy
- Violation of International Law

Case Discussion: The Virtual Professor

Life History

"… In 1978 …"

"Like most people with real jobs, I had to travel a lot which meant that I was frequently out of town on the nights I was supposed to be teaching. So I had to develop some strategies for coping with this problem. One was to participate in the class via an audio conference. I would have someone set up a speakerphone in my classroom and call in from wherever I happened to be. I could give lectures and engage students in discussions. It worked quite well."

"Another strategy was to ask colleagues to "cover" for me by going to the class and giving a guest lecture. I would pick out people with expertise on the topic to be taught so their contribution to the class was usually very worthwhile and enjoyed by the students. I found that having 5 or 6 guest lectures in a course made it much more interesting to the students than being taught by a single person."

"The third strategy involved the use of computer bulletin board systems to provide a way for students to contact me and each other via email and online conferences. Not only was this useful for me, but it was helpful to the students as well since many of them also traveled a lot and this gave them a way to keep up to date with course work."

Questions:

a) Was the Professor or the classroom virtual? Or the students? Give your opinion.

b) How would you help the Professor if he had the same problem now? What about online education?

c) If you implement the concept of the information superhighway in this case, what would be the implications for the professor and the students?

Case Discussion: Internet Use

Around 2000, The Atlantic Monthly and other publications revealed a distinct issue: The Internet allows people who exhibit or wish to practice abnormal behavior to find one another easily, due to anonymous search engines and online forums or services. As sparse subpopulations, it was often unlikely or difficult to find willing partners or like-minded individuals prior to the Internet.

A small number of these subcultures promote self-destructive or mutually destructive behavior. Websites and mailing lists exist that explicitly promote anorexia, apotemnophilia, necrophilia, and suicide. While these activities are easily recognized as abnormal and self-destructive by most adults, many people fear that children or mentally ill persons visiting such sites would lack the maturity necessary to make that discrimination.

In rare cases, people have used the Internet to find willing partners for abnormal activities, but with disastrous or fatal results. In one case, a German named Armin Meiwes (the "German cannibal'') made an online arrangement with Bernd Jurgen Armando Brandes to kill and eat him. Meiwes was later convicted of manslaughter.

Questions

1. What do you conclude from the above passage?
2. Do you think this kind of issue can be highly sensitive for minors? How?
3. What are the possible solutions that you see to the above problem?
4. What are some of the major impacts of Internet you have seen in Nepal?

Case Discussion: Philosophical Ethics

You are a German citizen working for a large computer company. You are in charge of the sales division and you personally handle all large orders. You are contacted by representatives of the German government. The German government has not yet fully automated its operations (computers are still relatively new) and it wants now to purchase several large computers and several hundred smaller computers to be networked.

You read the newspapers and know how the war is proceeding so you have a pretty good idea of how the German government will use the computers. It is quite likely they will use the computers to help keep track of their troops and equipment, to identify Jews and monitor their activities, to build more efficient gas chambers, and so on.

Questions

1. Would it be permissible for you to sell the computers to Hitler and his government? Explain in context to the above passage
2. What do you think being a Relativist in the above scenario?
3. Does utilitarian theory and deontological theory will help you to decide in selling computers to Hitler and its government? Explain
4. What is an ethical argument? What is the argument raised in the above passage? Expalin

Case Discussion: Control over the Internet

Chinese web users are being denied access to a range of religious sites based abroad as the Chinese government's "Golden Shield" firewall is used to censor the Internet. While blocking undesirable sites promoting pornography and violence, the firewall also limits religious web content such as sites related to the Dalai Lama, the Falun Gong cult and various Buddhist and Muslim movements. Also blocked are sites covering persecution of religious communities in China and a number of Catholic sites, including the website of the Hong Kong diocese and the Divine Word Missionaries in Taiwan. Not blocked are sites in European languages covering religious freedom issues, even those covering repression within China. While overall Internet usage in China may be low by developed country standards, it has been rapidly growing, especially in the capital of Beijing and the coastal region. The official China Internet Network Information Center puts the number of Chinese with access to the Internet in June 2003 at 79.5 million

This number has been doubling every six months. Internet censorship is part of a comprehensive attempt to censor all means of communication. While printed publications have long been censored in China, authorities also have tried to keep up with technological developments. The Global Internet Policy Initiative warned in June of new technology from a Chinese firm that monitors "subversive" text messages sent by mobile phone. (Forum 18 News Service)

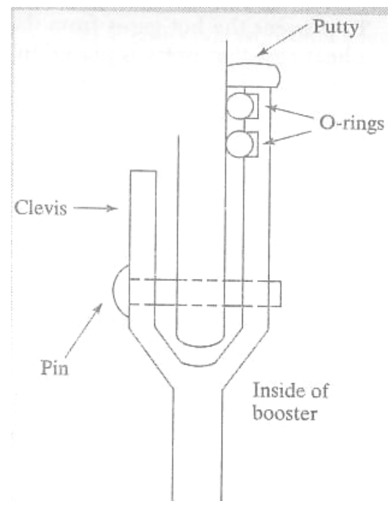
Questions

1. **What do you conclude from the above passage?**
2. **Do you think is it right to exercise control over internet? Justify**
3. **What is the scenario of Nepalese Government in context to the situation described above?**
4. **Write down some of the commercial implications of Internet in Nepal.**

Case Discussion: The Space Shuttle Challenger Accident

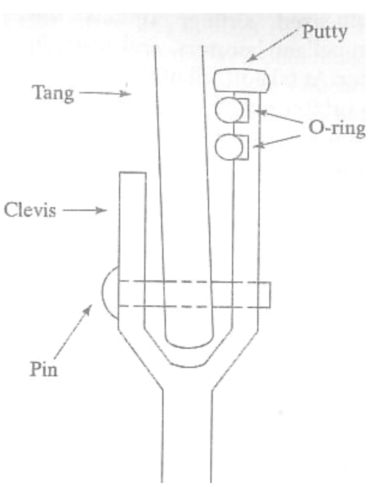
BACKGROUND

The space shuttle was designed to be a reusable launch vehicle. The vehicle consists of an orbiter, which looks much like a medium-sized airliner (minus the engines!), two solid-propellant boosters, and a single liquid-propellant booster. At takeoff, all of the boosters are ignited and lift the orbiter out of the earth's atmosphere. The solid rocket boosters are only used early in the flight and are jettisoned soon after takeoff, parachute back to earth, and are recovered from the ocean. They are subsequently repacked with fuel and are reused. The liquid-propellant booster is used to finish lifting the shuttle into orbit, at which point the booster is jettisoned and burns up during reentry. The liquid booster is the only part of the shuttle vehicle that is not reusable. After completion of the mission, the orbiter uses its limited thrust capabilities to reenter the atmosphere and glides to a landing.



The accident on January 28, 1986 was blamed on a failure of one of the solid rocket boosters. Solid rocket boosters have the advantage that they deliver far more thrust per pound of fuel than do their liquid-fueled counterparts, but have the disadvantage that once the fuel is lit, there is no way to turn the booster off or even to control the amount of thrust produced. In contrast, a liquid-fuel rocket can be controlled by throttling the supply of fuel to the combustion chamber or can be shut off by stopping the flow of fuel entirely.

In 1974, the National Aeronautics and Space Administration (NASA) awarded the contract to design and build the solid rocket boosters for the shuttle to Morton Thiokol. The design that was submitted by Thiokol was a scaled-up version of the Titan missile, which had been used successfully for many years to launch satellites. This design was accepted by NASA in 1976. The solid rocket consists of several cylindrical pieces that are filled with solid propellant and stacked one on top of the other to form the completed booster. The assembly of the propellant-filled cylinders was performed at Thiokol's plant in Utah. The cylinders were then shipped to the Kennedy Space Center in Florida for assembly into a completed booster.



A key aspect of the booster design are the joints where the individual cylinders come together, known as the field joints, illustrated schematically in Figure 1.la. These are tang and clevis joints, fastened with 177 clevis pins. The joints are sealed by two O-rings, a primary and a secondary. The O-rings are designed to prevent hot gases from the combustion of the solid propellant from escaping. The O-rings are made from a type of synthetic rubber and so are not particularly heat resistant.

To prevent the hot gases from damaging the O-rings, a heat-resistant putty is placed in the joint. The Titan booster had only one O-ring in the field joint. The second O-ring was added to the booster for the shuttle to provide an extra margin of safety since, unlike the Titan, this booster would he used for a manned space craft.

EARLY PROBLEMS WITH THE SOLID ROCKET BOOSTERS

Problems with the field-joint design had been recognized long before the launch of the Challenger. When the rocket is ignited, the internal pressure causes the booster wall to expand outward, putting pressure on the field joint. This pressure causes the joint to open slightly; a process called "joint rotation,' illustrated in Figure 1.lb. The joint was designed so that the internal pressure pushes on the putty, displacing the primary O-ring into this gap, helping to seal it. During testing of the boosters in 1977, Thiokol became aware that this joint-rotation problem was more severe than on the Titan and discussed it with NASA, Design changes were made, including an increase in the thickness of the O-ring, to try to control this problem.

Further testing revealed problems with the secondary seal, and more changes were initiated to correct that problem. In November of 1981, after the second shuttle flight, a post-launch examination of the booster field joints indicated that the O-rings were being eroded by hot gases during the launch. Although there was no failure of the joint, there was some concern about this situation, and Thiokol looked into the use of different types of putty and alternative methods for applying it to solve the problem. Despite these efforts, approximately half of the shuttle flights before the Challenger accident had experienced some degree of O-ring erosion. Of course, this type of testing and redesign is not unusual in engineering. Seldom do things work correctly the first time, and modifications to the original design are often required.

It should be pointed out that erosion of the O-rings is not necessarily a bad thing. Since the solid rocket boosters are only used for the first few minutes of the flight, it might be perfectly acceptable to design a joint in which O-rings erode in a controlled manner. As long as the O-rings don't completely burn through before the solid boosters run out of fuel and are jettisoned, this design should be fine. However, this was not the way the space shuttle was designed, and O-ring erosion was one of the problems that the Thiokol engineers were addressing.

The first documented joint failure came after the launch on January 24, 1985, which occurred during very cold weather. The post-flight examination of the boosters revealed black soot and grease on the outside of the booster, which indicated that hot gases from the booster had blown by the O-ring seals. This observation gave rise to concern about the resiliency of the O-ring materials at reduced temperatures Thiokol performed tests of the ability of the O-rings to compress to fill the joints and found that they were inadequate. In July of 1985, Thiokol engineers redesigned the field joints without O-rings. Instead, they used steel billets, which should have been better able to withstand the hot gases. Unfortunately, the new design was not ready in time for the Challenger flight in early 1986.

THE POLITICAL CLIMATE

To fully understand and analyze the decision making that took place leading to the fatal launch, it is important also to discuss the political environment under which NASA was operating at the time, NASA'S budget was determined by Congress, which was becoming increasingly unhappy with delays in the shuttle project and shuttle performance, which wasn't meeting initial promises. NASA had billed the shuttle as a reliable, inexpensive launch vehicle for a variety of scientific and commercial purposes, including the launching of commercial and military satellites. It had been promised that the shuttle would be capable of frequent flights (several per year) and quick turnarounds and would be competitively priced with more traditional non-reusable launch vehicles. NASA was feeling some urgency in the program because the European Space Agency was developing what seemed to be a cheaper alternative to the shuttle, which could potentially put the shuttle out of business.

These pressures led NASA to schedule a record number of missions for 1986 to prove to Congress that the program was on track. Launching a mission was especially important in January 1986, since the previous mission had been delayed numerous times by both weather and mechanical failures. NASA also felt pressure to get the Challenger launched on time so that the next shuttle launch, which was to carry a probe to examine Halley's Comet, would be launched before a Russian probe designed to do the same thing. There was additional political pressure to launch the Challenger before the upcoming state-of-the-union address, in which President Reagan hoped to mention the shuttle and a special astronaut—the first teacher in space, Christa McAuliffe - in the context of his comments on education.

THE DAYS BEFORE THE LAUNCH

Even before the accident, the Challenger Launch didn't go off without a hitch, as NASA had hoped. The first launch date had to be abandoned due to a cold front expected to move through the area. The front stalled and the launch could have taken place on schedule. But the launch had already been postponed in deference to Vice President George Bush, who was to attend. NASA did not want to antagonize Bush, a strong NASA supporter, by postponing the launch due to inclement weather after he had arrived. Launch of the shuttle was further delayed by a defective micro switch in the hatch-locking mechanism. When this problem was resolved, the front had changed course and was now moving through the area. The front was expected to bring extremely cold weather to the launch site, with temperatures predicted to be in the low 20's (ºF) by the new launch time

Given the expected cold temperatures, NASA checked with all of the shuttle contractors to determine if they foresee any problems with launching the shuttle in cold temperatures. Alan McDonald, the director of Thiokol's Solid Rocket Motor Project, was concerned about the cold weather problems that had been experienced with the solid rocket boosters. The evening before the scheduled launch, a teleconference was arranged between engineers and management from the Kennedy Space Center, NASA'S Marshall Space Flight Center in Huntsville, Alabama, and Thiokol in Utah to discuss the possible effects of cold temperatures on the performance of the solid rocket boosters. During this teleconference, Roger Boisjoly and Arnie Thompson, two Thiokol engineers, who had worked on the solid-propellant booster design, gave an hour-long presentation on how the cold weather would increase the problems of joint rotation and sealing of the joint by the O-rings.

Their point was that the lowest temperature at which the shuttle had previously been launched was 53°F, on January 24, 1985, when there was blow-by of the O-rings. The O-ring temperature at Challenger's expected launch time the following morning was predicted to be 29°F, far below the temperature at which NASA had previous experience. After the engineer's presentation, Bob Lund, the vice president for engineering at Morton Thiokol, presented his recommendations. He reasoned that since there had previously been severe O-ring erosion at 53°F and the launch would take place at significantly below this temperature where no data and no experience were available. NASA should delay the launch until the O-ring temperature could be at least 53°F. Interestingly, in the original design, it was specified that the booster should operate properly down to an outside temperature of 31°F.

Larry Mulloy, the Solid Rocket Booster project manager at Marshall and a NASA employee, correctly pointed out that the data were inconclusive and disagreed with the Thiokol engineers. After some discussion, Mulloy asked Joe Kilminster, an engineering manager working on the project, for his opinion. Kilminster backed up the recommendation of his fellow engineers. Others from Marshall expressed their disagreement with the Thiokol engineer's recommendation, which prompted Kilminster to ask to take the discussion off line for a few minutes. Boisjoly and other engineers reiterated to their management that the original decision not to launch was the correct one.

A key fact that ultimately swayed the decision was that in the available data, there seemed to be no correlation between temperature and the degree to which blow-by gasses had eroded the O-rings in previous launches. Thus, it could be concluded that there was really no trend in the data indicating that launch at the expected temperature would necessarily be unsafe. After much discussion, Jerald Mason, a senior manager with Thiokol, turned to Lund and said, "Take off your engineering hat and put on your management hat," a phrase that has become famous in engineering ethics discussions. Lund reversed his previous decision and recommended that the launch proceed. The new recommendation included an indication that there was a safety concern due to the cold weather, but that the data were inconclusive and the launch was recommended. McDonald, who was in Florida, was surprised by this recommendation and attempted to convince NASA to delay the launch, but to no avail.

THE LAUNCH

Contrary to the weather predictions, the overnight temperature was 8°F, colder than the shuttle had ever experienced before. In fact, there was a significant accumulation of ice on the launchpad from safety showers and fire hoses that had been left on to prevent the pipes from freezing. It has been estimated that the aft field joint of the right-hand booster was at 28°F.

NASA routinely documents as many aspects of launches as possible. One part of this monitoring is the extensive use of cameras focused on critical areas of the launch vehicle. One of these cameras, looking at the right booster, recorded puffs of smoke coming from the aft field joint immediately after the boosters were ignited. This smoke is thought to have been caused by the steel cylinder of this segment of the booster expanding outward and causing the field joint to rotate. But, due to the extremely cold temperature, the O-ring didn't seat properly. The heat-resistant putty was also so cold that it didn't protect the O-rings, and hot gases burned past both O-rings. It was later determined that this blow-by occurred over 70° of arc around the O-rings.

Very quickly, the field joint was sealed again by byproducts of the solid rocket-propellant combustion, which formed a glassy oxide on the joint. This oxide formation might have averted the disaster had it not been for a very strong wind shear that the shuttle encountered almost one minute into the flight. The oxides that were temporarily sealing the field joint were shattered by the stresses caused by the wind shear. The joint was now opened again, and hot gases escaped from the solid booster. Since the booster was attached to the large liquid-fuel booster, the flames from the solid-fuel booster blow-by quickly burned through the external tank. The liquid propellant was ignited and the shuttle exploded.

THE AFTERMATH

As a result of the explosion, the shuttle program was grounded as a thorough review of shuttle safety was conducted. Thiokol formed a failure-investigation team on January 31, 1986 which included Roger Boisjoly. There were also many investigations into the cause of the accident, both by the contractors involved (including Thiokol) and by various government bodies. As part of the governmental investigation, President Reagan appointed a blue-ribbon commission, known as the Rogers commission, after its chair. The commission consisted of distinguished scientists and engineers who were asked to look into the cause of the accident and to recommend changes in the shuttle program.

One of the commission members was Richard Feynman, a Nobel Prize winner in physics, who ably demonstrated to the country what had gone wrong. in a demonstration that was repeatedly shown on national news programs, he demonstrated the problem with the O-rings by taking a sample of the O-ring material and bending it, The flexibility of the material at room temperature was evident, He then immersed it in ice water. When Feynman again bent the O-ring, it was very clear that the resiliency of the material was severely reduced, a very clear demonstration of what happened to the O-rings on the cold launch date in Florida.

As part of the commission hearings, Boisjoly and other Thiokol engineers were asked to testify. Boisjoly handed over to the commission copies of internal Thiokol memos and reports detailing the design process and the problems that had already been encountered. Naturally, Thiokol was trying to put the best possible spin on the situation, and Boisjoly's actions hurt this effort. According to Boisjoly, after this action he was isolated within the company, his responsibilities for the redesign of the joint were taken away, and he was subtly harassed by Thiokol management.

Eventually; the atmosphere became intolerable for Boisjoly and he took extended sick leave from his position at Thiokol. The joint was redesigned, and the shuttle has since flown numerous successful missions. However, the ambitious launch schedule originally intended by NASA has never been met. It was reported in 2001 that NASA has spent $5 million to study the possibility of installing some type of escape system to protect the shuttle crew in the event of an accident. No decision has yet been made. Possibilities include ejection seats or an escape capsule that would work during the first three minutes of flight. These features were incorporated into earlier manned space vehicles and in fact were in place on the shuttle until 1982. Whether such a system would have saved the astronauts aboard the Challenger is unknown.

SPACE SHUTTLE CHALLENGER ACCIDENT: WHO'S WHO

ORGANIZATIONS

NASA	The National Aeronautics and Space Administration, responsible for space exploration. The space shuttle is one of NASA's programs.
Marshall Space Flight Center	A NASA facility that was In charge of the solid rocket booster development for the shuttle.
Morton Thiokol	A private company that won the contract from NASA for building the solid rocket boosters for the shuttle.

PEOPLE

NASA	
Larry Mulloy	Solid Rocket Booster Project manager at Marshall
Morton Thiokol	
Roger Boisjoly and Anile Johnson	Engineers who worked on the Solid Rocket Booster Development Program.
Joe Kilminster	Engineering manager on the Solid Rocket Booster Development Program
Alan McDonald	Director of the Solid Rocket Booster Project.
Bob Lund	Vice president for engineering
Jerald Mason	General manager.

ISSUES / Questions

1. The astronauts on the Challenger mission were aware of the dangerous nature of riding a complex machine such as the space shuttle, so they can be thought of as having given informed consent to participating in a dangerous enterprise. What role did informed consent play in this case? Do you think that the astronauts had enough information to give informed consent to launch the shuttle that day?
2. Can an engineer who has become a manager truly ever take off her engineer's hat? Should she?
3. Some say that the shuttle was really designed by Congress rather than NASA. What does this statement mean! What are the ramifications if this is true?
4. Aboard the shuttle for this flight was the first teacher in space. Should civilians be allowed in what is basically an experimental launch vehicle? At the time, many felt that the placement of a teacher on the shuttle was for purely political purposes. President Reagan was widely seen as doing nothing while the American educational system decayed. Cynics felt that the teacher-in-space idea was cooked up as a method of diverting attention from this problem and was to be seen as Reagan's doing something for education while he really wasn't doing anything. What are the ethical implications if this scenario is true?
5. Should a launch have been allowed when there was no test data for the expected conditions? Keep in mind that it is probably impossible to test for all possible operating conditions. More generally, should a product be released for even when It hasn't been tested over all expected operational conditions? When the data is inconclusive, which way should the decision go?
6. During the aftermath of the accident. Thiokol and NASA investigated possible causes of the explosion. Boisjoly accused Thiokol and NASA of intentionally downplaying the problems

with the O-rings while looking for other causes of the accident. If true, what are the ethical implications of this type of investigation?

7. It might be assumed that the management decision to launch was prompted in part by concerns for the health of the company and the space program as a whole. Given the political climate at the time of the launch, if problems and delays continued, ultimately Thiokol might have lost NASA contracts, or NASA budgets might have been severely reduced. Clearly, this scenario could have led to the loss of many jobs at Thiokol and NASA. How might these considerations ethically be factored into the decision?
8. Engineering codes of ethics require engineers to protect the safety and health of the public in the course of their duties. Do the astronauts count as the "public" in this context?
9. What should NASA management have done differently? What should Thiokol management have done differently?
10. What else could Boisjoly and the other engineers at Thiokol have done to prevent the launch from occurring?

Case Discussion: Conflict of Interest

Juan Roriguez makes a al a private consultant. Small businesses hire him to advise them about their computer needs. Typically, a company asks him to come in, examine the company's operations, evaluate their automation needs, and make recommendations about the kind of hardware and software that they should purchase.

Recently, Juan was hired by a small, private hospital. The hospital was interested in upgrading the hardware and software it uses for patient records and accounting. The hospital had already solicited proposals for upgrading their system. They hired Juan to evaluate the proposals they had received. Juan examined the proposals very carefully. He considered which system would best meet the hospital's needs, which company offered the best services in terms of training of staff and future updates, which offered the best price, and so on. He concluded that Tri-Star Systems was the best alternative for the hospital, and he recommended this in his report, explaining his reasons for drawing this conclusion.

What Juan failed to mention (at any time in his dealings with the hospital) was that he is a silent partner (a co-owner) in Tri-Star Systems.

Questions

1. Was Juan's behavior unethical? Was it a professional practice? Analyze.
2. Should Juan have disclosed the fact that he had ties to one of the companies that made a proposal? Give logic to your arguments.
3. Should he have declined the job once he learned that Tri-Star had made a bid? Why?

Case Discussion: The Intel Pentium Chip

In late 1994, the media began to report that there was a flaw in the new Pentium microprocessor produced by Intel. The microprocessor is the heart of a personal computer and controls all of the operations and calculations that take place. A flaw in the Pentium was especially significant, since it was the microprocessor used in 80% of the personal computers produced in the world at that time.

Apparently, flaws in a complicated integrated circuit such as the Pentium, which at the time contained over one million transistors, are common. However, most of the flaws are undetectable by the user and do not affect the operation of the computer. Many of the flaws are easily compensated for through software. The flaw that came to light in 1994 was different; it was detectable by the user. This particular flaw was in the floating-point unit (FPU) and caused wrong answer when double-precision arithmetic, a very common operation, was performed.

A standard test was widely published to determine whether a user's microprocessor was flawed. Using spreadsheet software the user was to take the number 4,195,835, multiply it by 3,145,727 and then divide that result by 3,145,727. As we all know from elementary math, when a number is multiplied and then divided by the same number the result should be the original number. In this example, the result should be 4,195,835. However, with the flawed FPU, the result of this calculation was 4,195,579. [InfoWorld, 1994] Depending on the application, this six-thousandths-of-a-percent error might be very significant.

At first, Intel's response to these reports was to deny that there was any problem with the chip. When it became clear that this assertion was not accurate, Intel switched it policy and stated that although there was indeed defect in the chip, it was insignificant and the vast majority of users would never even notice it. The chip would be replaced for free only for users who could demonstrate that they needed an unflawed version of the chip. [InfoWorld, 1994] There is some logic to this policy from Intel's point of view, since over two million computers had already been sold with the defective chip.

Of course, this approach did not satisfy most of the Pentium owners. After all, how can you predict whether you might have a future application where this flaw might be significant? IBM, a major Pentium user, cancelled the sales of all computers containing the flawed chip. Finally after much negative publicity in the popular personal computer literature and an outcry from Pentium users, Intel agreed to replace the flawed chip with and unflawed version for any customer who asked to have it replaced.

It should be noted that long before news of the flaw surfaced in the popular press, Intel was aware of the problem and had already corrected it on subsequent versions. It did, however, continue to sell the flawed version and, based on its early insistence that the flaw did not present a significant problem to users, seemingly planned to do so until the new version was available and the stocks of the flawed one were exhausted. Eventually, the damage caused by this case was fixed as the media reports of the problem died down and as customers were able to get unflawed chips into their computers. Ultimately, Intel had a write-off of 475 million dollars to solve this problem.

What did Intel learn from this experience? The early designs for new chips continue to have flaws, and sometimes these flaws are not detected until the product is already in use by consumers. However, Intel's approach to these problems has changed. It now seems to feel that problems need to be fixed immediately. In addition, the decision is now based on the consumer's perception of the significance of the flaw, rather than on Intel's opinion of its significance.

Indeed, similar flaws were found in 1997 in the early versions of the Pentium Pro processors. This time, Intel immediately confirmed that the flaw existed and offered customers software that would correct it. Other companies also seem to have benefited from Intel's experience. For example, Intuit, a leading manufacturer of tax preparation and financial software, called a news conference in March of 1995 to apologize for flaws in its TurboTax software that had become apparent earlier in that year. In addition to the apology, they offered consumers replacements for the defective software.

QUESTIONS

1. Was this case simply a customer-relations and PR problem, or are there ethical issues to be considered as well?
2. Use one of the codes of ethics (IEEE or ACM) to analyze this case. Especially, pay attention to issues of accurate representation of engineered products and to safety issues.
3. When a product is sold, is there an implication that it will work as advertised?
4. Should you reveal defects in a product to a consumer? Is the answer to this question different if the defect is a safety issue rather than simply a flaw? (*It might be useful to note in this discussion that although there is no apparent safety concern for someone using a computer with this flaw, PCs are often used to control a variety of instruments, such as medical equipment. For such equipment, a flaw might have a very real safety implication.*) Is the answer to this question different if the customer is a bank that uses the computer to calculate interest paid, loan payments, etc., for customers?
5. Should you replace defective products even if customers won't recognize the defect?
6. How thorough should testing be? Is it ever possible to say that no defect exists in a product or structure?
7. Do flaws that Intel found previously in the 386 and 486 chips have any bearing on these questions? In other words, if Intel got away the selling flawed chips before without informing consumers, does that fact have any bearing on this case?
8. C. Richard Thornan, an IBM senior vice president was quoted as saying, ‘Nobody should have to worry about the integrity of data calculated on an IBM machine.’ How does this statement by a major Intel customer change the answers to the previous questions?
9. Just prior to when this problem surfaced, Intel had begun a major advertising campaign to make Intel a household name. They had gotten computer manufacturers to place "Intel Inside" labels on their computers and had spent money on television advertising seeking to increase the public demand for computers with Intel processors, with the unstated message that Intel chips were of significantly higher quality than other manufacturers' chips. How might this campaign have affected what happened in this case?
10. What responsibilities did the engineers who were aware of the flaw have before the chip was sold? After the chips began to be sold? After the flaw became apparent?

Case Discussion: Therac-25 Accidents

Computers are increasingly being introduced into safety-critical systems and, as a consequenc e, have been involved in accidents. Some of the most widely cited software-related accidents in safety-critical systems involved a computerized radiation therapy machine called the Therac-25. Between June 1985 and January 1987, six known accidents involved massive overdoses by the Therac-25 – with resultant deaths and serious injuries. They have been described as the worst series of radiation accidents in the 35-year history of medical accelerators.

Genesis of the Therac-25

Medical linear accelerators (linacs) accelerate electrons to create high- energy beams that can destroy tumors with minimal impact on the surrounding healthy tissue. Relatively shallow tising healthy tissue. Relatively shallow tissue is treated with the accelerated electrons; to reach deeper tissue, the electron beam is converted into X-ray photons.

Atomic Energy Commission Limited (AECL) and a French company called CGR collaborated to build linear accelerators. (AECL is an arms-length entity, called a crown corporation, of the Canadian government. Since the time of the incidents related in this article, AECL Medical, a division of AECL, is in the process of being privatized and is now called Theratronics International Limited. Currently, AECL's primary business is the design and installation of nuclear reactors.) The products of AECL and CGR's cooperation were (1) the Therac-6, a 6 million electron volt (MeV) accelerator capable of producing X rays only and, later, (2) the Therac-20, a 20-MeV dual-mode (X rays or electrons) accelerator. B oth were versions of older CGR machines, the Neptune and Sagittaire, respectively, which were augmented with computer control using a DEC PDP 11 minicomputer.

Software functionality was limited in both machines: The computer merely added convenience to the existing hardware, which was capable of standing alone. Industry-standard hardware safety features and interlocks in the underlying machines were retained. We know that some old Therac-6 software routines were used in the Therac-20 and that CGR developed the initial software.

The business relationship between AECL and CGR faltered after the Therac-20 effort. Citing competitive pressures, the two companies did not renew their cooperative agreement when scheduled in 1981. In the mid-1970s, AECL developed a radical new "double-pass" concept for electron acceleration. A double-pass accelerator needs much less space to develop comparable energy levels because it folds the long physical mechanism required to accelerate the electrons, and it is more economic to produce (since it uses a magnetron rather than a klystron as the energy source).

Using this double-pass concept, AECL designed the Therac-25, a dual-mode linear accelerator that can deliver either photons at 25 MeV or electrons at various energy levels. Compared with the Therac-20, the Therac-25 is notably more compact, more versatile, and arguably easier to use. The higher energy takes advantage of the phenomenon of "depth dose": As the energy increases, the depth in the body at which maximum dose buildup occurs also increases, sparing the tissue above the target area. Economic advantages also come into play for the customer, since only one machine is required for both treatment modalities (electrons and photons).

Several features of the Therac-25 are important in understanding the accidents. First, like the Therac-6 and the Therac-20, the Therac-25 is controlled by a PDP 11. However, AECL designed the Therac-25 to take advantage of computer control from the outset; AECL did not build on a stand-alone machine. The Therac-6 and Therac-20 had been designed around machines that already had histories of clinical use without computer control.

In addition, the Therac-25 software has more responsibility for maintaining safety than the software in the previous machines. The Therac-20 has independent protective circuits for monitoring electron-beam scanning, plus mechanical interlocks for policing the machine and ensuring safe operation. The Therac-25 relies more on software for these functions. AECL took advantage of the computer's abilities to control and monitor the hardware and decided not to duplicate all the existing hardware safety mechanisms and interlocks. This approach is becoming more common as companies decide that hardware interlocks and backups are not worth the expense, or they put more faith (perhaps misplaced) on software than on hardware reliability.

Finally, some software for the machines was interrelated or reused. In a letter to a Therac-25 user, the AECL quality assurance manager said, "The same Therac-6 package was used by the AECL software people when they started the Therac-25 software. The Therac-20 and Therac-25 software programs were done independently, starting from a common base." Reuse of Therac-6 design features or modules may explain some of the problematic aspects of the Therac-25 software (see the sidebar "Therac-25 software development and design"). The quality assurance manager was apparently unaware that some Therac-20 routines were also used in the Therac-25; this was discovered after a bug related to one of the Therac-25 accidents was found in the Therac-20 software.

AECL produced the first hardwired prototype of the Therac-25 in 1976, and the completely computerized commercial version was available in late 1982.

In March 1983, AECL performed a safety analysis on the Therac-25. This analysis was in the form of a fault tree and apparently excluded the software. According to the final report, the analysis made several assumptions:

(1) Programming errors have been reduced by extensive testing on a hardware simulator and under field conditions on teletherapy units. Any residual software errors are not included in the analysis.

(2) Program software does not degrade due to wear, fatigue, or reproduction process.

(3) Computer execution errors are caused by faulty hardware components and by "soft" (random) errors induced by alpha particles and electromagnetic noise.

The fault tree resulting from this analysis does appear to include computer failure, although apparently, judging from these assumptions, it considers only hardware failures. For example, in one OR gate leading to the event of getting the wrong energy, a box contains "Computer selects wrong energy" and a probability of 10^11 is assigned to this event. For "Computer selects wrong mode," a probability of 4 x 10^9 is given. The report provides no justification of either number.

Accident history

Eleven Therac-25s were installed: five in the US and six in Canada. Six accidents involving massive overdoses to patients occurred between 1985 and 1987. The machine was recalled in 1987 for extensive design changes, including hardware safeguards against software errors.

Related problems were found in the Therac-20 software. These were not recognized until after the Therac-25 accidents because the Therac-20 included hardware safety interlocks and thus no injuries resulted.

One of the accidents is discussed here: -

Kennestone Regional Oncology Center, 1985. Details of this accident in Marietta, Georgia, are sketchy since it was never carefully investigated. There was no admission that the injury was caused by the Therac-25 until long after the occurrence, despite claims by the patient that she had been injured during treatment, the obvious and severe radiation burns the patient suffered, and the suspicions of the radiation physicist involved.

After undergoing a lumpectomy to remove a malignant breast tumor, a 61-year-old woman was receiving follow-up radiation treatment to nearby lymph nodes on a Therac-25 at the Kennestone facility in Marietta. The Therac-25 had been operating at Kennestone for about six months; other Therac-25s had been operating, apparently without incident, since 1983.

On June 3, 1985, the patient was set up for a 10-MeV electron treatment to the clavicle area. When the machine turned on, she felt a "tremendous force of heat . . . this red-hot sensation." When the technician came in, the patient said, "You burned me." The technician replied that that was not possible. Although there were no marks on the patient at the time, the treatment area felt "warm to the touch."

It is unclear exactly when AECL learned about this incident. Tim Still, the Kennestone physicist, said that he contacted AECL to ask if the Therac-25 could operate in electron mode without scanning to spread the beam. Three days later, the engineers at AECL called the physicist back to explain that improper scanning was not possible.

In an August 19, 1986, letter from AECL to the FDA, the AECL quality assurance manager said, "In March of 1986, AECL received a lawsuit from the patient involved. . . This incident was never reported to AECL prior to this date, although some rather odd questions had been posed by Tim Still, the hospital physicist." The physicist at a hospital in Tyler, Texas, where a later accident occurred, reported, "According to Tim Still, the patient filed suit in October 1985 listing the hospital, manufacturer, and service organization responsible for the machine. AECL was notified informally about the suit by the hospital, and AECL received official notification of a lawsuit in November 1985."

Because of the lawsuit (filed on November 13, 1985), some AECL administrators must have known about the Marietta accident -- although no investigation occurred at this time. Further comments by FDA investigators point to the lack of a mechanism in AECL to follow up reports of suspected accidents. The lack of follow-up in this case appears to be evidence of such a problem in the organization.

The patient went home, but shortly afterward she developed a reddening and swelling in the center of the treatment area. Her pain had increased to the point that her shoulder "froze" and she experienced spasms. She was admitted to West Paces Ferry Hospital in Atlanta, but her oncologists continued to send her to Kennestone for Therac-25 treatments. Clinical explanation was sought for the reddening of the skin, which at first her oncologist attributed to her disease or to normal treatment reaction.

About two weeks later, the physicist at Kennestone noticed that the patient had a matching reddening on her back as though a burn had gone through her body, and the swollen area had begun to slough off layers of skin. Her shoulder was immobile, and she was apparently in great pain. It was obvious that she had a radiation burn, but the hospital and her doctors could provide no satisfactory explanation. Shortly afterward, she initiated a lawsuit against the hospital and AECL regarding her injury.

The Kennestone physicist later estimated that she received one or two doses of radiation in the 15,000- to 20,000-rad (radiation absorbed dose) range. He does not believe her injury could have been caused by less than 8,000 rads. Typical single therapeutic doses are in the 200-rad range. Doses of 1,000 rads can be fatal if delivered to the whole body; in fact, the accepted figure for whole-body radiation that will cause death in 50 percent of the cases is 500 rads. The consequences of an overdose to a smaller part of the body depend on the tissue's radiosensitivity. The director of radiation oncology at the Kennestone facility explained their confusion about the accident as due to the fact that they had never seen an overtreatment of that magnitude before.

Eventually, the patient's breast had to be removed because of the radiation burns. She completely lost the use of her shoulder and her arm, and was in constant pain. She had suffered a serious radiation burn, but the manufacturer and operators of the machine refused to believe that it could have been caused by the Therac-25. The treatment prescription printout feature was disabled at the time of the accident, so there was no hard copy of the treatment data. The lawsuit was eventually settled out of court.

From what we can determine, the accident was not reported to the FDA until after the later Tyler accidents in 1986 (described in later sections). The reporting regulations for medical device incidents at that time applied only to equipment manufacturers and importers, not users. The regulations required that manufacturers and importers report deaths, serious injuries, or malfunctions that could result in those consequences. Health-care professionals and institutions were not required to report incidents to manufacturers. (The law was amended in 1990 to require health-care facilities to report incidents to the manufacturer and the FDA.) The comptroller general of the US Government Accounting Office, in testimony before Congress on November 6, 1989, expressed great concern about the viability of the incident-reporting regulations in preventing or spotting medical-device problems. According to a GAO study, the FDA knows of less than 1 percent of deaths, serious injuries, or equipment malfunctions that occur in hospitals.[3]

At this point, the other Therac-25 users were unaware that anything untoward had occurred and did not learn about any problems with the machine until after subsequent accidents. Even then, most of their information came through personal communication among themselves.

Case Discussion: The Trouble Ticketing System

The Trouble Ticketing System (TTS) was a mainframe-based system, developed in the early 1980s for use by telephone-company repair personnel, for scheduling, work routing, and record keeping. When trouble was reported, a job ticket was generated, and was sent to the appropriate telephone company office. There, a worker picked up the ticket and began work on the job. When the repair was completed, or when the worker had done all that she could do from that location, the ticket was sent back to the central TTS.

Before TTS, job tickets were generated, but work was more collaborative. Testers called one another, consulting with someone at the other end of a problematic line, or someone who knew a particular part of the system especially well, for help with troubleshooting. One of the motivations for the development of the TTS was to ensure that workers spent more time on repair tasks by eliminating conversations between workers, which were thought to be inefficient and "off task." With the TTS, each tester worked alone. If she could not complete a repair job, the tester recorded what had been done and sent the ticket back to the TTS for someone else to pick up and work on.

The need for conversation was eliminated, but the benefits of conversation (more information available to diagnose problems, and the chance to learn more about the system) were lost too. Because TTS also monitored the number of hours that each worker spent doing jobs, testers who spent time consulting with one another or training new workers (neither of which activities were accounted for in the system) were penalized.

While TTS was designed to make job performance more efficient, it has created the opposite effect: discouraging the training of new hands, breaking up the community of practice by eliminating troubleshooting conversations, and extending the time spent on a single job by segmenting coherent troubleshooting efforts into unconnected ticket-based tasks.

Case Discussion: Big Bank

At a large urban bank, teller operations were supported by a computer-based system, which was designed and modified over time by the bank's systems department. The system had to meet the bank's needs for accuracy, efficiency, security, and customer service, although, as this case illustrates, not all of these goals can maximized simultaneously. The Big Bank system embodied in a rigid form the rules and procedures of the bank, which before automation would have been enforced by people--generally supervisors--who were relatively close to the action and who exercised professional judgment based on experience with the day-to-day work of bank tellers. One of the security features built into the system was that, "under a specified set of exceptional situations (defined by management at the bank through the setting of software parameter settings), the teller's terminal will freeze with a message about the account on the screen." The transaction could be completed only after a bank officer's authorization card was passed through a card reader on the teller's terminal. The purpose of the freeze was to ensure supervisory oversight in circumstances that were deemed exceptional, such as more than three transactions on a single account on a single day.

It turned out, however, that at a large downtown branch of the bank, the exceptional happened every 5 or 10 minutes. To keep lines flowing and to avoid costly inefficiencies, the manager gave a bank officer's card to the tellers, which they passed among themselves to unfreeze their terminals. Only once or twice an hour did the tellers judge that a supervisor was needed--when a customer was unknown, or when an unusual situation arose. Of course, overcoming this one feature undermined all security provisions in the system, because the officer card was now freely available to the tellers. "This implementation of the system increases the responsibility of the teller although, at the same time, the design of the system is reminding the teller that, formally, the bank management does not trust him or her to make even routine decisions. The result of the system's design is the worst of both worlds."

Case Discussion: HELP System

In a large machining area in a production plant, a major U.S. aircraft manufacturer installed a new computer-based system known as the HELP (Help Employees Locate People) system. The system had two principal functions. The first--which was the source of its official name--was to enable machinists to signal for assistance when they needed replacement tooling, wanted consultation, or were due for a break. At the push of a button, a machinist could indicate his location and the nature of the request. This aspect of the system got good reviews from both machinists and shop management, because it enabled operations to run more smoothly.

The second function of the system, not acknowledged in the official name, was the monitoring of 66 machines in the shop. A panel in a control room above the shop floor displayed the status of each machine with colored lights. A supervisor could check these lights, and could gain further information by glancing out at the floor below. Daily reports told supervisors not only about production levels, but also about how each worker spent his time. Upper management received weekly and monthly reports.

The purpose of the monitoring system was to gain greater managerial control over how machinists spent their time, as an aid to increasing productivity. The problem was that the information captured by the system and reported to the managers provided only a partial view of the work of machinists--a view so one-sided as to be nearly useless. The very concreteness of the statistics, however, invited unwarranted conclusions. For example, managers wanted to discourage machinists from slowing down production unnecessarily, so the system was designed to report all the time that a computer-controlled machine tool spent halted or operating at less than 80 percent of the programmed feed rate (the rate at which the cutting tool moves across the surface of the metal). What the system did not report, however, was whether the programmer who wrote the program for that particular part had set the feed rate correctly. It is common for feed rates to be set incorrectly, and one of the skills of the machinist is to judge whether conditions require slowing down or allow speeding up. Information about the appropriate feed rate is crucial to the proper interpretation of the data generated by the monitoring system.

The data generated by the HELP system could mislead managers into thinking that any machine status other than "running" was an indication of unproductive activity. "One machinist...had to work long and furiously to set up a particularly intricate part to be cut. As a result, his machine sat idle most of the day. While he felt that he had never worked harder, his supervisor reprimanded him because the system reported that his machine was idle."